

Renewal Spectral Geometry and the Emergence of Lorentzian Spacetime

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Abstract

We show that Lorentzian spacetime can emerge from predictive memory rather than being imposed as prior geometry. Starting from a finite renewal process of completely positive channels, with no assumed metric, causal order, time direction, or spacetime dimension, we construct a positive spectral geometry and an intrinsic order of predictive continuation. We prove that positivity alone cannot produce Lorentzian signature and that the minimal required extension is a signed double cover. Reversible predictive transformations then generate a Clifford sector with a distinguished temporal direction. Under nondegenerate coarse-graining, the resulting signed geometry converges to a Lorentzian Dirac continuum. The discrete Hamiltonians converge globally in strong resolvent sense and in norm on fixed momentum bands. For slowly varying renewal statistics, channel composition reconstructs torsion-free Levi-Civita spin transport while allowing nonzero curvature. The primitive revision algebra further implies that (3+1) dimensions are the minimal spatially nondegenerate Lorentzian endpoint. Thus predictive renewal memory contains the causal, spectral, temporal, and spinorial structures required for Lorentzian spacetime, without assuming gravitational field equations or matter content. A `Lean` formalization is available at <https://github.com/AI-MathPhys/renewal-geometry>.

Contents

1	Introduction	3
2	Renewal memories and the positive predictive geometry	6
2.1	Spectral triples and predictive noncommutative spaces	6
2.2	Renewal memories and predictive quotient	7
2.3	The predictive-length regularity criterion	9
2.4	The positive predictive spectral triple	11
2.5	Predictive order	13
2.6	The positivity boundary	15
3	The signed Krein sector	16
3.1	Minimal indefinite enrichments over a fixed component	16
3.2	A record-theoretic origin of the signed class	20
3.3	Concrete signed covers and Krein symmetry	21
3.4	Real signed quotients and amplitude lifts	23
3.5	Checkable conditions for bounded twisted commutators	26
3.6	Characterisation of the modular signed Dirac	28
3.7	Pressure selection of the modular scale	30
3.8	Signed predictive reconstruction	31

4	Reversible revision dynamics and dimensional minimality	33
4.1	Flat holonomy does not produce the Clifford twist	34
4.2	The temporal row is forced by the signed cover	34
4.3	The modular revision algebra from reversible predictive dynamics	36
4.4	Quadratic refinement and the finite central extension	39
4.5	Canonical symplectic reduction of the derived revision algebra	40
4.6	Intrinsic generation by a graded Clifford renewal memory	43
4.7	Unconditional minimality of the $3 + 1$ endpoint	45
5	The Lorentzian Dirac continuum	48
5.1	Signed Clifford symbols and Lorentzian frame universality	48
5.2	Operator-level convergence of the rounded signed Dirac	53
5.3	Channel plaquettes and Levi–Civita reconstruction	61
5.4	Adiabatic curved Dirac limit, discrete Cartan reconstruction, and stability	67
6	Interpretation, scope, and conclusion	73
6.1	From predictive memory to Lorentzian geometry	73
6.2	What lies upstream of renewal memory?	75
6.3	Conclusion	77
A	Logical status and reduction of the hypotheses	78
B	Regularity criteria and auxiliary positive invariants	80
B.1	Finite fibres and bounded increments	80
B.2	Metric dimension and renewal calibration	82
C	Renewal-tail singularities	85
C.1	Renewal tail and the analytic type of the resolvent	85
D	The deterministic product-order continuum	89
D.1	The renewal continuum ladder	89
E	Internal holonomy and conditional dimension selectors	92
E.1	Internal holonomy realization of the spatial block	92
E.2	The common cycle-rank invariant	94
E.3	Survival of the volume-dual response as a primitive first-order observable	98
E.4	An explicit block-renewal coarse-graining map	100
E.5	From interference postulate to the canonical volume-dual response	102
E.6	Alternative conditional selection mechanisms	108
E.7	Why isotropy cannot make the selection unconditional	109
E.8	Alternative conditional uniqueness: the access fixed point	109
E.9	A secondary optional functional: access efficiency	111
E.10	Why decimation does not discharge the access principle	113
E.11	Interfaces with downstream constructions	114
E.12	The canonicity theorem	116

F	Further continuum, propagation, and global-topology results	117
F.1	A least-distortion counterexample	117
F.2	Joint convergence of propagation geometry	118
F.3	Universality and metric reconstruction	122
F.4	A marked flat-torus correspondence carrying spin structure	125
G	Examples and sharp model cases	128
G.1	Positive predictive geometry	129
G.2	Signed covers and holonomy	129
G.3	Polyhedral order cones and Lorentzian Clifford symbols	130
G.4	Primitive modular dimension patterns	132

1 Introduction

A persistent ambition of mathematical physics is to *derive* spacetime geometry rather than to postulate it. Ideally, dimension, causal order, Lorentzian signature, and relativistic propagation should arise as different aspects of a common underlying structure, rather than being introduced as independent kinematic assumptions. This is difficult because the mathematical frameworks that most naturally encode geometry tend to presuppose precisely the structures one would like to explain. A manifold already carries a topology and a differentiable structure; a causal set begins with an order; and a Lorentzian spectral geometry is ordinarily supplied with an indefinite or temporal datum from the outset.

The tension is particularly sharp in noncommutative geometry. In Connes’ spectral formulation, the metric information of a Riemannian spin space is encoded by a spectral triple $((\mathcal{A}, \mathcal{H}, D))$, consisting of an algebra of observables, a Hilbert-space representation, and a Dirac operator [1]. This operator-theoretic formulation is well adapted to a setting in which geometry and quantum observables should be described in a common language. Length-function spectral triples further show that nontrivial metric and dimensional information can be carried by discrete or totally disconnected structures [2, 3, 4, 5]. Yet the ordinary Hilbert-space framework is positive-definite and therefore naturally Riemannian. Physical spacetime, by contrast, is Lorentzian.

Krein, pseudo-Riemannian, temporal, and twisted spectral triples address this signature problem by enriching the spectral data with an indefinite inner product, a fundamental symmetry, a temporal operator, or a modular twist [6, 7, 8, 9, 10, 11, 12, 13, 14]. Temporal Lorentzian triples isolate a distinguished time direction through an exact time operator and lapse, while spectral-spacetime formulations promote time orientation to structural data [15, 16, 17]. Twisted almost-commutative constructions provide another route from Euclidean to pseudo-Riemannian spectral geometry [18, 19, 20, 21]. These theories explain how Lorentzian geometry is encoded once suitable signed or temporal data are supplied. The question pursued here is more primitive: can the need for such data, their minimal form, and the associated Dirac structure be recovered from a process that is not geometric at the outset?

We propose predictive renewal memory as such a primitive object. Renewal theory studies processes that repeatedly forget and restart, and its resolvents control systems with memory throughout probability, statistical mechanics, reliability, and information theory [22]. Here renewal is not placed inside a pre-existing spacetime. Instead, a finite alphabet of reset operations acts on an internal observable algebra through unital completely positive maps. A history is merely an ordered sequence of such operations. It is not yet a trajectory in space or time.

The basic reduction is predictive rather than geometric. Two histories are identified when they induce the same accumulated channel and therefore have the same future predictive action. This resembles causal-state reduction in computational mechanics, where histories are grouped according to their predictive equivalence [23, 24]. Continuous-time causal-state models and their labelled transition operators provide the closest probabilistic comparison [25, 26, 27, 28, 29]. The distinction here is that predictive labels are completely positive operators rather than scalar transition probabilities.

This quotient already has geometric content. Its minimal history length defines a positive spectral geometry under explicit regularity assumptions, and the growth of distinguishable predictive classes becomes a spectral dimension. At the same time, continuation of one accumulated channel by another defines predictive divisibility. After quotienting mutual reachability, this becomes a partial order; under the stated monotonicity condition, the quotient length becomes a time function and grading. Thus the order is not inserted as causal-set kinematics. It is the order induced by the possibility of future predictive continuation. In this sense, geometry is what prediction remembers.

This positive construction reaches a sharp boundary. Complete positivity produces a stable Hilbert-positive geometry, but that same positivity cannot supply a nontrivial Lorentzian fundamental symmetry in the natural irreducible representation. More generally, any compatible nonconstant diagonal involution is equivalent to introducing additional $(\mathbb{Z}/2)$ -valued cocycle data on the predictive graph. The signed sector is therefore not an optional decoration of an already Lorentzian geometry. It is the minimal structure required to pass beyond positivity.

The form of this enrichment is rigid even though its cohomology class is not selected by the graph-general classification alone. Preservation of the distinguished renewal cone forces local unitary transports to permute its extremal rays. Fibrewise indefiniteness requires at least two local directions, and at minimal rank the only possible transports are the identity and the sheet exchange. Consequently every objectwise-minimal renewal-positive indefinite enrichment is a principal double cover, classified by a class in $H^1(G, \mathbb{Z}/2)$. In record-regular operational realizations, this abstract class can be selected by the determinant parity of stable future-distinct record transport; the present paper classifies the geometry resulting from the selected class. The corresponding sheet-sign operator supplies the Krein fundamental symmetry. If the same signed holonomy is retained at the level of unitary amplitudes, a nontrivial sign must admit a square root; the minimal finite amplitude completion is then a $\mathbb{Z}/4$ lift. The double cover carries the real/Krein signature data, while the lift retains the additional fourth-root phase information.

A second structural ingredient comes from reversibility. Reversible predictive transformations are not assumed to be unitary at the microscopic level. Rather, a unital completely positive map with a unital completely positive inverse is automatically a $(*)$ -automorphism. On the primitive finite factor these automorphisms are inner, and their unitary implementers therefore compose projectively. The resulting commutator pairing supplies the symplectic and Clifford structure needed for a Dirac geometry. Thus the Clifford sector is not attached independently to the signed cover: it arises from the reversible algebra of predictions.

The renewal grading also provides a natural modular clock. When elementary resets transform homogeneously under a faithful modular flow, the modular Hamiltonian is forced to be affine in the renewal length. On a recurrent weighted component, the remaining exponent is selected by a Perron–Frobenius pressure equation: under supercriticality, the spectral radius of the length-weighted transfer matrix decreases through one at a unique positive value β . Together with deck oddness, this determines the unbounded signed Dirac magnitude up to its overall physical-unit conversion and a bounded Krein-self-adjoint perturbation. This gives a renewal realization of the

modular-time perspective of Connes and Rovelli [30], within the broader theory of twisted and modular spectral triples [14, 31, 32].

The continuum problem is then not to manufacture an indefinite metric from positive averages, which is impossible, but to understand the scaling limit of the signed Clifford geometry. A finite deterministic reset frame generally produces a polyhedral order cone. At the level of the signed principal symbol, however, a nondegenerate spatial second moment yields a Lorentzian quadratic cone with a single possible timelike direction. No microscopic rotational invariance is required: a positive-definite limiting second moment defines a spatial vielbein, and the corresponding flat Hamiltonian is unitarily equivalent to the standard massless Dirac Hamiltonian. Stationary ergodic reset statistics therefore produce the flat Minkowski Dirac isometry class almost surely whenever the spatial reset directions span.

The curved regime reveals why channel composition is geometrically significant. For slowly varying renewal statistics, the defect between the two orders of traversing an elementary plaquette separates into a translational and a rotational component. The former is discrete torsion; the latter is curvature. Metric compatibility need not be imposed independently: covariance of neighbouring spatial second moments makes the normalised comparison transport orthogonal, and preservation of the selected orientation places it in $SO(d_{\text{Cl}})$. Torsion freedom is likewise derived within a distinguished phase. For a primitive exchange-covariant stationary microscopic diamond process, the unique stationary law is exchange invariant, so every exchange-odd translational plaquette defect has vanishing expectation, while the rotational record remains unconstrained. The resulting translational closure, together with metric compatibility, allows the discrete fundamental theorem to select the Levi–Civita spin connection without forcing rotational holonomy to vanish. The corresponding covariant Hamiltonians converge to the variable-coefficient Lorentzian Dirac Hamiltonian through subprincipal order. Curvature survives the construction, although no equation governing that curvature is imposed.

Dimension enters through the primitive revision algebra rather than through a chosen number of coordinates. Its reduced commutator pairing has even rank, and one of the corresponding Clifford directions is distinguished as temporal. The spatial rank is therefore odd. The spatially one-dimensional branch is degenerate for the nontrivial continuum mechanism considered here, so three spatial directions give the first nondegenerate Lorentzian Dirac endpoint. The conclusion is an unconditional *minimality* result: $(3+1)$ is the lowest nondegenerate dimension in the primitive modular class. Parity alone does not exclude higher odd spatial ranks, and their elimination requires the additional closure condition stated explicitly later. Related parity and single-time constraints for first-order relativistic equations have classical precedents [33, 34, 35]; the role of renewal memory is to generate the relevant projective Clifford factor rather than to postulate it.

The construction is related to, but distinct from, other programmes of emergent spacetime. Thermodynamic approaches seek gravitational dynamics from entropy and horizon laws [36, 37], while holographic approaches relate spacetime connectivity to entanglement [38]. Causal-set theory begins with a locally finite order [39, 40], and graph, simplicial, and group-field models begin with combinatorial structures whose collective phases approximate geometry [41, 42, 43]. Here the primitive datum is neither a metric nor an order, but predictive memory. Order arises from divisibility of accumulated channels, the necessity of a signed extension is exposed by a positivity obstruction, and the continuum endpoint is a controlled Krein–Clifford Dirac evolution.

The result is kinematic and operator-theoretic. We construct a positive predictive spectral geometry, an intrinsic causal-order skeleton, a minimal signed/Krein enrichment, a projective Clifford revision sector, and controlled flat and adiabatic Lorentzian Dirac limits. We do not derive Einstein’s equations, a gravitational variational principle, matter multiplets, gauge interactions,

conformal completion, or phenomenological predictions. The claim is instead that predictive renewal memory already contains enough structure to support the causal, spectral, temporal, and spinorial architecture of Lorentzian spacetime.

The hypotheses used below do not all have the same logical status. The finite UCP renewal memory is the mathematical starting object, while finite fibres, bounded increments, and continuum regularity are analytic domain conditions for particular spectral or convergence results. Nontrivial signed holonomy, a realised reversible projective sector, pressure supercriticality, elliptic spatial scaling, and exchange-symmetric stationary transport are phase conditions selecting the Lorentzian regime. Regular variation, exact isotropy, metric coding, and the higher-rank uniqueness conditions are optional strengthenings rather than premises of the core emergence theorem. A detailed condition audit and the canonical reductions of global connectivity, faithfulness, irreducibility, and projective degeneracy are recorded in Appendix [Section A](#).

The paper is organized as follows. [Section 2](#) constructs the positive predictive spectral geometry and order and establishes the positivity boundary. [Section 3](#) classifies the minimal signed enrichment, constructs the associated Krein and modular Dirac structures, and proves the signed reconstruction theorem. [Section 4](#) derives the projective revision algebra from reversible predictive dynamics, performs its canonical symplectic reduction to the primitive Clifford factor, and proves the (3+1)-dimensional minimality theorem. With this Clifford structure in place, [Section 5](#) establishes the flat and slowly varying Lorentzian Dirac limits, including operator convergence, self-averaging, and the reconstruction of Levi-Civita spin transport from channel plaquettes. [Section 6.3](#) discusses the interpretation, precise scope, and limitations of the construction. The appendices collect regularity criteria, auxiliary dimension results, renewal-tail singularities, deterministic product-order limits, conditional dimension-selection mechanisms, propagation and global-topology results, and worked examples.

2 Renewal memories and the positive predictive geometry

The logic of this section is one-way and will be used throughout the paper. Histories are first reduced to their future predictive action. The resulting quotient is then tested for the operator-theoretic regularity needed to carry a length spectral triple. The same quotient also carries a divisibility order. Only after these positive structures have been constructed do we ask whether they can determine a Lorentzian sign. The final no-go result answers that question negatively and identifies the precise datum required in the next section.

We recall briefly the part of the spectral-triple formalism that will be used in this section. The reader familiar with noncommutative geometry may pass directly to [Section 2.2](#); the purpose of the present subsection is only to fix conventions and to explain why the quotient constructed below should be viewed as a positive noncommutative geometry rather than merely as a labelled counting problem.

2.1 Spectral triples and predictive noncommutative spaces

In noncommutative geometry a compact Riemannian spin geometry is encoded by a spectral triple. In the form used here this consists of a unital involutive algebra $\mathcal{A}$ represented by bounded operators on a separable Hilbert space $\mathcal{H}$, together with a self-adjoint, usually unbounded, operator D such that $(1 + D^2)^{-1/2}$ is compact and $[D, a]$ is bounded for all a in a dense $*$ -subalgebra of $\mathcal{A}$ [\[1\]](#). For a compact spin manifold X the model example is

$$(C^\infty(X), L^2(X, S), D_X),$$

where S is the spinor bundle and D_X is the classical Dirac operator. In this commutative case the metric can be recovered from the spectral data; in the noncommutative case the same conditions are taken as the definition of a “space” whose metric information is carried by the operator D .

Two analytic features of this definition are important below. First, compact resolvent is the replacement for compactness of the underlying space: it means that the spectrum of D escapes to infinity with finite multiplicities. Second, the boundedness of commutators is the Lipschitz condition: the algebra elements are precisely the observables whose variation is controlled by the metric measured by D . Thus, once an algebra of creation operators and a length operator are specified, the question of whether they form a spectral triple is not formal. It is an operator-theoretic regularity condition, and it is exactly this condition that will force finite fibres and bounded length increments in [Theorem 2.8](#).

The dimension attached to a spectral triple is also spectral. One considers zeta functions of the form

$$\zeta_D(s) = \text{Tr}(|D|^{-s})$$

(or, more generally, localized zeta functions $\text{Tr}(a|D|^{-s})$), and the abscissa of convergence gives the metric growth exponent of the geometry. In regular cases the poles form the dimension spectrum and enter the spectral action expansion [\[44\]](#). We shall not use the spectral action in this paper. What we use is the same principle at the kinematic level: a length Dirac on the predictive quotient turns the growth of distinguishable predictive states into a spectral dimension. Length-function spectral triples of this discrete kind go back to Connes [\[2\]](#) and were developed for AF algebras and Cantor sets by Christensen–Ivan [\[3\]](#) and, for ultrametric and aperiodic media, by Pearson–Bellissard [\[4\]](#) and Kellendonk–Savinien [\[5\]](#); the predictive-length triple constructed below is the renewal instance of this lineage.

The adjective *positive* in this section is meant literally. The Hilbert space is an ordinary Hilbert space, the length Dirac is self-adjoint for a positive definite scalar product, and the channels used to define the quotient are completely positive maps. Consequently the geometry constructed in this section is Riemannian in the sense relevant to spectral triples. It will supply length, dimension, volume residues, and an order skeleton. It will not supply a Lorentzian fundamental symmetry. The latter appears only in the signed sector, where the Hilbert-positive construction is replaced by a Krein datum.

2.2 Renewal memories and predictive quotient

We now introduce the primitive object. It is deliberately smaller than a manifold, a metric graph, or a causal set: it is only a finite family of completely positive reset operations and the histories obtained by composing them. The predictive quotient identifies histories that have the same accumulated channel. It is the space on which the positive spectral triple will be built.

Definition 2.1 (Renewal memory). A finite completely positive renewal memory consists of a finite alphabet E , a finite-dimensional unital C^* -algebra $\mathcal{A}_{\text{int}}$, and a family of unital completely positive maps

$$\Phi_\sigma : \mathcal{A}_{\text{int}} \longrightarrow \mathcal{A}_{\text{int}}, \quad \sigma \in E.$$

A history is a word $w = \sigma_1 \cdots \sigma_k \in E^*$. Its accumulated channel is

$$\Phi_w = \Phi_{\sigma_k} \circ \cdots \circ \Phi_{\sigma_1}.$$

Definition 2.2 (Predictive quotient). Two histories are predictively equivalent, written $w \sim_{\text{CP}} w'$, if $\Phi_w = \Phi_{w'}$. The predictive quotient is

$$\mathcal{W}_{\text{CP}} = E^* / \sim_{\text{CP}}.$$

The quotient length is

$$\Lambda_{\min}([w]) = 1 + \min\{|v| : \Phi_v = \Phi_w\}.$$

Definition 2.3 (Predictive channel monoid and its units). The accumulated channels form the composition monoid

$$\mathcal{M}_{\text{pred}} := \{\Phi_w : w \in E^*\}.$$

Its unit group is

$$\mathcal{U}_{\text{pred}} := \{\Psi \in \mathcal{M}_{\text{pred}} : \exists \Theta \in \mathcal{M}_{\text{pred}}, \quad \Theta\Psi = \Psi\Theta = \text{id}\}.$$

This definition depends only on the predictive channel quotient and not on a choice of Kraus representation.

Theorem 2.4 (Predictive-unit theorem). *Let Ψ be a unital completely positive map on a finite-dimensional C^* -algebra. If Ψ has a unital completely positive inverse, then Ψ is a $*$ -automorphism. Consequently*

$$\mathcal{U}_{\text{pred}} \subseteq \text{Aut}(\mathcal{A}_{\text{int}}).$$

Proof. Let Θ be the UCP inverse of Ψ . Kadison–Schwarz gives

$$\Psi(x)^* \Psi(x) \leq \Psi(x^* x).$$

Applying Θ yields

$$\Theta(\Psi(x)^* \Psi(x)) \leq x^* x.$$

Kadison–Schwarz for Θ , applied to $\Psi(x)$, gives the reverse inequality,

$$x^* x = \Theta(\Psi(x))^* \Theta(\Psi(x)) \leq \Theta(\Psi(x)^* \Psi(x)).$$

Thus equality holds. Applying the same argument to xx^* shows that every x belongs to the multiplicative domain of Ψ . Hence Ψ is a unital $*$ -homomorphism, and bijectivity makes it a $*$ -automorphism. $\square$

Remark 2.5 (Sharp scope). A generic UCP renewal tuple need not possess nontrivial units. For example, a completely depolarising channel annihilates every traceless observable and is not invertible. Reversible revisions are therefore a property of the signed recurrent phase, not a consequence of bare complete positivity.

The predictive-unit theorem is recorded at this early stage for later use in [Section 4](#). It is not needed for the existence of the positive spectral geometry: irreversible channels contribute to the predictive quotient just as well. Its role is instead to identify the special reversible part of the quotient from which the projective Clifford sector will eventually be derived.

Remark 2.6 (The channel family controls the predictive phase). The maps $\{\Phi_\sigma\}_{\sigma \in E}$ are not decorative input data: they determine the phase of the predictive quotient. If the channels erase all distinctions, for instance if every non-empty history induces the same accumulated channel, then $N_{\text{CP}}(R) = O(1)$ and the algebraic predictive dimension is $q_{\text{alg}} = 0$. If, at the opposite extreme,

the channels preserve essentially all symbolic distinctions, then $N_{\text{CP}}(R)$ grows exponentially and the algebraic predictive dimension is infinite. The finite-dimensional geometric regime lies between these extremes: enough memory must survive to produce growth, but enough predictive identification must occur to reduce the free symbolic explosion to polynomial order.

Lemma 2.7 (Shift congruence). *The equivalence relation $\sim_{\text{CP}}$ is a left shift congruence: if $w \sim_{\text{CP}} w'$, then $\sigma w \sim_{\text{CP}} \sigma w'$ for all $\sigma \in E$. Consequently the creation operators descend to*

$$\widehat{S}_\sigma e_{[w]} = e_{[\sigma w]}$$

on $\ell^2(\mathcal{W}_{\text{CP}})$.

Proof. With the convention of Definition 2.1, $\Phi_{\sigma w} = \Phi_w \circ \Phi_\sigma$. Therefore $\Phi_w = \Phi_{w'}$ implies $\Phi_{\sigma w} = \Phi_{\sigma w'}$. $\square$

Assumption 2.1 (Finite fibres and bounded increments). The descended shifts have uniformly finite fibres. Moreover, for every $\sigma \in E$ there exists $C_\sigma < \infty$ such that

$$|\Lambda_{\min}([\sigma w]) - \Lambda_{\min}([w])| \leq C_\sigma$$

for all $[w] \in \mathcal{W}_{\text{CP}}$.

Let $\mathcal{B}_{\text{CP},0}$ denote the $*$ -algebra generated on $\ell^2(\mathcal{W}_{\text{CP}})$ by the descended shifts, their adjoints, and finite-support diagonal multipliers. Let $\mathcal{H}_{\text{int}}$ be a faithful Hilbert-space representation of $\mathcal{A}_{\text{int}}$.

2.3 The predictive-length regularity criterion

For the quotient length to define a spectral geometry, the descended renewal shifts must be bounded and their commutators with the length operator must be bounded. In the present setting these conditions are equivalent to uniformly finite shift fibres and uniformly bounded length increments. Several useful finite-combinatorial criteria verifying these hypotheses are collected in Section B; the main text records the sharp operator-theoretic dichotomy.

Theorem 2.8 (Finite-fibre dichotomy for the predictive triple). *Let $\mathcal{W}_{\text{CP}}$ be the predictive quotient of a finite CP renewal memory, with descended shifts $\widehat{S}_\sigma$ (Theorem 2.7) and canonical length Dirac $D_{\text{CP}} e_{[w]} = \Lambda_{\min}([w]) e_{[w]}$. Exactly one of the following holds.*

- (1) (regular) *every $\widehat{S}_\sigma$ has uniformly finite fibres and every $\Lambda_{\min}$ -increment is bounded (Assumption 2.1). Then each $\widehat{S}_\sigma$ is bounded with $\|\widehat{S}_\sigma\| = (\sup_{[u]} \#\text{fibre}_\sigma[u])^{1/2}$, each $[D_{\text{CP}}, \widehat{S}_\sigma]$ is bounded, and $(\mathcal{B}_{\text{CP},0}, \ell^2(\mathcal{W}_{\text{CP}}) \otimes \mathcal{H}_{\text{int}}, D_{\text{CP}})$ is a spectral triple (Theorem 2.11); it is θ -summable, and finitely summable if and only if $N_{\text{CP}}(R)$ grows polynomially.*
- (2) (singular) *some $\widehat{S}_\sigma$ has an infinite fibre, or some $\Lambda_{\min}$ -increment is unbounded. Then the canonical data fail to be a spectral triple, and the failure is not reparable within the predictive-length geometry:*
 - (2a) *an infinite fibre makes $\widehat{S}_\sigma$ unbounded ($\|\widehat{S}_\sigma\| = \infty$), so no choice of diagonal Dirac can make $\widehat{S}_\sigma$ a bounded algebra element;*
 - (2b) *an unbounded increment makes $[D_{\text{CP}}, \widehat{S}_\sigma]$ unbounded, and more generally $[f(D_{\text{CP}}), \widehat{S}_\sigma]$ is unbounded for every f that is bi-Lipschitz on the length spectrum (every length-comparable reparametrisation).*

Proof. The two alternatives are the negation of one another, so exactly one holds.

(1). On basis vectors $\widehat{S}_\sigma^* \widehat{S}_\sigma e_{[u]} = (\#\text{fibre}_\sigma[u]) e_{[u]}$, so $\|\widehat{S}_\sigma\|^2 = \sup_{[u]} \#\text{fibre}_\sigma[u]$, finite by hypothesis. By [Theorem 2.11](#), $[D_{\text{CP}}, \widehat{S}_\sigma] e_{[w]} = (\Lambda_{\min}([\sigma w]) - \Lambda_{\min}([w])) e_{[\sigma w]}$ has norm $\leq \text{sup-increment} < \infty$, and D_{CP} has compact resolvent because each length shell is finite (finite alphabet, finite fibres, bounded increments) and $\Lambda_{\min} \rightarrow \infty$. Thus the triple exists; $\text{Tr } e^{-tD_{\text{CP}}^2} = \dim(\mathcal{H}_{\text{int}}) \sum_R h(R) e^{-tR^2} < \infty$ gives θ -summability, and finite summability $\sum_R h(R) R^{-s} < \infty$ for some s holds iff the abscissa $q_{\text{alg}} = \limsup \log N_{\text{CP}}(R) / \log R$ is finite, i.e. iff N_{CP} is polynomial.

(2a). If $\#\text{fibre}_\sigma[u] = \infty$ for some $[u]$, then $\widehat{S}_\sigma^* \widehat{S}_\sigma$ has the eigenvalue $+\infty$ on $e_{[u]}$, so $\widehat{S}_\sigma$ is unbounded. Boundedness of an algebra element is a property of $\widehat{S}_\sigma$ alone and does not depend on the choice of D ; hence no diagonal Dirac repairs it.

(2b). If $\Lambda_{\min}([\sigma w_n]) - \Lambda_{\min}([w_n]) \rightarrow \infty$ along a sequence, then for f bi-Lipschitz on the length spectrum with lower slope $c > 0$, $|f(\Lambda_{\min}([\sigma w_n])) - f(\Lambda_{\min}([w_n]))| \geq c |\Lambda_{\min}([\sigma w_n]) - \Lambda_{\min}([w_n])| \rightarrow \infty$, so $[f(D_{\text{CP}}), \widehat{S}_\sigma]$ is unbounded. $\square$

Remark 2.9 (Why the dichotomy is sharp, and its one genuine limitation). The two cases are exhaustive and mutually exclusive at the level of hypotheses; the content is that each forces a definite operator-theoretic verdict for the *predictive-length* geometry — the object whose growth exponent is the predictive dimension q_{alg} . The infinite-fibre obstruction (2a) is absolute: no diagonal Dirac can rescue an unbounded shift. The unbounded-increment obstruction (2b) is sharp *within length-comparable Diracs*, and this is the precise limitation of the statement: it does *not* assert that no diagonal Dirac whatsoever can be summable in case (2b). The graph-distance function $\lambda_{[w]} = d_G(x_0, [w])$ has unit increments and tends to infinity whenever balls are finite, so $D' = \text{diag}(\lambda)$ gives bounded commutators even when $\Lambda_{\min}$ -increments are unbounded. But D' is no longer the quotient length: it measures the graph metric, a different dimension, not the predictive growth exponent. The dichotomy is therefore exactly a statement about the predictive-length Dirac; swapping in an unrelated metric changes the geometry being measured.

Corollary 2.10 (Sharp existence criterion for the predictive-length triple). *For the canonical length Dirac $D_{\text{CP}} e_{[w]} = \Lambda_{\min}([w]) e_{[w]}$,*

$$\begin{aligned} &(\mathcal{B}_{\text{CP},0}, \ell^2(\mathcal{W}_{\text{CP}}) \otimes \mathcal{H}_{\text{int}}, D_{\text{CP}}) \text{ is a spectral triple} \\ &\iff \text{bounded shift fibres } \mathbf{and} \text{ bounded } \Lambda_{\min}\text{-increments.} \end{aligned}$$

When the right-hand side holds the triple is θ -summable, and finitely summable if and only if $N_{\text{CP}}(R)$ grows polynomially. When it fails,

$$\text{either some } \widehat{S}_\sigma \text{ is unbounded, or some } [D_{\text{CP}}, \widehat{S}_\sigma] \text{ is unbounded.}$$

Both equivalences are statements about the canonical predictive-length geometry; the qualification of [Theorem 2.9](#) applies to the second box (an unrelated diagonal metric may still be summable, but it no longer measures the predictive growth dimension q_{alg}).

Proof. The forward implication of the first box and the failure dichotomy in the second box are, respectively, alternatives (1) and (2) of [Theorem 2.8](#); the converse of the first box is immediate, since a spectral triple requires the $\widehat{S}_\sigma$ to be bounded operators (whence finite fibres, $\|\widehat{S}_\sigma\|^2 = \sup_{[u]} \#\text{fibre}_\sigma[u] < \infty$) and the commutators $[D_{\text{CP}}, \widehat{S}_\sigma]$ to be bounded (whence bounded increments). The summability statements are [Theorem 2.8](#) (1). $\square$

2.4 The positive predictive spectral triple

The preceding dichotomy separates two questions that are easily conflated. Finite fibres and bounded length increments decide whether the canonical predictive-length geometry exists at all. Once that regular branch has been selected, growth of the quotient becomes geometric information: it determines summability, spectral dimension, and, under regular variation, a Weyl coefficient and residue volume.

We can now pass from the regularity criterion to the actual spectral triple. The algebra is generated by the predictive shifts and finite-rank diagonal observables; the Hilbert space is the square-summable space of predictive classes, with the internal representation carried along as a finite fibre; and the Dirac operator is the quotient length. The theorem identifies the spectral dimension with the polynomial growth exponent of the predictive quotient.

Theorem 2.11 (Predictive spectral triple). *Under [Assumption 2.1](#), define*

$$D_{\text{CP}} e_{[w]} \otimes \xi = \Lambda_{\min}([w]) e_{[w]} \otimes \xi.$$

Then

$$(\mathcal{B}_{\text{CP},0}, \ell^2(\mathcal{W}_{\text{CP}}) \otimes \mathcal{H}_{\text{int}}, D_{\text{CP}})$$

is a spectral triple. Its zeta function

$$\zeta_{\text{CP}}(s) = \dim(\mathcal{H}_{\text{int}}) \sum_{[w] \in \mathcal{W}_{\text{CP}}} \Lambda_{\min}([w])^{-s}$$

has abscissa of convergence

$$q_{\text{alg}} = \limsup_{R \rightarrow \infty} \frac{\log N_{\text{CP}}(R)}{\log R}, \quad N_{\text{CP}}(R) = \#\{[w] : \Lambda_{\min}([w]) \leq R\}.$$

Proof. Uniform finite fibres give boundedness of the descended shifts. Bounded increments give

$$[D_{\text{CP}}, \widehat{S}_{\sigma}] e_{[w]} = (\Lambda_{\min}([\sigma w]) - \Lambda_{\min}([w])) e_{[\sigma w]},$$

so $[D_{\text{CP}}, \widehat{S}_{\sigma}]$ is bounded. The adjoints are handled similarly, and finite-support diagonal multipliers commute with D_{CP} up to finite rank operators. Compactness of the resolvent follows from finiteness of rank balls. The statement about the zeta abscissa is the standard counting-function criterion for Dirichlet series with non-negative coefficients. $\square$

Remark 2.12 (Compact resolvent uses finite rank balls). The compact-resolvent assertion in [Theorem 2.11](#) uses the quotient-length definition in an essential way: for each R there are at most finitely many classes with $\Lambda_{\min} \leq R$, because each such class has a representative word of length at most $R - 1$ over the finite alphabet E . Thus the rank balls are finite even before imposing polynomial growth. Polynomial growth is needed for finite summability, not for compactness of the resolvent.

Definition 2.13 (Algebraic predictive dimension). The number q_{alg} of [Theorem 2.11](#) is the algebraic predictive dimension. It measures the polynomial growth rate of distinguishable accumulated channels as a function of quotient length.

Theorem 2.14 (Cancellative commuting regime). *Suppose the channels Φ_σ commute and the resulting predictive monoid is cancellative and embeds as a finitely generated affine semigroup in its Grothendieck group. Then its growth exponent is an integer*

$$q_{\text{alg}} = r_{\text{mon}} \in \{0, 1, \dots, |E|\},$$

and $N_{\text{CP}}(R) \asymp R^{r_{\text{mon}}}$. More precisely, r_{mon} is the rank of the abelian group generated by the image monoid of exponent vectors. Without the cancellative/affine-semigroup hypothesis, commutativity alone gives a finitely generated commutative monoid but not this lattice-cone counting theorem.

Proof. When the channels commute, Φ_w depends only on the exponent vector $n(w) \in \mathbb{N}^{|E|}$. The map $n \mapsto \Phi_n$ has image a finitely generated commutative monoid. Under the additional cancellative affine-semigroup hypothesis, this image embeds in its Grothendieck group, a finitely generated abelian group of rank r_{mon} . For $r_{\text{mon}} \geq 1$, counting elements of the image monoid of length at most R is then equivalent, up to constants, to counting lattice points in a rational cone of dimension r_{mon} , which gives $N_{\text{CP}}(R) \asymp R^{r_{\text{mon}}}$. In the degenerate case $r_{\text{mon}} = 0$ the image monoid is a finite group, so N_{CP} is eventually constant and $N_{\text{CP}}(R) \asymp R^0$ holds trivially. $\square$

Assumption 2.2 (Regularly varying predictive multiplicities). Let $h(R) = \#\{[w] \in \mathcal{W}_{\text{CP}} : \Lambda_{\min}([w]) = R\}$ be the number of predictive classes of quotient length R . Assume h is regularly varying of index $q - 1$ with $q = q_{\text{alg}} > 0$: there is a constant $a > 0$ and a slowly varying $L(R) \rightarrow 1$ with

$$h(R) = a R^{q-1} L(R), \quad R \rightarrow \infty.$$

In the free commutative case on r_{mon} generators this holds with $L \equiv 1$ and $h(R) = \binom{R+r_{\text{mon}}-1}{r_{\text{mon}}-1} \sim \frac{1}{(r_{\text{mon}}-1)!} R^{r_{\text{mon}}-1}$, so $q = r_{\text{mon}}$ and $a = 1/(r_{\text{mon}}-1)!$. More generally, affine-semigroup regimes with a positive rational degree have Ehrhart-type cumulative asymptotics; the shell regular-variation hypothesis is automatic only in the standard aperiodic/unimodular cases, or should be checked as an explicit hypothesis.

Theorem 2.15 (Renewal Weyl law and residue volume). *Under [Assumptions 2.1](#) and [2.2](#), the positive predictive triple has a genuine Weyl asymptotic governed by the single constant a of [Assumption 2.2](#), with $q_{\text{alg}} = q$:*

(i) (counting / Weyl asymptotics)

$$N_{\text{CP}}(R) \sim C_W R^q, \quad C_W = \frac{a}{q};$$

(ii) (zeta boundary residue) *the predictive zeta function $\zeta_{\text{CP}}(s) = \dim(\mathcal{H}_{\text{int}}) \sum_{[w]} \Lambda_{\min}([w])^{-s}$ has abscissa of convergence q , and*

$$\lim_{s \downarrow q} (s - q) \zeta_{\text{CP}}(s) = \dim(\mathcal{H}_{\text{int}}) a = \dim(\mathcal{H}_{\text{int}}) q C_W;$$

(iii) (heat trace)

$$\text{Tr}(e^{-tD_{\text{CP}}^2}) \sim \dim(\mathcal{H}_{\text{int}}) \frac{a \Gamma(q/2)}{2} t^{-q/2}, \quad t \downarrow 0;$$

(iv) (residue volume) *the operator $|D_{\text{CP}}|^{-q}$ lies in the weak trace ideal $\mathcal{L}^{1,\infty}$, is measurable, and its Dixmier trace equals the Weyl constant,*

$$\text{Tr}_\omega(|D_{\text{CP}}|^{-q}) = \dim(\mathcal{H}_{\text{int}}) C_W.$$

Thus the renewal volume $C_{\text{ren}} := \text{Tr}_\omega(|D_{\text{CP}}|^{-q})$ is the leading Weyl coefficient $\dim(\mathcal{H}_{\text{int}}) C_W$; the predictive triple obeys a genuine Weyl law rather than merely possessing a finite zeta abscissa.

Proof. Write $m = \dim(\mathcal{H}_{\text{int}})$; each predictive class contributes the eigenvalue $\Lambda_{\min}([w]) = R$ with internal multiplicity m , so the eigenvalue R of D_{CP} has total multiplicity $m h(R)$.

(i) By Karamata's theorem for regularly varying sequences [22], partial sums of a regularly varying sequence of index $q - 1$ satisfy $\sum_{r \leq R} h(r) \sim aR^q/q$. Hence $N_{\text{CP}}(R) \sim (a/q)R^q$, which also identifies the abscissa of Theorem 2.11 as $q_{\text{alg}} = q$.

(ii) For $s > q$, $\zeta_{\text{CP}}(s) = m \sum_{R \geq 1} h(R)R^{-s}$. The Abelian/Karamata theorem for Dirichlet series with regularly varying coefficients gives

$$\sum_{R \geq 1} h(R)R^{-s} \sim \frac{a}{s - q}, \quad s \downarrow q,$$

so $(s - q)\zeta_{\text{CP}}(s) \rightarrow ma$. This is a boundary-residue statement; by itself regular variation does not imply meromorphic continuation of ζ_{CP} across $s = q$.

(iii) $\text{Tr}(e^{-tD_{\text{CP}}^2}) = m \sum_{R \geq 1} h(R)e^{-tR^2}$. By the Tauberian/Karamata correspondence applied to the Laplace–Stieltjes transform of the regularly varying h ,

$$\sum_{R \geq 1} h(R)e^{-tR^2} \sim a \int_0^\infty r^{q-1} e^{-tr^2} dr = \frac{a}{2} \Gamma\left(\frac{q}{2}\right) t^{-q/2},$$

giving the stated constant after multiplying by m .

(iv) The counting asymptotic in (i) says that the singular-value counting function of $|D_{\text{CP}}|^{-q}$ has logarithmic Cesàro mean mC_W . Equivalently, $|D_{\text{CP}}|^{-q} \in \mathcal{L}^{1,\infty}$ is measurable and $\text{Tr}_\omega(|D_{\text{CP}}|^{-q}) = mC_W$ by the standard Weyl-asymptotic form of Connes' trace theorem [1]. $\square$

Remark 2.16 (Meromorphic strengthening). If the regular variation hypothesis is strengthened to an error estimate such as $h(R) = aR^{q-1} + O(R^{q-1-\delta})$ for some $\delta > 0$, or to the exact polynomial shell formula available in the free commutative case, then the zeta function has a genuine simple pole at $s = q$ and

$$\text{Res}_{s=q} \zeta_{\text{CP}}(s) = \dim(\mathcal{H}_{\text{int}}) a, \quad \text{Tr}_\omega(|D_{\text{CP}}|^{-q}) = \frac{1}{q} \text{Res}_{s=q} \zeta_{\text{CP}}(s).$$

Without such additional analytic control, Theorem 2.15 asserts the Tauberian boundary residue and Dixmier-trace volume, not meromorphic continuation.

Remark 2.17 (One constant, four readings). Theorem 2.15 ties the counting measure, the zeta boundary residue, the small-time heat kernel, and the noncommutative volume to the *same* leading coefficient. This is the spectral-geometric content that a bare length triple lacks: the predictive dimension q_{alg} is not only the zeta abscissa but a genuine Weyl exponent, and the renewal volume $C_{\text{ren}} = \text{Tr}_\omega(|D_{\text{CP}}|^{-q})$ is its Dixmier-trace volume. The regular-variation hypothesis Assumption 2.2 is the precise condition converting the lim sup of Theorem 2.11 into a genuine asymptotic law; meromorphic continuation is the separate strengthening recorded in Theorem 2.16.

2.5 Predictive order

So far the quotient has been read metrically, through the length operator and its spectrum. We now read the same object dynamically. Composition supplies a notion of future continuation, and quotienting mutual reachability removes the reversible cycling that would otherwise obstruct

antisymmetry. The result is an order skeleton on exactly the same predictive states; no second set of events is introduced.

The same quotient that carries the positive spectral triple also carries a natural order. This order is not imposed externally. It is the divisibility order of accumulated predictive channels: one predictive state precedes another when the latter can be obtained from the former by further renewal evolution. Thus the positive sector supplies both a spectral dimension and the order skeleton that will later be used as the causal substrate.

Let $\mathcal{M}$ be the monoid generated by the channels Φ_σ .

Definition 2.18 (Predictive divisibility order). For $a, b \in \mathcal{M}$ write $a \preceq b$ if $b = \Psi a$ for some $\Psi \in \mathcal{M}$. Write $a \approx b$ if $a \preceq b$ and $b \preceq a$. The predictive poset is

$$\mathfrak{P} = \mathcal{M}/\approx$$

with the induced order.

Lemma 2.19 (Poset reflection). *The relation on $\mathfrak{P}$ induced by Definition 2.18 is a partial order.*

Proof. Left divisibility is a preorder on $\mathcal{M}$. Quotienting a preorder by its symmetric part gives a partial order. $\square$

Assumption 2.3 (Aperiodic elementary grading). On the condensation $\mathfrak{P} = \mathcal{M}/\approx$, the quotient length satisfies:

- (i) it is constant on every mutual-reachability class, and therefore descends to a function $\Lambda_{\mathfrak{P}} : \mathfrak{P} \rightarrow \mathbb{N}$;
- (ii) it is strictly increasing on strict divisibility classes;
- (iii) every Hasse cover $x < y$ is elementary in the sense that

$$\Lambda_{\mathfrak{P}}(y) = \Lambda_{\mathfrak{P}}(x) + 1;$$

- (iv) every length ball $\{x \in \mathfrak{P} : \Lambda_{\mathfrak{P}}(x) \leq R\}$ is finite.

The first two items supply a time function; the third is the additional hypothesis that promotes that time function to a genuine rank function.

Proposition 2.20 (Causal-order skeleton). *Under Assumption 2.3, $(\mathfrak{P}, \preceq)$ is a locally finite graded poset with rank function*

$$T(x) = \Lambda_{\mathfrak{P}}(x).$$

In particular it is a causal-set-like order skeleton. If only Assumption 2.3(i),(ii),(iv) are assumed, the same conclusion holds with “graded rank” replaced by “strictly increasing integer-valued time function.”

Proof. The quotient by mutual reachability is a partial order by Theorem 2.19. Items (i)–(ii) make T well defined and strictly increasing on strict comparable pairs. Item (iii) says exactly that every Hasse cover raises T by one, so T is a rank function and the poset is graded. Finally, an interval $[x, y]$ lies in the finite length ball of radius $T(y)$, hence is finite by (iv). This proves local finiteness. $\square$

Proposition 2.21 (Chain-counting renewal resolvent). *Let Φ denote the cover transfer operator of the Hasse diagram of $\mathfrak{P}$. In the completed path-counting incidence algebra [45] graded by rank,*

$$\mathcal{R}_{\mathfrak{P}} := \sum_{k \geq 0} \Phi^k = (1 - \Phi)^{-1}.$$

The kernel $\mathcal{R}_{\mathfrak{P}}(x, y)$ counts saturated chains from x to y . It coincides with the ordinary incidence zeta $\zeta_{\mathfrak{P}}(x, y) = \mathbf{1}_{x \preceq y}$ only when each interval has a unique saturated chain, or after passing to the Boolean incidence algebra in which all positive multiplicities are identified.

Proof. The k th power Φ^k counts saturated chains of length k . Summing over all k therefore gives the chain-counting renewal resolvent. In a general poset this is not the ordinary incidence zeta: for example, in $\mathbb{N}^2$ there are two saturated chains from $(0, 0)$ to $(1, 1)$, while the incidence zeta takes the value 1. The two notions agree precisely in the unique-chain case, or after Booleanising multiplicities. $\square$

Remark 2.22 (Markov and renewal singularities). This identification makes the analytic type of memory visible in the order. Markov holding times give rational renewal resolvents and pole singularities. Heavy-tailed renewal laws give branch-point singularities and fractional exponents. Thus genuine renewal memory can appear as a fractional singularity of the chain-counting renewal resolvent, not merely as a probabilistic feature of the source.

2.6 The positivity boundary

At this point the positive sector already supplies the main Riemannian ingredients: points, length, dimension, volume, and a causal-order skeleton. The remaining question is qualitative rather than quantitative: can these positive data also distinguish one negative direction? The following results show that they cannot. Moreover, they identify every compatible attempt to do so with additional $\mathbb{Z}/2$ -valued holonomy.

The completely positive construction supplies a Hilbert-positive spectral geometry, residue data, and a predictive order. It does not, by itself, distinguish a negative subspace. The following two statements isolate the precise obstruction and identify the extra information carried by any compatible nontrivial fundamental symmetry.

Assumption 2.4 (Irreducible natural positive sector). Let $\mathfrak{B}_+ \subset \mathcal{B}(\ell^2(\mathcal{W}_{\text{CP}}) \otimes \mathcal{H}_{\text{int}})$ be the unital $*$ -algebra generated by the represented renewal shifts, pointwise internal observables, and admissible relabellings. Assume

$$\mathfrak{B}'_+ = \mathbb{C}\mathbf{1}.$$

A fundamental symmetry extracted functorially from the positive sector is required to commute with $\mathfrak{B}_+$ and may not use an added sign, phase, orientation, or sheet label.

Theorem 2.23 (Irreducible positive-sector no-go). *Under Assumption 2.4, every functorially extracted self-adjoint involution is scalar,*

$$J = +\mathbf{1} \quad \text{or} \quad J = -\mathbf{1}.$$

Its associated form is definite. Hence the natural irreducible positive representation does not determine a Lorentzian splitting.

Proof. Naturality places J in $\mathfrak{B}'_+ = \mathbb{C}\mathbf{1}$, so $J = c\mathbf{1}$. Self-adjointness and $J^2 = \mathbf{1}$ force $c = \pm 1$. $\square$

Theorem 2.24 (Compatible fundamental symmetries are signed data). *Let a recurrent positive predictive component be fixed. Suppose a self-adjoint involution J commutes with the atomic diagonal algebra and, for every descended generator $\hat{S}_\sigma$, satisfies*

$$J\hat{S}_\sigma = c(\sigma)\hat{S}_\sigma J, \quad c(\sigma) \in \{\pm 1\}.$$

Then the edgewise ratios of the diagonal signs of J define a $\mathbb{Z}/2$ -valued cocycle on the predictive graph. Conversely every such cocycle produces the corresponding exchange relations on its signed double cover. Thus every compatible nonconstant diagonal fundamental symmetry contains precisely the additional signed information classified in [Section 3](#).

Proof. Write $Je_x = \eta(x)e_x$, with $\eta(x) \in \{\pm 1\}$. Along a generator edge $x \rightarrow \sigma x$, the exchange relation gives $\eta(\sigma x) = c(\sigma)\eta(x)$. These edge ratios form a $\mathbb{Z}/2$ -valued one-cochain; on a graph every one-cochain is a cocycle, and a change of vertex signs changes it by a coboundary. The converse is the signed cover construction of [Theorem 3.16](#). $\square$

Corollary 2.25 (CP positivity boundary). *The completely positive sector canonically supplies positive spectral geometry, dimension, volume, and order. A nontrivial Lorentzian/Krein splitting requires additional signed data; in the irreducible natural representation it cannot be extracted from the positive sector alone.*

Section 2 therefore ends with both a construction and an obstruction. Predictive memory already carries a positive geometry and an intrinsic order, but Lorentzian signature is absent for a structural reason, not because a numerical parameter has yet to be tuned. The next section adjoins the minimal signed information singled out by the obstruction and classifies it up to gauge.

3 The signed Krein sector

This section converts the obstruction of [Section 2.6](#) into a canonical signed geometry. The argument proceeds from abstract to concrete: we first classify all objectwise-minimal indefinite enrichments, then explain one operational origin of their cohomology class, realise the class as a two-sheeted cover, and distinguish its real/Krein content from an optional amplitude-level $\mathbb{Z}/4$ lift. The signed cover is then combined with the renewal clock to obtain a modular Dirac operator, whose scale is selected by a pressure equation. A reconstruction theorem closes the section by showing that these data retain the original predictive geometry together with its sign holonomy.

The positivity boundary identifies what must be added without determining a particular nonzero class. We first characterize the objectwise-minimal renewal-positive indefinite enrichment, then give its concrete double-cover realization, distinguish the real signed layer from its amplitude lift, and finally construct and characterize the modular signed Dirac operator. The section closes with a reconstruction theorem for the graded signed predictive geometry.

3.1 Minimal indefinite enrichments over a fixed component

We now strengthen the signed-cover construction from existence to categorical minimality. Throughout this subsection $G = (V, E_G, \Lambda)$ is one fixed finite connected recurrent predictive component. The edge count $|E_{\text{geom}}|$ below counts unoriented geometric edges, equivalently one chosen orientation from each pair $e, \bar{e}$. Thus

$$H^1(G, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^{b_1(G)}, \quad b_1(G) = |E_{\text{geom}}| - |V| + 1.$$

The categorical level is important. We classify enrichments *over this fixed* G and then show that the classification itself is contravariantly natural under renewal isomorphisms. We do not require one global enrichment functor to choose a class on every object simultaneously; such a requirement would restrict the value at G to automorphism-invariant classes.

Definition 3.1 (Renewal-positive indefinite enrichment over G). An enrichment over the fixed positive predictive component G consists of:

- (i) a finite Hermitian local system $F = (F_x)_{x \in V}$ with a distinguished orthonormal renewal basis and its simplicial cone $\mathcal{C}_x$;
- (ii) unitary edge transports $T_e : F_x \rightarrow F_y$, with $T_{\bar{e}} = T_e^{-1}$, augmenting the renewal shifts;
- (iii) fibrewise self-adjoint involutions J_x , commuting with the full atomic diagonal algebra generated by the distinguished renewal-basis projections, as well as with the length grading;
- (iv) a homogeneous degree $\epsilon(e) \in \mathbb{Z}/2$ for each edge,

$$J_y T_e = (-1)^{\epsilon(e)} T_e J_x;$$

- (v) structural recovery of the untwisted positive base by the deck quotient.

It is *renewal-positive* when $T_e(\mathcal{C}_x) = \mathcal{C}_y$ for every edge, and *fibrewise indefinite* when both eigenspaces of every J_x are nonzero. Its fibre rank is $\text{rank } F := \dim_{\mathbb{C}} F_x$, independent of x because G is connected and the transports are unitary. It is *objectwise minimal* when this constant rank is least among all renewal-positive fibrewise-indefinite enrichments over G .

Definition 3.2 (Gauge equivalence and pullback). An isomorphism over the identity of G is a vertexwise cone-preserving unitary which intertwines the transports and transforms the local fundamental symmetries covariantly. On a rank-two fibre a sheet trivialisation writes

$$J_x = (-1)^{j(x)} J_0, \quad J_0 = \text{diag}(1, -1), \quad T_e = S^{c(e)}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A change of sheet orientation $a : V \rightarrow \mathbb{Z}/2$ acts by

$$j \mapsto j + a, \quad c \mapsto c + \delta a.$$

The homogeneity degree ϵ is invariant and satisfies

$$\epsilon = c + \delta j.$$

For a renewal isomorphism $\varphi : G \rightarrow G'$, pullback transports, cones, and local symmetries define

$$\varphi^* : \text{Enr}_{\min}(G') \longrightarrow \text{Enr}_{\min}(G).$$

Lemma 3.3 (Cone-positive unitaries are permutations). *A unitary $T : F_x \rightarrow F_y$ satisfying $T(\mathcal{C}_x) = \mathcal{C}_y$ permutes the distinguished renewal basis.*

Proof. A linear isomorphism carrying a simplicial cone onto a simplicial cone maps extremal rays bijectively to extremal rays. Hence T sends each renewal basis vector to a positive scalar multiple of a renewal basis vector. Unitarity makes that scalar equal to one. $\square$

Lemma 3.4 (Minimal rank and reduction of the structure group). *Every fibrewise-indefinite enrichment has $\text{rank } F \geq 2$. The signed cover realises rank two. On an objectwise-minimal enrichment,*

$$\text{rank } F = 2, \quad T_e \in \{I, S\}.$$

In particular all $U(1)$ phases and residual signs are excluded and the structure group is $\{I, S\} \cong \mathbb{Z}/2$.

Proof. Fibrewise indefiniteness gives $\dim F_x^+ \geq 1$ and $\dim F_x^- \geq 1$, hence $\text{rank } F \geq 2$. The two-sheeted signed cover has one positive and one negative line in each fibre, so rank two is attainable. Commutation with the full atomic diagonal algebra makes every J_x diagonal in the distinguished renewal basis. At rank two, fibrewise indefiniteness therefore forces $J_x = \pm \text{diag}(1, -1)$. Moreover, [Theorem 3.3](#) leaves exactly the two permutation matrices I and S as edge transports. They are respectively even and odd relative to J_0 . Matrices $-I$, $-S$, and arbitrary phases fail cone positivity. $\square$

Lemma 3.5 (Two cochains, one cohomology class). *For an objectwise-minimal enrichment, choose a sheet trivialisation and write $T_e = S^{c(e)}$, $J_x = (-1)^{j(x)} J_0$. Then:*

- (i) $c \in C^1(G; \mathbb{Z}/2)$ is the transition cochain of a principal $\mathbb{Z}/2$ -cover;
- (ii) because G is a 1-complex, $C^2(G; \mathbb{Z}/2) = 0$, so every 1-cochain is a cocycle;
- (iii) a sheet change $a \in C^0(G; \mathbb{Z}/2)$ sends $c \mapsto c + \delta a$;
- (iv) the gauge-invariant homogeneity degree is

$$\epsilon(e) = c(e) + j(x) + j(y) = c(e) + (\delta j)(e),$$

and therefore $[\epsilon] = [c] \in H^1(G, \mathbb{Z}/2)$.

Proof. The edge transport either preserves or exchanges the two sheets, producing $c(e)$. Since the graph has no 2-cells, $\delta^1 = 0$, so $Z^1 = C^1$. Conjugating by $S^{a(x)}$ at the vertices gives the standard coboundary action. Finally,

$$(-1)^{j(y)} J_0 S^{c(e)} = (-1)^{j(y)+c(e)} S^{c(e)} J_0 = (-1)^{c(e)+j(x)+j(y)} T_e J_x,$$

which proves the formula for ϵ and the equality of classes. $\square$

Remark 3.6 (The invariant is the cover). The principal cover $P \rightarrow G$, equivalently $[c]$, is the intrinsic invariant. The representatives c , j , and J are gauge-dependent realization data. For a fixed cover all sheet-sign choices form one gauge orbit, so $[J]$ contributes no independent invariant. The deck involution and the fundamental symmetry are distinct: in a sheet trivialisation the deck swap anticommutes with J_0 .

Theorem 3.7 (Categorical minimality of the signed enrichment). *Let G be a finite connected recurrent predictive component. Let $\text{Enr}_{\min}(G)$ denote the groupoid of objectwise-minimal renewal-positive fibrewise-indefinite enrichments over G , with morphisms covering the identity. Then*

$$\pi_0 \text{Enr}_{\min}(G) \cong H^1(G, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^{b_1(G)}.$$

Every such enrichment is the signed cover $G^\times$, its constant fibre rank is two, and its fundamental symmetry is gauge-covariant realization data of the cover. These bijections are contravariantly natural under renewal isomorphisms.

Proof. By [Theorem 3.4](#), every minimal enrichment has rank two and every edge transport is $S^{c(e)}$. By [Theorem 3.5](#), these transitions define a principal $\mathbb{Z}/2$ -cover and gauge equivalence is precisely quotienting C^1 by coboundaries. Thus

$$\pi_0 \text{Enr}_{\min}(G) \cong C^1(G; \mathbb{Z}/2) / \delta C^0(G; \mathbb{Z}/2) = H^1(G, \mathbb{Z}/2).$$

Conversely every principal double cover gives the rank-two signed enrichment, so the map is bijective. For a connected finite graph, the cohomology dimension is $|E_{\text{geom}}| - |V| + 1$. Pullback along a renewal isomorphism pulls back the transition cochain and all associated local data, so the class of the pulled-back enrichment is $\varphi^*[c]$. $\square$

Corollary 3.8 (Removable and connected sectors). *There are $2^{b_1(G)}$ gauge classes. The zero class gives the split cover $G \sqcup G$. The $2^{b_1(G)} - 1$ nonzero classes are exactly the non-removable sectors, and each corresponding double cover is connected.*

Proof. A principal double cover is classified by the homomorphism $\pi_1(G) \rightarrow \mathbb{Z}/2$ represented by its cohomology class. The zero map gives the split cover. Every nonzero map has image all of $\mathbb{Z}/2$, hence its cover is connected. Triviality of the class is exactly removability by a sheet gauge. $\square$

The classification has a simple geometric reading. A nonzero class in $H^1(G, \mathbb{Z}/2)$ records a sign that cannot be removed consistently around all loops. Locally one may always relabel the two sheets, but globally the loop holonomy remains. Thus the theorem does not merely exhibit one convenient double cover: it proves that every minimal renewal-positive indefinite enrichment is of this form.

Remark 3.9 (Edge-locality, label-locality, and multiplicativity). The theorem uses independent edge-local transitions. If signs are required to be constant on reset-label classes, the realised classes form the subgroup

$$H_{\text{loc}}^1(G, \mathbb{Z}/2) := \{[c] : c \text{ has a label-local representative}\}.$$

Literal multiplicativity is not a hypothesis of the classification. Tensoring two rank-two minimal enrichments gives rank four on the product component, although that component itself admits a rank-two signed cover. Thus global objectwise minimality and literal tensor-product multiplicativity are incompatible. Multiplicativity may be imposed only as optional downstream coherence on chosen monoidal generators, producing rank 2^r on an r -fold product and the usual Koszul/super structure.

Remark 3.10 (Role of cone preservation). Without preservation of the distinguished renewal cone, rank-two homogeneous unitaries carry arbitrary phase holonomy and the classification enlarges beyond $H^1(G, \mathbb{Z}/2)$. Without fibrewise indefiniteness, one-dimensional vertex-sign assignments survive but realise only the trivial class. Without structural base recovery, the quotient need not return the positive predictive component. Thus positivity, locality, base recovery, and fibrewise indefiniteness are the essential hypotheses; naturality belongs to the family of classifications, not to the definition of one enrichment over a fixed G .

Remark 3.11 (Holonomy, not deficiency). The classifying invariant is the holonomy class $[c]$, not a spectral-radius deficiency. Deficiency depends on a chosen transfer operator and may fail to detect half-period or otherwise spectrally silent classes.

3.2 A record-theoretic origin of the signed class

The classification above treats $[\chi]$ as the intrinsic datum of the minimal indefinite enrichment. In a record-regular operational realisation, the class can itself be generated by the orientation of the stable future-distinct record bundle.

Proposition 3.12 (Stable-record determinant orientation). *Let G be a connected recurrent predictive graph. Assign to every vertex x a finite-dimensional real vector space $\mathcal{R}_x$ of stable future-distinct records, of constant rank, and to every oriented edge $e : x \rightarrow y$ an invertible record transport*

$$L_e : \mathcal{R}_x \longrightarrow \mathcal{R}_y, \quad L_{\bar{e}} = L_e^{-1}.$$

After choosing local orientations, define

$$\chi_{\text{rec}}(e) := \text{sgn det } L_e \in \{\pm 1\}.$$

Then a change of local record orientation changes χ_{rec} by a vertex coboundary, while the product around every loop is invariant. Hence

$$[\chi_{\text{rec}}] = w_1(\text{det } \mathcal{R}) \in H^1(G, \mathbb{Z}/2)$$

is well defined, and the principal double cover classified by this class is exactly the orientation cover of the stable-record determinant line.

Proof. Choose an oriented basis in each $\mathcal{R}_x$. Reversing the chosen orientation at a vertex x multiplies the determinant sign of every incident transport by -1 ; if $g(x) \in \{\pm 1\}$ records that choice, then

$$\chi_{\text{rec}}(e : x \rightarrow y) \longmapsto g(y)\chi_{\text{rec}}(e)g(x),$$

which is the standard coboundary action. The vertex factors cancel in the product around a closed loop, so the loop sign depends only on the cohomology class. A lift of a path to the orientation torsor records whether its transport preserves or reverses the determinant orientation; its monodromy is therefore precisely the loop product of χ_{rec} . This is the principal cover of [Theorem 3.17](#). $\square$

Remark 3.13 (Affinity and orientation are different classes). The same transports also define the real one-cochain

$$A_{\text{rec}}(e) := \log |\text{det } L_e|.$$

Under a positive rescaling of local determinant frames it changes by a real coboundary and therefore defines an additive class $[A_{\text{rec}}] \in H^1(G; \mathbb{R})$. This class measures determinant magnitude, whereas $w_1(\text{det } \mathcal{R})$ measures determinant parity. Neither determines the other. The present paper classifies the signed geometry produced by the second class; a companion upstream construction derives stable record spaces and their transports from operational prediction.

The record-theoretic proposition supplies a possible source of the class, but the downstream geometry depends only on the resulting cohomology element. Accordingly, the remainder of the section fixes a representative χ , constructs its cover explicitly, and tracks which statements are gauge-dependent and which depend only on $[\chi]$.

3.3 Concrete signed covers and Krein symmetry

Fix a recurrent predictive component, represented by a connected directed graph $G = (V, E_G)$. Its vertices are predictive classes, and an edge $e : x \rightarrow y$ records a generator transition $y = \sigma \cdot x$. The quotient length is denoted by $\Lambda : V \rightarrow \mathbb{N}$.

Definition 3.14 (Sign cocycle). A sign cocycle on G is a map

$$\chi : E_G \longrightarrow \{\pm 1\}.$$

For an oriented path $p = e_k \cdots e_1$ we write

$$\chi(p) = \prod_{j=1}^k \chi(e_j).$$

Since a graph has no 2-cells, every $\{\pm 1\}$ -valued 1-cochain is a 1-cocycle. A flat gauge is a vertex function $g : V \rightarrow \{\pm 1\}$, acting by

$$\chi^g(e : x \rightarrow y) = g(y) \chi(e) g(x).$$

The gauge-invariant content of χ is its holonomy class

$$[\chi] \in H^1(G, \mathbb{Z}/2).$$

Equivalently, $[\chi]$ is the homomorphism from the cycle space of G to $\{\pm 1\}$ given by the product of signs around loops.

Remark 3.15 (Relation with history signs). If the original renewal memory carries a multiplicative history sign $\varepsilon : E^* \rightarrow \{\pm 1\}$, then it defines a sign cocycle on the predictive graph provided it is compatible with the predictive quotient: whenever two generator transitions define the same edge of G , they have the same sign. In that case $\chi(e)$ is the sign of any history segment representing the edge e . This compatibility is the precise form of “internality” needed for the sign to descend to predictive geometry.

Construction 3.16 (Signed predictive cover). The signed predictive cover associated with χ is the graph $G^\chi = (V^\chi, E_G^\chi)$ with

$$V^\chi = V \times \{\pm 1\}.$$

For every edge $e : x \rightarrow y$ in G and every sheet label $\eta \in \{\pm 1\}$, there is an edge

$$(e, \eta) : (x, \eta) \longrightarrow (y, \eta \chi(e)).$$

The projection

$$\pi : G^\chi \longrightarrow G, \quad \pi(x, \eta) = x,$$

is a two-sheeted graph cover, and the deck transformation is

$$\tau(x, \eta) = (x, -\eta).$$

We write $\mathcal{W}_{\text{CP}}^\chi$ for the corresponding signed predictive quotient.

Theorem 3.17 (Principal $\mathbb{Z}/2$ -cover and holonomy). *The projection $\pi : G^\chi \rightarrow G$ is a principal $\mathbb{Z}/2$ -cover. Its monodromy around a loop ℓ is multiplication by $\chi(\ell)$. The cover is gauge-equivalent to the trivial double cover if and only if*

$$[\chi] = 0 \in H^1(G, \mathbb{Z}/2).$$

Thus the non-removable content of the signed cover is precisely the holonomy class $[\chi]$.

Proof. The deck transformation τ acts freely and transitively on each fibre $\pi^{-1}(x) = \{(x, +1), (x, -1)\}$, so π is a principal $\mathbb{Z}/2$ -cover. Lifting a loop $\ell = e_k \cdots e_1$ from (x, η) gives the endpoint $(x, \eta\chi(\ell))$, hence the monodromy is the cycle-sign homomorphism.

If $[\chi] = 0$, then $\chi(e : x \rightarrow y) = g(y)g(x)$ for some $g : V \rightarrow \{\pm 1\}$, and the map $(x, \eta) \mapsto (x, \eta g(x))$ identifies G^χ with the trivial cover. Conversely, a trivialisation chooses one sheet over each vertex; comparing the chosen sheet across each edge gives a vertex gauge whose coboundary is χ . Hence χ is cohomologically trivial exactly when the cover is removable. $\square$

Definition 3.18 (Canonical fundamental symmetry on the signed cover). On

$$\mathcal{K}_\chi = \ell^2(V^\chi) \otimes \mathcal{H}_{\text{int}}$$

define

$$J_\chi e_{(x, \eta)} = \eta e_{(x, \eta)}.$$

The quotient length lifts to the cover by

$$N e_{(x, \eta)} = \Lambda(x) e_{(x, \eta)}.$$

For an edge $e : x \rightarrow y$, the lifted shift is

$$\tilde{S}_e e_{(x, \eta)} = e_{(y, \eta\chi(e))},$$

on its natural domain, and zero away from the source fibre.

Proposition 3.19 (Krein datum). *The operator J_χ is a self-adjoint unitary. The form*

$$[u, v]_\chi = \langle u, J_\chi v \rangle$$

is a Krein form whenever both sheets occur. Moreover the lifted shifts satisfy the exchange relation

$$J_\chi \tilde{S}_e = \chi(e) \tilde{S}_e J_\chi.$$

A flat gauge changes the representative cocycle but leaves the loop holonomies, and hence the class $[\chi]$, unchanged.

Proof. Since $\eta = \pm 1$, $J_\chi^2 = 1$ and $J_\chi^* = J_\chi$. The form is positive on the $+1$ sheet and negative on the -1 sheet, giving a fundamental decomposition. Finally,

$$J_\chi \tilde{S}_e e_{(x, \eta)} = J_\chi e_{(y, \eta\chi(e))} = \eta\chi(e) e_{(y, \eta\chi(e))} = \chi(e) \tilde{S}_e J_\chi e_{(x, \eta)}.$$

Gauge transformations multiply edge signs by coboundaries, which do not change products around closed loops. $\square$

Corollary 3.20 (Removability is cohomology). *The signed sector is removable by a flat gauge if and only if*

$$[\chi] = 0 \in H^1(G, \mathbb{Z}/2).$$

Equivalently, nonzero classes in $H^1(G, \mathbb{Z}/2)$ are precisely the non-removable signed/Krein sectors.

Proof. This is the triviality criterion in [Theorem 3.17](#), translated from covers to the associated fundamental symmetries. $\square$

3.4 Real signed quotients and amplitude lifts

The classification above is intentionally a real/Krein classification. It answers the question: what is the objectwise-minimal indefinite enrichment of a positive renewal component that still preserves the renewal cone and recovers the positive base? The answer is a principal $\mathbb{Z}/2$ -cover. A different question appears only when the same holonomy is asked to act on amplitudes or charged local systems: a sign is then not the transported quantity itself, but the square of a unitary phase. The next elementary facts record this distinction without changing the signed-cover theorem.

Definition 3.21 (Amplitude lift of a signed cohomology class). Let $[c] \in H^1(G, \mathbb{Z}/2)$ be the signed-cover class of [Theorem 3.7](#). A finite amplitude lift of $[c]$ is a class $[\tilde{c}] \in H^1(G, \mathbb{Z}/4)$ whose reduction modulo two is $[c]$:

$$\rho([\tilde{c}]) = [c], \quad \rho : \mathbb{Z}/4 \longrightarrow \mathbb{Z}/2.$$

Equivalently, if a cycle has lifted exponent $k \in \mathbb{Z}/4$, the associated unitary phase is i^k and its real signed shadow is

$$(i^k)^2 = (-1)^k.$$

Thus the $\mathbb{Z}/2$ class records the square of the amplitude transport, while the $\mathbb{Z}/4$ lift records a choice of square root.

Lemma 3.22 (Surjectivity of the amplitude-lift map on graphs). *For every finite connected graph G , reduction modulo two gives a surjection*

$$\rho : H^1(G, \mathbb{Z}/4) \twoheadrightarrow H^1(G, \mathbb{Z}/2).$$

If $b_1(G) = r$, then every signed class has exactly 2^r amplitude lifts.

Proof. A connected graph is homotopy equivalent to a wedge of $r = b_1(G)$ circles. Hence

$$H^1(G, \mathbb{Z}/4) \cong (\mathbb{Z}/4)^r, \quad H^1(G, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^r.$$

Under these identifications ρ is coordinatewise reduction $\mathbb{Z}/4 \rightarrow \mathbb{Z}/2$, which is surjective and has a two-point fibre in each coordinate. Therefore the fibre over every element of $(\mathbb{Z}/2)^r$ has size 2^r . $\square$

Theorem 3.23 (Minimal finite square-root completion). *Let a signed renewal sector contain a loop whose real holonomy is -1 . Any finite phase group acting on amplitudes and containing a unitary square root of that sign contains a cyclic subgroup of order four. Consequently the minimal finite amplitude completion of a nontrivial signed sector is $\mathbb{Z}/4$. Its reduction modulo two is exactly the principal $\mathbb{Z}/2$ -cover of [Theorem 3.7](#).*

Proof. Let u be the amplitude phase assigned to a loop with real signed holonomy -1 . The square-root condition is $u^2 = -1$. Then $u^4 = 1$, while $u^2 \neq 1$; hence u has order exactly four. Therefore the subgroup $\langle u \rangle$ is cyclic of order four, so any finite phase group carrying such a square root contains a copy of $\mathbb{Z}/4$. Conversely $\mathbb{Z}/4$ works: its generator may be represented by i , whose square is -1 . Reducing the exponent of i^k modulo two records precisely whether the square is $+1$ or -1 , which is the original $\mathbb{Z}/2$ signed-cover class. $\square$

Corollary 3.24 (Conservativity over the real/Krein layer). *Replacing a signed class $[c] \in H^1(G, \mathbb{Z}/2)$ by any amplitude lift $[\tilde{c}] \in H^1(G, \mathbb{Z}/4)$ with $\rho([\tilde{c}]) = [c]$ does not alter any construction depending only on the principal double cover, the Krein fundamental symmetry, or the real commutator polar form. The $\mathbb{Z}/4$ datum is extra amplitude information above the real quotient, not a replacement of the $\mathbb{Z}/2$ Lorentzian-signature mechanism.*

Proof. The principal double cover, sheet swap, and fundamental symmetry are obtained by applying ρ to the amplitude lift. If two amplitude lifts have the same reduction, they define the same $\mathbb{Z}/2$ -cover and therefore the same signed sheet Hilbert space, the same diagonal fundamental symmetry J_χ , and the same exchange signs used in [Theorem 4.4](#) and [Theorem 4.5](#). Likewise the polar commutator form is the antisymmetric real quotient of the projective multiplier; scalar rephasings and choices of square root may change the lifted fourth-root phases, but not their reduction to the $\{\pm 1\}$ -valued commutator. Thus the real/Krein theorems are unchanged. $\square$

Remark 3.25 (Where the lift is used). The $\mathbb{Z}/2$ layer is sufficient for Lorentzian signature and for the temporal Krein exchange. The $\mathbb{Z}/4$ layer becomes relevant only when one retains amplitude-level local systems: charged holonomy, square-root phases of projective transports, or finite matter-sector refinements. In particular, a proof carried out purely in the real category should be stated as a real-category statement; it need not control all amplitude-level splittings after a $\mathbb{Z}/4$ lift is retained.

The amplitude lift will not be used to change the Lorentzian conclusions. Those conclusions depend on the real shadow $[\chi]$, the principal double cover, and the corresponding fundamental symmetry. The $\mathbb{Z}/4$ lift records additional square-root phases for later amplitude-sensitive sectors while leaving this real/Krein layer unchanged.

The cover and the fundamental symmetry provide the indefinite inner product. To obtain the signed Dirac operator we combine the symmetry with the renewal clock. The clock is the same quotient length as in the positive sector, but it is inserted through an exponential modular weight.

The three pieces of the operator have distinct jobs. The factor J_χ carries the indefinite sign, $e^{\beta N}$ turns renewal depth into an unbounded modular scale, and the bounded term B carries the nontrivial finite-propagation differential content. Keeping these roles separate will be important when the finite signed geometry is later converted into a continuum Hamiltonian.

Definition 3.26 (Signed modular twist). Let

$$\Delta = e^{\beta N}, \quad \beta > 0.$$

For an operator a for which $\Delta a \Delta^{-1}$ extends boundedly, set

$$\sigma_\beta(a) = \Delta a \Delta^{-1}, \quad \rho_\chi(a) = \text{Ad}(J_\chi) \sigma_\beta(a) = J_\chi \Delta a \Delta^{-1} J_\chi.$$

The signed modular Dirac operator is

$$D_\chi = J_\chi \Delta + B,$$

where B is a bounded J_χ -self-adjoint perturbation, i.e. $J_\chi B^* J_\chi = B$.

Theorem 3.27 (Canonical signed modular Dirac). *Assume that the length shells of N are finite and that $N \rightarrow \infty$ along the signed quotient. Let $\mathcal{A}_\chi$ be a $*$ -algebra of bounded operators on $\mathcal{K}_\chi$ such that, for every $a \in \mathcal{A}_\chi$,*

$$\rho_\chi(a) \text{ is bounded,} \quad [B, a]_{\rho_\chi} := Ba - \rho_\chi(a)B \text{ is bounded.}$$

Then

$$D_\chi = J_\chi e^{\beta N} + B$$

is Krein-self-adjoint with respect to J_χ and defines a twisted Krein spectral triple

$$(\mathcal{A}_\chi, \mathcal{K}_\chi, D_\chi; \rho_\chi).$$

Its resolvent is compact. If the shell multiplicities have subexponential growth, and in particular if they have the polynomial growth of [Assumption 2.2](#), then

$$\mathrm{Tr}(1 + D_\chi^* D_\chi)^{-s/2} < \infty \quad \text{for every } s > 0.$$

Equivalently, the regularised singular-value summability abscissa is zero. If D_χ is invertible, the unregularised statement

$$\mathrm{Tr} |D_\chi|^{-s} < \infty \quad \text{for every } s > 0$$

also holds. The summability is signature-blind and depends only on the growth of length shells, not on the sign holonomy.

Proof. Put $A := J_\chi \Delta$, where $\Delta = e^{\beta N}$. Since J_χ commutes with N , the operator A is Hilbert-self-adjoint on $\mathrm{Dom} \Delta$, satisfies $|A| = \Delta$, and has compact resolvent. Moreover

$$J_\chi A^* J_\chi = A.$$

Together with $J_\chi B^* J_\chi = B$, this gives

$$J_\chi D_\chi^* J_\chi = D_\chi,$$

so D_χ is Krein-self-adjoint.

Compactness of the resolvent can be verified without invoking an unproved operator inequality. Choose $\eta > \|B\|$. Since A is self-adjoint, $\|(A - i\eta)^{-1}\| \leq \eta^{-1}$, and

$$D_\chi - i\eta = (\mathbf{1} + B(A - i\eta)^{-1})(A - i\eta).$$

The first factor is invertible by a Neumann series. Hence

$$(D_\chi - i\eta)^{-1} = (A - i\eta)^{-1}(\mathbf{1} + B(A - i\eta)^{-1})^{-1},$$

which is compact because $(A - i\eta)^{-1}$ is compact.

For $a \in \mathcal{A}_\chi$, compute on the common core of finitely supported vectors:

$$\begin{aligned} [J_\chi \Delta, a]_{\rho_\chi} &= J_\chi \Delta a - \rho_\chi(a) J_\chi \Delta \\ &= J_\chi \Delta a - (J_\chi \Delta a \Delta^{-1} J_\chi) J_\chi \Delta = 0. \end{aligned}$$

Therefore

$$[D_\chi, a]_{\rho_\chi} = [B, a]_{\rho_\chi},$$

which is bounded by hypothesis.

It remains to justify the summability statement. The bounded perturbation preserves the operator domain and gives equivalent graph norms:

$$\begin{aligned} \|D_\chi u\| + \|u\| &\leq \|\Delta u\| + (1 + \|B\|)\|u\|, \\ \|\Delta u\| + \|u\| &\leq \|D_\chi u\| + (1 + \|B\|)\|u\|, \quad u \in \mathrm{Dom} \Delta. \end{aligned}$$

Consequently the compact embeddings

$$(\text{Dom } D_\chi, \|\cdot\|_{D_\chi}) \hookrightarrow \mathcal{K}_\chi, \quad (\text{Dom } \Delta, \|\cdot\|_\Delta) \hookrightarrow \mathcal{K}_\chi$$

have comparable approximation numbers. Equivalently, the singular values of $(\mathbf{1} + D_\chi^* D_\chi)^{-1/2}$ are comparable, up to fixed constants, with those of $(\mathbf{1} + \Delta^2)^{-1/2}$. The latter have shell multiplicity $h(R)$ and size asymptotic to $e^{-\beta R}$. Thus

$$\text{Tr}(\mathbf{1} + D_\chi^* D_\chi)^{-s/2} < \infty$$

is governed by

$$\sum_R h(R) e^{-s\beta R},$$

which converges for every $s > 0$ under subexponential shell growth. If D_χ is invertible, replacing the regularised modulus by $|D_\chi|^{-1}$ changes only finitely many low singular values and gives the last assertion. $\square$

The preceding theorem establishes analytic viability: the signed operator is Krein-self-adjoint, has compact resolvent, and is summable under the same shell growth that controlled the positive quotient. It does not yet explain why the exponential dependence on N is forced or how β is chosen. Those are the two logical questions addressed by the characterisation and pressure subsections below.

Remark 3.28 (Position relative to modular spectral triples). The exponential weight $\Delta = e^{\beta N}$ is a Gibbs weight, and the modular automorphism $\sigma_\beta = \text{Ad } \Delta$ makes D_χ an instance of a *modular* (twisted, type III) spectral triple in the sense of Connes–Moscovici [14] and of Carey–Phillips–Rennie and Carey–Neshveyev–Nest–Rennie [31, 32]; its all-orders finite summability under subexponential shell growth lies in an entire/JLO-type regime [46]. The descended renewal shifts generate a Cuntz–Krieger/graph algebra [47, 48], and signed graph weights of the kind used here—weights allowed to carry a sign rather than being positive—already appear in the modular index theory of Carey–Marcolli–Rennie [49]. None of this machinery is claimed as new. What is new is the *role* assigned to the sign: it is the carrier of Lorentzian signature, represented by the Krein fundamental symmetry J_χ and classified by $[\chi] \in H^1(G, \mathbb{Z}/2)$, rather than a device for computing a modular index.

3.5 Checkable conditions for bounded twisted commutators

The previous theorem isolates the analytic condition needed for the signed modular triple. We now give a concrete sufficient condition in terms of the renewal graph. The following graph-theoretic condition gives an explicit sufficient criterion for bounded twisted commutators.

Assumption 3.1 (Bounded geometry and bounded clock increments). The recurrent predictive graph has bounded geometry: for each r there is a uniform bound on the number of vertices in a graph ball of radius r . Moreover there is $L < \infty$ such that every generator edge $e : x \rightarrow y$ satisfies

$$|\Lambda(y) - \Lambda(x)| \leq L.$$

Definition 3.29 (Finite-propagation renewal algebra). An operator a on $\mathcal{K}_\chi$ has propagation $\leq r$ if its matrix coefficient $\langle e_{(y,\eta')}, a e_{(x,\eta)} \rangle$ vanishes whenever the graph distance between x and y is greater than r . Let $\mathcal{A}_{\text{fp}}^\chi$ be the algebra generated by lifted shifts, their adjoints, and finite-propagation diagonal multipliers for which the matrix entries are uniformly bounded. Under bounded geometry these operators are bounded, and products remain finite-propagation.

Lemma 3.30 (Modular conjugation of finite-propagation operators). *Assume [Assumption 3.1](#). If $a \in \mathcal{A}_{\text{fp}}^X$ has propagation $\leq r$, then $\sigma_\beta(a) = \Delta a \Delta^{-1}$ extends to a bounded operator. More explicitly, for a lifted generator $\tilde{S}_e$ one has*

$$\|\sigma_\beta(\tilde{S}_e)\| \leq e^{\beta L} \|\tilde{S}_e\|.$$

For general propagation- r elements, boundedness follows with a constant depending only on r , βL , and the bounded-geometry constants of G . Consequently, if B is bounded and $B \in \mathcal{A}_{\text{fp}}^X$, then

$$[D_\chi, a]_{\rho_\chi} = [B, a]_{\rho_\chi}$$

is bounded for every $a \in \mathcal{A}_{\text{fp}}^X$.

Proof. For a lifted generator $e : x \rightarrow y$,

$$\sigma_\beta(\tilde{S}_e)e_{(x,\eta)} = e^{\beta(\Lambda(y)-\Lambda(x))} e_{(y,\eta\chi(e))},$$

and the multiplier is bounded in modulus by $e^{\beta L}$. This gives the stated bound.

If a has propagation $\leq r$, then every nonzero matrix coefficient connects vertices whose lengths differ by at most rL . Thus conjugation by Δ multiplies each nonzero coefficient by a scalar bounded by $e^{\beta r L}$. Bounded geometry gives a uniform row and column bound for the propagation- r matrix pattern, so the Schur test yields boundedness of $\sigma_\beta(a)$, with a constant depending only on the propagation radius and the bounded-geometry constants.

By [Theorem 3.27](#),

$$[D_\chi, a]_{\rho_\chi} = [B, a]_{\rho_\chi} = Ba - \rho_\chi(a)B.$$

The two terms on the right are bounded because B , a , and $\rho_\chi(a)$ are bounded. $\square$

Corollary 3.31 (Checkable signed modular triple). *Under [Assumption 3.1](#), if $B \in \mathcal{A}_{\text{fp}}^X$ is bounded and J_χ -self-adjoint, then*

$$(\mathcal{A}_{\text{fp}}^X, \mathcal{K}_\chi, D_\chi; \rho_\chi), \quad D_\chi = J_\chi e^{\beta N} + B,$$

is a twisted Krein spectral triple. If the length-shell multiplicities are subexponential, it is s -summable for every $s > 0$.

Proof. This is [Theorem 3.27](#) with the boundedness of the modular conjugates and twisted commutators supplied by [Theorem 3.30](#). $\square$

Remark 3.32 (The degenerate unperturbed case). For $B = 0$ the matched modular twist makes the twisted commutator of $J_\chi \Delta$ vanish. Thus the unperturbed signed modular triple is analytically valid but differentially degenerate. A nontrivial signed differential geometry requires a genuine finite-propagation Dirac-type perturbation B ; the role of the exponential clock is to supply modular time and summability, not by itself to create a nonzero twisted differential.

Remark 3.33 (Holonomy and deficiency). The holonomy class and spectral-radius deficiency are distinct invariants. The class $[\chi]$ controls removability of the signed cover. A deficiency ratio such as

$$r_\chi = \frac{\rho(A_\chi)}{\rho(A)}$$

measures cancellation in a chosen weighted transfer operator. Nontrivial holonomy need not imply a strict spectral-radius drop; periodic or half-period classes can be cohomologically nontrivial but spectrally silent. The present paper uses the holonomy class, the fundamental symmetry, and the signed Dirac operator, not a strict deficiency hypothesis.

3.6 Characterisation of the modular signed Dirac

The preceding construction introduces $D_- = J_\chi e^{\beta N} + B$ and verifies it is Krein-self-adjoint. We show the form $e^{\beta N}$ is not a choice: it is the unique positive operator implementing the renewal modular flow on one-reset operations. The only residual input is the scalar β , which we then tie to a faithful stationary weight.

3.6.1 The deck-odd diagonal normal form

Recall $J_\chi e_{(x,\eta)} = \eta e_{(x,\eta)}$, $N e_{(x,\eta)} = \Lambda(x) e_{(x,\eta)}$, $\tau e_{(x,\eta)} = e_{(x,-\eta)}$. The atomic (diagonal) algebra $\mathcal{D}_\chi$ is generated by the rank-one projections onto the basis vectors $e_{(x,\eta)}$.

Lemma 3.34 (Deck-odd diagonal operators are signed magnitudes). *Let D_0 be affiliated to $\mathcal{D}_\chi$ (i.e. diagonal in the signed-cover basis) and deck-odd, $\tau D_0 \tau^{-1} = -D_0$. Then there is a sheet-independent diagonal operator g , with $g e_{(x,\eta)} = h(x) e_{(x,\eta)}$ for some real $h : V \rightarrow \mathbb{R}$, such that*

$$D_0 = J_\chi g.$$

If in addition D_0 has trivial kernel and $|D_0|$ is its magnitude, then $h > 0$ can be chosen and $g = |D_0| > 0$.

Proof. Diagonality gives $D_0 e_{(x,\eta)} = d(x,\eta) e_{(x,\eta)}$ with $d(x,\eta) \in \mathbb{C}$. Deck-oddness:

$$\tau D_0 \tau^{-1} e_{(x,\eta)} = \tau D_0 e_{(x,-\eta)} = d(x,-\eta) e_{(x,\eta)} \stackrel{!}{=} -d(x,\eta) e_{(x,\eta)},$$

so $d(x,-\eta) = -d(x,\eta)$. Writing $h(x) := d(x, +1)$ gives $d(x, -1) = -h(x)$, hence $d(x,\eta) = \eta h(x)$ and

$$D_0 e_{(x,\eta)} = \eta h(x) e_{(x,\eta)} = J_\chi (h(x) e_{(x,\eta)}),$$

which is $J_\chi g$ with $g e_{(x,\eta)} = h(x) e_{(x,\eta)}$, manifestly sheet-independent. Self-adjointness of D_0 forces h real; trivial kernel forces $h(x) \neq 0$, and $g = J_\chi D_0 = |D_0| > 0$ after fixing the sign $h > 0$. $\square$

So a diagonal, deck-odd unbounded part is *automatically* of the form $J_\chi g$ with g a sheet-independent positive diagonal magnitude. This fixes the factor J_χ from deck oddness.

3.6.2 The renewal modular flow scales elementary resets

Lemma 3.35 (Graded clock scaling). *For every elementary reset edge $e : x \rightarrow y$ (so $\Lambda(y) = \Lambda(x) + 1$),*

$$\text{Ad}(e^{\beta N})(\tilde{S}_e) = e^{\beta N} \tilde{S}_e e^{-\beta N} = e^\beta \tilde{S}_e.$$

Proof. On $e_{(x,\eta)}$,

$$e^{\beta N} \tilde{S}_e e^{-\beta N} e_{(x,\eta)} = e^{-\beta \Lambda(x)} e^{\beta N} \tilde{S}_e e_{(x,\eta)} = e^{-\beta \Lambda(x)} e^{\beta N} e_{(y,\eta \chi(e))} = e^{\beta(\Lambda(y) - \Lambda(x))} e_{(y,\eta \chi(e))} = e^\beta \tilde{S}_e e_{(x,\eta)},$$

using $\Lambda(y) - \Lambda(x) = 1$. As this holds on every basis vector and $\tilde{S}_e$ annihilates vectors off its source fibre, the operator identity follows. $\square$

Thus the modular automorphism $\sigma_\beta := \text{Ad}(e^{\beta N})$ scales every elementary reset by the *single* constant e^β – the renewal incarnation of the Connes–Rovelli thermal-time flow.

3.6.3 The characterisation theorem

Condition 3.1 (Renewal modular Dirac). A closed operator D on $\mathcal{K}_\chi$ is a *renewal modular Dirac* with parameter $\beta > 0$ if:

- (M1) [decomposition] $D = D_0 + B$ with B bounded and J_χ -self-adjoint ($J_\chi B^* J_\chi = B$), and D_0 unbounded and affiliated to the atomic algebra $\mathcal{D}_\chi$;
- (M2) [deck-odd] $\tau D_0 \tau^{-1} = -D_0$;
- (M3) [modular implementation] the magnitude $g := |D_0|$ implements the renewal modular flow on one-reset operations: $\text{Ad}(g)(\tilde{S}_e) = e^\beta \tilde{S}_e$ for every elementary reset edge e ;
- (M4) [recurrence] G^χ is connected (automatic when $[\chi] \neq 0$; otherwise applied componentwise).

Theorem 3.36 (The exponential weight is forced). *Every renewal modular Dirac with parameter β has the form*

$$D = J_\chi e^{\beta N} + B$$

after normalising the magnitude scale to $g e_{(x_0, \eta)} = e_{(x_0, \eta)}$ at a base vertex x_0 with $\Lambda(x_0) = 0$. Conversely $J_\chi e^{\beta N} + B$ is a renewal modular Dirac and is Krein-self-adjoint with respect to J_χ .

Proof. By (M1)–(M2) and [Theorem 3.34](#), $D_0 = J_\chi g$ with $g > 0$ sheet-independent and diagonal. Consider $m := g e^{-\beta N}$, positive and diagonal. For any elementary reset e , [Theorem 3.35](#) and (M3) give, since $g, e^{\beta N}$ commute,

$$\text{Ad}(m)(\tilde{S}_e) = \text{Ad}(g) \text{Ad}(e^{-\beta N})(\tilde{S}_e) = \text{Ad}(g)(e^{-\beta} \tilde{S}_e) = e^{-\beta} e^\beta \tilde{S}_e = \tilde{S}_e.$$

Hence m commutes with every lifted shift. Writing $m e_{(x, \eta)} = \mu(x, \eta) e_{(x, \eta)}$, the relation $m \tilde{S}_e = \tilde{S}_e m$ on $e_{(x, \eta)}$ reads

$$\mu(y, \eta \chi(e)) = \mu(x, \eta) \quad \text{for every edge } e : x \rightarrow y.$$

By (M4) the signed cover is connected, so μ is constant: $\mu \equiv c > 0$, i.e. $m = c \mathbf{1}$ and $g = c e^{\beta N}$. Normalising at x_0 gives $c = 1$, so $g = e^{\beta N}$ and $D_0 = J_\chi e^{\beta N}$.

Conversely J_χ commutes with N , hence with $e^{\beta N}$, so $J_\chi e^{\beta N}$ is Hilbert-self-adjoint and $J_\chi (J_\chi e^{\beta N})^* J_\chi = J_\chi e^{\beta N}$; with $J_\chi B^* J_\chi = B$ this gives $J_\chi D^* J_\chi = D$. Deck-oddness holds since $\tau J_\chi \tau^{-1} = -J_\chi$ and τ commutes with the sheet-independent $e^{\beta N}$; modular implementation is [Theorem 3.35](#). $\square$

Remark 3.37 (Equivalent recurrence formulation). If one keeps shell-scalarly of g as an explicit hypothesis, $g e_{(x, \eta)} = f(\Lambda(x)) e_{(x, \eta)}$, then (M3) reduces on each edge to the linear recurrence $f(n+1) = e^\beta f(n)$, hence $f(n) = f(0) e^{\beta n}$. The proof above is sharper: it *derives* shell-scalarly from (M3) plus connectivity rather than assuming it.

Condition 3.2 (Modular homogeneity of elementary resets). There is a faithful normal weight φ whose modular automorphism group σ_t^φ leaves the recurrent renewal algebra invariant. Every elementary reset shift $\tilde{S}_e$ is analytic and is a modular eigenoperator with one common eigenvalue:

$$\sigma_t^\varphi(\tilde{S}_e) = e^{it\beta} \tilde{S}_e \quad (t \in \mathbb{R}, \text{ all elementary } e),$$

for some $\beta \in \mathbb{R}$. On the represented analytic reset algebra, the commutant of all elementary reset shifts is assumed to be $\mathbb{C} \mathbf{1}$.

Theorem 3.38 (Affine modular Hamiltonian). *Under [Condition 3.2](#), write $\sigma_t^\varphi = \text{Ad}(\Delta_\varphi^{it})$ and $\Delta_\varphi = e^H$ in standard form. Then, on the analytic one-reset algebra,*

$$[H, \tilde{S}_e] = \beta \tilde{S}_e = \beta [N, \tilde{S}_e].$$

Consequently $H - \beta N$ belongs to the reset-shift commutant and hence

$$\boxed{H = \beta N + c\mathbf{1}, \quad \Delta_\varphi = e^{\beta N + c\mathbf{1}}.}$$

After normalising the weight at a base vertex, $c = 0$. Thus the modular Hamiltonian is affine in the renewal clock, and β is the common modular eigenvalue of the elementary resets.

Proof. The eigenoperator relation reads $\text{Ad}(e^{itH})(\tilde{S}_e) = e^{it\beta} \tilde{S}_e$. Differentiating at $t = 0$ gives $[H, \tilde{S}_e] = \beta \tilde{S}_e$. Since every elementary reset raises the grading by one, $[N, \tilde{S}_e] = \tilde{S}_e$, and therefore $[H - \beta N, \tilde{S}_e] = 0$ for every e . The scalar-commutant hypothesis in [Condition 3.2](#) gives $H - \beta N = c\mathbf{1}$. Exponentiation and base normalisation give the asserted formula. $\square$

Remark 3.39 (Sign convention). We define β by $\sigma_t^\varphi(\tilde{S}_e) = e^{it\beta} \tilde{S}_e$, so the corresponding modular operator is $e^{\beta N}$ after normalisation. A thermodynamic convention written with $e^{-N/T}$ describes the inverse modular operator and therefore has $\beta = -1/T$ in this convention. No imaginary-time continuation is used in the theorem.

Remark 3.40 (Scope of the modular characterisation). The functional form $J_\chi e^{\beta N} + B$ is forced by (M1)–(M4), and the value of β is fixed by the common modular eigenvalue of the elementary reset shifts under [Condition 3.2](#). The bounded J_χ -self-adjoint term B remains free and carries the differential content of the geometry; $B = 0$ is differentially degenerate. Thus the signed modular Dirac is intrinsically characterised up to its bounded differential part.

3.7 Pressure selection of the modular scale

The preceding characterisation forces the functional form $e^{\beta N}$. The remaining dimensionless exponent is selected by the recurrent weighted renewal transfer whenever the selected component is supercritical.

Theorem 3.41 (Perron pressure selects the modular exponent). *Let G_0 be a finite strongly connected recurrent component and let*

$$K(s)_{xy} := \sum_{e:x \rightarrow y} q_e e^{-s\ell_e}, \quad q_e > 0, \quad \ell_e \geq \ell_0 > 0.$$

Write $r(s) = \rho(K(s))$. Then r is continuous and strictly decreasing on $[0, \infty)$, and $r(s) \rightarrow 0$ as $s \rightarrow \infty$. Consequently:

- (i) if $r(0) > 1$, there is a unique $\beta > 0$ satisfying

$$\boxed{\rho(K(\beta)) = 1};$$

- (ii) if $r(0) = 1$, the unique nonnegative pressure root is $\beta = 0$;

- (iii) if $r(0) < 1$, no nonnegative pressure root exists.

At the root, the Perron eigenvalue is simple and has strictly positive left and right eigenvectors. Thus, in the supercritical recurrent phase, the renewal transfer selects the parameter appearing in $\Delta = e^{\beta N}$; conversion of this dimensionless modular scale into seconds or energy units is a separate metrological calibration.

Proof. For $s_2 > s_1$, every positive matrix entry associated with an edge is strictly smaller in $K(s_2)$ than in $K(s_1)$, and the two matrices have the same irreducible zero pattern. Perron–Frobenius strict monotonicity therefore gives $r(s_2) < r(s_1)$. Continuity follows from finite dimensionality. Since $\ell_e \geq \ell_0 > 0$ and there are finitely many edges, all entries of $K(s)$ tend to zero, hence $r(s) \rightarrow 0$. The three alternatives follow from the intermediate value theorem and strict monotonicity. Perron–Frobenius also gives simplicity of the spectral radius and strict positivity of its left and right eigenvectors. $\square$

The modular analysis has now removed two apparent choices. Homogeneity forces the functional form $e^{\beta N}$, and supercritical Perron pressure selects the dimensionless value of β . What remains free in the signed Dirac is the bounded differential part B and the separate conversion from renewal units to physical units. We now verify that, despite these enrichments, the signed spectral data still reconstruct the underlying predictive object.

3.8 Signed predictive reconstruction

The classification theorem counts the possible signature sectors over a fixed predictive graph. We next record the reconstruction statement: in a fixed gauge, the signed spectral data recover the predictive graph, its length grading, and the sign cocycle. Modulo flat gauge, the recovered sign invariant is the cohomology class $[\chi]$.

A fixed-gauge signed predictive geometry is the tuple

$$\mathfrak{G}_\chi = (\mathcal{A}_\chi, \mathcal{K}_\chi, N, J_\chi, \tau, \{\tilde{S}_e\}_{e \in E_G}),$$

where $\mathcal{K}_\chi = \ell^2(V^\chi) \otimes \mathcal{H}_{\text{int}}$, N is the lifted length operator, J_χ is the sheet fundamental symmetry, τ is the deck transformation, and the $\tilde{S}_e$ are the lifted renewal shifts.

Lemma 3.42 (Atomic-diagonal rigidity). *Let $\mathcal{D}_\chi$ and $\mathcal{D}_{\chi'}$ be the atomic diagonal algebras generated by the rank-one projections onto the basis vectors of V^χ and $V^{\chi'}$. If a unitary $U : \mathcal{K}_\chi \rightarrow \mathcal{K}_{\chi'}$ satisfies $U\mathcal{D}_\chi U^* = \mathcal{D}_{\chi'}$, then there is a bijection*

$$\psi : V^\chi \longrightarrow V^{\chi'}$$

and phases $\lambda_z \in \mathbb{T}$ such that

$$Ue_z = \lambda_z e'_{\psi(z)}.$$

Proof. The atomic diagonal algebras are maximal abelian with rank-one minimal projections. Conjugation by U carries minimal projections to minimal projections, giving the bijection ψ . Each corresponding one-dimensional range is then mapped by a phase. $\square$

Theorem 3.43 (Signed predictive reconstruction in a fixed gauge). *Let $\mathfrak{G}_\chi$ and $\mathfrak{G}'_{\chi'}$ be fixed-gauge signed predictive geometries. Suppose a unitary $U : \mathcal{K}_\chi \rightarrow \mathcal{K}'_{\chi'}$ satisfies*

$$(R1) \quad U\mathcal{D}_\chi U^* = \mathcal{D}'_{\chi'};$$

$$(R2) \quad U\tau U^* = \tau';$$

(R3) $U\tilde{S}_eU^* = \tilde{S}'_{e'}$, under an induced relabelling of lifted generator transitions;

(R4) $UNU^* = N'$;

(R5) $UJ_\chi U^* = J'_{\chi'}$.

Then U descends to an isomorphism $\varphi : G \rightarrow G'$ of predictive graphs which preserves the length grading:

$$\Lambda'(\varphi(x)) = \Lambda(x),$$

intertwines the generated predictive monoids and orders,

$$(\mathcal{M}, \preceq, \Lambda) \cong (\mathcal{M}', \preceq', \Lambda'),$$

and identifies the fixed-gauge cocycles:

$$\chi(e) = \chi'(\varphi(e)).$$

In particular,

$$[\chi] = \varphi^*[\chi'] \in H^1(G, \mathbb{Z}/2).$$

Proof. By [Theorem 3.42](#), U is a phase-weighted permutation of the signed-cover basis. The condition $U\tau U^* = \tau'$ implies that this permutation maps deck orbits to deck orbits, hence descends to a bijection $\varphi : V \rightarrow V'$ of base vertices. Intertwining the lifted shifts gives

$$\varphi(y) = \varphi(e \cdot x) = e' \cdot \varphi(x)$$

whenever $e : x \rightarrow y$, so φ is an isomorphism of the predictive transition graphs, with the stated relabelling of generator transitions. The equality $UNU^* = N'$ gives $\Lambda'(\varphi(x)) = \Lambda(x)$, so the grading is preserved. Since the predictive order is divisibility in the generated transition monoid, the graph isomorphism gives an isomorphism of graded ordered predictive monoids.

It remains to identify the sign. Let $e : x \rightarrow y$ and lift it from (x, η) to $(y, \eta\chi(e))$. The equality $UJ_\chi U^* = J'_{\chi'}$ fixes the sheet label in the chosen gauge, while the intertwining of lifted shifts sends this lifted edge to the lifted edge over $\varphi(e)$. Therefore the endpoint sheet must be

$$\eta\chi(e) = \eta\chi'(\varphi(e)),$$

for both choices of η , and hence $\chi(e) = \chi'(\varphi(e))$. Taking loop products gives $[\chi] = \varphi^*[\chi']$. $\square$

Corollary 3.44 (Gauge-invariant complete invariant). *Two signed predictive geometries are equivalent up to unitary equivalence and flat gauge if and only if there is a predictive graph isomorphism $\varphi : G \rightarrow G'$ preserving Λ and satisfying*

$$[\chi] = \varphi^*[\chi'] \in H^1(G, \mathbb{Z}/2).$$

Equivalently, the pair

$$\left((\mathcal{M}, \preceq, \Lambda) \text{ up to graded-ordered-monoid isomorphism, } [\chi] \in H^1(G, \mathbb{Z}/2) \right)$$

is a complete invariant of signed predictive geometries modulo flat gauge.

Proof. The forward implication follows from [Theorem 3.43](#), with the observation that composing with a flat gauge changes χ by a coboundary and therefore leaves $[\chi]$ unchanged. Conversely, if $[\chi] = \varphi^*[\chi']$, choose a gauge representative of χ' for which the equality holds as cochains. Then the map

$$e_{(x,\eta)} \longmapsto e'_{(\varphi(x),\eta)}$$

intertwines the diagonal algebra, deck transformation, lifted shifts, length operator, and sheet fundamental symmetry in that gauge. $\square$

Remark 3.45 (Position relative to Connes reconstruction). This is a discrete reconstruction theorem. Connes' reconstruction theorem recovers a Riemannian spin manifold from a commutative spectral triple via orientability, regularity, finiteness, and Poincaré-duality hypotheses [50]. Here the recovered object is not a manifold but a graded ordered predictive monoid decorated by a $\mathbb{Z}/2$ holonomy class. The mechanism is correspondingly simpler: the atomic diagonal recovers the predictive points, the lifted shifts recover the transition monoid, the length operator recovers the grading, and the signed cover recovers the holonomy.

Remark 3.46 (Transition to the continuum endpoint). The result of this section is finite and operator-theoretic. Starting from the positive predictive geometry of [Section 2](#), a sign cocycle produces a principal double cover, a Krein fundamental symmetry, and a signed modular Dirac operator. The non-removable signature sectors are exactly the classes in $H^1(G, \mathbb{Z}/2)$, and the signed spectral data reconstruct the predictive monoid together with this holonomy. The next section derives the projective Clifford factor from reversible predictive dynamics before taking any continuum limit.

The outcome of the signed section is finite and kinematic: it supplies the global sign obstruction, the Krein fundamental symmetry, and the modular clock. It does not yet supply enough noncommutativity for a spatial Clifford algebra. The next section isolates exactly where that missing anticommutation comes from—reversible predictive dynamics on the recurrent internal factor.

4 Reversible revision dynamics and dimensional minimality

The section has six logical steps. Flat signed holonomy is first shown to be insufficient for a Clifford twist. The signed cover nevertheless fixes the entire temporal row of the commutator form. Reversible predictive units then produce the missing spatial projective multiplier automatically. Quotienting its radical removes central spectator directions and leaves a canonical Clifford factor. An explicit graded renewal memory demonstrates that the abstract mechanism is realised. Finally, the rank and spatial nondegeneracy conditions give the $3 + 1$ minimality result and an intrinsic timelike Clifford generator.

The signed cover supplies a flat holonomy class and a distinguished temporal exchange relation, but flat graph topology alone cannot generate a noncommutative Clifford algebra. The missing spatial commutator form is derived from reversible predictive dynamics. Predictive units are automorphisms; on a finite recurrent factor they are inner, so their implementers compose projectively. Factoring the radical of the resulting alternating form gives a canonical symplectic quotient and its Clifford matrix factor. Only after this finite operator-algebraic derivation do we address dimensional minimality.

4.1 Flat holonomy does not produce the Clifford twist

We begin by separating two roles that are easy to conflate. The signed cover of [Section 3](#) provides holonomy classes, hence fundamental symmetries modulo flat gauge. A Clifford algebra, however, requires noncommuting generators, or equivalently a nontrivial commutator bicharacter. Flat graph holonomy by itself has only 1-dimensional topology and therefore cannot supply this twist.

Let G be a recurrent predictive component and write $H = H^1(G, \mathbb{Z}/2)$ for its group of signed holonomy sectors. The classification result of [Theorem 3.7](#) shows that H classifies compatible fundamental symmetries modulo flat gauge. It is tempting to identify H directly with the spatial Clifford grading group. The following observation explains why that identification cannot be a consequence of graph topology alone.

Proposition 4.1 (Flat-holonomy obstruction). *The signed predictive cover is a flat $\mathbb{Z}/2$ -cover. Since a graph is a 1-complex, $H^2(G, \mathbb{Z}/2) = 0$. Consequently any twist obtained only from the topology of the cover, for example by a cup product or a commutator 2-cell, is cohomologically trivial. The topological holonomy sector algebra is therefore the commutative group algebra $\mathbb{C}[H]$, not a Clifford algebra.*

Proof. A connected graph is homotopy equivalent to a wedge of circles, hence has no cohomology in degree 2. The signed cover is classified by a class in $H^1(G, \mathbb{Z}/2)$, so it is flat: it has holonomy around loops but no curvature 2-class. A Clifford exchange sign is a projective commutator bicharacter, equivalently a twist of the *group algebra of the holonomy group*. Such a twist may exist after extra dynamical input, but it is not produced by the topology of the graph cover itself: the graph has no 2-cells on which a curvature class could live. In the flat topological sector the fundamental symmetries multiply as $J_\alpha J_\beta = J_{\alpha+\beta} = J_\beta J_\alpha$, so their algebra is commutative. The spatial-spatial noncommutative Clifford twist must therefore be supplied by modular/internal dynamics rather than by flat topology. The temporal row is an exception: [Theorem 4.5](#) shows that it is forced by the Krein exchange relation of the signed cover. $\square$

Thus topology supplies the signed sectors, but not their Clifford anticommutation. The missing sign must be dynamical: it must come from how revision operations are ordered by the modular renewal clock.

4.2 The temporal row is forced by the signed cover

The flat-cover obstruction does not mean that every entry of the modular commutator form is additional. The temporal row is already fixed by the signed cover.

Definition 4.2 (Realized reversible sign subgroup). Fix the recurrent algebra on which revisions are to act. A class $\alpha \in H = H^1(G, \mathbb{Z}/2)$ is *realized reversibly* if it is represented by a predictive unit preserving that recurrent algebra and admits a clock-preserving signed lift whose sheet action is the geometric sheet shift associated with α . The set of such classes is denoted

$$K_{\text{rev}} \subseteq H.$$

Lemma 4.3 (Realized reversible classes form a subgroup). *One has*

$$K_{\text{rev}} \leq H.$$

Proof. The identity channel realizes the zero class. If revisions realizing α and β are composed, their predictive units compose and their sheet shifts add, so the composite realizes $\alpha + \beta$. The inverse of a predictive unit is again a predictive unit and reverses the same elementary-two sheet class. Hence the realized classes are closed under identity, addition, and inversion. $\square$

Let $\tilde{S}_e$ denote the lifted shift associated with an oriented edge $e : x \rightarrow y$ of the recurrent predictive graph. With

$$J_\chi e_{(x,\eta)} = \eta e_{(x,\eta)}, \quad \tilde{S}_e e_{(x,\eta)} = e_{(y,\eta\chi(e))},$$

the defining exchange relation is

$$J_\chi \tilde{S}_e = \chi(e) \tilde{S}_e J_\chi. \quad (1)$$

For a recurrent cycle representing $\alpha \in H = H^1(G, \mathbb{Z}/2)$, write $\tilde{S}_\alpha$ for the ordered product of its lifted edge shifts and $\chi(\alpha) = (-1)^{\langle [\chi], \alpha \rangle}$ for its sign holonomy.

Lemma 4.4 (Krein exchange along a recurrent cycle). *For every recurrent cycle class α ,*

$$J_\chi \tilde{S}_\alpha = (-1)^{\langle [\chi], \alpha \rangle} \tilde{S}_\alpha J_\chi.$$

The sign depends only on the cohomology–homology pairing and not on the chosen cycle representative.

Proof. Move J_χ through the ordered edge product using (1). The accumulated sign is the product of the edge signs, namely $\chi(\alpha)$. Representative independence is exactly the invariance of the pairing of the cocycle $[\chi]$ with the cycle class. $\square$

Theorem 4.5 (Canonical temporal row of the modular form). *Let $R_t = J_\chi$ and let R_α be any clock-preserving revision operator whose action on the signed sheet is the geometric sheet shift associated with $\alpha \in K_{\text{rev}}$. Then*

$$\omega(\alpha, t) = \langle [\chi], \alpha \rangle \quad (\alpha \in K_{\text{rev}}).$$

Equivalently,

$$J_\chi R_\alpha = (-1)^{\langle [\chi], \alpha \rangle} R_\alpha J_\chi.$$

Thus the temporal row of ω is forced by the signed cover and is not a freely chosen part of the modular revision datum.

Proof. The internal and base-preserving factors of R_α commute with the diagonal sheet operator J_χ , while its sheet-shift factor obeys Theorem 4.4. Hence the commutator of R_α with $R_t = J_\chi$ is the scalar $(-1)^{\langle [\chi], \alpha \rangle}$. Comparing with the definition of the modular commutator form gives the identity. $\square$

Corollary 4.6 (Canonical hyperbolic core). *If the restriction of $[\chi]$ to K_{rev} is nonzero, there is $\alpha_0 \in K_{\text{rev}}$ with $\omega(\alpha_0, t) = 1$. The restriction of ω to $\langle t, \alpha_0 \rangle$ is the standard rank-two alternating form, and the noncentral revision algebra contains a canonical factor*

$$\text{Cl}_2(\mathbb{C}) \cong M_2(\mathbb{C}).$$

This rank-two temporal Clifford core is supplied by the signed cover before any spatial–spatial commutator is specified.

Proof. Nontriviality of the restriction means that the functional $\alpha \mapsto \langle [\chi], \alpha \rangle$ is nonzero on K_{rev} , so some $\alpha_0 \in K_{\text{rev}}$ pairs nontrivially with t . The resulting two-dimensional alternating form is nondegenerate. Its twisted group algebra has noncentral factor $\text{Cl}_2(\mathbb{C}) \cong M_2(\mathbb{C})$. $\square$

Proposition 4.7 (Flat sheet transport has trivial spatial block). *For the undressed flat realization $R_\alpha = \tilde{S}_\alpha$ on the abelian signed cover, restricted to realized reversible classes and with reorderable recurrent base transport,*

$$\omega(\alpha, \beta) = 0 \quad (\alpha, \beta \in K_{\text{rev}}).$$

Consequently the signed cover alone fixes the temporal row but produces no spatial–spatial Clifford twist.

Proof. The sheet contribution to the group commutator is $\chi(\alpha)\chi(\beta)\chi(\alpha)^{-1}\chi(\beta)^{-1} = 1$, and the reorderable base transport contributes no scalar phase. Hence the spatial commutator is trivial. $\square$

Remark 4.8 (The remaining cocycle content). After [Theorem 4.5](#), the only portion of the modular form not fixed by the signed cover is

$$\omega|_{K_{\text{rev}} \times K_{\text{rev}}}.$$

The cover supplies the temporal Clifford pair. Ranks beyond the canonical rank-two core require noncommutative internal transport. Thus only the spatial–spatial block remains to be generated dynamically.

4.3 The modular revision algebra from reversible predictive dynamics

The preceding subsection has separated what is already fixed from what still has to be generated. The signed cover determines how every realised spatial revision commutes with the temporal grading J_χ , but flat sheet transport leaves spatial revisions mutually commuting. We now retain their action on the recurrent matrix factor. Inner implementation then turns ordinary composition of automorphisms into projective composition of unitaries, producing precisely the missing spatial commutator data.

The signed cover supplies the topologically available spatial label module

$$H = H^1(G, \mathbb{Z}/2),$$

while the dynamics realizes only the reversible subgroup $K_{\text{rev}} \leq H$ of [Theorem 4.2](#). The Krein grading supplies the temporal label t . Put

$$H_{\text{full}} = K_{\text{rev}} \oplus \langle t \rangle.$$

The full-access phase is the special case $K_{\text{rev}} = H$. The projective revision law will not be assumed. It follows from reversible predictive channels: units of the predictive channel monoid are automorphisms by [Theorem 2.4](#); on a finite irreducible matrix block those automorphisms are inner; equality of the labelled automorphism products then forces their unitary implementers to compose up to a scalar cocycle.

Assumption 4.1 (Signed recurrent revision phase). The recurrent signed phase has the following properties.

(R1) every $\alpha \in K_{\text{rev}}$ is represented by a predictive unit $\Phi_\alpha \in \mathcal{U}_{\text{pred}}$ preserving the fixed recurrent algebra;

(R2) the chosen labels are compatible with the subgroup law, so labelled concatenation is additive,

$$\Phi_\alpha \Phi_\beta = \Phi_{\alpha+\beta};$$

(R3) the revisions preserve the minimal sufficient recurrent algebra;

(R4) after quotienting revision labels that act trivially, the preserved algebra is a finite matrix factor $\mathcal{A}_{\text{rev}}$, and the induced action is faithful and irreducible.

The temporal revision is represented separately by $R_t = J_\chi$, with the canonical exchange relation of [Theorem 4.5](#).

Theorem 4.9 (Reversible predictive revision theorem). *Under [Assumption 4.1](#), every spatial revision $\alpha \in K_{\text{rev}}$ acts on $\mathcal{A}_{\text{rev}}$ by an involutive $*$ -automorphism,*

$$\Phi_\alpha \in \text{Aut}(\mathcal{A}_{\text{rev}}), \quad \Phi_\alpha^2 = \text{id}, \quad \Phi_\alpha \Phi_\beta = \Phi_{\alpha+\beta}.$$

The action is intrinsic to the predictive channel quotient.

Proof. Each Φ_α is a unit of the predictive channel monoid, hence a $*$ -automorphism by [Theorem 2.4](#). The elementary-two label law gives $\Phi_\alpha^2 = \text{id}$, and labelled concatenation gives the additive composition rule. Preservation of the sufficient algebra and passage to the faithful finite factor give the stated restricted action. $\square$

Theorem 4.10 (Automatic projective lift). *Choose unitary implementers on the matrix factor,*

$$\Phi_\alpha = \text{Ad}(R_\alpha), \quad R_0 = \mathbf{1}.$$

Then there is a normalised scalar two-cocycle $\sigma_s \in Z^2(K_{\text{rev}}, U(1))$ such that

$$R_\alpha R_\beta = \sigma_s(\alpha, \beta) R_{\alpha+\beta}.$$

Changing the implementing section by phases changes σ_s by a coboundary. Its antisymmetric ratio is an intrinsic alternating form

$$\sigma_s(\alpha, \beta) \sigma_s(\beta, \alpha)^{-1} = (-1)^{\omega_s(\alpha, \beta)}, \quad \omega_s \in \text{Alt}^2(K_{\text{rev}}, \mathbb{F}_2).$$

No preferred Kraus section or preferred unitary section is required.

Proof. Every automorphism of a complex matrix factor is inner. The unitaries $R_\alpha R_\beta$ and $R_{\alpha+\beta}$ implement the same automorphism, so $R_\alpha R_\beta R_{\alpha+\beta}^{-1}$ lies in the centre of $\mathcal{A}_{\text{rev}}$ and is a scalar phase. Associativity is exactly the scalar cocycle identity. A scalar change of section produces the usual coboundary. For a scalar cocycle on an abelian group the antisymmetric ratio is a bicharacter; since $2\alpha = 0$, it takes values in $\{\pm 1\}$ and therefore defines the displayed $\mathbb{F}_2$ -valued alternating form. $\square$

Proposition 4.11 (Canonical extension to the full revision group). *Let $R_t = J_\chi$ and write*

$$J_\chi R_\alpha = (-1)^{\langle [\chi], \alpha \rangle} R_\alpha J_\chi \quad (\alpha \in K_{\text{rev}})$$

as in [Theorem 4.5](#). Define

$$R_{\alpha+\epsilon t} := R_\alpha J_\chi^\epsilon, \quad \alpha \in K_{\text{rev}}, \quad \epsilon \in \mathbb{F}_2.$$

Then

$$R_{\alpha+\epsilon t} R_{\beta+\delta t} = \sigma((\alpha, \epsilon), (\beta, \delta)) R_{\alpha+\beta+(\epsilon+\delta)t},$$

where

$$\sigma((\alpha, \epsilon), (\beta, \delta)) = \sigma_s(\alpha, \beta) (-1)^{\epsilon \langle [\chi], \beta \rangle}. \quad (2)$$

This is a normalised scalar cocycle on H_{full} , with alternating form

$$\omega((\alpha, \epsilon), (\beta, \delta)) = \omega_s(\alpha, \beta) + \epsilon \langle [\chi], \beta \rangle + \delta \langle [\chi], \alpha \rangle. \quad (3)$$

Thus the spatial block and the canonical temporal row assemble intrinsically.

Proof. Move J_χ^ϵ through R_β using the exchange relation and then apply [Theorem 4.10](#). Associativity proves the cocycle identity, and taking the antisymmetric ratio gives (3). $\square$

Definition 4.12 (Modular revision datum). The modular revision datum is the derived scalar projective family $\{R_\gamma : \gamma \in H_{\text{full}}\}$ with multiplier $\sigma \in Z^2(H_{\text{full}}, U(1))$. Its finite-dimensional revision algebra is

$$\mathcal{A}_{\text{rev}} := \text{span}_{\mathbb{C}}\{R_\gamma : \gamma \in H_{\text{full}}\} \cong \mathbb{C}_\sigma[H_{\text{full}}].$$

Its gauge-invariant commutator bicharacter is

$$(-1)^{\omega(\alpha, \beta)} = \sigma(\alpha, \beta) \sigma(\beta, \alpha)^{-1}.$$

Remark 4.13 (The datum is derived, not imposed). On the signed recurrent revision phase, neither the scalar multiplier, the projective law, nor bilinearity of the commutator is assumed. They follow from predictive reversibility, the finite-factor reduction, innerness, and associativity. Geometric verticality and exact recurrent covariance may be used as sufficient criteria for identifying the same finite factor, but are not additional hypotheses in the main projective-lift theorem.

Definition 4.14 (Modular commutator form). The modular commutator form is the alternating form ω characterised by

$$R_\alpha R_\beta R_\alpha^{-1} R_\beta^{-1} = (-1)^{\omega(\alpha, \beta)} \mathbf{1}.$$

For a length-shell projection P_N , the same identity holds after compression to every nonempty shell.

Theorem 4.15 (Exact-bicharacter stabilisation). *For the derived modular revision datum, the compressed commutator form is shell-independent:*

$$\omega_N(\alpha, \beta) = \omega(\alpha, \beta)$$

on every nonempty shell.

Proof. The group commutator is the scalar operator $(-1)^{\omega(\alpha, \beta)} \mathbf{1}$ before compression, so compression to any nonzero shell leaves the same scalar. $\square$

4.4 Quadratic refinement and the finite central extension

The alternating form records which revisions anticommute, but it does not record the squares of chosen projective lifts. The quadratic refinement adds this information. For the reader, the key distinction is that the polar form ω is gauge invariant and controls the Clifford algebra, whereas the individual lift squares depend on a projective section and may carry the additional amplitude-level fourth-root phases.

The scalar cocycle determines more than the alternating form. Its lift squares form a quadratic refinement, and the corresponding projective representation has a canonical finite extraspecial realisation in the primitive case.

Theorem 4.16 (Quadratic refinement of the modular form). *After scalar rephasing, the cocycle σ may be chosen $\{\pm 1\}$ -valued:*

$$\sigma(\alpha, \beta) = (-1)^{B(\alpha, \beta)},$$

where B is bilinear and $B - B^\top = \omega$. The lift squares are

$$R_\gamma^2 = (-1)^{q(\gamma)} \mathbf{1}, \quad q(\gamma) := B(\gamma, \gamma),$$

and satisfy

$$q(\alpha + \beta) = q(\alpha) + q(\beta) + \omega(\alpha, \beta).$$

Thus q is a quadratic refinement of ω . Basis lifts may be chosen involutive, but when $\omega \neq 0$ not every lift can square to $\mathbf{1}$ while the cocycle remains $\{\pm 1\}$ -valued.

Proof. For an elementary abelian 2-group, $H^2(H_{\text{full}}, U(1))$ is 2-torsion and every class admits a $\{\pm 1\}$ -valued bilinear representative. The square and polarisation identities follow from $2\gamma = 0$ and bilinearity of B . If q vanished identically, its polar form ω would vanish. $\square$

Theorem 4.17 (Central extension and projective kernel). *Choose projectively equivalent representatives $\tilde{R}_\gamma$ with the $\{\pm 1\}$ -valued cocycle of [Theorem 4.16](#), and let*

$$\Gamma := \langle \tilde{R}_\gamma : \gamma \in H_{\text{full}} \rangle, \quad A := \Gamma \cap U(1) \mathbf{1}.$$

The canonical map

$$\pi_{\text{proj}} : H_{\text{full}} \longrightarrow \Gamma/A, \quad \gamma \longmapsto [\tilde{R}_\gamma],$$

is surjective, with kernel

$$K_{\text{proj}} := \{\gamma : \tilde{R}_\gamma \in A\} \subseteq \text{rad}(\omega).$$

Hence

$$\Gamma/A \cong H_{\text{full}}/K_{\text{proj}}, \quad A \leq \{\pm 1\}.$$

If ω is nondegenerate, then $K_{\text{proj}} = 0$, $A = \{\pm 1\}$, and Γ is an extraspecial 2-group of order

$$|\Gamma| = 2^{1 + \dim_{\mathbb{F}_2} H_{\text{full}}}.$$

The projectively equivalent section may be chosen with $\tilde{R}_t = R_t = J$.

Proof. The quotient map is well defined because products of the rephased representatives differ from the representative of the sum by a sign. Its kernel is exactly K_{proj} , giving the quotient by the first isomorphism theorem. A projectively scalar lift commutes with every other lift, hence lies in $\text{rad}(\omega)$. If ω is nondegenerate, the projective kernel is trivial and some commutator equals $-\mathbf{1}$, so $A = \{\pm 1\}$. Every group element has a unique normal form up to the central sign, giving the stated order and the extraspecial property. The rephasing cochain is determined up to a character; using that freedom one normalises its value at t to one, preserving $R_t = J$. $\square$

Proposition 4.18 (Projective revision law). *The commutator bicharacter*

$$\chi_{\text{rev}}(\alpha, \beta) := \sigma(\alpha, \beta)\sigma(\beta, \alpha)^{-1} = (-1)^{\omega(\alpha, \beta)}$$

is alternating, bilinear over $\mathbb{F}_2$, and invariant under scalar rephasing of the projective section. The quadratic refinement changes by an affine linear term under rephasing, while its polar form ω is gauge invariant.

Proof. The bicharacter identities follow from the scalar cocycle identity on the abelian group H_{full} . A coboundary is symmetric, so it cancels in the antisymmetric ratio. Rephasing changes the lift squares by a character, hence changes q by a linear function without changing its polar form. $\square$

Corollary 4.19 (Amplitude square-root lift of the revision squares). *Let $q : H_{\text{full}} \rightarrow \mathbb{F}_2$ be the lift-square function of Theorem 4.16. A choice of amplitude square roots is equivalently a lift $\tilde{q} : H_{\text{full}} \rightarrow \mathbb{Z}/4$ with $\tilde{q}(\gamma) \bmod 2 = q(\gamma)$, assigning the phase $i^{\tilde{q}(\gamma)}$ to the lift of R_γ . If $q(\gamma) = 1$ for some γ , the corresponding amplitude phase has order four. Hence the nontrivial quadratic-square sector has minimal finite amplitude coefficient group $\mathbb{Z}/4$. The polar commutator form ω is unchanged by the choice of $\tilde{q}$.*

Proof. The identity $R_\gamma^2 = (-1)^{q(\gamma)}\mathbf{1}$ says that the real square of the amplitude lift is determined by $q(\gamma)$. Choosing $\tilde{q}(\gamma) \in \mathbb{Z}/4$ with parity $q(\gamma)$ is exactly choosing a phase whose square is $(-1)^{q(\gamma)}$. If $q(\gamma) = 1$, this phase squares to -1 , so by Theorem 3.23 it generates a cyclic order-four subgroup and no smaller finite phase group can carry it. The commutator form ω is the antisymmetric ratio of products in the real signed quotient; multiplying individual lifts by square-root phases changes lift squares but cancels from the antisymmetric ratio except through the already fixed $\{\pm 1\}$ -valued polar form. $\square$

4.5 Canonical symplectic reduction of the derived revision algebra

Not every revision label contributes to noncommutative geometry. A label in $\text{rad } \omega$ commutes with every other label and therefore acts as central superselection data rather than as a Clifford direction. The canonical operation is to quotient these invisible directions. This is why “primitive” below is a reduction of derived data, not an extra nondegeneracy axiom.

The alternating form is already determined by the automatic projective lift. Primitivity is therefore not an assumption on a postulated form: it is the canonical passage from the derived twisted group algebra to its noncentral symplectic quotient.

Lemma 4.20 (The real polar commutator is $\mathbb{Z}/2$ -valued). *For the derived scalar projective family,*

$$R_a R_b R_a^{-1} R_b^{-1} = \sigma(a, b)\sigma(b, a)^{-1}\mathbf{1} = (-1)^{\omega(a, b)}\mathbf{1},$$

where $\omega \in \text{Alt}^2(H_{\text{full}}, \mathbb{F}_2)$. *The form is independent of the implementing section.*

Proof. The cocycle ratio is a bicharacter on the abelian group H_{full} . Since $2a = 0$, its square is one, so its values lie in $\{\pm 1\}$. Coboundaries are symmetric and cancel from the ratio. $\square$

Definition 4.21 (Primitive revision sector). Set

$$\text{rad } \omega := \{a \in H_{\text{full}} : \omega(a, b) = 0 \text{ for all } b \in H_{\text{full}}\},$$

$$H_{\text{prim}} := H_{\text{full}} / \text{rad } \omega.$$

The induced form is denoted $\bar{\omega}$, and the reduced twisted group algebra by $\mathcal{A}_{\text{rev}}^{\text{prim}}$.

Theorem 4.22 (Radical–centre dictionary). *The centre of the derived revision algebra is*

$$Z(\mathcal{A}_{\text{rev}}) = \text{span}_{\mathbb{C}}\{R_r : r \in \text{rad } \omega\}.$$

It is a commutative unital finite-dimensional C^ -algebra and hence*

$$Z(\mathcal{A}_{\text{rev}}) \cong \mathbb{C}^{2^{\dim_{\mathbb{F}_2} \text{rad } \omega}}.$$

In particular, $\mathcal{A}_{\text{rev}}$ is a factor if and only if $\text{rad } \omega = 0$.

Proof. A homogeneous generator R_r is central exactly when its commutator with every R_b is trivial, equivalently when $r \in \text{rad } \omega$. Linear independence of the homogeneous basis gives the displayed span. It is closed under multiplication and adjoint, and its homogeneous generators commute because ω vanishes on the radical. Its complex dimension is $|\text{rad } \omega| = 2^{\dim \text{rad } \omega}$, so the finite-dimensional commutative C^* -algebra has the stated form. It is one-dimensional exactly when the radical is zero. $\square$

Theorem 4.23 (Block decomposition and factor quotients). *Let*

$$2m := \dim_{\mathbb{F}_2}(H_{\text{full}}/\text{rad } \omega).$$

Then

$$\boxed{\mathcal{A}_{\text{rev}} \cong \bigoplus_{\psi \in \widehat{\text{rad } \omega}} M_{2^m}(\mathbb{C}).}$$

For each character $\psi \in \widehat{\text{rad } \omega}$, the ideal

$$I_\psi := \langle R_r - \psi(r)\mathbf{1} : r \in \text{rad } \omega \rangle$$

gives a factor quotient

$$\mathcal{A}_{\text{rev}}/I_\psi \cong M_{2^m}(\mathbb{C}) \cong \text{Cl}_{2m}(\mathbb{C}).$$

The quotients are pairwise isomorphic but their quotient maps are not canonical. The canonical invariant is the symplectic quotient $(H_{\text{prim}}, \bar{\omega})$ and the common matrix/Morita type; choosing a specific factor representation additionally requires a central character ψ .

Proof. By [Theorem 4.22](#), the centre decomposes into minimal central projections indexed by $\widehat{\text{rad } \omega}$. On each central block the radical acts by the corresponding character, while the induced form on H_{prim} is nondegenerate. A symplectic basis identifies the resulting twisted group algebra with the tensor product of m rank-two Weyl pairs, namely $M_{2^m}(\mathbb{C})$. Summing the central blocks gives the decomposition. Different characters select different central projections, so the abstract factor type is canonical but a quotient map is not. $\square$

Theorem 4.24 (Canonical primitive Clifford factor). *The induced form $\bar{\omega}$ on H_{prim} is nondegenerate and therefore has even rank $2m$. Every factor representation of $\mathcal{A}_{\text{rev}}$ has the canonical matrix type*

$$\boxed{\mathcal{A}_{\text{rev}}^{\text{prim}} \cong M_{2^m}(\mathbb{C}) \cong \text{Cl}_{2m}(\mathbb{C}).}$$

It carries the unique faithful irreducible projective representation of the nondegenerate quotient, of dimension 2^m , up to unitary equivalence.

Proof. Well-definedness of $\bar{\omega}$ follows from quotienting by the radical. If $[a]$ lies in the radical of $\bar{\omega}$, then $a \in \text{rad } \omega$, so $[a] = 0$; thus $\bar{\omega}$ is nondegenerate. A nondegenerate alternating form has even rank. Choosing a symplectic basis gives m finite Weyl pairs, and finite Stone–von Neumann gives the unique irreducible projective representation of dimension 2^m , with full matrix algebra $M_{2^m}(\mathbb{C}) \cong \text{Cl}_{2m}(\mathbb{C})$. $\square$

The preceding theorem is the algebraic turning point of the paper. Before the radical quotient, the revision algebra may contain several central blocks. After the quotient, every factor has the same full matrix type $M_{2^m}(\mathbb{C})$, equivalently the same complex Clifford algebra. Choosing a central character selects a representation block, but it does not change the primitive Clifford geometry.

Theorem 4.25 (Internal revision twirl). *If the primitive quotient has rank $2m$ and acts irreducibly on $M_{2^m}(\mathbb{C})$ through projective implementers R_α , then*

$$\frac{1}{|H_{\text{prim}}|} \sum_{\alpha \in H_{\text{prim}}} R_\alpha X R_\alpha^* = \frac{\text{Tr } X}{2^m} \mathbf{1}.$$

Consequently the internal fixed algebra is $\mathbb{C}\mathbf{1}$.

Proof. The range of the twirl is the commutant of the irreducible projective representation, hence is scalar by Schur’s lemma. Trace preservation fixes the scalar. $\square$

Theorem 4.26 (Sheet ergodicity of the maximal recurrent cover). *For the maximal elementary-two signed cover with tautological surjective holonomy, recurrent first-return loops act transitively on each sheet fibre. The only recurrently invariant sheet functions are constants.*

Proof. First-return loops generate the recurrent fundamental group, and surjectivity of the tautological holonomy makes their deck transformations generate the full sheet group, which acts transitively on its torsor. $\square$

Corollary 4.27 (Double ergodicity is derived in the primitive phase). *After passage to an irreducible primitive factor and the maximal recurrent signed cover, internal and sheet ergodicity follow from the construction. They need not be imposed as independent scalar-centralisation assumptions.*

Corollary 4.28 (Primitive data are not an input). *The primitive modular revision sector is the canonical symplectic reduction of the projective action derived in [Theorems 4.10](#) and [4.11](#). “Primitive” means passage to the noncentral factor type, not an assumed nondegeneracy of a separately postulated form. Its Clifford rank is determined by $\dim_{\mathbb{F}_2} H_{\text{prim}}$, independently of the central character used to realise one factor block.*

Theorem 4.29 (Structure and rigidity of the revision sector). *Under [Assumption 4.1](#), the spatial projective multiplier and its alternating form are automatic, the temporal grading extends them canonically to H_{full} , and the commutator form is shell-independent. After canonical symplectic reduction, if $\dim_{\mathbb{F}_2} H_{\text{prim}} = 1 + d_{\text{Cl}}$, then*

$$\mathcal{A}_{\text{rev}}^{\text{prim}} \cong \text{Cl}_{1+d_{\text{Cl}}}(\mathbb{C}) \cong M_{2^{(1+d_{\text{Cl}})/2}}(\mathbb{C}), \quad d_{\text{Cl}} = \dim_{\mathbb{F}_2} H_{\text{prim}} - 1 \text{ is odd.}$$

The irreducible primitive revision representation is unique up to unitary equivalence; a realization of one factor block is parametrised only by a central character of the radical.

Proof. Automatic projectivity is [Theorem 4.10](#); the temporal extension is [Theorem 4.11](#); shell independence is [Theorem 4.15](#); and the factor/Clifford statement is [Theorems 4.23](#) and [4.24](#). Evenness of the nondegenerate alternating rank makes $1 + d_{\text{Cl}}$ even. $\square$

Corollary 4.30 (Full-access revision phase). *If*

$$K_{\text{rev}} = H^1(G, \mathbb{Z}/2),$$

then

$$H_{\text{full}} = H^1(G, \mathbb{Z}/2) \oplus \langle t \rangle,$$

and all projective-lift, radical-reduction, and primitive Clifford conclusions of this section apply to the complete signed cohomology module. Thus the earlier full- H^1 formulation is recovered as a corollary, while the main theorems require only the dynamically realized reversible subgroup.

Proof. This is the specialization of [Theorems 4.10](#), [4.11](#) and [4.29](#) to $K_{\text{rev}} = H^1(G, \mathbb{Z}/2)$. $\square$

Corollary 4.31 (Universality of the Clifford datum). *On the reversible, faithful finite-factor signed phase, neither the scalar projective law nor the primitive Clifford representation is an additional hypothesis. The graded Clifford renewal memory constructed below is an explicit representative of the unique primitive matrix type.*

Remark 4.32 (Sharp scope). The predictive-unit theorem does not assert that every CP renewal tuple has a nontrivial reversible phase, nor does it derive full access to the elementary-two label module. The remaining hypotheses are precisely the existence of a realized reversible subgroup, preservation of a finite sufficient factor, and faithfulness after quotienting neutral labels. Once those hold, projectivity, the commutator form, its radical, and the primitive Clifford type are derived. The $\mathbb{Z}/4$ amplitude lift records square-root phases but does not alter the real polar form.

4.6 Intrinsic generation by a graded Clifford renewal memory

The preceding results still allow the projective revision datum to be external to the renewal memory. We now exhibit a fully specified completely positive memory for which the signed reset operators themselves generate the required projective law and its shell-stable commutator form.

Construction 4.33 (Graded Clifford renewal memory). Fix an odd integer $d_{\text{Cl}} \geq 1$, put $n = 1 + d_{\text{Cl}}$, and let $\Gamma_0, \dots, \Gamma_{d_{\text{Cl}}}$ be Hermitian generators of $\text{Cl}_n(\mathbb{C})$ on its irreducible spinor module S , so that

$$\Gamma_\mu \Gamma_\nu + \Gamma_\nu \Gamma_\mu = 2\delta_{\mu\nu} \mathbf{1}.$$

Take

$$E = \{a_0, \dots, a_{d_{\text{Cl}}}\}, \quad \mathcal{A}_{\text{int}} = \text{Cl}_n(\mathbb{C}), \quad \Phi_{a_\mu} = \text{Ad}(\Gamma_\mu).$$

Each reset channel is a unital $*$ -automorphism and hence completely positive. On the signed lift retain the implementing operator,

$$\hat{S}_{a_\mu} = \tilde{S}_\mu \otimes \Gamma_\mu.$$

For $\alpha = (\alpha_0, \dots, \alpha_{d_{\text{Cl}}}) \in \mathbb{F}_2^n$, define

$$\Gamma^\alpha = \Gamma_0^{\alpha_0} \dots \Gamma_{d_{\text{Cl}}}^{\alpha_{d_{\text{Cl}}}}, \quad R_\alpha = \mathbf{1} \otimes \Gamma^\alpha.$$

We identify the rank- n noncentral revision block with $H_{\text{full}} = \langle t, e_1, \dots, e_{d_{\text{Cl}}} \rangle$, with $R_t = \Gamma_0$.

Remark 4.34 (Grading generator versus timelike Clifford generator). In the primitive factor, $R_t = \Gamma_0$ represents the image of the Krein grading J_χ . It is a Hermitian involution and is not itself the anti-Hermitian Lorentzian generator. Choosing, for example, $\alpha_0 = e_1$, one has

$$\omega(\alpha_0, t) = 1, \quad R_{\alpha_0} = \Gamma_1,$$

and the physical timelike Clifford generator is

$$\gamma^0 = R_t R_{\alpha_0} = \Gamma_0 \Gamma_1, \quad (\gamma^0)^* = -\gamma^0, \quad (\gamma^0)^2 = -\mathbf{1}.$$

Thus R_t carries the indefinite grading, while the canonical hyperbolic pair (R_t, R_{α_0}) carries the Lorentzian temporal square.

The completely positive quotient sees only the conjugation channels. Those channels commute because the central sign in $\Gamma_\mu \Gamma_\nu = -\Gamma_\nu \Gamma_\mu$ disappears under Ad. The signed lift retains precisely the sign that the positive quotient forgets.

Lemma 4.35 (Clifford commutator form). *For $\alpha, \beta \in \mathbb{F}_2^n$,*

$$\Gamma^\alpha \Gamma^\beta = (-1)^{\omega(\alpha, \beta)} \Gamma^\beta \Gamma^\alpha, \quad \omega(\alpha, \beta) = \left(\sum_\mu \alpha_\mu \right) \left(\sum_\nu \beta_\nu \right) - \sum_\mu \alpha_\mu \beta_\mu \pmod{2}.$$

In the generator basis, the Gram matrix is $\Omega = J_n - I_n$. It is nonsingular exactly when n is even.

Proof. Reordering $\Gamma^\alpha \Gamma^\beta$ requires one sign for each pair of distinct occupied indices and no sign for coincident indices, giving the displayed exponent. It is bilinear and alternating over $\mathbb{F}_2$. On basis vectors it equals $1 - \delta_{\mu\nu}$, hence $\Omega = J_n - I_n$. Over $\mathbb{F}_2$, $(J_n - I_n)^2 = I_n$ when n is even; when n is odd the all-ones vector lies in its kernel. $\square$

Theorem 4.36 (Intrinsic primitive and shell-stable revision block). *For the graded Clifford renewal memory, the operators R_α satisfy the projective law*

$$R_\alpha R_\beta = \tau(\alpha, \beta) R_{\alpha+\beta}, \quad \tau(\alpha, \beta) = (-1)^{\sum_{\mu > \nu} \alpha_\mu \beta_\nu},$$

and are linearly independent. Their commutator is the exact operator identity

$$R_\alpha R_\beta R_\alpha^{-1} R_\beta^{-1} = (-1)^{\omega(\alpha, \beta)} \mathbf{1}.$$

Consequently $\omega_N \equiv \omega$ on every nonempty shell. For every odd d_{Cl} , the rank- $(1 + d_{\text{Cl}})$ noncentral revision block is primitive and

$$\mathcal{A}_{\text{rev}} \cong \text{Cl}_{1+d_{\text{Cl}}}(\mathbb{C}), \quad \text{rank}_{\mathbb{F}_2} \Omega = 1 + d_{\text{Cl}}.$$

In particular, for $d_{\text{Cl}} = 3$, $\Omega = J_4 - I_4$ has rank four and $\mathcal{A}_{\text{rev}} \cong \text{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C})$.

Proof. The cocycle formula follows by putting each Clifford monomial back into the fixed ordered basis. The monomials form a vector-space basis of $\text{Cl}_n(\mathbb{C})$, hence are linearly independent. They act only on the internal signed fibre, so they commute with the renewal clock; the temporal grading generator is the Hermitian involution $R_t = \Gamma_0$, while the physical timelike generator is the composite described in [Theorem 4.34](#). The exact commutator identity is [Theorem 4.35](#), and [Theorem 4.15](#) gives shell independence. Since $n = 1 + d_{\text{Cl}}$ is even, $J_n - I_n$ is nonsingular, so the radical vanishes on the noncentral block and [Theorem 4.24](#) identifies the twisted group algebra with the full Clifford matrix algebra. $\square$

Remark 4.37 (What has and has not been derived). For this named model, existence of the projective operators, their linear independence, primitivity, and shell stabilisation are direct calculations. By [Theorem 4.29](#), however, the resulting primitive block is not special: every local, vertical, spatially exactly covariant, doubly ergodic primitive revision sector is unitarily equivalent to it. The full predictive graph may still contain additional central cohomology directions; these are factored through the radical before applying the primitive rank theorem. Universality therefore applies to the noncentral primitive factor, while sectors outside the doubly ergodic phase may retain operator-valued or sheet-dependent commutator data.

Corollary 4.38 (Unconditional minimality in the graded class). *Within the graded Clifford class, $2 + 1$ is impossible, $1 + 1$ is spatially degenerate, and $3 + 1$ is the minimal spatially nondegenerate Lorentzian Dirac endpoint, with no separate existence or stabilisation hypothesis. Higher odd spatial ranks remain admissible, so this is a minimality theorem rather than a uniqueness theorem.*

Proof. By [Theorem 4.36](#), the primitive revision block exists and stabilises for every odd d_{Cl} . The parity and spatial nondegeneracy argument of [Theorem 4.40](#) then gives the claim. The same construction for $d_{\text{Cl}} = 5, 7, \dots$ proves that no higher odd rank is excluded without an additional selection principle. $\square$

The rank criterion tells us which Clifford ranks are possible. It does not yet distinguish a spatially meaningful endpoint from a one-axis toy model. We make that distinction explicit before stating the minimality theorem.

Definition 4.39 (Spatially nondegenerate endpoint). A Lorentzian renewal endpoint is *spatially nondegenerate* if its spatial revision module has a nonzero two-direction sector,

$$\bigwedge^2 V_{\text{sp}} \neq 0.$$

Equivalently, the endpoint supports a nontrivial spatial plaquette or antisymmetric two-reset interference observable. This excludes the one-axis case $d_{\text{Cl}} = 1$ for an algebraic reason intrinsic to the renewal construction. Since a primitive alternating form forces d_{Cl} to be odd, the first spatially nondegenerate primitive rank is $d_{\text{Cl}} = 3$.

4.7 Unconditional minimality of the $3 + 1$ endpoint

We can now state the unconditional dimension result. The only inputs are primitivity of the modular revision algebra and the spatial nondegeneracy condition just defined. No access principle is used in this subsection.

The parity consequence should be read in relation to existing dimension-signature arguments. Mankoč Borštnik and Nielsen proved that linear-in-momentum equations for nonzero spin in even spacetime dimension, subject to Hermiticity and Lorentz irreducibility, allow only signatures with an odd number of time directions; van Dam and Ng gave a Wigner-representation argument against multiple time directions; and Nielsen and Rugh observed that Weyl-matrix counting singles out $d = 4$ through the equality $n_{\text{Weyl}}^2 = d$ among nontrivial even-dimensional Weyl systems [\[33, 34, 35\]](#). The theorem below uses a different microscopic input: the nondegenerate alternating $\mathbb{F}_2$ -commutator form is generated by renewal holonomy itself, and the classical parity conclusion is recovered as a consequence of that generated symplectic rank.

Theorem 4.40 (Minimal primitive endpoint with nontrivial spatial bivector sector). *Let G be a primitive signed renewal component with*

$$H_{\text{full}} = K_{\text{rev}} \oplus \langle t \rangle, \quad d_{\text{Cl}} := \dim_{\mathbb{F}_2} K_{\text{rev}}.$$

In the full-access phase, $d_{\text{Cl}} = b_1(G)$. Then:

- (i) $1 + d_{\text{Cl}}$ is even, hence d_{Cl} is odd;
- (ii) $2 + 1$ is impossible in the primitive class, since $1 + d_{\text{Cl}} = 3$ carries no nondegenerate alternating $\mathbb{F}_2$ -form;
- (iii) $1 + 1$ is primitive but spatially degenerate in the sense of [Theorem 4.39](#);
- (iv) the minimal spatially nondegenerate primitive endpoint is

$$\boxed{d_{\text{Cl}} = 3},$$

hence the minimal nondegenerate Lorentzian Dirac endpoint is $3 + 1$.

This statement uses only primitivity and the nontrivial spatial-bivector requirement; no access or cost selection principle is used.

Proof. A nondegenerate alternating bilinear form over $\mathbb{F}_2$ exists only in even dimension. Primitivity is nondegeneracy of ω on H_{full} , so $1 + d_{\text{Cl}} = \dim_{\mathbb{F}_2} H_{\text{full}}$ must be even and d_{Cl} must be odd. Thus $d_{\text{Cl}} = 2$ is forbidden. For $d_{\text{Cl}} = 1$, $H_{\text{full}} \cong (\mathbb{Z}/2)^2$ admits the standard symplectic form, so primitivity is possible; but $\bigwedge^2 \mathbb{R} = 0$, so it has no nonzero spatial plaquette or two-reset interference sector and is spatially degenerate by [Theorem 4.39](#). The next odd spatial rank is $d_{\text{Cl}} = 3$, for which $\bigwedge^2 \mathbb{R}^3 \neq 0$. Hence $3 + 1$ is the minimal spatially nondegenerate primitive endpoint. $\square$

Proposition 4.41 (Primitive revision data exist in every odd spatial rank). *For every odd d_{Cl} there exists a recurrent signed graph with $b_1 = d_{\text{Cl}}$ and a modular revision datum on $H_{\text{full}} = H^1(G, \mathbb{Z}/2) \oplus \langle t \rangle$ whose commutator form is nondegenerate. Consequently the primitive revision-algebra axioms do not select $d_{\text{Cl}} = 3$ over $d_{\text{Cl}} = 5, 7, \dots$*

Proof. Take a recurrent component whose graph is the bouquet of d_{Cl} loops on one vertex, so $b_1 = d_{\text{Cl}}$. Since d_{Cl} is odd, $1 + d_{\text{Cl}}$ is even, and $H_{\text{full}} \cong (\mathbb{Z}/2)^{1+d_{\text{Cl}}}$ carries the standard symplectic form

$$\Omega = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

after choosing a symplectic basis. Define a twisted group algebra $\mathbb{C}^\tau[H_{\text{full}}]$ whose commutator bicharacter is $(-1)^\Omega$; equivalently, take the finite Heisenberg representation associated with this symplectic $\mathbb{F}_2$ -space. This is a modular revision datum in the sense of [Theorem 4.12](#), and its commutator form is nondegenerate. Thus no algebraic obstruction in the primitive revision axioms excludes higher odd Clifford ranks. $\square$

Remark 4.42 (Minimality, not unconditional uniqueness). [Theorem 4.40](#) proves that $3 + 1$ is the first nondegenerate primitive endpoint and that $2 + 1$ is impossible. It does not prove that primitive dynamics alone chooses $d_{\text{Cl}} = 3$. The higher odd primitive endpoints are algebraically admissible by [Theorem 4.41](#); a selection statement requires an additional principle.

The minimality theorem counts the admissible Clifford directions, but a Lorentzian continuum also requires an explicit generator whose square has the timelike sign. The final two results of this section construct that generator from the canonical hyperbolic pair and show that it admits the required spatial Clifford completion. No arbitrary phase choice is used to manufacture the negative square.

Proposition 4.43 (Intrinsic timelike square from a hyperbolic pair). *Let $\alpha_0 \in K_{\text{rev}}$ satisfy $\omega(\alpha_0, t) = 1$. Normalize the order-two projective implementers so that*

$$R_t = J_\chi = R_t^* = R_t^{-1}, \quad R_{\alpha_0} = R_{\alpha_0}^* = R_{\alpha_0}^{-1}.$$

Such a normalization is unique up to independent signs. Then

$$\boxed{\gamma^0 := J_\chi R_{\alpha_0}}$$

is anti-Hermitian, Krein-odd, and has the forced square

$$(\gamma^0)^* = -\gamma^0, \quad J_\chi \gamma^0 J_\chi = -\gamma^0, \quad (\gamma^0)^2 = -\mathbf{1}.$$

Thus the Lorentzian temporal square is fixed by the normalized hyperbolic pair and is independent of every scalar rephasing compatible with the Hermitian involution normalization.

Proof. The temporal-row theorem gives $J_\chi R_{\alpha_0} = -R_{\alpha_0} J_\chi$. Hence

$$(J_\chi R_{\alpha_0})^* = R_{\alpha_0} J_\chi = -J_\chi R_{\alpha_0}, \quad (J_\chi R_{\alpha_0})^2 = -J_\chi^2 R_{\alpha_0}^2 = -\mathbf{1}.$$

Conjugation by J_χ gives Krein oddness. An order-two inner automorphism has an implementing unitary whose square is scalar; scalar normalization makes it a Hermitian involution, after which the residual phase freedom is only ± 1 , which does not alter the displayed square. $\square$

Lemma 4.44 (Clifford completion of the normalized temporal generator). *Suppose the primitive quotient has rank $1 + d_{\text{Cl}} = 2m$, so that its factor is $M_{2m}(\mathbb{C}) \cong \text{Cl}_{1+d_{\text{Cl}}}(\mathbb{C})$. The operator $\gamma^0 = J_\chi R_{\alpha_0}$ of Theorem 4.43 can be completed by Hermitian involutions $\gamma^1, \dots, \gamma^{d_{\text{Cl}}}$ satisfying*

$$J_\chi \gamma^i J_\chi = \gamma^i, \quad \{\gamma^0, \gamma^i\} = 0, \quad \{\gamma^i, \gamma^j\} = 2\delta^{ij} \mathbf{1}.$$

Any two such completions determine unitarily equivalent representations of the full complex Clifford algebra.

Proof. The anticommuting Hermitian involutions J_χ and R_{α_0} form a hyperbolic pair. Their $+1$ and -1 eigenspaces have equal dimension, and after a unitary change of basis they take the form

$$J_\chi = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \mathbf{1}, \quad R_{\alpha_0} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \mathbf{1}.$$

Consequently

$$\gamma^0 = J_\chi R_{\alpha_0} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \otimes \mathbf{1}.$$

Since $d_{\text{Cl}} = 2m - 1$, choose Hermitian generators $\eta_1, \dots, \eta_{d_{\text{Cl}}}$ of the irreducible complex $\text{Cl}_{d_{\text{Cl}}}(\mathbb{C})$ -module on $\mathbb{C}^{2^{m-1}}$, and set

$$\gamma^i = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \eta_i.$$

These operators are Hermitian involutions, commute with J_χ , anticommute with γ^0 , and satisfy the required spatial Clifford relations. The resulting $1 + d_{\text{Cl}} = 2m$ generators give an irreducible representation of the simple complex algebra $\text{Cl}_{2m}(\mathbb{C}) \cong M_{2^m}(\mathbb{C})$. Unitary equivalence of any two full completions follows from uniqueness of its irreducible representation. $\square$

Sections 3 and 4 have now produced all finite algebraic ingredients needed for propagation: a fundamental symmetry, one normalized timelike generator, and a compatible spatial Clifford family. What has not yet been shown is that these finite data possess a controlled large-scale geometric limit. This is the task of the next section.

5 The Lorentzian Dirac continuum

The main line of the section separates four levels of the continuum claim. First, the derived Clifford generators and spatial second moments determine a Lorentzian principal symbol. Second, finite-difference Hamiltonians converge to the corresponding flat Dirac evolution. Third, stationary ergodic reset statistics make the flat coframe self-averaging. Fourth, when the statistics vary slowly, channel plaquettes reconstruct Levi–Civita spin transport and the covariant Hamiltonians converge through subprincipal order. This separation keeps signature, operator convergence, probabilistic selection, and curvature conceptually distinct.

The preceding section has already produced the finite Clifford factor from reversible predictive dynamics. We now pass from that derived operator algebra to its continuum Dirac symbol. The deterministic product-order continuum and its polyhedral obstruction are recorded separately in [Section D](#); the main line begins with the signed spectral cone, then proves operator convergence, self-averaging, and the slowly varying Levi–Civita limit.

5.1 Signed Clifford symbols and Lorentzian frame universality

The continuum ladder has identified the obstruction: a deterministic finite product order gives a polyhedral cone. We now move from order alone to the signed Dirac symbol. The guiding idea is simple. The signed sector supplies the timelike Clifford generator, while a large family of reset directions supplies the spatial second moment. A nondegenerate second moment supplies a spatial vielbein; literal isotropy is only a convenient representative of the resulting flat isometry class.

There is, however, an important structural boundary. Positive averaging by itself cannot create Lorentzian signature. The tensor $\sum_a p_a v_a \otimes v_a$ formed from real reset directions and positive weights is positive semidefinite. Thus isotropisation supplies roundness, but the fundamental symmetry J_χ supplies the indefinite sign. We first record this positivity obstruction, then introduce the Krein–Clifford reset datum, then prove the model isotropic limit and its general form.

5.1.1 Positive averaging is Riemannian

Proposition 5.1 (Symbol-level positivity obstruction). *Let $v_1, \dots, v_m \in \mathbb{R}^n$ be real vectors and $p_1, \dots, p_m \geq 0$. Then the tensor*

$$G = \sum_{a=1}^m p_a v_a \otimes v_a$$

is positive semidefinite, and positive definite if $\text{span}\{v_a\} = \mathbb{R}^n$. In particular G is never of Lorentzian signature, and the quadratic form $\xi \mapsto G(\xi, \xi)$ has trivial null cone $\{0\}$ when G is definite.

Proof. For every covector ξ , $G(\xi, \xi) = \sum_a p_a (v_a \cdot \xi)^2 \geq 0$, with equality forcing $v_a \cdot \xi = 0$ for all a with $p_a > 0$. If the v_a span, this forces $\xi = 0$. A positive semidefinite form has no negative eigenvalue, hence cannot have signature $(1, n-1)$. $\square$

Theorem 5.1 is the principal-symbol shadow of **Theorem 2.25**: completely positive data, averaged with positive weights, remain Riemannian. The remedy is the same as in **Section 3**, and we now make it explicit at the level of the Clifford symbol.

5.1.2 The derived Krein–Clifford datum

Definition 5.2 (Derived Krein–Clifford reset datum). Let $J = J^* = J^{-1}$ be the fundamental symmetry of the signed cover, and let α_0 be a revision class with $\omega(\alpha_0, t) = 1$. With the Hermitian involutive normalization of **Theorem 4.43**, define

$$\gamma^0 := JR_{\alpha_0}.$$

Choose the compatible Clifford completion of **Theorem 4.44**, consisting of Hermitian spatial generators $\gamma^1, \dots, \gamma^{d_{\text{Cl}}}$ satisfying

$$J\gamma^i J = \gamma^i, \quad \{\gamma^0, \gamma^i\} = 0, \quad \{\gamma^i, \gamma^j\} = 2\delta^{ij}\mathbf{1}.$$

Then

$$(\gamma^0)^* = -\gamma^0, \quad (\gamma^0)^2 = -\mathbf{1}, \quad J\gamma^0 J = -\gamma^0.$$

For $\hat{\theta} \in \mathbb{S}^{d_{\text{Cl}}-1}$, write $\gamma(\hat{\theta}) = \sum_i \hat{\theta}^i \gamma^i$, so that $\gamma(\hat{\theta})^2 = \mathbf{1}$.

Definition 5.3 (Direction measure and coarse-grained Dirac symbol). A direction system is a family $\{(\hat{\theta}_a, p_a)\}_{a=1}^m$ with $\hat{\theta}_a \in \mathbb{S}^{d_{\text{Cl}}-1}$, $p_a > 0$, and $\sum_a p_a = 1$. Its empirical measure and second moment are

$$\nu = \sum_a p_a \delta_{\hat{\theta}_a}, \quad M(\nu)^{ij} = \int_{\mathbb{S}^{d_{\text{Cl}}-1}} \hat{\theta}^i \hat{\theta}^j d\nu(\hat{\theta}).$$

For $\kappa > 0$, define

$$\sigma_\nu(\xi) = i \left(\gamma^0 \xi_0 + \kappa \int_{\mathbb{S}^{d_{\text{Cl}}-1}} \gamma(\hat{\theta}) (\hat{\theta} \cdot \vec{\xi}) d\nu(\hat{\theta}) \right).$$

Equivalently, the spatial term is $\kappa \gamma^i M(\nu)^{ij} \xi_j$.

Lemma 5.4 (Forced Lorentzian symbol square). *For every direction measure ν ,*

$$\sigma_\nu(\xi)^2 = \left(\xi_0^2 - \kappa^2 \vec{\xi}^{\top} M(\nu)^2 \vec{\xi} \right) \mathbf{1} = g_\nu^{\mu\nu} \xi_\mu \xi_\nu \mathbf{1},$$

where

$$g_\nu^{-1} = \text{diag}(1, -\kappa^2 M(\nu)^2).$$

Proof. The temporal square contributes $-i^2 \xi_0^2 = +\xi_0^2$, the spatial square contributes $-\kappa^2 \vec{\xi}^{\top} M(\nu)^2 \vec{\xi}$, and the mixed terms vanish by $\{\gamma^0, \gamma^i\} = 0$. $\square$

Theorem 5.5 (Forced single-time Lorentzian signature). *If the reset directions span the spatial module, equivalently $M(\nu) \succ 0$, then g_ν^{-1} has signature $(1, d_{\text{Cl}})$. Its unique positive eigenvalue is the temporal slot generated by the canonical hyperbolic pair (J, R_{α_0}) ; the positive spatial averaging contributes only the negative-definite block $-\kappa^2 M(\nu)^2$. In particular, no Euclidean branch or multiple-time branch remains after the normalized derived temporal generator is used.*

Proof. Positive definiteness of $M(\nu)$ makes $-\kappa^2 M(\nu)^2$ negative definite, while the temporal coefficient is $+1$ by [Theorems 4.43](#) and [5.4](#). Hence the inertia is exactly $(1, d_{\text{Cl}})$. $\square$

The signature theorem makes the division of labour transparent. The positive second moment determines only the shape and scale of the spatial quadratic form. The single opposite sign is inherited from the Krein–Clifford temporal generator constructed in the preceding section. Thus no negative probability, indefinite covariance, or analytic continuation is used to create time.

Remark 5.6 (Phase independence of the temporal sign). The metric sign is not the square of an arbitrarily phased projective lift. It is the square of the composite JR_{α_0} after the canonical Hermitian-involution normalization of the order-two factors. The residual ± 1 choices leave $(\gamma^0)^2 = -1$ unchanged.

5.1.3 Full generality

The isotropic theorem is the round special case of a more general statement. A non-isotropic but nondegenerate second moment still gives a Lorentzian quadric; it is simply an anisotropic one until the spatial frame is rescaled.

Theorem 5.7 (General Clifford rounding). *Let $\{\nu_R\}$ be any sequence of direction measures with weak-* limit ν on $\mathbb{S}^{d_{\text{Cl}}-1}$ whose second moment $M(\nu) \succ 0$ (equivalently, $\text{supp } \nu$ is not contained in the sphere of any proper subspace), and let any Krein–Clifford reset datum of signature $(1, d_{\text{Cl}})$ be given. Then:*

- (i) $g_{\nu_R} \rightarrow g_\nu = \text{diag}(1, -\kappa^2 M(\nu)^2)$, a nondegenerate Lorentzian inverse metric of signature $(1, d_{\text{Cl}})$;
- (ii) the characteristic variety of σ_ν is the smooth quadric Lorentz cone of g_ν , hence is linearly equivalent to the standard round Lorentz cone;
- (iii) if, in addition, the reset rays used for the order cone become dense in the null directions of this limiting quadric, then the corresponding order cones converge in Hausdorff distance on compact ray sections to the solid causal cone dual to g_ν ;
- (iv) in the isotropic normalisation $M(\nu) = d_{\text{Cl}}^{-1} I_{d_{\text{Cl}}}$ and $\kappa = d_{\text{Cl}}$, the preceding item reduces to the round-cone convergence of [Theorem 5.8](#);
- (v) the signature of g_ν is fixed by the Krein–Clifford datum and is independent of the direction measure, as long as $M(\nu) \succ 0$.

Proof. (i) As in [Theorem 5.8](#) (i), $M(\nu_R) \rightarrow M(\nu)$ by weak-* convergence against the bounded continuous integrands $\hat{\theta}^i \hat{\theta}^j$. Since $M(\nu) \succ 0$, the spatial block $-\kappa^2 M(\nu)^2$ is negative definite, so g_ν has signature $(1, d_{\text{Cl}})$ by [Theorem 5.5](#).

(ii) By [Lemma 5.4](#) the characteristic variety is $\{g_\nu^{\mu\nu} \xi_\mu \xi_\nu = 0\}$, the zero set of a nondegenerate Lorentzian quadratic form. Any two nondegenerate Lorentzian quadratic forms of the same signature are linearly equivalent, so a spatial frame change diagonalising $M(\nu)$ followed by rescaling carries the quadric to the standard round cone.

(iii) The order cone is determined by the conical hull of the reset rays, not merely by the second moment. Therefore density of the reset rays in the null directions of the limiting quadratic cone is the required hypothesis. Under this hypothesis, convergence of closed convex conical hulls on compact ray sections is the standard Hausdorff convergence of convex hulls of dense subsets.

(iv) In the isotropic normalisation, the null directions of the limiting quadric are exactly $(1, \hat{\theta})$ with $\hat{\theta} \in \mathbb{S}^{d_{\text{Cl}}-1}$, so (iii) is precisely [Theorem 5.8](#) (iii).

(v) The form $g_\nu = \text{diag}(1, -\kappa^2 M(\nu)^2)$ has one positive and d_{Cl} negative directions whenever $M(\nu) \succ 0$. The single positive slot is the one created by the Krein-odd timelike Clifford generator, as in [Theorem 5.5](#); changing ν changes the spatial scale and shape, not the inertia. $\square$

5.1.4 Model case: isotropic rounding

The cleanest case is the isotropic one. Here the spatial second moment becomes a scalar multiple of the identity, so the Lorentzian quadric is round after the normalisation $\kappa = d_{\text{Cl}}$.

Theorem 5.8 (Isotropic Clifford rounding). *Let $\nu_R \rightarrow \nu_{\text{unif}}$ weak-* on $\mathbb{S}^{d_{\text{Cl}}-1}$, where ν_{unif} is the rotation-invariant probability measure; for instance let ν_R be normalised counting measure on a net that equidistributes on $\mathbb{S}^{d_{\text{Cl}}-1}$. Take $\kappa = d_{\text{Cl}}$. Then:*

- (i) $M(\nu_R) \rightarrow \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}$, hence $g_{\nu_R} \rightarrow \text{diag}(1, -I_{d_{\text{Cl}}})$, the standard Minkowski inverse metric of signature $(1, d_{\text{Cl}})$;
- (ii) the spectral null cones $\{g_{\nu_R}^{\mu\nu} \xi_\mu \xi_\nu = 0\}$ converge, in Hausdorff distance on the unit sphere of covectors, to the round null cone $\{\xi_0^2 = |\vec{\xi}|^2\}$;
- (iii) the order cones $\text{cone}\{v_a^{(R)}\}$ converge, in Hausdorff distance on the unit sphere of directions, to the solid round future cone $\{x_0 \geq |\vec{x}|\}$.

Consequently both the spectral null cone and the causal order cone round to the same standard Lorentz cone, and the polyhedral obstruction of [Theorem D.6](#) is removed in the limit.

Proof. (i) By rotation invariance $\int \hat{\theta}^i \hat{\theta}^j d\nu_{\text{unif}} = \lambda \delta^{ij}$ for some λ , and taking the trace gives $d_{\text{Cl}} \lambda = \int |\hat{\theta}|^2 d\nu_{\text{unif}} = 1$, so $\lambda = 1/d_{\text{Cl}}$ and $M(\nu_{\text{unif}}) = \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}$. The entries $\hat{\theta} \mapsto \hat{\theta}^i \hat{\theta}^j$ are bounded and continuous on the compact sphere, so weak-* convergence $\nu_R \rightarrow \nu_{\text{unif}}$ gives $M(\nu_R) \rightarrow \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}$. With $\kappa = d_{\text{Cl}}$, $\kappa^2 M(\nu_R)^2 \rightarrow d_{\text{Cl}}^2 \cdot \frac{1}{d_{\text{Cl}}^2} I_{d_{\text{Cl}}} = I_{d_{\text{Cl}}}$, so $g_{\nu_R} \rightarrow \text{diag}(1, -I_{d_{\text{Cl}}})$.

(ii) The null cone of a nondegenerate quadratic form depends continuously, in Hausdorff distance on the covector sphere, on the coefficients of the form; apply this to $g_{\nu_R} \rightarrow \text{diag}(1, -I_{d_{\text{Cl}}})$.

(iii) The future order cone is the closed convex conical hull of the null rays $v = (1, \hat{\theta})$. As the directions $\hat{\theta}_a^{(R)}$ become dense in $\mathbb{S}^{d_{\text{Cl}}-1}$, the conical hull of $\{(1, \hat{\theta}_a^{(R)})\}$ converges in Hausdorff distance (on rays) to the conical hull of $\{(1, \hat{\theta}) : \hat{\theta} \in \mathbb{S}^{d_{\text{Cl}}-1}\}$, which is exactly the solid cone $\{x_0 \geq |\vec{x}|\}$. This is the standard convergence of convex hulls of χ -nets. $\square$

Example 5.9 (Spectral roundness with finitely many directions, $d_{\text{Cl}} = 2$). The two cones round under different conditions. Take $d_{\text{Cl}} = 2$ and the three equally spaced directions $\hat{\theta}_a = (\cos \frac{2\pi a}{3}, \sin \frac{2\pi a}{3})$, $a = 0, 1, 2$, with $p_a = \frac{1}{3}$. Then $\sum_a \hat{\theta}_a \hat{\theta}_a^\top = \frac{3}{2} I_2$, so $M = \frac{1}{2} I_2$ is already isotropic, and with $\kappa = 2$ the spectral null cone is the round circular cone $\{\xi_0^2 = |\vec{\xi}|^2\}$ *exactly*, with only three directions. Yet the order cone $\text{cone}\{(1, \hat{\theta}_a)\}$ is a three-faced pyramid: polyhedral, not round. Thus a finite tight frame rounds the Dirac symbol but not the causal hull; rounding the order cone genuinely requires the equidistributing limit $m_R \rightarrow \infty$ of [Theorem 5.8](#) (iii).

Example 5.10 (The octahedral 3 + 1 isotropic symbol). Take $d_{\text{Cl}} = 3$, $n_{\text{Cl}} = 1 + 3$, and let the reset directions be the six vertices of the spatial cross-polytope,

$$\hat{\theta}_a \in \{\pm e_1, \pm e_2, \pm e_3\} \subset \mathbb{S}^2, \quad p_a = \frac{1}{6}.$$

Then the second moment is exactly isotropic:

$$M(\nu) = \sum_a p_a \hat{\theta}_a \hat{\theta}_a^\top = \frac{1}{6} \left(2 \sum_{i=1}^3 e_i e_i^\top \right) = \frac{1}{3} I_3.$$

With the normalisation $\kappa = d_{\text{Cl}} = 3$, this gives

$$\kappa^2 M(\nu)^2 = 9 \left(\frac{1}{3} I_3 \right)^2 = I_3, \quad g_\nu = \text{diag}(1, -I_3).$$

Hence, by Lemma 5.4, the characteristic variety of σ_ν is

$$\xi_0^2 = \xi_1^2 + \xi_2^2 + \xi_3^2,$$

the standard 3 + 1-dimensional Minkowski light cone. Thus, once the primitive endpoint has $d_{\text{Cl}} = 3$, the round Lorentzian principal symbol is realised exactly by a finite isotropic reset frame. The single timelike slot is still the Krein-odd generator γ^0 ; the octahedral averaging supplies only the round spatial symbol.

Remark 5.11 (Symbol roundness versus order roundness). The preceding example is exact at the level of the Dirac principal symbol, but not at the level of the order cone. The order cone generated by the six rays

$$v_a = (1, \hat{\theta}_a) \in \mathbb{R}^{1+3}$$

has cross-section

$$\text{conv}\{\pm e_1, \pm e_2, \pm e_3\},$$

the octahedron. It is therefore polyhedral, with six extreme null rays, not a round solid cone. Rounding the order cone requires the equidistributing limit $\nu_R \rightarrow \nu_{\text{unif}}$ of Theorem 5.8; only then do the causal cones converge in Hausdorff distance to

$$\{x_0 \geq |\vec{x}|\}.$$

Thus finite tight frames can make the spectral cone round exactly, while rounding the causal hull is a genuine continuum-limit statement.

5.1.5 Scope of the theorem

Remark 5.12 (Consistency with the obstruction). Theorems 5.7 and 5.8 do not contradict Theorems D.6 and D.7. Those concern the deterministic finite c -generator product *order* cone, which is polyhedral with c extreme rays. Here the number of directions tends to infinity and the cone is read from the signed Dirac symbol. The hypotheses are disjoint, and the escape has a definite cost: one leaves the finite deterministic model and commits to the spectral cone of Definition 5.3.

Remark 5.13 (What is and is not proved). The theorems in this section are statements about the principal symbol and its characteristic cone. The flat constant-coefficient operator-level upgrade is proved separately in [Theorem 5.20](#). The following are still *not* proved at this stage: (a) a position-dependent curved metric $g(x)$, which would require the direction measure $\nu = \nu_x$ to vary slowly in x and a variable-coefficient/adiabatic analysis ; (b) a fully discrete spacetime evolution with a discretised timelike direction, which requires a separate stability/CFL hypothesis; (c) field equations, curvature dynamics, or matter coupling. The Lorentzian signature is carried by the minimal signed cover and the forced temporal Clifford generator; it is not produced by positive averaging.

Remark 5.14 (Dimension count). With $n_{\text{Cl}} = 1 + d_{\text{Cl}}$, the case $d_{\text{Cl}} = 3$ gives a $(3+1)$ -dimensional round Lorentz cone, consistent with [Theorem D.12](#): the emergent geometry is $1 + (c - 1)$ -dimensional with $c = n$, the single timelike slot being the Krein-odd generator γ^0 rather than an added coordinate.

5.2 Operator-level convergence of the rounded signed Dirac

The preceding subsection was a statement about principal symbols and cones. We now upgrade it to operators. The goal is to show that the rescaled signed renewal Dirac operators converge to a constant-coefficient Lorentzian Dirac evolution. This requires one care point: the covariant Lorentzian Dirac operator itself is hyperbolic, not elliptic, and has empty resolvent set in the flat L^2 spacetime picture. The operator limit therefore has two layers: graph convergence for the full spacetime Dirac operator, and strong resolvent convergence for the associated self-adjoint Dirac Hamiltonian.

Throughout this subsection the coefficients are constant. Curved and slowly-varying reset statistics are treated in [Section 5.4](#).

5.2.1 Why the Lorentzian Dirac operator forces the evolution picture

Let $g = g_\nu$ be a normalised emergent inverse metric of [Lemma 5.4](#) in the isotropic case $\kappa^2 M(\nu)^2 = I_{d_{\text{Cl}}}$, so $g = \text{diag}(1, -I_{d_{\text{Cl}}})$, and let D_g be the constant-coefficient operator on $L^2(\mathbb{R}^{1+d_{\text{Cl}}}, V)$ with Fourier multiplier $\sigma_g(\xi) = i(\gamma^0 \xi_0 + \gamma^i \xi_i)$, i.e. $D_g = \gamma^0 \partial_0 + \gamma^i \partial_i$.

Proposition 5.15 (The Lorentzian Dirac operator has empty resolvent set). *The operator D_g is closed and densely defined on $L^2(\mathbb{R}^{1+d_{\text{Cl}}}, V)$, but $\text{spec}(D_g) = \mathbb{C}$; its resolvent set is empty. Consequently strong (and a fortiori norm) resolvent convergence to D_g is not the right notion, and the operator-level limit cannot be phrased as resolvent convergence of the covariant Lorentzian Dirac operator itself.*

Proof. The operator is the Fourier multiplier by the matrix $\sigma_g(\xi)$ with domain $\{u : \sigma_g \hat{u} \in L^2\}$, hence is closed and densely defined. We show that no $z \in \mathbb{C}$ lies in the resolvent set.

First take $z = 0$. The multiplier $\sigma_g(\xi)$ fails to be bounded below in any neighbourhood of the nonzero null cone. Indeed, after writing $Q(\xi) = g^{\mu\nu} \xi_\mu \xi_\nu$, choose a nonzero null covector η and a transverse covector ζ with $dQ_\eta(\zeta) \neq 0$. For large R and $|s| < c/R$, the covectors $\xi = R\eta + s\zeta$ form positive-measure tubes on which $|Q(\xi)|$ remains bounded while $|\xi| \simeq R$. Since $\det \sigma_g(\xi)$ vanishes on $Q = 0$ and the smallest singular value of $\sigma_g(\xi)$ tends to zero along these tubes as they approach the null cone, $\sigma_g(\xi)^{-1}$ is not essentially bounded. Hence 0 cannot belong to the resolvent set.

Now let $z \neq 0$. Since

$$\sigma_g(\xi)^2 = -Q(\xi)\mathbf{1},$$

where the sign depends only on the convention for the factor i in σ_g , the inverse, wherever it exists, has the form

$$(\sigma_g(\xi) - z)^{-1} = \frac{\sigma_g(\xi) + z\mathbf{1}}{-Q(\xi) - z^2}.$$

Choose a constant $C > 0$ such that the denominator stays bounded away from zero whenever $|Q(\xi)| \leq C$; this is possible after taking $C < |z|^2/2$. The same positive-measure tubular neighbourhoods of the null cone contain covectors with $|Q(\xi)| \leq C$ and $|\xi| \rightarrow \infty$. On these sets the denominator is uniformly bounded above and below, whereas $\|\sigma_g(\xi) + z\mathbf{1}\|$ grows linearly in $|\xi|$ on a subregion of positive measure. Thus the inverse multiplier is not essentially bounded. A Fourier multiplier is bounded on L^2 only when its multiplier is essentially bounded. Therefore $(D_g - z)^{-1}$ is unbounded for every $z \neq 0$. The resolvent set is empty, i.e. $\text{spec}(D_g) = \mathbb{C}$. $\square$

Remark 5.16 (The fix is the Dirac Hamiltonian). [Theorem 5.15](#) reflects the hyperbolicity of the wave operator $D_g^2 = g^{\mu\nu} \partial_\mu \partial_\nu$, which is not elliptic and not sign-definite. The standard remedy is to read $D_g \psi = 0$ as an evolution equation in the timelike direction. Writing $D_g \psi = 0$ as $\gamma^0 \partial_0 \psi = -\gamma^i \partial_i \psi$ and multiplying by $-\gamma^0 = (\gamma^0)^{-1}$ gives $i \partial_0 \psi = H_g \psi$ with the Dirac *Hamiltonian*

$$H_g = A^i (-i \partial_i), \quad A^i := \gamma^0 \gamma^i.$$

The matrices A^i are the analogues of the Dirac α -matrices; the Krein-odd timelike generator γ^0 is exactly what makes them self-adjoint. This passage to the self-adjoint generator is where resolvent and group convergence legitimately live.

Lemma 5.17 (The Dirac Hamiltonian is self-adjoint). *With the Hilbert adjoints induced by the fundamental symmetry J of Definition 3.18 (so $\gamma^{0\dagger} = -\gamma^0$, $\gamma^{i\dagger} = \gamma^i$), the matrices $A^i = \gamma^0 \gamma^i$ are Hermitian and satisfy $\{A^i, A^j\} = 2\delta^{ij}\mathbf{1}$. Hence for any positive-definite symmetric M the multiplier $h_g(\vec{\xi}) = \kappa A^i M^{ij} \xi_j$ is a Hermitian matrix with $h_g(\vec{\xi})^2 = \kappa^2 \vec{\xi}^\top M^2 \vec{\xi} \mathbf{1}$, and the operator $H_g = \kappa A^i M^{ij} (-i \partial_j)$ is self-adjoint on the Sobolev domain $H^1(\mathbb{R}^{d_{\text{Cl}}}, V)$, essentially self-adjoint on $\mathcal{S}(\mathbb{R}^{d_{\text{Cl}}}, V)$.*

Proof. $(A^i)^\dagger = (\gamma^i)^\dagger (\gamma^0)^\dagger = \gamma^i (-\gamma^0) = -\gamma^i \gamma^0 = \gamma^0 \gamma^i = A^i$, using $\{\gamma^0, \gamma^i\} = 0$. Next $A^i A^j = \gamma^0 \gamma^i \gamma^0 \gamma^j$; since $\gamma^0 \gamma^i \gamma^0 = -(\gamma^0)^2 \gamma^i = \gamma^i$, this equals $\gamma^i \gamma^j$, whence $\{A^i, A^j\} = \{\gamma^i, \gamma^j\} = 2\delta^{ij}\mathbf{1}$. Contracting the symmetric tensor $(M\xi)_i (M\xi)_j$ against $A^i A^j$ replaces $A^i A^j$ by $\frac{1}{2}\{A^i, A^j\}$, giving $h_g(\vec{\xi})^2 = \kappa^2 |M\vec{\xi}|^2 \mathbf{1}$. A constant-coefficient first-order operator with Hermitian symbol of linear growth is self-adjoint on H^1 and essentially self-adjoint on $\mathcal{S}$ by the standard Fourier-multiplier criterion. $\square$

5.2.2 The discrete renewal Dirac and its finite-difference multipliers

We now define the discrete operators whose limit will be taken. The symmetric difference is chosen so that the spatial momentum multipliers are real and the Hamiltonians are self-adjoint.

Definition 5.18 (Symmetric reset differences). Fix mesh scales $h_R \downarrow 0$. For a spatial direction $\hat{\theta} \in \mathbb{S}^{d_{\text{Cl}}-1}$ let

$$(\delta_{\hat{\theta}, R} f)(x) = \frac{f(x + h_R \hat{\theta}) - f(x - h_R \hat{\theta})}{2h_R},$$

whose Fourier multiplier is $i s_{\hat{\theta}, R}(\vec{\xi})$, with $s_{\hat{\theta}, R}(\vec{\xi}) = h_R^{-1} \sin(h_R \hat{\theta} \cdot \vec{\xi})$. The self-adjoint lattice momentum is

$$P_{\hat{\theta}, R} := -i \delta_{\hat{\theta}, R},$$

with real multiplier $s_{\hat{\theta},R}(\vec{\xi})$. One has $s_{\hat{\theta},R}(\vec{\xi}) \rightarrow \hat{\theta} \cdot \vec{\xi}$ as $R \rightarrow \infty$, uniformly on compact $\vec{\xi}$ -sets, with the uniform bound $|s_{\hat{\theta},R}(\vec{\xi})| \leq |\hat{\theta} \cdot \vec{\xi}| \leq |\vec{\xi}|$.

For a direction system $\nu_R = \sum_a p_a^{(R)} \delta_{\hat{\theta}_a^{(R)}}$ as in Definition 5.3, define the discrete renewal Dirac Hamiltonian

$$H_R = \kappa \sum_a p_a^{(R)} A(\hat{\theta}_a^{(R)}) P_{\hat{\theta}_a^{(R)},R}, \quad A(\hat{\theta}) = \gamma^0 \gamma(\hat{\theta}) = \sum_i \hat{\theta}^i A^i,$$

with spatial multiplier $h_R(\vec{\xi}) = \kappa \sum_a p_a^{(R)} A(\hat{\theta}_a^{(R)}) s_{\hat{\theta}_a^{(R)},R}(\vec{\xi})$. Each $A(\hat{\theta})$ is Hermitian (Lemma 5.17), so $h_R(\vec{\xi})$ is Hermitian and H_R is self-adjoint by Fourier multiplication. The full spacetime operator D_R discretises the timelike direction and the spatial Dirac part by

$$m_R(\xi) = i\gamma^0 s_{0,R}(\xi_0) + i\kappa \sum_a p_a^{(R)} \gamma(\hat{\theta}_a^{(R)}) s_{\hat{\theta}_a^{(R)},R}(\vec{\xi}),$$

which is the finite-difference analogue of the Lorentzian Dirac multiplier.

5.2.3 Strong convergence of the full operator

For the full spacetime Dirac operator the correct convergence statement is convergence on a common core. This is enough to justify the symbol limit at the operator level, while respecting the empty-resolvent obstruction above.

Theorem 5.19 (Graph convergence of the full signed Dirac). *Assume the symbol convergence hypotheses of Theorem 5.7 ($\nu_R \rightarrow \nu$ weak-*, $M(\nu) \succ 0$) and $h_R \downarrow 0$. Then for every $u \in \mathcal{S}(\mathbb{R}^{1+d_{\text{Cl}}}, V)$,*

$$D_R u \longrightarrow D_g u \quad \text{in } L^2(\mathbb{R}^{1+d_{\text{Cl}}}, V),$$

where D_g has multiplier σ_ν . Since $\mathcal{S}$ is a core for both operators, this is convergence on a common core (graph convergence). By Theorem 5.15 this is the strongest operator-level statement available for D_g itself; resolvent convergence is not.

Proof. Let $m_R(\xi)$ be the multiplier of D_R and $\sigma_\nu(\xi)$ that of D_g . By Definition 5.18, $s_{0,R}(\xi_0) \rightarrow \xi_0$ and

$$\kappa \sum_a p_a^{(R)} \gamma(\hat{\theta}_a^{(R)}) s_{\hat{\theta}_a^{(R)},R}(\vec{\xi}) \longrightarrow \kappa \gamma^i M(\nu)^{ij} \xi_j$$

pointwise. Hence $m_R(\xi) \rightarrow \sigma_\nu(\xi)$ pointwise, with the uniform domination $\|m_R(\xi)\| \leq C(1 + |\xi|)$, C independent of R , because $|s_{\hat{\theta},R}| \leq |\vec{\xi}|$, $|s_{0,R}| \leq |\xi_0|$ and $\sum_a p_a^{(R)} \|A(\hat{\theta}_a)\| \leq \max_{\hat{\theta}} \|A(\hat{\theta})\| < \infty$. By Plancherel,

$$\|D_R u - D_g u\|_{L^2}^2 = \int_{\mathbb{R}^{1+d_{\text{Cl}}}} \|(m_R(\xi) - \sigma_\nu(\xi)) \hat{u}(\xi)\|^2 d\xi.$$

The integrand tends to 0 pointwise and is dominated by $(\|m_R(\xi)\| + \|\sigma_\nu(\xi)\|)^2 \|\hat{u}(\xi)\|^2 \leq C'(1 + |\xi|)^2 \|\hat{u}(\xi)\|^2$, which is integrable for $u \in \mathcal{S}$. Dominated convergence gives the claim. $\square$

5.2.4 Strong resolvent and propagator convergence of the Hamiltonian

After passing to the Hamiltonian picture, the limiting operator is self-adjoint. Standard functional calculus then gives the usual strong resolvent and propagator convergence.

Theorem 5.20 (Operator-level Clifford continuum limit). *Under the hypotheses of Theorem 5.19, the self-adjoint discrete Hamiltonians H_R converge to the self-adjoint continuum Dirac Hamiltonian $H_g = \kappa A^i M(\nu)^{ij}(-i\partial_j)$ in the strong resolvent sense:*

$$(H_R - z)^{-1} \longrightarrow (H_g - z)^{-1} \quad \text{strongly, for all } z \in \mathbb{C} \setminus \mathbb{R}.$$

Consequently:

- (i) *the Dirac propagators converge, $e^{-itH_R} \rightarrow e^{-itH_g}$ strongly, uniformly for t in compact intervals;*
- (ii) *$\varphi(H_R) \rightarrow \varphi(H_g)$ strongly for every bounded continuous φ , so spectral projections onto intervals whose endpoints are not atoms of the limit converge strongly;*
- (iii) *the limiting dispersion is $\text{spec}(H_g) = \mathbb{R}$, absolutely continuous, with energies $\pm \kappa |M(\nu)\vec{\xi}|$, recovering the Lorentzian light cone $\xi_0^2 = \kappa^2 \vec{\xi}^\top M(\nu)^2 \vec{\xi}$ of Theorem 5.7.*

Proof. By Lemma 5.17 both H_R and H_g are self-adjoint with common core $\mathcal{S}(\mathbb{R}^{d_{\text{Cl}}}, V)$. The argument of Theorem 5.19, applied to the Hermitian spatial multipliers $h_R(\vec{\xi}) \rightarrow h_g(\vec{\xi})$ with domination $\|h_R(\vec{\xi})\| \leq \kappa \|A\| |\vec{\xi}|$, gives $H_R u \rightarrow H_g u$ in L^2 for every $u \in \mathcal{S}$. Strong convergence on a common core of self-adjoint operators implies strong resolvent convergence by the Trotter–Kato criterion [51, Thm. VIII.25(a)]. Item (i) is the Trotter–Kato theorem relating strong resolvent convergence to strong convergence of the unitary groups, uniformly on compact time intervals [51, Thm. VIII.21]; item (ii) is the bounded-continuous functional-calculus form of strong resolvent convergence. For (iii), $h_g(\vec{\xi})$ is Hermitian with $h_g(\vec{\xi})^2 = \kappa^2 \vec{\xi}^\top M^2 \vec{\xi} \mathbf{1}$ by Lemma 5.17, so its eigenvalues are $\pm \kappa |M\vec{\xi}|$; as $\vec{\xi}$ ranges over $\mathbb{R}^{d_{\text{Cl}}}$ these fill $\mathbb{R}$, with purely absolutely continuous spectrum by the standard Fourier-multiplier (limiting absorption) argument, and the energy relation $\xi_0 = \pm \kappa |M\vec{\xi}|$ is the null condition $g^{\mu\nu} \xi_\mu \xi_\nu = 0$. $\square$

The operator-limit theorem upgrades the previous symbol calculation to an actual convergence of dynamics. The full covariant spacetime operator is handled on a common core because its resolvent is empty, while the self-adjoint Hamiltonian admits strong-resolvent, propagator, and functional calculus convergence. The distinction is analytic rather than physical: both descriptions encode the same limiting dispersion relation.

Theorem 5.21 (Frame universality of the nondegenerate flat endpoint). *Let $M \in \text{Sym}_{d_{\text{Cl}}}(\mathbb{R})$ be positive definite and set $V_M = \kappa M$. Define*

$$(U_M \psi)(y) := (\det V_M)^{1/2} \psi(V_M y), \quad \psi \in L^2(\mathbb{R}^{d_{\text{Cl}}}, V).$$

Then U_M is unitary, acts trivially on the spinor factor, and

$$U_M [\kappa A^i M^{ij}(-i\partial_j)] U_M^{-1} = A^i(-i\partial_i) =: H_*.$$

Consequently every nondegenerate constant-coefficient endpoint

$$g_M^{-1} = \text{diag}(1, -\kappa^2 M^2)$$

is spatially linearly isometric to standard Minkowski space, and its Hamiltonian is unitarily equivalent to the standard isotropic massless Dirac Hamiltonian. The spatial change of variables leaves the Krein-odd timelike generator, fundamental symmetry, Krein form, and single-time-slot conclusion unchanged.

Proof. With $x = V_M y$, one has $dx = (\det V_M) dy$, so U_M is unitary. For $\phi = U_M \psi$,

$$(-i\partial_{y_k} \phi)(y) = (\det V_M)^{1/2} (V_M)_{jk} (-i\partial_{x_j} \psi)(V_M y).$$

Since $V_M = \kappa M$ is symmetric,

$$A^k (V_M)_{jk} = \kappa A^k M^{kj},$$

and hence $H_* U_M = U_M H_g$ on $C_c^\infty(\mathbb{R}^{d_{C1}}, V)$. Essential self-adjointness on this common core extends the identity to the closures. On covectors, the dual change $\eta = V_M^\top \xi$ sends

$$\xi_0^2 = \kappa^2 \xi^\top M^2 \xi \quad \text{to} \quad \xi_0^2 = |\eta|^2.$$

Because U_M acts only on the spatial argument, it commutes with the constant spinor-space operators carrying the Krein and temporal data. $\square$

For the isolated flat endpoint, this theorem removes exact isotropy from the list of necessary assumptions. A positive-definite M already gives the standard massless Dirac operator after a fixed linear change of spatial coordinates. Isotropy is therefore a convenient normal form; nondegeneracy is the load-bearing condition.

Remark 5.22 (Scope of frame universality). The theorem removes constant anisotropy from the isometry class of the *isolated, single-sector, flat, free Dirac endpoint*. It does not assert that every microscopic tensor becomes physically irrelevant once additional structures are retained. If two matter or propagation sectors carry inequivalent positive tensors M_1 and M_2 , one spatial frame can normalise one of them but not generally both; the relative tensor $M_1^{-1} M_2$ can then be invariant data. Likewise, interactions, higher-order dispersion, boundaries, or a separately fixed microscopic frame may retain relative anisotropy. No such independent spatial tensor is part of the flat kinematic endpoint proved here.

Corollary 5.23 (Explicit $(3+1)$ free Dirac endpoint). *Take $d_{C1} = 3$, $\kappa = 3$, and $\nu_R \rightarrow \nu_{\text{unif}}$ on $\mathbb{S}^2$. Then $M(\nu_R) \rightarrow \frac{1}{3} I_3$, the continuum Hamiltonian is the free massless $(3+1)$ -dimensional Dirac Hamiltonian*

$$H_g = \sum_{j=1}^3 A^j (-i\partial_j), \quad \{A^i, A^j\} = 2\delta^{ij},$$

and e^{-itH_R} converges strongly, uniformly on compact time intervals, to the free Dirac propagator e^{-itH_g} . The emergent light cone is the round cone $\xi_0^2 = |\vec{\xi}|^2$, i.e. a $(3+1)$ -Minkowski signature, consistent with the dimension count $1 + d_{C1} = 4$.

Proof. By [Theorem 5.8](#) (i), $M(\nu_R) \rightarrow \frac{1}{3} I_3$; with $\kappa = 3$, $\kappa M(\nu) = I_3$, so $h_g(\vec{\xi}) = A^j \xi_j$ and $H_g = A^j (-i\partial_j)$. The conclusions are [Theorem 5.20](#) specialised to this case. $\square$

5.2.5 Scope and sharpness

The preceding theorem gives the correct global convergence statement. The next remarks explain why it should not be upgraded to global norm-resolvent convergence for the symmetric finite-difference scheme, and why a norm estimate does hold after imposing a fixed momentum cutoff.

Remark 5.24 (Strong, not norm, resolvent convergence). The convergence in [Theorem 5.20](#) is strong resolvent and no better: for symmetric finite differences, norm resolvent convergence of lattice Dirac operators to their continuum counterpart *fails* in spatial dimension $d_{C1} \geq 2$, even in the elliptic (Riemannian) case. For Euclidean lattice Dirac operators this is established by

Cornean–Garde–Jensen and by Schmidt–Umeda [52, 53]; the same obstruction applies here. Norm resolvent convergence can be restored by a non-standard modification of the mesh operator (a mass-type correction of Wilson type [54]), which we do not pursue.

Remark 5.25 (Why norm-resolvent convergence of the mesh limit is unavailable). It is natural to ask whether the strong-resolvent convergence of Theorem 5.20 can be upgraded to norm-resolvent convergence. For the symmetric finite-difference scheme used here this is *provably false* in spatial dimension $d_{\text{Cl}} \geq 2$: the obstruction of [52, 53], recalled in Theorem 5.24, rules it out already in the flat Euclidean case, and our signed scheme inherits it. The only norm-resolvent statement the construction supports is the *stability* estimate of Theorem 5.53: it compares two continuum operators at fixed mesh-free limit, where both lie in a common uniformly elliptic class, and is therefore not a discretisation limit at all (Theorem 5.55). Restoring norm-resolvent convergence of the *mesh* approximation would require a non-standard mesh correction of Wilson/mass type [54], which modifies the renewal difference operator away from the signed reset dynamics; we do not adopt it. This item is thus recorded as a boundary of the method, not a gap to be filled.

Theorem 5.26 (Band-limited norm-resolvent convergence). *Let $\mathbf{H}_R$ and $\mathbf{H}_g$ be the self-adjoint Dirac Hamiltonians of Theorem 5.20, with Hermitian Fourier multipliers $h_R(\vec{\xi}) = \kappa \sum_a p_a^{(R)} A(\hat{\theta}_a^{(R)}) s_{\hat{\theta}_a^{(R)}, R}(\vec{\xi})$ and $h_g(\vec{\xi}) = \kappa A^i M(\nu)^{ij} \xi_j$ of Theorems 5.7 and 5.18. For $\Lambda_0 > 0$ let P_{Λ_0} be the momentum band projection on $L^2(\mathbb{R}^{d_{\text{Cl}}}, V)$ with multiplier $\mathbf{1}_{\{|\vec{\xi}| \leq \Lambda_0\}} \otimes I_V$; being a Fourier multiplier it commutes with $\mathbf{H}_R$ and $\mathbf{H}_g$. Set*

$$\chi_R(\Lambda_0) := \sup_{|\vec{\xi}| \leq \Lambda_0} \|h_R(\vec{\xi}) - h_g(\vec{\xi})\|.$$

For the symmetric reset differences one has the explicit second-order bound

$$\chi_R(\Lambda_0) \leq \kappa \|A\| \left(\frac{1}{6} h_R^2 \Lambda_0^3 + \|M(\nu_R) - M(\nu)\| \Lambda_0 \right) \xrightarrow{R \rightarrow \infty} 0.$$

Then for every $z \in \mathbb{C} \setminus \mathbb{R}$,

$$\left\| [(\mathbf{H}_R - z)^{-1} - (\mathbf{H}_g - z)^{-1}] P_{\Lambda_0} \right\| \leq \frac{\chi_R(\Lambda_0)}{|\text{Im } z|^2} = O(h_R^2),$$

the $O(h_R^2)$ rate holding whenever $M(\nu_R) = M(\nu)$ (e.g. an exactly isotropic frame at every R , as in Theorem 5.10). Consequently, for every bounded continuous φ ,

$$\left\| [\varphi(\mathbf{H}_R) - \varphi(\mathbf{H}_g)] P_{\Lambda_0} \right\| \xrightarrow{R \rightarrow \infty} 0.$$

Thus on any fixed momentum band the convergence improves from strong resolvent (Theorem 5.20) to norm resolvent, at second order in the mesh.

Proof. Since P_{Λ_0} , $\mathbf{H}_R$, $\mathbf{H}_g$ are all Fourier multipliers they commute, and on the range of P_{Λ_0} the resolvent difference is the multiplication operator with symbol $(h_R(\vec{\xi}) - z)^{-1} - (h_g(\vec{\xi}) - z)^{-1}$ restricted to $|\vec{\xi}| \leq \Lambda_0$. The norm of a multiplier operator is the essential supremum of the pointwise matrix norm, so

$$\left\| [(\mathbf{H}_R - z)^{-1} - (\mathbf{H}_g - z)^{-1}] P_{\Lambda_0} \right\| = \sup_{|\vec{\xi}| \leq \Lambda_0} \|(h_R - z)^{-1} - (h_g - z)^{-1}\|.$$

By the resolvent identity $(h_R - z)^{-1} - (h_g - z)^{-1} = (h_R - z)^{-1}(h_g - h_R)(h_g - z)^{-1}$, and since h_R, h_g are Hermitian, $\|(h_\bullet - z)^{-1}\| \leq |\operatorname{Im} z|^{-1}$; hence the pointwise norm is $\leq |\operatorname{Im} z|^{-2} \|h_R - h_g\|$, giving the boxed bound after taking the supremum.

For χ_R : each scalar component obeys $|s_{\hat{\theta}, R}(\vec{\xi}) - \hat{\theta} \cdot \vec{\xi}| = h_R^{-1} |\sin(h_R \hat{\theta} \cdot \vec{\xi}) - h_R \hat{\theta} \cdot \vec{\xi}| \leq h_R^{-1} \frac{1}{6} |h_R \hat{\theta} \cdot \vec{\xi}|^3 \leq \frac{1}{6} h_R^2 |\vec{\xi}|^3 \leq \frac{1}{6} h_R^2 \Lambda_0^3$ on the band, using $|\sin t - t| \leq |t|^3/6$ and $|\hat{\theta} \cdot \vec{\xi}| \leq |\vec{\xi}|$; summing the Hermitian matrices $A(\hat{\theta}_a)$ against $p_a^{(R)}$ contributes $\kappa \|A\|$. The replacement of $M(\nu_R)$ by $M(\nu)$ in the limit symbol adds $\kappa \|A\| \|M(\nu_R) - M(\nu)\| |\vec{\xi}| \leq \kappa \|A\| \|M(\nu_R) - M(\nu)\| \Lambda_0$. This proves the displayed bound on $\chi_R(\Lambda_0)$, which tends to 0 since $h_R \downarrow 0$ and $M(\nu_R) \rightarrow M(\nu)$ (Theorem 5.7).

For the functional-calculus statement, φ is uniformly continuous on the compact set $\{h_g(\vec{\xi}) : |\vec{\xi}| \leq \Lambda_0\}$ enlarged by an χ_R neighbourhood; uniform convergence $h_R \rightarrow h_g$ on the band then gives $\sup_{|\vec{\xi}| \leq \Lambda_0} \|\varphi(h_R(\vec{\xi})) - \varphi(h_g(\vec{\xi}))\| \rightarrow 0$, which is the multiplier norm of $[\varphi(H_R) - \varphi(H_g)]P_{\Lambda_0}$. $\square$

Remark 5.27 (Momentum band, not energy window; consistency with the no-go). Two points fix the scope of Theorem 5.26 and reconcile it with Theorems 5.24 and 5.25.

(a) *The band is in momentum, and this is essential.* The norm-resolvent failure in $d_{\text{Cl}} \geq 2$ is caused by lattice fermion doublers [55, 56]: the discrete symbol $h_R(\vec{\xi})$ has spurious near-zeros at high momentum $|\vec{\xi}| \sim \pi/h_R$, where $h_R(\vec{\xi}) \approx 0$ while $h_g(\vec{\xi})$ is large. At such a covector the resolvent symbols differ by $\|(h_R - z)^{-1} - (h_g - z)^{-1}\| \rightarrow |z|^{-1} \neq 0$, so the full-line supremum does not vanish. These doublers sit at *low energy* but *high momentum*; hence an energy-window truncation $\varphi(H_R)$ with φ supported near 0 does *not* excise them, whereas the fixed momentum band P_{Λ_0} does, since $\pi/h_R \rightarrow \infty$. This is why the theorem is stated with a momentum projection and why the analogous energy-window statement is false.

(b) *No contradiction with the no-go.* Norm-resolvent convergence of the full operator is exactly the statement that $\sup_{\vec{\xi} \in \mathbb{R}^{d_{\text{Cl}}}} \|(h_R - z)^{-1} - (h_g - z)^{-1}\| \rightarrow 0$; it fails because of the doubler region $|\vec{\xi}| \sim \pi/h_R$, precisely as in [52, 53]. Theorem 5.26 restores norm convergence by removing that region, and says nothing about it: the symmetric scheme is second-order consistent *uniformly on compact momentum sets* but not *uniformly in $\vec{\xi}$* . The band-limited statement is thus the sharp positive complement of Theorem 5.25: full norm-resolvent convergence needs a Wilson/mass mesh correction [54] (there discussed and not adopted), while on any fixed momentum band the unmodified renewal scheme already converges in norm resolvent at order h_R^2 .

Remark 5.28 (Scope of the flat operator theorem). The following are proved here, in the flat constant-coefficient regime: graph convergence of the full signed Dirac (Theorem 5.19) and strong resolvent, propagator, and spectral-projection convergence of the self-adjoint Dirac Hamiltonian (Theorem 5.20). Section 5.4 upgrades the Hamiltonian convergence to a variable-coefficient adiabatic setting and proves stability under perturbation of the reset statistics. The curved refinement now reconstructs the Levi–Civita spin connection from the renewal-generated coframe by the discrete torsion-free Cartan rule and proves global strong-resolvent convergence through subprincipal order. What is still not claimed is different: (a) derivation of the exchange-symmetric diamond phase from bare channel associativity (closure itself is derived from that phase in Theorem 5.39); (b) a fully discrete spacetime unitary with the timelike direction also discretised, which would require a separate stability/CFL hypothesis; (c) full norm-resolvent convergence of the mesh approximation, which fails by Theorem 5.24, although fixed momentum bands converge in norm resolvent in the flat case by Theorem 5.26; and (d) curvature dynamics, field equations, or matter coupling. As in Theorem 5.13, the Lorentzian signature is carried by the signed/Krein sector and the derived temporal generator, not by positive averaging.

Remark 5.29 (Relation to the modular signed Dirac). The continuum operators above are the coarse-grained, spatially isotropic counterparts of the modular signed Dirac $D_- = J_\chi e^{\beta N} + B$ of [Theorem 3.27](#): the modular factor governs the timelike generator, while the equidistributed reset directions of [Definition 5.18](#) furnish the spatial Dirac matrices A^i . The present theorems show that, after rescaling, the spatial sector of the signed renewal dynamics converges to a genuine self-adjoint Lorentzian Dirac evolution.

5.2.6 Self-averaging from ergodic reset statistics

The preceding operator theorem is already anisotropic: it requires only convergence of the second-moment tensors to a positive-definite limit. Ergodicity therefore need not create a rotationally invariant stationary law. It is enough that the stationary directions span the spatial slot. The limiting tensor then defines a flat vielbein, and [Theorem 5.21](#) removes its constant frame anisotropy exactly.

Theorem 5.30 (Self-averaged flat Dirac endpoint under nondegeneracy). *Let $(\hat{\theta}_a)_{a \geq 1}$ be a stationary ergodic sequence of spatial reset directions in $\mathbb{S}^{d_{\text{Cl}}-1}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with one-dimensional marginal law ζ_* . Let $m_R \rightarrow \infty$ and $h_R \downarrow 0$, form*

$$\nu_R = \frac{1}{m_R} \sum_{a=1}^{m_R} \delta_{\hat{\theta}_a}, \quad M_* := \int_{\mathbb{S}^{d_{\text{Cl}}-1}} \hat{\theta} \hat{\theta}^\top d\zeta_*(\hat{\theta}),$$

and let $\mathbf{H}_R$ be the discrete renewal Dirac Hamiltonian of [Theorem 5.18](#) with equal weights $p_a^{(R)} = 1/m_R$. Assume only

$$M_* \succ 0,$$

equivalently that $\text{supp } \zeta_*$ spans $\mathbb{R}^{d_{\text{Cl}}}$, and fix $\kappa > 0$. Then for $\mathbb{P}$ -almost every realisation:

- (i) $\nu_R \rightarrow \zeta_*$ weak-* and $M(\nu_R) \rightarrow M_*$;
- (ii) the random signed principal symbols converge pointwise to the deterministic Lorentzian symbol with inverse metric

$$g_*^{-1} = \text{diag}(1, -\kappa^2 M_*^2),$$

of signature $(1, d_{\text{Cl}})$;

- (iii) $\mathbf{H}_R$ converges in strong resolvent sense to

$$\mathbf{H}_g[M_*] = \kappa A^i M_*^{ij} (-i\partial_j),$$

and $e^{-it\mathbf{H}_R} \rightarrow e^{-it\mathbf{H}_g[M_*]}$ strongly, uniformly for t in compact intervals;

- (iv) on every fixed momentum band P_{Λ_0} , the resolvents converge in norm with the estimate

$$\begin{aligned} & \|[(\mathbf{H}_R - z)^{-1} - (\mathbf{H}_g[M_*] - z)^{-1}]P_{\Lambda_0}\| \\ & \leq \frac{\kappa \|A\|}{|\text{Im } z|^2} \left(\frac{1}{6} h_R^2 \Lambda_0^3 + \|M(\nu_R) - M_*\| \Lambda_0 \right) \rightarrow 0; \end{aligned}$$

- (v) the fixed unitary U_{M_*} of [Theorem 5.21](#) intertwines the limit with the standard massless Dirac Hamiltonian:

$$U_{M_*} \mathbf{H}_g[M_*] U_{M_*}^{-1} = A^i (-i\partial_i).$$

Thus stationary ergodicity and the nondegeneracy already required for Lorentzian signature suffice almost surely to produce the standard flat Minkowski Dirac isometry class; rotational invariance of ζ_* is not an additional hypothesis.

Proof. For each i, j , the function $\hat{\theta} \mapsto \hat{\theta}_i \hat{\theta}_j$ is bounded and continuous on the compact sphere. Birkhoff's ergodic theorem gives

$$M(\nu_R)^{ij} = \frac{1}{m_R} \sum_{a=1}^{m_R} (\hat{\theta}_a)_i (\hat{\theta}_a)_j \longrightarrow \int \hat{\theta}_i \hat{\theta}_j d\zeta_* = M_*^{ij}$$

almost surely. Applying the same theorem to a countable convergence-determining family of continuous functions gives $\nu_R \rightarrow \zeta_*$ weak-* almost surely. Fix a realisation in this probability-one event. Since $M_* \succ 0$, the deterministic anisotropic operator limit [Theorem 5.20](#) applies pathwise and proves (ii)–(iii). The fixed-band estimate is exactly [Theorem 5.26](#) with $M(\nu) = M_*$, proving (iv). Finally, (v) is [Theorem 5.21](#). $\square$

This result supplies the probabilistic selection missing from the deterministic flat theorem. The limiting coframe need not be assumed sample by sample: ergodicity produces a deterministic second moment almost surely. It does not force that moment to be scalar, but frame universality shows that scalarity is unnecessary for the single-sector flat isometry class.

Proposition 5.31 (Literal isotropy is not implied by ergodic renewal dynamics). *For every $d_{\text{Cl}} \geq 2$ there exists a stationary ergodic finite reset law with $M_* \succ 0$ but $M_* \notin \mathbb{R}I_{d_{\text{Cl}}}$.*

Proof. Choose the directions $e_1, \dots, e_{d_{\text{Cl}}}$ independently with probabilities $p_1, \dots, p_{d_{\text{Cl}}} > 0$, not all equal. The resulting i.i.d. sequence is stationary and ergodic, while

$$M_* = \sum_{i=1}^{d_{\text{Cl}}} p_i e_i e_i^\top = \text{diag}(p_1, \dots, p_{d_{\text{Cl}}})$$

is positive definite and non-scalar. Such a finite label process can be coupled to any primitive finite family of unital completely positive reset channels, so it lies within the renewal class considered here. $\square$

Remark 5.32 (What self-averaging removes, and what it does not). Three qualifications remain. (a) Self-averaging produces the deterministic tensor M_* ; it does not force M_* to be scalar, and [Theorem 5.31](#) shows that no such implication follows from stationarity and ergodicity. Scalarity is unnecessary because the isolated flat endpoint is frame-universal. (b) Global convergence is strong resolvent only; the doubler obstruction of [Theorem 5.24](#) persists, while fixed-band norm-resolvent convergence holds pathwise by [Theorem 5.26](#). (c) The conclusion concerns the single-sector flat free endpoint. Relative anisotropy may remain when independent sectors carry inequivalent second-moment tensors, as explained in [Theorem 5.22](#). The balancing layer of [Section E.11](#) is not needed to impose isotropy; its role is instead to determine which nondegenerate stationary tensor is dynamically selected.

5.3 Channel plaquettes and Levi–Civita reconstruction

The proof strategy of this subsection is worth stating before the formulas. The discrete fundamental theorem says that a coframe has a unique Levi–Civita connection once two defects vanish: the metricity defect Q and the torsion T . We therefore derive them separately. Covariance of

neighbouring second moments gives $Q = 0$; the translational part of a channel plaquette measures T ; and exchange symmetry of the stationary diamond law makes that translational defect vanish. The rotational part is deliberately left free, so curvature is not removed together with torsion.

The renewal coframe determines the principal metric, while the composed-channel plaquette separates translational torsion from rotational curvature. Covariance of transported second moments yields isometric comparison and hence metric compatibility. In the exchange-symmetric stationary diamond phase, the translational first moment vanishes, and the discrete fundamental theorem selects the unique Levi–Civita connection without suppressing rotational holonomy.

5.3.1 Setup and the two defects

Fix a smooth nondegenerate coframe field $E(x) = (e_j^a(x)) \in GL(d_{\text{Cl}})$, C_b^4 , with metric $g_{ij} = \delta_{ab} e_i^a e_j^b = (E^\top E)_{ij}$ and frame Gram $P := EE^\top \succ 0$. Let $h \downarrow 0$ be the mesh and δ_i the symmetric difference (so $\delta_i e_j^a = \partial_i e_j^a + O(h^2)$). A discrete connection is an assignment $\gamma_{iab}(x)$; its near-identity frame transport is $U_i(x) = \exp(-h \gamma_i(x))$ with $(\gamma_i)^a_b = \delta^{ac} \gamma_{icb}$.

Two tensors measure the failure of the two structural requirements:

$$Q_{iab} := \gamma_{iab} + \gamma_{iba} \quad (\text{metricity defect}), \quad T_{ij}^a := \delta_i e_j^a - \delta_j e_i^a + \gamma_i^a_b e_j^b - \gamma_j^a_b e_i^b \quad (\text{discrete torsion}).$$

Metric compatibility is $Q \equiv 0$ (the connection is $\mathfrak{so}(d_{\text{Cl}})$ -valued); torsion-freeness is $T \equiv 0$.

5.3.2 The discrete fundamental theorem

Write $C_{ij}^a := \delta_i e_j^a - \delta_j e_i^a$ for the coframe curl and $C_{abc} := \delta_{ad} e_b^i e_c^j C_{ij}^d$ (antisymmetric in b, c), with e_a^j the inverse coframe.

Theorem 5.33 (Discrete metric / torsion-free uniqueness). *For every nondegenerate coframe there is a unique connection with $Q \equiv 0$ and $T \equiv 0$. In frame components $\gamma_{d;ae} := e_d^i \gamma_{iae}$ it is the discrete Koszul formula*

$$\gamma_{d;ae} = \frac{1}{2}(C_{dea} - C_{ade} - C_{ead}),$$

which satisfies $\gamma_{d;ae} = -\gamma_{d;ea}$. Moreover $\gamma_{iab} = \omega_{iab}^{\text{LC}} + O(h^2)$ locally uniformly, where ω^{LC} is the Levi–Civita spin connection of g .

Proof. Existence. Contracting $T_{ij}^a = 0$ with $e_d^i e_e^j$ gives the frame equation $C_{ade} + \gamma_{d;ae} - \gamma_{e;ad} = 0$. Using $C_{abc} = -C_{acb}$,

$$\gamma_{d;ae} - \gamma_{e;ad} = \frac{1}{2}(C_{dea} - C_{ade} - C_{ead}) - \frac{1}{2}(C_{eda} - C_{aed} - C_{dae}) = -C_{ade},$$

since $C_{dea} = -C_{dae}$ and $C_{ead} = -C_{eda}$ cancel pairwise. Antisymmetry in a, e ($Q = 0$) is immediate: swapping $a \leftrightarrow e$ negates each term.

Uniqueness. If γ, γ' both satisfy $Q = 0$, $T = 0$, the difference $\Delta_{d;ae}$ is metric, $\Delta_{d;ae} = -\Delta_{d;ea}$, and obeys the homogeneous torsion equation $\Delta_{d;ae} = \Delta_{e;ad}$. Then

$$\Delta_{d;ae} = \Delta_{e;ad} = -\Delta_{e;da} = -\Delta_{a;de} = \Delta_{a;ed} = \Delta_{d;ea} = -\Delta_{d;ae},$$

so $\Delta = 0$. Nondegeneracy of the coframe carries this back to coordinate indices.

Convergence. $\delta_i e_j^a = \partial_i e_j^a + O(h^2)$ for C_b^4 coframes, so $C_{abc} = C_{abc}^{\text{cont}} + O(h^2)$; the formula is linear in C with coefficients bounded by the uniformly bounded frame and inverse frame, and its continuum value is the Levi–Civita spin connection, whence $\gamma = \omega^{\text{LC}} + O(h^2)$. $\square$

5.3.3 Isometric comparison forces metric compatibility

Assumption 5.1 (Covariance of transported second moments). Let $M_x, M_y \succ 0$ be the spatial second moments of neighbouring local direction-record laws, and let $T_e : \mathbb{R}^{d_{\text{Cl}}} \rightarrow \mathbb{R}^{d_{\text{Cl}}}$ be the invertible transport of those records along an edge $e : x \rightarrow y$. Their transported quadratic statistics agree:

$$T_e M_x T_e^\top = M_y.$$

The stable-record determinant orientation is preserved along the selected oriented branch.

Proposition 5.34 (Second-moment covariance derives isometric comparison). *Under [Assumption 5.1](#), the comparison in the orthonormal frames selected by the two moments,*

$$U_e := M_y^{-1/2} T_e M_x^{1/2},$$

lies in $SO(d_{\text{Cl}})$. Thus isometric comparison is a consequence of covariance of renewal second moments, rather than an independent metric postulate.

Proof. The covariance identity gives

$$U_e U_e^\top = M_y^{-1/2} T_e M_x T_e^\top M_y^{-1/2} = I.$$

Hence $U_e \in O(d_{\text{Cl}})$. Preservation of the selected determinant orientation gives $\det U_e > 0$, and therefore $U_e \in SO(d_{\text{Cl}})$. $\square$

Proposition 5.35 (Isometric comparison $\Leftrightarrow$ metric compatibility). *A near-identity transport $U_i = \exp(-h\gamma_i)$ lies in $SO(d_{\text{Cl}})$ to first order in h iff $\gamma_i \in \mathfrak{so}(d_{\text{Cl}})$, i.e. $Q \equiv 0$. Consequently [Assumption 5.1](#) and [Theorem 5.34](#) imply metric compatibility of the induced connection.*

Proof. $\exp(-h\gamma_i) \in SO(d_{\text{Cl}})$ for all small h iff $\gamma_i^\top = -\gamma_i$, i.e. $\gamma_{iab} = -\gamma_{iba}$, i.e. $Q_{iab} = 0$. An $\mathfrak{so}$ -valued connection annihilates the constant frame metric δ_{ab} under transport, which is metric compatibility. $\square$

The coframe is $e_a^j = \kappa M^{aj}$, with M the renewal second moment. The proposition shows that covariance of the transported record law canonically produces its orthogonal frame comparison. Among such comparisons, the orthogonal Procrustes factor remains the distinguished least-distortion representative.

Remark 5.36 (The Procrustes minimiser and why it is not the whole story). The unique distortion-minimising isometry is the orthogonal polar factor: among $R \in SO(d_{\text{Cl}})$, the minimiser of $\|R E(x) - E(x + h e_i)\|_{\text{HS}}$ is the orthogonal factor of $E(x + h e_i) E(x)^\top$. At first order, writing $R = \mathbf{1} - h\omega + O(h^2)$ with $\omega \in \mathfrak{so}$ and $E(x + h e_i) = E + h E_{,i} + O(h^2)$,

$$\|R E - E'\|_{\text{HS}}^2 = h^2 \|\omega E + E_{,i}\|_{\text{HS}}^2 + O(h^3),$$

and the minimiser solves the Sylvester equation $P\omega + \omega P = 2 \text{Asym}(E E_{,i}^\top)$ with $P = E E^\top \succ 0$, which has a unique antisymmetric solution. This selects a *particular* metric connection – but, as the plaquette analysis below shows, it is generally *not* torsion-free, hence not Levi–Civita. Thus least distortion delivers $Q = 0$ (metric compatibility), not $T = 0$. The two requirements are logically distinct: second-moment covariance supplies the first, while the exchange-symmetric stationary phase will supply the second.

5.3.4 Reset channels as Euclidean motions of the developed frame

A coarse-grained elementary reset transports the renewal frame and advances the developed position. The natural carrier of these two data is the Euclidean motion group $E(d_{\text{Cl}}) = \text{SO}(d_{\text{Cl}}) \ltimes \mathbb{R}_{\text{Cl}}^d$, acting on the developed position $q \in \mathbb{R}_{\text{Cl}}^d$ (recorded in the orthonormal frame) by $(R, t) \cdot q = Rq + t$, with composition $(R_2, t_2)(R_1, t_1) = (R_2 R_1, R_2 t_1 + t_2)$.

Definition 5.37 (Elementary reset motion and first moment). The coarse-grained elementary reset channel in coordinate direction i at base x , $\Phi_i^{(h)}(x)$, transports the local direction records by $T_i(x)$. Under [Assumption 5.1](#), its *developed-frame data* are the Euclidean motion

$$g_i^{(h)}(x) = (R_i(x), h e_i(x)) \in E(d_{\text{Cl}}), \quad R_i(x) = M_{x+he_i}^{-1/2} T_i(x) M_x^{1/2} = \exp(-h \gamma_i(x)) \in \text{SO}(d_{\text{Cl}}),$$

where $e_i(x) = (e_i^a(x))_a$ is the coframe column (the mean reset displacement in the frame, a renewal datum) and $\gamma_i(x)$ is the (a priori unknown) internal frame transport generated by the channel. The *first spatial moment* of a channel with data (R, t) is its translation part,

$$\mathbf{m}_1(\Phi) := t \in \mathbb{R}_{\text{Cl}}^d,$$

i.e. the mean developed displacement of a localised renewal state.

The displacement e_i is fixed by the renewal second moment, while the raw record transport T_i is supplied by the channel. The rotation R_i and hence the metric-compatible part of the connection are not adopted separately: they are the orthonormalised transport derived in [Theorem 5.34](#). An arbitrary Stinespring dilation need not satisfy the covariance hypothesis; the load-bearing requirement is agreement of the transported quadratic record statistics, together with determinant orientation preservation.

5.3.5 Translational and rotational parts of the plaquette defect

For $i \neq j$ form the two opposite plaquette histories and their channel defect

$$\mathcal{P}_{ij}^{(h)}(x) := \Phi_j^{(h)}(x + h e_i) \Phi_i^{(h)}(x) - \Phi_i^{(h)}(x + h e_j) \Phi_j^{(h)}(x),$$

read through the developed-frame data, with translational projection $\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)}$ (difference of first moments) and rotational projection $\Pi_{\text{spin}} \mathcal{P}_{ij}^{(h)}$ (difference of frame rotations).

Lemma 5.38 (Torsion is the translational defect, curvature the rotational defect). *With $g_i^{(h)} = (R_i, h e_i)$ as in [Theorem 5.37](#) and $\gamma_i \in C^1$,*

$$\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)}(x) = h^2 T_{ij}^a(x) + O(h^3), \quad \Pi_{\text{spin}} \mathcal{P}_{ij}^{(h)}(x) = -h^2 F_{ij}(x) + O(h^3),$$

where $T_{ij}^a = \delta_i e_j^a - \delta_j e_i^a + \gamma_i^a e_j^b - \gamma_j^a e_i^b$ is the discrete torsion and $F_{ij} = \partial_i \gamma_j - \partial_j \gamma_i + [\gamma_i, \gamma_j]$ the discrete curvature.

Proof. Translation. In $E(d_{\text{Cl}})$, the composite $g_j(x + h e_i) g_i(x)$ has translation part $R_j(x + h e_i) h e_i(x) + h e_j(x + h e_i)$. Expanding with $R_j = \mathbf{1} - h \gamma_j + O(h^2)$ and $e_j(x + h e_i) = e_j(x) + h \partial_i e_j(x) + O(h^2)$,

$$R_j(x + h e_i) h e_i(x) = h e_i(x) - h^2 \gamma_j(x) e_i(x) + O(h^3), \quad h e_j(x + h e_i) = h e_j(x) + h^2 \partial_i e_j(x) + O(h^2),$$

so the path-A translation is $he_i + he_j + h^2(\partial_i e_j - \gamma_j e_i) + O(h^3)$. The path-B translation is the $i \leftrightarrow j$ image, $he_j + he_i + h^2(\partial_j e_i - \gamma_i e_j) + O(h^3)$. Subtracting, the $O(h)$ terms cancel and the connection terms combine as $-\gamma_j e_i - (-\gamma_i e_j) = \gamma_i e_j - \gamma_j e_i$, giving

$$\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)} = h^2 \left[(\partial_i e_j^a - \partial_j e_i^a) + (\gamma_i^a e_j^b - \gamma_j^a e_i^b) \right] + O(h^3) = h^2 T_{ij}^a + O(h^3),$$

using $\delta_i e_j^a = \partial_i e_j^a + O(h^2)$.

Rotation. The composite rotations are $R_j(x + he_i)R_i(x)$ and $R_i(x + he_j)R_j(x)$. With $R_i = \mathbf{1} - h\gamma_i + O(h^2)$,

$$R_j(x + he_i)R_i(x) = \mathbf{1} - h\gamma_i - h\gamma_j - h^2\partial_i\gamma_j + h^2\gamma_j\gamma_i + O(h^3),$$

and subtracting the $i \leftrightarrow j$ image gives

$$\begin{aligned} \Pi_{\text{spin}} \mathcal{P}_{ij}^{(h)} &= h^2 (-\partial_i\gamma_j + \partial_j\gamma_i + \gamma_j\gamma_i - \gamma_i\gamma_j) + O(h^3) \\ &= -h^2 (\partial_i\gamma_j - \partial_j\gamma_i + [\gamma_i, \gamma_j]) + O(h^3) = -h^2 F_{ij} + O(h^3). \end{aligned}$$

□

5.3.6 The torsion theorem

Assumption 5.2 (Exchange-covariant stationary diamond phase). For each $i \neq j$, let $\mathcal{D}_{ij}$ be a finite set of resolved two-reset diamond records, equipped with an involution $\mathbf{s}_{ij}$ exchanging the histories ij and ji . Let P_{ij} be a primitive Markov kernel satisfying

$$P_{ij}(\mathbf{s}_{ij}a, \mathbf{s}_{ij}b) = P_{ij}(a, b),$$

and let $d_{ij}^{\text{tr}} : \mathcal{D}_{ij} \rightarrow \mathbb{R}^{d_{\text{Cl}}}$ be the developed translational defect, with

$$d_{ij}^{\text{tr}}(\mathbf{s}_{ij}a) = -d_{ij}^{\text{tr}}(a).$$

Writing π_{ij} for the stationary law of P_{ij} , the coarse-grained translational plaquette defect has the expansion

$$\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)} = h^2 \sum_{a \in \mathcal{D}_{ij}} \pi_{ij}(a) d_{ij}^{\text{tr}}(a) + O(h^3).$$

No exchange constraint is imposed on the rotational record.

Theorem 5.39 (Stationary exchange theorem). *Under [Assumption 5.2](#), the stationary law is exchange invariant,*

$$\pi_{ij}(\mathbf{s}_{ij}a) = \pi_{ij}(a),$$

and therefore

$$\sum_a \pi_{ij}(a) d_{ij}^{\text{tr}}(a) = 0, \quad \boxed{\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)} = O(h^3)}.$$

The rotational part of the plaquette is not constrained and may retain nonzero curvature.

Proof. Define $\pi_{ij}^s(a) := \pi_{ij}(\mathbf{s}_{ij}a)$. Exchange covariance of P_{ij} shows that π_{ij}^s is stationary. Primitivity gives a unique stationary law, hence $\pi_{ij}^s = \pi_{ij}$. Pairing every record with its exchanged partner and using exchange oddness gives

$$\sum_a \pi_{ij}(a) d_{ij}^{\text{tr}}(a) = - \sum_a \pi_{ij}(\mathbf{s}_{ij}a) d_{ij}^{\text{tr}}(a) = - \sum_a \pi_{ij}^s(a) d_{ij}^{\text{tr}}(a),$$

so the sum vanishes. The asserted closure follows from the expansion in [Assumption 5.2](#). Since no exchange condition was placed on the rotational record, no analogous cancellation is implied there. $\square$

Proposition 5.40 (One-step detailed balance does not imply torsion freedom). *Ordinary reversibility of the microscopic diamond chain is insufficient to derive the conclusion of [Theorem 5.39](#).*

Proof. Take two records $+, -$, exchanged by $\mathbf{s}(+) = -$, and the primitive kernel

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}.$$

Its stationary law is $\pi = (2/3, 1/3)$, and it satisfies detailed balance because $\pi_+ P_{+-} = \pi_- P_{-+} = 1/6$. For any nonzero vector v , set $d^{\text{tr}}(+) = v$ and $d^{\text{tr}}(-) = -v$. Then

$$\mathbb{E}_\pi d^{\text{tr}} = \frac{1}{3}v \neq 0.$$

The kernel is not exchange covariant, since $P_{++} \neq P_{--}$. Thus ordinary one-step detailed balance does not force the exchange-invariant stationary law needed for translational cancellation. $\square$

Theorem 5.41 (Channel-composition torsion theorem). *Suppose the elementary reset channels admit the smooth Euclidean first-moment data of [Theorem 5.37](#) with $\gamma_i \in C^1$, their transported second moments satisfy [Assumption 5.1](#), and their resolved two-reset records lie in the stationary diamond phase of [Assumption 5.2](#). Then the renewal coframe satisfies*

$$T_{ij}^a \equiv 0.$$

The induced connection is the unique discrete Levi-Civita connection of [Theorem 5.33](#) and converges to $\omega^{\text{LC}} + O(h^2)$. The curvature F_{ij} is unconstrained and survives as rotational holonomy.

Proof. By [Theorems 5.34](#) and [5.35](#), covariance of the transported second moments gives $Q \equiv 0$. By [Theorem 5.39](#), the translational plaquette satisfies $\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)} = O(h^3)$. Comparing this with [Theorem 5.38](#),

$$\Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)} = h^2 T_{ij}^a + O(h^3),$$

forces $T_{ij}^a = 0$. The discrete fundamental theorem then gives the unique metric-compatible torsion-free connection and its $\omega^{\text{LC}} + O(h^2)$ limit. The rotational projection remains $-h^2 F_{ij} + O(h^3)$ and is untouched. $\square$

The logical chain can therefore be read without the intermediate notation:

second-moment covariance $\implies Q = 0$, stationary exchange $\implies \Pi_{\text{trans}} \mathcal{P}_{ij}^{(h)} = O(h^3) \implies T = 0$,

and the discrete fundamental theorem then gives the Levi-Civita connection. At no point is the rotational defect set to zero. The same plaquette that eliminates translational torsion may therefore retain nontrivial curvature.

Remark 5.42 (Why only the translational projection, and why exact commutativity over-kills). The decomposition of [Theorem 5.38](#) is the whole point: torsion is the translational component of the channel plaquette defect and curvature is the rotational component. Requiring the *full* channel defect to vanish, $\|\mathcal{P}_{ij}^{(h)}\|_{\text{cb}} = O(h^3)$ – equivalently full predictive equivalence of the two histories, or exact commutativity of the elementary channels – would force *both* $T = 0$ and $F = 0$, collapsing to a flat (teleparallel-trivial) geometry and discarding curvature and gauge information. This is the case of the commuting-CP regime of the preceding construction, where opposite plaquette histories have identical accumulated channels and the geometry is flat. The curved reconstruction therefore needs exactly the weaker [Theorem 5.39](#): stationary predictive coincidence of *positions*, not full equivalence of *channels*.

Remark 5.43 (Scope). Given transported second-moment covariance and the exchange-covariant stationary diamond phase, the curved Cartan reconstruction is unconditional within this class. What is not claimed is that torsion freedom follows from bare channel associativity or from ordinary one-step detailed balance: [Theorem 5.40](#) gives a counterexample. Torsion-free and torsional phases are both admissible; the exchange-symmetric translational phase is the genuine selector, while the closure equation itself is derived by [Theorem 5.39](#).

5.4 Adiabatic curved Dirac limit, discrete Cartan reconstruction, and stability

The flat operator limit assumed that the second-moment tensor of the reset statistics was constant in space. We now allow the reset statistics to vary slowly. The signed principal symbol then determines the variable Lorentzian inverse metric

$$g^{\mu\nu}(x) = \text{diag}(1, -\kappa^2 M(x)^2).$$

A variable principal symbol alone does not determine the lower-order spin transport. The preceding plaquette theorem identifies torsion with the translational channel defect. Second-moment covariance gives metric compatibility, while stationary exchange symmetry derives the vanishing translational first moment. The resulting connection is therefore the unique discrete Levi–Civita connection. We now prove convergence of the corresponding covariant renewal Hamiltonians.

Throughout, the Krein–Clifford datum of [Definition 5.2](#) is fixed. Write $A^i = \gamma^0 \gamma^i$ for the Hermitian Dirac matrices of [Lemma 5.17](#), with $\{A^i, A^j\} = 2\delta^{ij}\mathbf{1}$, and fix $\kappa > 0$. For a mesh $h_R \downarrow 0$, translations and symmetric differences act on the common Hilbert space $L^2(\mathbb{R}^{d_{\text{Cl}}}, V)$, as in [Definition 5.18](#). This common-space realization is sufficient for the strong-resolvent argument and avoids any additional sampling or interpolation hypothesis.

5.4.1 The emergent coframe and the discrete Cartan connection

Definition 5.44 (Smooth nondegenerate reset field). A *smooth reset field* is a finite family of unit directions $\hat{\theta}_1, \dots, \hat{\theta}_m \in \mathbb{S}^{d_{\text{Cl}}-1}$ and weights $p_a \in C_b^4(\mathbb{R}^{d_{\text{Cl}}})$, $p_a \geq 0$, $\sum_a p_a = 1$, with second moment

$$M(x) = \sum_{a=1}^m p_a(x) \hat{\theta}_a \hat{\theta}_a^\top.$$

We assume uniform nondegeneracy: $M(x) \succeq m_0 I_{d_{\text{Cl}}}$ for some $m_0 > 0$. Set

$$e_a^j(x) := \kappa M^{aj}(x), \quad g^{ij}(x) = \delta^{ab} e_a^i(x) e_b^j(x) = \kappa^2 (M(x)^2)^{ij},$$

and let e_j^a denote the inverse coframe. Then $g^{\mu\nu} = \text{diag}(1, -g^{ij})$ has Lorentzian signature $(1, d_{\text{Cl}})$ at every point.

Let

$$\delta_i^{(R)} f(x) = \frac{f(x + h_R e_i) - f(x - h_R e_i)}{2h_R}$$

and define the discrete coframe structure coefficients

$$C_R^a{}_{bc} := e_b^i e_c^j (\delta_i^{(R)} e_j^a - \delta_j^{(R)} e_i^a), \quad C_{Rabc} := \delta_{ad} C_R^d{}_{bc}.$$

Define the discrete Levi-Civita coefficients by the Koszul formula

$$\gamma_{d;ae}^{(R)} := \frac{1}{2} (C_{Rdea} - C_{Rade} - C_{Read}), \quad \gamma_{i,Rae} := e_i^d \gamma_{d;ae}^{(R)}.$$

Their spin lift and the corresponding link transport are

$$\Omega_{i,R} := \frac{1}{4} \gamma_{i,Rab} A^a A^b, \quad U_{i,R}(x) := \exp(-h_R \Omega_{i,R}(x)).$$

By [Theorem 5.33](#),

$$\gamma_{i,Rab} = \omega_{iab}^{\text{LC}} + O(h_R^2)$$

locally uniformly.

5.4.2 Midpoint spin transport and the curved Hamiltonian

Evaluate the link at edge midpoints and define

$$\nabla_{i,R} \psi(x) := \frac{U_{i,R}(x + \frac{1}{2} h_R e_i)^{-1} \psi(x + h_R e_i) - U_{i,R}(x - \frac{1}{2} h_R e_i) \psi(x - h_R e_i)}{2h_R}.$$

Lemma 5.45 (Second-order covariant consistency). *For every $\psi \in C_c^\infty(\mathbb{R}^{d_{C_1}}, V)$,*

$$\nabla_{i,R} \psi = (\partial_i + \Omega_i^{\text{LC}}) \psi + O_{L^2}(h_R^2), \quad \Omega_i^{\text{LC}} = \frac{1}{4} \omega_{iab}^{\text{LC}} A^a A^b.$$

More precisely, on each compact set containing the support of ψ ,

$$\|\nabla_{i,R} \psi - (\partial_i + \Omega_i^{\text{LC}}) \psi\|_{L^2} \leq C_\psi h_R^2.$$

Proof. Taylor-expand the forward and backward spinors and the midpoint link exponentials around x . The order- h_R^2 brackets in the two transported terms agree and cancel in the central difference; the first surviving truncation term is order h_R^3 , hence order h_R^2 after division by $2h_R$. Replacing γ_R by ω^{LC} costs another $O(h_R^2)$ by [Theorem 5.33](#). The orientation convention in the definition transports both neighbouring spinors back to the fibre over x , yielding the displayed sign of Ω_i^{LC} . $\square$

Set $B^j(x) = A^a e_a^j(x) = \kappa A^a M^{aj}(x)$. On flat measure define

$$\mathsf{H}_g^{\text{LC}} := \frac{1}{2} \sum_j \left[B^j (-i(\partial_j + \Omega_j^{\text{LC}})) + (-i(\partial_j + \Omega_j^{\text{LC}}))^\dagger B^j \right].$$

This is the half-density realization of the geometric spatial Dirac Hamiltonian. Its principal symbol is $B^j \xi_j$, and

$$(B^j \xi_j)^2 = \kappa^2 \xi^\top M(x)^2 \xi \mathbf{1}.$$

Lemma 5.46 (Self-adjointness of the reconstructed curved Hamiltonian). *The operator H_g^{LC} is essentially self-adjoint on $C_c^\infty(\mathbb{R}^{d_{\text{Cl}}}, V)$. Its closure is self-adjoint and satisfies*

$$\|u\|_{H^1} \leq C(\|H_g^{\text{LC}}u\|_{L^2} + \|u\|_{L^2}).$$

The same conclusions hold for $H_g^{\text{LC}} + V_{\text{ren}}$ whenever $V_{\text{ren}} \in L^\infty(\mathbb{R}^{d_{\text{Cl}}}, \mathbb{R})$.

Proof. The principal coefficients B^j are bounded C_b^1 Hermitian matrices, the propagation speed is uniformly bounded, and the Weyl symmetrisation makes the operator symmetric. Essential self-adjointness follows from the same Chernoff argument used in the flat-ordering formulation. Uniform nondegeneracy of M gives first-order ellipticity and the graph estimate. A bounded real scalar potential is a bounded self-adjoint perturbation. $\square$

Definition 5.47 (Covariant discrete renewal Hamiltonian). Let $V_{\text{ren},R}$ be real bounded multiplication operators satisfying

$$\sup_R \|V_{\text{ren},R}\|_\infty < \infty, \quad V_{\text{ren},R} \longrightarrow V_{\text{ren}} \quad \text{locally uniformly.}$$

Define

$$H_R^{\text{LC}} = \frac{1}{2} \sum_j \left[B^j (-i\nabla_{j,R}) + (-i\nabla_{j,R})^\dagger B^j \right] + V_{\text{ren},R}.$$

Each H_R^{LC} is bounded and self-adjoint for fixed R . The purely geometric model is recovered by setting $V_{\text{ren},R} = V_{\text{ren}} = 0$.

Proposition 5.48 (Common-core consistency). *For every $u \in C_c^\infty(\mathbb{R}^{d_{\text{Cl}}}, V)$,*

$$\|H_R^{\text{LC}}u - (H_g^{\text{LC}} + V_{\text{ren}})u\|_{L^2} \longrightarrow 0.$$

If $V_{\text{ren},R} - V_{\text{ren}} = O(h_R^2)$ locally uniformly, then the left-hand side is $O_u(h_R^2)$.

Proof. Lemma 5.45 controls the covariant differences. Multiplication by the C_b^4 coefficients B^j , their discrete adjoints, and the Weyl symmetrisation preserve the estimate and reproduce the half-density divergence term in H_g^{LC} . The potential contribution converges by local uniform convergence on the compact support of u . Summing over the finitely many coordinate directions gives the claim. $\square$

Theorem 5.49 (Curved Levi–Civita strong-resolvent limit). *Under Definitions 5.44 and 5.47,*

$$(H_R^{\text{LC}} - z)^{-1} \longrightarrow (H_g^{\text{LC}} + V_{\text{ren}} - z)^{-1} \quad \text{strongly}$$

for every $z \in \mathbb{C} \setminus \mathbb{R}$. Consequently,

$$e^{-itH_R^{\text{LC}}} \longrightarrow e^{-it(H_g^{\text{LC}} + V_{\text{ren}})}$$

strongly, uniformly for t in compact intervals, and

$$\varphi(H_R^{\text{LC}}) \longrightarrow \varphi(H_g^{\text{LC}} + V_{\text{ren}}) \quad \text{strongly}$$

for every $\varphi \in C_0(\mathbb{R})$.

The limiting operator is the full geometric Dirac Hamiltonian through subprincipal order: its principal symbol determines

$$\xi_0^2 = \kappa^2 \xi^\top M(x)^2 \xi,$$

its spin subprincipal term is the Levi–Civita connection, and the only additional term retained here is the explicitly separated real scalar V_{ren} .

Proof. Proposition 5.48 gives convergence on C_c^∞ , which is a core for the limiting self-adjoint operator by Lemma 5.46. Every H_R^{LC} is self-adjoint. By the strong graph-convergence criterion for self-adjoint operators, equivalently the common-core Trotter–Kato theorem, convergence

$$H_R^{LC} u \longrightarrow (H_g^{LC} + V_{\text{ren}})u, \quad u \in C_c^\infty(\mathbb{R}^{d_{\text{Cl}}}, V),$$

on a core of the limiting self-adjoint operator implies strong-resolvent convergence. The unitary-group statement follows uniformly on compact time intervals by Stone’s theorem and the Trotter–Kato equivalence, and the $C_0(\mathbb{R})$ functional-calculus statement follows from the bounded-continuous functional-calculus form of strong-resolvent convergence. $\square$

The curved limit is the endpoint of the reconstruction chain. The renewal statistics determine the principal metric, their transported covariance and stationary exchange determine the Levi–Civita spin term, and the discrete covariant operators converge to the resulting variable-coefficient Hamiltonian. The optional scalar V_{ren} is displayed separately so that it cannot be confused with either the metric or the spin connection.

Proposition 5.50 (Worked reversible reset-diffusion residue). *Let $\rho_* \in C_b^2(\mathbb{R}^{d_{\text{Cl}}})$ satisfy $0 < \rho_- \leq \rho_* \leq \rho_+ < \infty$, and consider the reversible reset-diffusion generator*

$$L_{\text{ren}} = D\rho_*^{-1} \nabla \cdot (\rho_* \nabla) = D(\Delta + \nabla \log \rho_* \cdot \nabla).$$

Under the ground-state transform $\Theta\psi = \rho_^{1/2}\psi$,*

$$\Theta L_{\text{ren}} \Theta^{-1} = D\Delta - V_{\text{ren}}, \quad V_{\text{ren}} = D\rho_*^{-1/2} \Delta \rho_*^{1/2}.$$

Thus the gradient drift is removed and the residual renewal correction is a bounded real scalar, with

$$\|V_{\text{ren}}\|_\infty \leq D \left(\frac{\|\Delta \rho_*\|_\infty}{2\rho_-} + \frac{\|\nabla \rho_*\|_\infty^2}{4\rho_-^2} \right).$$

A consistent discrete realization is

$$V_{\text{ren},R} = D\rho_{*,R}^{-1/2} \Delta_R \rho_{*,R}^{1/2},$$

for positive $\rho_{,R} \rightarrow \rho_*$ with the corresponding discrete derivatives converging locally uniformly.*

Proof. The product rule applied to $\rho_*^{1/2} L_{\text{ren}}(\rho_*^{-1/2} \psi)$ cancels the first-order terms and gives the displayed scalar potential. Expanding $\Delta \rho_*^{1/2}$ yields

$$V_{\text{ren}} = D \left(\frac{1}{2} \frac{\Delta \rho_*}{\rho_*} - \frac{1}{4} \frac{|\nabla \rho_*|^2}{\rho_*^2} \right),$$

from which the bound follows. This is a worked reversible diffusion model, not a claim that every finite CP reset memory produces the same scalar. $\square$

Remark 5.51 (Scope of the curved closure). The curved result has the following precise scope: the renewal second moment determines the coframe and Lorentzian principal symbol; covariance of transported second moments derives metric compatibility; exchange symmetry of the stationary diamond law cancels the translational plaquette defect while leaving its rotational part free to retain curvature; the resulting discrete Cartan system uniquely reconstructs the Levi–Civita spin connection; and the corresponding self-adjoint covariant Hamiltonians converge globally in strong-resolvent sense. A smooth-cutoff, variable-coefficient norm-resolvent estimate remains a separate microlocal problem, while full norm-resolvent convergence is obstructed by lattice doublers.

Remark 5.52 (Why uniform ellipticity is needed). Uniform ellipticity $M(x) \succeq m_0 I$ is essential: it bounds the propagation speed away from degeneration and yields the elliptic estimate of Lemma 5.46. Where $M(x)$ degenerates, the cone closes and the geometry becomes horizon-like; such loci are excluded here and would require a genuinely different (degenerate-hyperbolic) analysis. As in Theorem 5.28, the signature is carried by the signed/Krein datum and the normalized temporal generator, convergence is strong (not norm) resolvent, and no curvature dynamics, field equation, or matter coupling is asserted.

5.4.3 Stability under perturbation of the reset statistics

We now show the construction is robust: a small change of the reset directions produces a small change of the metric, the cone, and the Hamiltonian. We treat the flat (constant- M) case in full and record the curved variant.

Theorem 5.53 (Stability of the rounded Lorentzian endpoint). *Fix the Krein-Clifford datum. Let two direction measures $\nu, \tilde{\nu}$ on $\mathbb{S}^{d_{\text{Cl}}-1}$ have constant second moments $M = M(\nu)$, $\tilde{M} = M(\tilde{\nu})$ with*

$$\|M - \tilde{M}\| \leq \chi, \quad M \succeq m_0 I, \quad \chi < m_0.$$

Then, with all constants explicit:

- (i) (signature is rigid) $\tilde{M} \succeq (m_0 - \chi)I \succ 0$, so $g_{\tilde{\nu}}$ has the same Lorentzian signature $(1, d_{\text{Cl}})$ as g_{ν} ;
- (ii) (metric) $\|g_{\nu} - g_{\tilde{\nu}}\| = \kappa^2 \|M^2 - \tilde{M}^2\| \leq 2\kappa^2 \chi$;
- (iii) (cone) on the section $\{|\vec{\xi}| = 1\}$ the null cones $\xi_0 = \pm \kappa |M\vec{\xi}|$ and $\xi_0 = \pm \kappa |\tilde{M}\vec{\xi}|$ are uniformly $\kappa\chi$ -close, so the cones are $O(\chi)$ -close in Hausdorff distance on compact sections;
- (iv) (Hamiltonian) $H_{g_{\nu}} - H_{g_{\tilde{\nu}}} = \kappa A^i (M - \tilde{M})^{ij} (-i\partial_j)$ is first order with

$$\|(H_{g_{\nu}} - H_{g_{\tilde{\nu}}})u\|_{L^2} \leq \kappa \|A\| \chi \|u\|_{H^1} \quad (u \in H^1);$$

- (v) (norm resolvent) for every $z \in \mathbb{C} \setminus \mathbb{R}$, the two operators have the common graph domain H^1 and

$$\|(H_{g_{\nu}} - z)^{-1} - (H_{g_{\tilde{\nu}}} - z)^{-1}\| \leq \frac{\kappa \|A\| C(z, \kappa, m_0)}{|\text{Im } z|} \chi = O(\chi);$$

- (vi) (propagator) for $u \in H^1$, $\|(e^{-itH_{g_{\nu}}} - e^{-itH_{g_{\tilde{\nu}}}})u\|_{L^2} \leq |t| \kappa \|A\| \chi \|u\|_{H^1}$.

Proof. (i) $\tilde{M} = M - (M - \tilde{M}) \succeq m_0 I - \chi I = (m_0 - \chi)I \succ 0$, so by Theorem 5.5 $g_{\tilde{\nu}} = \text{diag}(1, -\kappa^2 \tilde{M}^2)$ has signature $(1, d_{\text{Cl}})$.

(ii) $g_{\nu} - g_{\tilde{\nu}} = \text{diag}(0, -\kappa^2(M^2 - \tilde{M}^2))$ and $M^2 - \tilde{M}^2 = M(M - \tilde{M}) + (M - \tilde{M})\tilde{M}$, so $\|M^2 - \tilde{M}^2\| \leq (\|M\| + \|\tilde{M}\|)\chi \leq 2\chi$ since second moments on the unit sphere satisfy $\|M\|, \|\tilde{M}\| \leq 1$.

(iii) For $|\vec{\xi}| = 1$, $||M\vec{\xi}| - |\tilde{M}\vec{\xi}|| \leq |(M - \tilde{M})\vec{\xi}| \leq \|M - \tilde{M}\| \leq \chi$, so the two cone graphs over the spatial sphere are within $\kappa\chi$.

(iv) The principal parts differ by $\kappa A^i (M - \tilde{M})^{ij} (-i\partial_j)$ (the constant- M symmetrisation term vanishes); for $u \in H^1$, $\|\kappa A^i (M - \tilde{M})^{ij} (-i\partial_j)u\| \leq \kappa \|A\| \|M - \tilde{M}\| \|\nabla u\| \leq \kappa \|A\| \chi \|u\|_{H^1}$.

(v) By Lemma 5.46 (flat case) both Hamiltonians have common graph domain H^1 and are uniformly elliptic, so $(H - z)^{-1}$ maps $L^2 \rightarrow H^1$ with norm bounded by $C(z, \kappa, m_0)$: indeed

$\|\nabla v\| \leq (\kappa m_0)^{-1} \|Hv\|$ and $\|H(H - z)^{-1}\| \leq 1 + |z|/|\operatorname{Im} z|$. The perturbation in (iv) is therefore bounded from the common graph domain H^1 to L^2 . The second resolvent identity

$$(H_{g_\nu} - z)^{-1} - (H_{g_{\bar{\nu}}} - z)^{-1} = (H_{g_\nu} - z)^{-1} (H_{g_{\bar{\nu}}} - H_{g_\nu}) (H_{g_{\bar{\nu}}} - z)^{-1},$$

together with $\|(H_{g_\nu} - z)^{-1}\|_{L^2 \rightarrow L^2} \leq |\operatorname{Im} z|^{-1}$, the $H^1 \rightarrow L^2$ bound (iv) for the middle factor, and the $L^2 \rightarrow H^1$ bound for the right factor, gives the stated norm bound.

(vi) Duhamel's formula $e^{-itH_{g_\nu}} - e^{-itH_{g_{\bar{\nu}}}} = -i \int_0^t e^{-i(t-s)H_{g_\nu}} (H_{g_\nu} - H_{g_{\bar{\nu}}}) e^{-isH_{g_{\bar{\nu}}}} ds$, applied to $u \in H^1$ (which $e^{-isH_{g_{\bar{\nu}}}}$ preserves, the propagator commuting with the elliptic $H_{g_{\bar{\nu}}}$), and the bound (iv), give the estimate after taking norms under the integral. $\square$

Corollary 5.54 (Curved stability). *If instead $M, \tilde{M} \in C_b^1$ are space-dependent with $\|M - \tilde{M}\|_{C^1} \leq \chi$ and both $\succeq m_0 I$, then conclusions (i)–(iii) hold pointwise in x , and (iv)–(vi) hold with the same $O(\chi)$ rate, the only change being that the difference of Hamiltonians now carries the bounded zeroth-order term $-\frac{i\kappa}{2} A^t \partial_j (M - \tilde{M})^{ij}$, of norm $\leq \frac{\kappa}{2} \|A\| \chi$, which is absorbed into the $H^1 \rightarrow L^2$ bound.*

Proof. Identical to [Theorem 5.53](#), with the extra bounded term estimated by $\|\partial(M - \tilde{M})\|_\infty \leq \chi$; ellipticity holds for both operators by [Lemma 5.46](#). $\square$

Remark 5.55 (Stability is stronger than the discretisation limit). The discretisation limits of [Theorems 5.20](#) and [5.49](#) are *strong* resolvent only, the symmetric-difference obstruction [\[52, 53\]](#) ruling out norm resolvent convergence in $d_{\text{Cl}} \geq 2$. By contrast, [Theorem 5.53](#) (v) is *norm* resolvent stability: perturbing M at fixed continuum keeps both operators in the same uniformly elliptic class with common H^1 graph domain, so the difference is a genuinely first-order $H^1 \rightarrow L^2$ perturbation with no mesh pathology. Robustness of the emergent geometry is therefore quantitatively sharper than its emergence.

Remark 5.56 (Novelty). The analytic ingredients are classical: Chernoff's essential self-adjointness for first-order symmetric hyperbolic systems [\[57\]](#), the strong-resolvent core criterion and Trotter–Kato group convergence [\[51\]](#), and second-resolvent/Duhamel perturbation bounds. The discrete-to-continuum Dirac literature [\[52, 53\]](#) treats a *constant* principal part (flat metric) with at most a bounded matrix potential; it does not address a position-dependent principal symbol $M(x)$, which is exactly the curved case here. The contribution of this section is therefore the *assembly*: position-dependent reset statistics yield a self-adjoint variable-coefficient Dirac Hamiltonian whose principal symbol is an emergent curved Lorentzian metric field, together with a quantitative norm-resolvent stability of that field under perturbation of the reset data. The present statement is the renewal/Krein counterpart of these stability principles, with an explicit $O(\chi)$ rate for perturbations of the reset statistics.

Taken together, [Sections 2–5](#) establish a single dependency chain:

predictive quotient $\implies$ positive spectral geometry and order $\implies$ signed double cover
 $\implies$ projective Clifford factor $\implies$ Lorentzian Dirac symbol and evolution $\implies$ Levi–Civita curved limit.

Each arrow has a distinct logical status recorded in the preceding sections: some are canonical constructions, some are no-go implications, and some require selection of the stated recurrent, reversible, elliptic, or exchange-symmetric phase. The next section interprets this theorem chain and states its physical limitations; it adds no further mathematical input.

6 Interpretation, scope, and conclusion

6.1 From predictive memory to Lorentzian geometry

The results of the preceding sections can be understood as a single conceptual progression. The construction begins with no manifold, metric, causal cone, preferred time direction, or prescribed spacetime dimension. Its primitive data consist only of renewal events and the completely positive channels generated by their composition. A history is therefore not yet a trajectory through spacetime, but a sequence of updates distinguished only by the predictive action it retains. The decisive step is to identify histories that are indistinguishable by every future continuation:

$$w \sim_{\text{CP}} w' \quad \Longleftrightarrow \quad \Phi_w = \Phi_{w'}.$$

This predictive quotient transforms a collection of symbolic histories into a positive spectral geometry. Distinguishable predictive actions become its points; minimal renewal depth becomes length; continuation becomes order; and the growth of distinguishable classes becomes dimension. Positivity nevertheless imposes a sharp boundary. It can organise what prediction retains, but it cannot by itself distinguish a negative sector or produce a Lorentzian signature. Crossing that boundary requires the minimal signed enrichment, a principal double cover of the positive predictive geometry. Reversible predictive transformations then generate a projective commutator structure, whose noncentral reduction yields a Clifford factor. After nondegenerate coarse-graining, this signed Clifford geometry admits controlled flat and slowly varying Lorentzian Dirac limits. In this precise sense, the paper does not place renewal dynamics inside an already existing spacetime. It shows instead how the kinematic architecture normally attributed to spacetime can arise as the stable large-scale organisation of predictive renewal memory.

This interpretation should not be confused with the claim that every renewal process produces a universe, or that the complete dynamics of gravity has already been derived. The theorem chain distinguishes the finite UCP renewal starting object and its analytic domain conditions from the phase conditions selecting nontrivial signed holonomy, reversible projective transport, nondegenerate spatial scaling, and an exchange-symmetric stationary translational regime. What it establishes is more specific: once a renewal process enters the stated predictive, signed, reversible, and coarse-grained phase, the resulting operator geometry is not merely analogous to Lorentzian geometry. Its principal symbol, causal orientation, Clifford structure, and continuum Hamiltonian have the corresponding Lorentzian form. The significance of the construction is therefore structural rather than cosmological. It identifies a route by which geometry can be secondary to memory without becoming metaphorical.

Renewal, opposite orientation, and cancellation. Renewal is ordinarily understood as a restart: part of the past is discarded, a new local state is formed, and evolution continues from the retained information. In the present setting, renewal is more abstract. It is the operation by which a history is extended while only its future predictive content remains relevant. The predictive quotient formalises this economy of memory. Two microscopically different histories become one state when no future continuation can distinguish them. Identity is therefore not defined by sameness of origin, but by sameness of consequence. A predictive state is what remains invariant after all details that make no difference to the future have been removed.

The signed cover introduces a second possibility that is invisible to the positive quotient. Every renewal continuation has a deck-related partner carrying the opposite Krein orientation. This partner is not an inverse completely positive channel, a retrodictive dual, a literal reversal of

microscopic time, or the destruction of a renewal event. It is the oppositely oriented realisation of the same positive predictive content once the minimal signed layer is retained. The two sheets therefore agree on the positive facts of prediction while differing in the sign with which those facts are embedded into the indefinite geometry.

This distinction makes cancellation possible. Positive probabilities do not cancel: the completely positive sector only adds and averages positive data. By contrast, signed amplitudes or signed observables can receive equal contributions from the two sheets with opposite orientation. When the two renewal orientations are exactly balanced, their signed contribution disappears, even though their common positive predictive content remains. The positive quotient records what both orientations agree upon; the signed sector records the difference that their agreement suppresses. Cancellation is thus not the erasure of reality but the removal of oriented excess. What survives perfect cancellation is the unsigned geometry of distinguishability, continuation, and volume. What survives imperfect cancellation is an orientation-sensitive residue capable of supporting a temporal Clifford direction.

This gives the signed construction a clear metaphysical reading. The positive sector describes being insofar as it is stably reproducible in prediction. The signed sector describes a difference of orientation that cannot be reduced to additional positive information. In that sense, Lorentzian signature does not arise because one more positive degree of freedom has been appended to the system. It arises because the same predictive content admits two oppositely oriented realisations whose relation is irreducibly signed. The negative direction is not another object among the objects. It is a structural opposition within the way predictive content can be carried.

From balance to temporal orientation. At the level of the double cover, the two renewal orientations are formally symmetric. The deck transformation exchanges the two sheets, and the positive quotient forgets which sheet was used. If the full state and dynamics were exactly invariant under this exchange, neither orientation would be physically preferred. The signed geometry would still exist, and its algebraic relations would still distinguish opposite orientations, but there would be no net arrow selecting one as the realised direction of becoming.

A temporal orientation appears when this balance is no longer exact. The mathematical construction identifies two ingredients of such an imbalance. First, a nontrivial signed holonomy makes the opposition between the two orientations globally non-removable: no local choice of sheet can eliminate it around every recurrent cycle. Second, modular homogeneity ties elementary renewal steps to a common scaling flow, making renewal depth into an oriented modular clock. Together with the canonical hyperbolic pair in the reversible revision sector, this produces the distinguished timelike Clifford generator. The resulting orientation is not inserted as an external coordinate. It is the operator-theoretic expression of a persistent mismatch between the two signed ways of carrying the same predictive process.

One may therefore interpret temporal becoming as the failure of cancellation between the two renewal orientations to remain exact across the recurrent structure. Under perfect balance, the signed excess vanishes and no orientation is selected at the level of states. Under broken balance, the difference accumulates coherently and becomes macroscopic. The emergence of time is then not the appearance of an additional substance called “time,” but the stabilization of an oriented asymmetry within renewal itself. The future is the direction in which predictive continuation is modularly organised; the opposite sheet represents the formally available anti-orientation against which this direction is defined.

This interpretation must be stated with care. The manuscript proves a distinguished modular orientation and an intrinsic timelike Clifford slot. It does not prove that a particular physical state

spontaneously breaks the deck symmetry, nor does it derive thermodynamic irreversibility, entropy production, or the observed cosmological arrow of time. Those would require additional state-selection and dynamical results. The present construction instead isolates the exact structural location at which such an arrow could arise: not in the positive renewal data alone, but in a persistent imbalance between the two signed orientations of that data.

Spacetime as stabilised predictive relation. The broader metaphysical conclusion is relational. Spacetime is not treated as a container that exists before events and gives them somewhere to occur. The primitive notion is continuation: one predictive state can be prolonged into another, some histories remain distinguishable, and others collapse to the same future action. Order expresses possible continuation, distance expresses the renewal depth required to realise a predictive distinction, and dimension expresses the rate at which new distinctions become available. The signed layer adds the irreducible opposition needed for Lorentzian signature, while reversible projective dynamics supplies the spinorial structure through which those distinctions propagate.

Smooth geometry appears only after coarse-graining. At finite resolution the order cone may be polyhedral, the reset frame anisotropic, and the empirical second moments history-dependent. Nothing at this scale requires the system to resemble a manifold. Stationary self-averaging explains how these fluctuating local frames can nevertheless produce a deterministic macroscopic coframe. The continuum metric is therefore not a microscopic object that has been sampled imperfectly. It is a stable law of organisation that emerges when the variable details of renewal cease to matter at the scale of propagation. Constant anisotropy is absorbed into the choice of spatial frame, while slowly varying statistics appear as a curved coframe whose plaquette defects separate torsion from curvature.

The same perspective clarifies the dimensional result. Dimension is not chosen by declaring how many coordinates reality possesses. It is constrained by the rank of the noncentral projective revision sector and by the requirement that the spatial part support a genuine two-direction interference or plaquette structure. One temporal direction leaves an odd spatial rank; the first nontrivial spatial bivector sector occurs in three dimensions. Thus $3 + 1$ is the first primitive Lorentzian endpoint of the mechanism, although higher odd spatial ranks remain algebraically possible without an additional selection principle.

The resulting picture is neither substance metaphysics nor pure instrumentalism. Predictive equivalence is operational, but the structures reconstructed from it are objective invariants of the renewal process: cohomology classes, spectral growth, projective commutators, Clifford rank, and continuum operator limits. The proposal is therefore that spacetime is real as a stable relational form, rather than fundamental as a prior arena. Its causal, spectral, temporal, and spinorial features are what remain when renewal histories are quotiented by predictive equivalence, their hidden orientation difference is retained, and their fluctuations are organised into a coherent large-scale limit.

6.2 What lies upstream of renewal memory?

The construction developed in this paper begins with renewal memory, but it does not follow that renewal memory is the final ontological layer. Even the primitive renewal object already contains several structural commitments. It assumes that elementary updates can be distinguished, that they can be composed into histories, that their effects act on an algebra of possible observables, and that two histories can be compared through the future action they induce. Later stages assume that some recurrent transformations are reversible, that a nontrivial signed sector is realised, that

the renewal grading is compatible with a modular flow, and that large-scale statistics enter a nondegenerate coarse-grained phase. These are not assumptions of an ambient spacetime, but they are nevertheless assumptions about what the pregeometric world is capable of doing.

It is therefore useful to distinguish the mathematical starting point of the manuscript from the most upstream metaphysical possibility suggested by it. The mathematical starting point is a finite family of completely positive renewal channels and the histories generated by their composition. The more primitive possibility is not a set of histories already given, but a capacity for distinctions to be made, retained, combined, and forgotten. On this view, a history is not a fundamental object. It is the record produced when elementary distinctions are composed in an order. Renewal is the rule by which some part of that record ceases to matter, while its consequences remain available for further composition.

The most economical upstream structure would therefore contain three ingredients: possible distinctions, composability, and retained consequence. One may represent them schematically as

distinction; $\longrightarrow$; update; $\longrightarrow$; history; $\longrightarrow$; predictive action.

The predictive quotient then removes everything in a history that makes no difference to its future action. What becomes geometrically real is consequently not the full microscopic record, but the structure of distinctions that remain effective under future continuation. The primitive “points” of the emergent geometry are not events considered in isolation. They are equivalence classes of pasts that retain different powers over what can happen next.

This suggests that the most fundamental entity is neither a spacetime point nor a history in the ordinary sense. A spacetime point is already a geometric notion, while a history presupposes a criterion for distinguishing successive updates. More fundamental than either is a relational capacity: the capacity of one effective distinction to condition another. The basic reality is then not a collection of things situated in a container, but a structured field of possible continuations. Histories are paths through this field; predictive states are the equivalence classes of paths that have the same remaining consequences; and spacetime is the stable geometry of those equivalence classes when viewed at large scale.

From this perspective, the observable universe is a particular phase of a deeper compositional structure. The positive predictive geometry expresses the stable content shared by the two renewal orientations. The signed cover retains the opposition between their two orientations. Reversible revision dynamics converts that opposition into a projective Clifford structure, while modular imbalance selects a temporal orientation. Coarse-graining then turns the finite and fluctuating renewal frame into the smooth Lorentzian geometry within which macroscopic propagation occurs. What we call spacetime is thus not the primitive arena of the process; it is the regular form assumed by the process once predictive distinctions, signed orientation, and recurrent composition have stabilised together.

The hypotheses of the present construction do not all have the same logical status. The finite UCP renewal memory is the mathematical starting object; finite fibres, bounded increments, and continuum regularity are analytic domain conditions. Nontrivial signed holonomy, a realised reversible projective sector, pressure supercriticality, elliptic spatial scaling, and the exchange-symmetric stationary regime are phase conditions selecting the Lorentzian universality class. The choice of one oriented extremal state is a physical or boundary-condition selection. Regular variation, exact isotropy, metric coding, and higher-rank uniqueness conditions are optional strengthenings used only for sharper spectral or selection statements. Within the exchange-symmetric stationary phase, translational plaquette closure is derived rather than independently imposed,

and the torsion-free branch follows from the resulting vanishing of the exchange-odd translational defect.

Some of these conditions may themselves arise dynamically. Complete positivity may be the effective remnant of a more primitive reversible evolution after inaccessible degrees of freedom have been discarded. Renewal may arise when only a finite sufficient record can be retained. Predictive equivalence may emerge because no internal observer can access differences that leave all subsequent interactions unchanged. The signed sector may originate in a deeper duality whose two branches are identical at the level of positive probabilities but distinct at the level of coherent orientation. Modular homogeneity may be the fixed-point behaviour of a renormalisation process rather than a microscopic law. The axioms would then describe the universality class in which Lorentzian spacetime appears, not the absolute beginning of the ontology.

The deepest candidate suggested by the construction is therefore not “history” but *effective difference under continuation*. Something is real at the pregeometric level insofar as its presence changes the space of possible futures. Two records that never differ in their future action are one predictive state; two orientations that agree positively but differ coherently form the signed pair; and a persistent asymmetry between those orientations becomes temporal direction. Geometry emerges when the network of such effective differences acquires stable growth, order, orientation, and propagation laws.

Under this interpretation, the universe is fundamentally processual but not merely sequential. Sequence alone would give a list of events, whereas the renewal construction requires equivalence, composition, recurrence, and orientation. The fundamental reality is a compositional structure in which distinctions may be created, erased, or retained according to their future effects. Our Lorentzian spacetime is the macroscopic fixed form of that structure: the geometry seen when predictive memory has become sufficiently regular that its relations can be represented as distance, causal order, spinorial propagation, curvature, and a single oriented time.

6.3 Conclusion

We have shown that a finite renewal process of completely positive channels can carry substantially more geometric structure than its microscopic definition appears to contain. Predictive quotienting turns update histories into a positive spectral geometry and a divisibility order: distinguishable predictive actions become points, minimal renewal depth becomes length, continuation becomes causal order, and the growth of distinguishable classes becomes dimension and residue volume. The same construction also exposes a sharp boundary. Positivity can organise what prediction retains, but it cannot by itself produce a Lorentzian fundamental symmetry.

The minimal passage beyond this boundary is a principal $\mathbb{Z}/2$ -cover, with a $\mathbb{Z}/4$ -lift only when the signed holonomy is retained at amplitude level. On the dynamically realised reversible subgroup, predictive units act by automorphisms, their inner implementers compose projectively, and the noncentral reduction of the resulting commutator form is a Clifford matrix factor. The signed cover fixes the temporal row, while a normalized hyperbolic pair gives the intrinsic timelike generator

$$\gamma^0 = J_\chi R_{\alpha_0}, \quad (\gamma^0)^2 = -\mathbf{1}.$$

Under subexponential shell growth, the associated modular signed operator has compact resolvent and regularized all-orders summability. The dimensionless modular exponent is not left arbitrary: on a supercritical recurrent weighted component it is the unique positive root of the associated Perron–Frobenius pressure equation. After nondegenerate coarse-graining, the derived Clifford symbols converge to Lorentzian Dirac symbols with one distinguished timelike direction, and the

corresponding Hamiltonians converge globally in strong resolvent sense and in norm resolvent sense on every fixed momentum band.

The slowly varying regime extends this reconstruction from flat propagation to curved transport. Covariance of neighbouring second moments makes the normalised frame comparison orthogonal and, after orientation selection, special orthogonal, so metric compatibility is derived rather than separately postulated. Channel composition separates translational torsion from rotational curvature. In the primitive exchange-covariant stationary diamond phase, exchange invariance forces the exchange-odd translational defect to vanish, while leaving the rotational record free. The resulting translational closure selects the torsion-free branch, and the discrete fundamental theorem yields the Levi–Civita spin connection without suppressing curvature. The covariant renewal Hamiltonians then converge through subprincipal order to the corresponding variable-coefficient Lorentzian Dirac Hamiltonian. Primitive rank parity, together with the requirement of a nontrivial spatial bivector sector, makes $3 + 1$ dimensions the first nondegenerate Lorentzian endpoint of the mechanism, although higher odd spatial ranks remain algebraically admissible without an additional selection principle.

These results are kinematic and operator-theoretic. They do not yet provide a field equation governing the reconstructed curvature, and they do not derive Einstein’s equations, a gravitational variational principle, matter multiplets, gauge interactions, black-hole thermodynamics, or phenomenological predictions. Such structures lie downstream of the present construction. It is equally important to distinguish theorem from interpretation. Predictive order, the positivity obstruction, the minimal signed-cover classification, modular affineness under its stated covariance hypothesis, the projective Clifford factor, the intrinsic temporal generator, and the continuum convergence results are mathematical statements. The description of imbalance between the two renewal orientations as a physical arrow of time, or of finite-scale renewal fluctuations as quantum spacetime fluctuations, is a metaphysical and physical interpretation of where further dynamics may enter; it is not required by the theorem chain.

The central conclusion is therefore not that a complete theory of the universe has already been obtained. It is that the basic kinematic architecture of Lorentzian spacetime need not be placed beneath predictive dynamics as a prior arena. Under the hypotheses isolated here, causal order, spectral dimension, temporal orientation, spinorial propagation, and curved Levi–Civita transport arise as different aspects of one renewal structure. Spacetime can thus be understood as the stable large-scale form of predictive memory: not the container in which renewal occurs, but the geometry that renewal acquires when distinguishability, signed orientation, reversibility, and coarse-grained propagation become coherent.

A Logical status and reduction of the hypotheses

The mathematical starting object of this paper is the finite UCP renewal memory of [Theorem 2.1](#); no claim is made here that bare operational composition forces every renewal system into the geometric regime studied below. To keep the logical status of the assumptions explicit, four distinct roles should not be conflated.

- (i) *Operational antecedents* concern composability, testing, predictive identification, and stability under ancillary extension. They lie upstream of the present paper and are not used as hidden hypotheses in its theorem proofs.
- (ii) *Microscopic model specifications* are the concrete alphabet, channels, record transports, branch weights, and local propagation laws selecting one renewal model inside the admissible

class.

- (iii) *Phase conditions* select the signed Lorentzian universality class: a recurrent component with nontrivial signed holonomy, a realised reversible projective sector of the required rank, a supercritical pressure transfer, a positive spatial second moment, and an exchange-symmetric translational diamond phase.
- (iv) *Physical selections* choose a realised component or extremal orientation when several stationary phases coexist. Such a selection is needed to speak of a realised arrow or state, but not to establish the kinematic operator identities common to the deck-related phases.

The remaining hypotheses have a separate analytic status. Finite shift fibres, bounded length increments, bounded propagation, and the stated smoothness bounds are domain conditions for the particular spectral or convergence theorem in which they occur. Regular variation is required for the sharp Weyl coefficient and residue volume, not for the existence of the predictive triple or the Lorentzian continuum. Exact isotropy is only a convenient representative of the flat isometry class. Lipschitz metric coding, uniform separation, and the dimension-uniqueness criteria of the appendices are optional strengthenings, not premises of the core Lorentzian result.

Proposition A.1 (Canonical reduction of global standing conditions). *Let G be a finite resolved predictive graph, let a finite group K act by automorphisms on a finite-dimensional complex C^* -algebra $\mathcal{A}$, and let ω be the alternating commutator form of any scalar projective implementation on a selected simple block. Then:*

- (a) *every stationary recurrent branch law is supported on terminal strongly connected components, so all recurrent constructions may be performed on one selected terminal component;*
- (b) *after selecting one invariant central orbit of $\mathcal{A}$, choose one simple block and restrict to its stabiliser in K ; quotienting the stabiliser's action kernel makes the action faithful, and every resulting automorphism of the selected matrix block is inner;*
- (c) *quotienting the projective label space by $\text{rad } \omega$ gives a nondegenerate alternating space and hence the canonical primitive matrix factor.*

Thus global strong connectivity, global faithfulness, global irreducibility, and nondegeneracy of the unreduced projective form are not independent phase axioms.

Proof. The condensation graph of a finite directed graph is acyclic. A stationary law assigns zero mass to a component from which probability can leave without return, and every recurrent path eventually remains in a terminal component, which proves (a). A finite-dimensional complex C^* -algebra is a finite direct sum of matrix blocks. The action permutes its minimal central projections. After selecting one orbit and one block in that orbit, its stabiliser acts on the block; quotienting the stabiliser's action kernel makes that action faithful, and every automorphism of the resulting matrix block is inner. This proves (b). A homogeneous projective implementer is central exactly when its label lies in $\text{rad } \omega$. The induced form on the quotient has zero radical, and finite Stone–von Neumann identifies its twisted group algebra with a full matrix algebra, proving (c). $\square$

Remark A.2 (Residual genuine phase content). After [Theorem A.1](#), the genuinely selective inputs are the existence of a terminal component carrying nontrivial stable-record sign holonomy,

a nontrivial realised reversible projective sector, pressure supercriticality, selection of a deck-oriented extremal state when a realised orientation is claimed, an elliptic spatial scaling phase, an exchange-symmetric translational phase rather than a torsional phase, and the continuum regularity required by the desired operator-convergence statement. Constant anisotropy is removed by [Theorem 5.21](#); regular variation and the optional dimension selectors do not enter this list.

B Regularity criteria and auxiliary positive invariants

The following results provide concrete classes satisfying the main-text regularity criterion and record auxiliary notions of predictive dimension that are not needed for the Lorentzian continuum theorem.

B.1 Finite fibres and bounded increments

The definitions above give a quotient set and descended creation maps, but not yet a spectral geometry. For the length operator to play the role of a Dirac operator, the creation maps must be bounded algebra elements and their commutators with the length Dirac must be bounded. These two requirements have an elementary form in the present setting: the shifts must have uniformly finite fibres, and the quotient length must change by a bounded amount under each reset.

We record several regular classes in which this condition holds with explicit constants. This is useful because [Assumption 2.1](#) is not intended as an opaque analytic assumption; in the examples that behave like finite automata, commuting cancellative monoids, or bounded quotients of such objects, it is a checkable finite-combinatorial condition. After these examples we prove a sharp dichotomy: either the predictive-length triple exists, or its failure has a visible operator-theoretic cause.

Proposition B.1 (Graded finite-type automata). *Suppose the predictive quotient is presented by a graded labelled graph*

$$Q = \bigsqcup_{R \geq 0} Q_R, \quad \partial|_{Q_R} = R,$$

whose transitions labelled by $\sigma \in E$ send Q_R to Q_{R+1} , and suppose $\Lambda_{\min}$ is the depth ∂ of the class. Assume the graph has uniformly bounded backward branching: for every vertex q and every label σ , there are at most B vertices p with a σ -edge $p \rightarrow q$. Then the descended shifts have fibres of size $\leq B$ and

$$|\Lambda_{\min}([\sigma w]) - \Lambda_{\min}([w])| = 1.$$

Hence [Assumption 2.1](#) holds with $C_\sigma = 1$.

Proof. The fibre of $\widehat{S}_\sigma$ over a class $[u]$ is its set of σ -predecessors, of cardinality $\leq B$ by hypothesis. Since labelled edges raise the grading by one, prepending σ raises $\Lambda_{\min}$ by exactly one. $\square$

Proposition B.2 (Finitely generated cancellative commuting monoids). *Let the predictive monoid $\mathcal{M}$ be finitely generated, commutative, and cancellative, with the additive grading $\Lambda_{\min} = \deg$ induced by a strictly positive linear functional on $\text{Gr}(\mathcal{M}) \cong \mathbb{Z}^{r_{\text{mon}}}$ (as in [Theorem C.7](#)). Then the descended shift for the generator σ has fibres of size ≤ 1 , and $\Lambda_{\min}([\sigma w]) - \Lambda_{\min}([w]) = \deg(a_\sigma)$ is constant, where $a_\sigma \in \mathcal{M}$ is the image of σ . Hence [Assumption 2.1](#) holds with $C_\sigma = \deg(a_\sigma)$.*

Proof. Cancellativity makes the equation $a_\sigma + m = u$ have at most one solution $m = u - a_\sigma$ in $\mathcal{M}$, so each class has at most one σ -predecessor. Additivity of $\deg$ gives $\deg(a_\sigma + m) - \deg(m) = \deg(a_\sigma)$. In the free case $\mathcal{M} = \mathbb{N}^c$ with unit weights this is the word-length grading, $C_\sigma = 1$. $\square$

Proposition B.3 (Commuting CP channels with finite relation ideal). *Suppose the channels $\{\Phi_\sigma\}$ commute and the channel monoid $\mathcal{M} = \{\Phi_w\}$ is finitely presented (the kernel congruence of $E^* \rightarrow \mathcal{M}$ is generated by finitely many relations) and cancellative. Then $\mathcal{M}$ is a finitely generated cancellative commutative monoid, and [Assumption 2.1](#) holds by [Theorem B.2](#).*

Proof. A finitely presented commutative monoid is finitely generated; with cancellativity it satisfies the hypotheses of [Theorem B.2](#). $\square$

Remark B.4 (Cancellativity is the real hypothesis). Composition of completely positive maps is not cancellative in general, so [Theorem B.3](#) genuinely requires the cancellativity of the channel monoid; it holds, for example, when the Φ_σ act faithfully and invertibly on the closure of their joint range, or are conjugations by a commuting family of unitaries. Without cancellativity the predictive shifts may have infinite fibres and the positive triple need not be finitely summable.

Proposition B.5 (Finite graph quotients with bounded rank distortion). *Let $\pi : G \rightarrow G' = G/\sim$ be a quotient of a predictive graph satisfying [Assumption 2.1](#). Assume the equivalence is shift-compatible, the fibres of π have size $\leq K$, and the rank functions satisfy*

$$|\Lambda'_{\min}(\pi(x)) - \Lambda_{\min}(x)| \leq D$$

(bounded rank distortion). Then G' satisfies [Assumption 2.1](#), with shift fibres bounded by $K \cdot \sup_\sigma |\text{fibre}_\sigma(G)|$ and increments bounded by $C_\sigma + 2D$.

Proof. Shift-compatibility ensures that every σ -predecessor in G' lifts to a σ -predecessor in G after choosing a lift of the target. There are at most K choices of lift and, for each lift, at most $\sup_\sigma |\text{fibre}_\sigma(G)|$ predecessors, giving the stated fibre bound. For the increment,

$$|\Lambda'_{\min}(\pi(\sigma w)) - \Lambda'_{\min}(\pi(w))| \leq |\Lambda_{\min}(\sigma w) - \Lambda_{\min}(w)| + 2D \leq C_\sigma + 2D.$$

$\square$

Corollary B.6 (Automatic finite-fibre spectral triples in the regular classes). *Let a finite completely positive renewal memory lie in any of the following classes:*

- (A) *graded finite-state automata with bounded backward branching* ([Theorem B.1](#));
- (B) *finitely generated cancellative commuting monoids* ([Theorem B.2](#));
- (C) *commuting CP channels with finite relation ideal and cancellative channel monoid* ([Theorem B.3](#));
- (D) *finite graph quotients of (A)–(C) with bounded rank distortion* ([Theorem B.5](#)).

Then [Assumption 2.1](#) holds, with the explicit constants of the cited propositions, and consequently the predictive quotient carries the positive predictive spectral triple of [Theorem 2.11](#): the descended shifts $\widehat{S}_\sigma$ have bounded commutators $[D_{\text{CP}}, \widehat{S}_\sigma]$, the length operator D_{CP} has compact resolvent, and the zeta abscissa is the algebraic predictive dimension q_{alg} . In the commuting classes (B)–(C) the multiplicities are regularly varying with integer $q_{\text{alg}} = r_{\text{mon}}$, so [Assumption 2.2](#) holds and the renewal Weyl law of [Theorem 2.15](#) applies, exhibiting q_{alg} as a genuine Weyl exponent. Thus, in classes (A)–(D), the spectral triple exists with no standing finite-fibre hypothesis.

Proof. Each class verifies [Assumption 2.1](#) by the cited proposition: [Theorem B.1](#) for (A) (fibres $\leq B$, $C_\sigma = 1$), [Theorem B.2](#) for (B) (fibres ≤ 1 , $C_\sigma = \deg a_\sigma$), [Theorem B.3](#) for (C) (via (B)), and [Theorem B.5](#) for (D) (transfer under bounded-fibre, bounded-distortion quotients). With [Assumption 2.1](#) in force, [Theorem 2.11](#) produces the spectral triple, its bounded commutators, and the compact resolvent, and identifies the abscissa as q_{alg} . For (B)–(C) the predictive monoid is a finitely generated commutative cancellative monoid, so by [Theorem 2.14](#) $q_{\text{alg}} = r_{\text{mon}} \in \mathbb{N}$ and the length- R multiplicities count lattice points in an r_{mon} -dimensional rational cone, hence are regularly varying of index $r_{\text{mon}} - 1$; [Assumption 2.2](#) holds and [Theorem 2.15](#) applies. $\square$

B.2 Metric dimension and renewal calibration

The preceding construction measures how many predictive classes are combinatorially distinguishable at length at most R . There is a second, independent notion of size: the metric size of the set of accumulated channels inside the completely bounded norm geometry of maps on $\mathcal{A}_{\text{int}}$. The two notions agree in manifold-like polynomial regimes, but they need not agree in contracting or fractal renewal systems. This distinction is useful, because it separates algebraic growth of predictions from metric resolution of the channel space.

Definition B.7 (Metric predictive dimension). Let $\mathcal{S} = \{\Phi_w : w \in E^*\}$ and let $\overline{\mathcal{S}}$ be its closure in completely bounded distance. The metric predictive dimension is the upper box dimension

$$q_{\text{met}} = \limsup_{\delta \downarrow 0} \frac{\log N_{\text{cb}}(\delta)}{\log(1/\delta)},$$

where $N_{\text{cb}}(\delta)$ is the minimal number of cb-balls of radius δ needed to cover $\overline{\mathcal{S}}$.

Remark B.8 (Relation to dimensions of causal-state presentations). Computational mechanics also assigns geometric dimensions to mixed-state and causal-state presentations. In the natural mixed-state realizations studied for discrete-time and continuous-time ε -machines, finite or suitably countable causal-state presentations can have vanishing box-counting dimension, whereas uncountable predictive-state attractors generically carry a nonzero fractal dimension [26, 25, 58]. This comparison is useful but not literally identical to q_{met} . Here the covering number is taken in completely bounded norm on the *closure* of accumulated channels. Countability of the raw predictive set alone therefore does not force $q_{\text{met}} = 0$: accumulation can leave a positive upper box dimension. The metric-collapse conclusion in [Theorem B.13](#) accordingly uses finite-scale stabilization, not merely a countability hypothesis.

Theorem B.9 (Contracting reset systems). *Suppose the accumulated channels are represented, in a finite affine coordinate chart, by similarities f_σ of ratios $0 < r_\sigma < 1$ satisfying the open set condition, and suppose the cb-metric on channels is bi-Lipschitz equivalent to the Euclidean metric on the attractor. Then*

$$q_{\text{met}} = s, \quad \sum_{\sigma \in E} r_\sigma^s = 1.$$

Thus the metric predictive dimension is the Moran–Falconer dimension and is generically non-integer.

Proof. The accumulated channel set is bi-Lipschitz to the self-similar attractor $K = \bigcup_\sigma f_\sigma(K)$. Under the open set condition, the Hausdorff and box dimensions of K are the unique solution of the Moran equation [59, 60]. Bi-Lipschitz equivalence preserves box dimension. $\square$

Remark B.10 (Provenance of the metric dimension). The identification of a box-counting dimension with the abscissa of a spectral-triple zeta function, and of the associated Dixmier-trace measure with the Hausdorff measure, is not new: it is due to Pearson–Bellissard for ultrametric Cantor sets [4] and was developed for subshifts, tilings, and aperiodic media by Kellendonk–Savinien [5], building on the length-module spectral triples of Connes [2] and Christensen–Ivan [3]; the Moran–Falconer equation itself is classical [59, 60, 61]. What is specific to the present setting is only that the self-similar attractor in question is the completely-bounded closure of accumulated renewal channels. The contracting-reset case is thus the renewal instance of an established mechanism, recorded here to fix the metric/algebraic dimension dictionary, not to claim the dimension formula.

Remark B.11 (Two dimensions). The two dimensions need not agree. The dimension q_{alg} is a growth exponent for exact distinguishability by word length. The dimension q_{met} is a resolution exponent for the cb-metric closure of accumulated channels. In algebraic commuting regimes q_{alg} is integer and q_{met} may collapse; in contracting free regimes q_{alg} can be infinite while q_{met} remains finite and fractal. A third quantity, the renewal exponent α of [Theorem C.3](#), measures neither growth nor resolution but the tail of calibrated first returns, equivalently the critical singularity of the predictive-order resolvent; the three invariants are logically independent without additional hypotheses.

Remark B.12 (Non-integer dimensions are metric, not Clifford ranks). The construction therefore contains two different routes to dimension. The algebraic dimension q_{alg} counts exact predictive classes by quotient length; in the commuting manifold-like regimes and in the primitive modular endpoint it is an integer rank. The metric dimension q_{met} , by contrast, measures the completely bounded metric closure of accumulated channels. In contracting reset systems it is the Moran–Falconer exponent s of [Theorem B.9](#), determined by

$$\sum_{\sigma \in E} r_{\sigma}^s = 1,$$

and it is generically non-integer. Thus non-integer dimension emerges as a resolution exponent of the CP-channel attractor, while the $3 + 1$ theorem concerns the integer Clifford rank of the primitive signed modular endpoint. The two numbers may interact through coding or balancing hypotheses, but they should not be identified.

Assumption B.1 (Lipschitz coding). Let $S_R = \{\Phi_v : \Lambda_{\min}([v]) \leq R\}$ be the length filtration of the channel set, and $\overline{S}$ its cb-closure. There is a constant $C > 0$ such that S_R is a (C/R) -net of $\overline{S}$ for every R , i.e. the Hausdorff cb-distance satisfies $\text{dist}_{\text{cb}}(S_R, \overline{S}) \leq C/R$.

Theorem B.13 (Metric dimension is dominated by algebraic dimension). *Under [Assumption B.1](#),*

$$N_{\text{cb}}(\delta) \leq N_{\text{CP}}(C/\delta) \quad (\delta > 0), \quad \text{hence} \quad \boxed{q_{\text{met}} \leq q_{\text{alg}}}.$$

The inequality is strict in the two collapsing/contracting regimes:

- (i) (metric collapse) *if the channels are exactly distinguishable by length (so $q_{\text{alg}} = c > 0$) but the cb-closure is finite, or the tail converges to a finite attractor faster than any power of the resolution scale, then $q_{\text{met}} = 0 < c$;*
- (ii) (contracting IFS) *in the self-similar setting of [Theorem B.9](#) with the words remaining exactly distinguishable, one has $q_{\text{alg}} = \infty$ while $q_{\text{met}} = s \in (0, \infty)$ is the finite Moran dimension, so $q_{\text{met}} < q_{\text{alg}}$.*

Proof. By [Assumption B.1](#) with $R = C/\delta$, the finitely many channels of $S_{C/\delta}$ form a δ -net of $\bar{\mathcal{S}}$, so a δ -cover needs at most $|S_{C/\delta}| = N_{\text{CP}}(C/\delta)$ balls: $N_{\text{cb}}(\delta) \leq N_{\text{CP}}(C/\delta)$. Taking logarithms and dividing by $\log(1/\delta)$, write $R = C/\delta$ so that $\log(1/\delta) = \log R - \log C$; then

$$q_{\text{met}} = \limsup_{\delta \downarrow 0} \frac{\log N_{\text{cb}}(\delta)}{\log(1/\delta)} \leq \limsup_{R \rightarrow \infty} \frac{\log N_{\text{CP}}(R)}{\log R - \log C} = \limsup_{R \rightarrow \infty} \frac{\log N_{\text{CP}}(R)}{\log R} = q_{\text{alg}}.$$

For (i), finite cb-closure has box dimension 0, and the same conclusion holds when the tail approaches a finite attractor faster than any power of the resolution scale; exact distinguishability gives $q_{\text{alg}} = c > 0$. This rate assumption is essential, since a general countable compact set can have positive upper box dimension. For (ii), [Theorem B.9](#) gives $q_{\text{met}} = s < \infty$, while exact distinguishability of words makes $N_{\text{CP}}(R)$ grow super-polynomially, $q_{\text{alg}} = \infty$. $\square$

Remark B.14 (The coding hypothesis is necessary). Without [Assumption B.1](#) the two dimensions are genuinely independent, consistent with [Theorem B.11](#): if a δ -separated family of channels is realised only at lengths $\gg 1/\delta$, then q_{met} can exceed q_{alg} . The Lipschitz coding is exactly the condition that metric resolution δ be achievable at length $O(1/\delta)$, which is what forces the metric dimension below the algebraic one. Thus q_{alg} bounds q_{met} precisely when short histories already resolve the channel attractor.

Assumption B.2 (Uniform coding regularity). In addition to the (C/R) -net property of [Assumption B.1](#), the length filtration is *uniformly cb-separated at the coding rate*: there exist $c > 0$ and R_0 such that any two distinct accumulated channels $\Phi_u \neq \Phi_v$ with $\Lambda_{\min}([u]), \Lambda_{\min}([v]) \leq R$ satisfy

$$\text{dist}_{\text{cb}}(\Phi_u, \Phi_v) \geq \frac{c}{R} \quad (R \geq R_0).$$

Equivalently, the $N_{\text{CP}}(R)$ distinct length- $\leq R$ channels are (c/R) -separated: short histories not only resolve the attractor ([Assumption B.1](#)) but fill it without collision.

Theorem B.15 (Metric and algebraic predictive dimensions coincide). *Under [Assumptions B.1](#) and [B.2](#),*

$$N_{\text{CP}}(R) \leq N_{\text{cb}}\left(\frac{c}{3R}\right) \quad \text{and} \quad N_{\text{cb}}(\delta) \leq N_{\text{CP}}(C/\delta),$$

and consequently

$q_{\text{met}} = q_{\text{alg}}.$

Proof. The upper bound $N_{\text{cb}}(\delta) \leq N_{\text{CP}}(C/\delta)$ and hence $q_{\text{met}} \leq q_{\text{alg}}$ are [Theorem B.13](#). For the reverse, set $\delta_R = c/(3R)$. A cb-ball of radius δ_R has diameter $2\delta_R = \frac{2c}{3R} < \frac{c}{R}$, strictly below the separation of [Assumption B.2](#), so it contains at most one of the $N_{\text{CP}}(R)$ distinct length- $\leq R$ channels. Hence any δ_R -cover of $\bar{\mathcal{S}} \supseteq S_R$ needs at least $N_{\text{CP}}(R)$ balls: $N_{\text{cb}}(\delta_R) \geq N_{\text{CP}}(R)$. Therefore

$$q_{\text{met}} = \limsup_{\delta \downarrow 0} \frac{\log N_{\text{cb}}(\delta)}{\log(1/\delta)} \geq \limsup_{R \rightarrow \infty} \frac{\log N_{\text{cb}}(\delta_R)}{\log(3R/c)} \geq \limsup_{R \rightarrow \infty} \frac{\log N_{\text{CP}}(R)}{\log R + \log(3/c)} = q_{\text{alg}},$$

the last equality because the additive constant $\log(3/c)$ does not affect the limsup. Combining the two inequalities gives $q_{\text{met}} = q_{\text{alg}}$. $\square$

Remark B.16 (Separation is exactly what promotes the inequality to equality). [Assumption B.2](#) is sharp in the following sense: it fails in the two strict regimes of [Theorem B.13](#), showing exactly why

the inequality need not be an equality. In the metric-collapse case (i) the channels cb-converge, so no fixed c keeps them (c/R) -separated, and indeed $q_{\text{met}} = 0 < q_{\text{alg}}$. In the contracting-IFS case (ii) the words stay exactly distinguishable, so $N_{\text{CP}}(R)$ grows super-polynomially while a bounded attractor admits only $O(R^{q_{\text{met}}})$ points at mutual distance c/R ; the separation is impossible and $q_{\text{met}} < q_{\text{alg}} = \infty$. Thus the algebraic and metric predictive dimensions coincide exactly when the predictive classes are *optimally distributed* — simultaneously a net ([Assumption B.1](#)) and a packing ([Assumption B.2](#)) at the common rate $1/R$ — and the gap $q_{\text{met}} < q_{\text{alg}}$ measures the failure of that optimality. The integer Clifford rank d_{Cl} of the signed endpoint is unrelated to either ([Theorem B.12](#)).

Proposition B.17 (Renewal calibration). *In the conformal contracting case, the equation defining q_{met} can be written as a pressure-zero condition. If $r_\sigma = e^{-\ell_\sigma}$, then $s = q_{\text{met}}$ is the unique value for which*

$$\Phi(s) := \sum_{\sigma \in E} e^{-s\ell_\sigma} = 1.$$

The renewal resolvent $(1 - \Phi(s))^{-1}$ has residue equal to the reciprocal of the mean calibrated length, whenever that mean is finite. For $|E| \geq 2$ this root is positive and interior; the one-letter alphabet $|E| = 1$ gives the boundary root $s = 0$ directly.

Proof. The identity $r_\sigma^s = e^{-s\ell_\sigma}$ rewrites the Moran equation as the Bowen pressure-zero condition $\Phi(s) = 1$ [[62](#), [61](#)]. Existence and uniqueness of the root are elementary for $|E| \geq 2$: Φ is continuous and strictly decreasing with $\Phi(0) = |E| > 1$ and $\Phi(s) \rightarrow 0$ as $s \rightarrow \infty$, so the intermediate value theorem gives a unique positive root. For $|E| = 1$ the equation $e^{-s\ell} = 1$ with $\ell > 0$ forces $s = 0$, and uniqueness is immediate. The residue statement is the elementary renewal theorem: if the calibrated probabilities are $p_\sigma = e^{-s\ell_\sigma}$ and $\mu = \sum p_\sigma \ell_\sigma$, then $1 - \Phi(s+t) = \mu t + o(t)$. $\square$

C Renewal-tail singularities

C.1 Renewal tail and the analytic type of the resolvent

The preceding remark can be made precise. This is the point at which the *renewal* character of the data—rather than only its automaton or transition-monoid structure—becomes load-bearing. Fix a recurrent predictive component and a base class x_0 . Using the calibration of [Theorem B.17](#) at the Bowen critical point, normalised to $z_c = 1$, let $(f_n)_{n \geq 1}$ be the first-return-chain probability law and set $F(z) = \sum_{n \geq 1} f_n z^n$. If u_n denotes the corresponding calibrated return-chain weight, the first-passage decomposition identifies the rank-graded diagonal of the chain-counting resolvent of [Theorem 2.21](#) as

$$\mathcal{R}_{\mathfrak{P}}(x_0, x_0; z) = \sum_{n \geq 0} u_n z^n = (1 - F(z))^{-1}, \quad u_0 = 1, \quad u_n = \sum_{k=1}^n f_k u_{n-k}.$$

Thus the diagonal spectral object constructed from the predictive order is a renewal generating function, and the tail of the first-return law measures the order of memory.

Theorem C.1 (Renewal singularity dichotomy). *Let $\mu = \sum_{n \geq 1} n f_n \in (0, \infty]$ be the calibrated mean return length.*

(i) (short memory) *If $\mu < \infty$, then*

$$\mathcal{R}_{\mathfrak{P}}(x_0, x_0; z) \sim \frac{1}{\mu(1-z)} \quad (z \uparrow 1).$$

If, in addition, the first-return law has an exponential moment—as for a finite irreducible aperiodic recurrent graph—then F is analytic in a neighbourhood of $z = 1$ and $\mathcal{R}_{\mathfrak{P}}(x_0, x_0; z)$ has a genuine simple pole there, with

$$\operatorname{Res}_{z=1} \mathcal{R}_{\mathfrak{P}}(x_0, x_0; z) = -\frac{1}{\mu}.$$

The singularity exponent is the integer 1.

(ii) (long memory) Suppose instead that, for some $0 < \alpha < 1$,

$$\bar{F}(n) := \sum_{k \geq n} f_k = n^{-\alpha} \ell(n),$$

where ℓ is slowly varying. Then $\mu = \infty$ and Karamata's Tauberian theorem gives

$$1 - F(z) \sim \Gamma(1 - \alpha)(1 - z)^{\alpha} \ell\left(\frac{1}{1 - z}\right), \quad z \uparrow 1.$$

Consequently

$$\mathcal{R}_{\mathfrak{P}}(x_0, x_0; z) \sim \frac{(1 - z)^{-\alpha}}{\Gamma(1 - \alpha) \ell\left(\frac{1}{1 - z}\right)} \quad (z \uparrow 1).$$

The critical singularity has non-integer exponent $-\alpha$. In particular, whenever this regularly varying equivalent is realised in a slit complex neighbourhood of 1, the critical point is a branch point and the resolvent cannot be meromorphic across it.

Proof. For (i),

$$\frac{1 - F(z)}{1 - z} = \sum_{n \geq 1} f_n(1 + z + \cdots + z^{n-1}) \longrightarrow \sum_{n \geq 1} n f_n = \mu$$

by monotone convergence as $z \uparrow 1$. Inversion gives the stated pole-type asymptotic. Under an exponential moment, F extends analytically past the unit circle near 1 and $F'(1) = \mu \neq 0$, so $1 - F$ has a simple zero and the residue formula follows. For (ii), the power-series form of Karamata's Tauberian theorem for non-negative coefficients gives the displayed equivalent for $1 - F(z)$ from the regularly varying tail [63, 22]; inversion gives the resolvent asymptotic. A non-integer algebraic singularity in a slit neighbourhood is not meromorphic. $\square$

Corollary C.2 (Renewal counting asymptotics). *Under the hypotheses of Theorem C.1(ii), the cumulative return weight satisfies the unconditional Tauberian asymptotic*

$$\sum_{k \leq n} u_k \sim \frac{\sin(\pi\alpha)}{\pi\alpha} \frac{n^{\alpha}}{\ell(n)}, \quad n \rightarrow \infty.$$

If, in addition, the strong renewal condition holds—automatically for $\alpha \in (\frac{1}{2}, 1)$ by Garsia–Lamperti, and under the sharp Caravenna–Doney criterion for $\alpha \in (0, \frac{1}{2}]$ —then the local asymptotic

$$u_n \sim \frac{\sin(\pi\alpha)}{\pi} \frac{n^{\alpha-1}}{\ell(n)}$$

also holds [64, 65, 66, 67].

Proof. The cumulative statement follows from the Hardy–Littlewood–Karamata theorem applied to the non-negative coefficients of $\mathcal{R}_{\mathfrak{P}}(x_0, x_0; z)$, using $\Gamma(1 + \alpha)\Gamma(1 - \alpha) = \pi\alpha/\sin(\pi\alpha)$. The local statement is precisely the strong renewal theorem under the cited range and criterion. $\square$

Definition C.3 (Renewal exponent). The number $\alpha \in (0, 1]$ of [Theorem C.1](#) is the *renewal exponent* of the recurrent predictive component: $\alpha = 1$ in the finite-mean pole regime and $\alpha \in (0, 1)$ in the regularly varying branch regime.

Remark C.4 (A genuinely renewal-theoretic invariant). The renewal exponent is not determined by either predictive dimension. Two recurrent components may have the same algebraic growth exponent q_{alg} ([Theorem 2.11](#)) and the same metric dimension q_{met} ([Theorem B.7](#)), while having different first-return tails and hence different values of α . The quantities q_{alg} and q_{met} measure, respectively, exact predictive growth and metric resolution, whereas α measures the order of contact of the predictive-order resolvent with its critical singularity. It is therefore a third invariant of specifically renewal-theoretic origin, integer in the short-memory regime and fractional in the heavy-tailed regime. This is the precise content of [Theorem 2.22](#).

Example C.5 (Parabolic renewal component realising $\alpha \in (0, 1)$). A finite recurrent-state graph has geometric return tails, so the long-memory regime requires a countable recurrent *induced component*; this is compatible with a finite reset alphabet. Consider the renewal structure obtained by inducing a Pomeau–Manneville intermittent source [\[68\]](#) on a base away from its indifferent fixed point. For an indifferent fixed point of order $1 + 1/\alpha$, the induced first-return law has

$$f_n \asymp n^{-1-\alpha}, \quad 0 < \alpha < 1,$$

in the standard parameterisation of the model [\[69, 70\]](#). The induced predictive component is countable and locally finite; whenever its calibrated shifts satisfy [Assumption 2.1](#), [Theorem C.1\(ii\)](#) applies and the chain-counting resolvent has critical exponent $-\alpha$. Varying the indifference order sweeps α through $(0, 1)$, so the fractional regime is realised rather than merely formally admissible.

Proposition C.6 (Commuting order and interval growth). *If the predictive monoid is freely generated by c commuting channels modulo no further relations, then $\mathfrak{P} \cong \mathbb{N}^c$ with product order. For balanced intervals of rank separation τ ,*

$$|[x, y]| \asymp \tau^c.$$

Consequently the interval-growth dimension of the predictive order agrees with the algebraic predictive dimension $q_{\text{alg}} = c$.

Proof. The divisibility order of the free commutative monoid on c generators is the product order on $\mathbb{N}^c$. A box with coordinate gaps $n_1, \dots, n_c$ has cardinality $\prod_i (n_i + 1)$. Along balanced intervals $n_i \simeq \tau/c$, this grows like τ^c . $\square$

Theorem C.7 (Ehrhart interval growth for normal affine commuting monoids). *Let the predictive monoid $\mathcal{M}$ be a finitely generated, commutative, cancellative, torsion-free, reduced affine semigroup. Write $L = \text{Gr}(\mathcal{M}) \cong \mathbb{Z}^{r_{\text{mon}}}$ for the lattice generated by $\mathcal{M}$, and let*

$$C = \mathbb{R}_{\geq 0} \cdot \mathcal{M} \subseteq L \otimes \mathbb{R}$$

be the salient rational polyhedral cone generated by $\mathcal{M}$. Assume, for this Ehrhart statement, that $\mathcal{M}$ is normal in its lattice:

$$\mathcal{M} = C \cap L.$$

Then:

(i) (geometric model) the order interval of $x \preceq y$ with $w := y - x \in \mathcal{M}$ is

$$[x, y] \cong \mathcal{M} \cap (w - \mathcal{M}) = (C \cap (w - C)) \cap L,$$

a finite set, since C is salient;

(ii) (Ehrhart asymptotics) fix an interior lattice direction $v_0 \in \text{int } C \cap L$ and consider balanced intervals $w = \tau v_0$, $\tau \in \mathbb{N}$. Then

$$|[x, y]| = \text{Vol}_L(P_{v_0}) \tau^{r_{\text{mon}}} + O(\tau^{r_{\text{mon}}-1}), \quad P_{v_0} := C \cap (v_0 - C),$$

where Vol_L is normalised so that a fundamental domain of L has volume 1.

Consequently, in this normal affine regime, the interval-growth dimension equals the algebraic predictive dimension $q_{\text{alg}} = r_{\text{mon}}$.

Proof. Because $\mathcal{M} = C \cap L$, divisibility $z \in [x, y]$ is equivalent to $z - x \in C \cap L$ and $y - z \in C \cap L$. Writing $w = y - x$, this gives

$$[x, y] - x = (C \cap (w - C)) \cap L.$$

The cone is salient, so $C \cap (w - C)$ is bounded and the interval is finite.

For $w = \tau v_0$, homogeneity gives

$$C \cap (\tau v_0 - C) = \tau(C \cap (v_0 - C)) = \tau P_{v_0}.$$

The set P_{v_0} is a rational polytope. Ehrhart's theorem [45, 71, 72], applied with respect to the intrinsic lattice L , gives

$$\#(\tau P_{v_0} \cap L) = \text{Vol}_L(P_{v_0}) \tau^{r_{\text{mon}}} + O(\tau^{r_{\text{mon}}-1}),$$

which is the stated interval asymptotic. The final assertion follows because r_{mon} is the rank of the affine semigroup, hence the polynomial growth exponent in the commuting affine regime. $\square$

Remark C.8 (Non-normal affine monoids). For a non-normal affine semigroup the same exponent r_{mon} is still expected from Hilbert–Serre theory, but the leading coefficient can depend on the semigroup module structure and may not be exactly the lattice volume of $C \cap (v_0 - C)$ without correction. This is why [Theorem C.7](#) is stated in the normal affine case. The paper only needs the exponent and the intrinsic-lattice normalisation; any non-normal refinement should be treated as a separate Hilbert-series calculation.

Corollary C.9 (Free case and consistency). *For $\mathcal{M} \cong \mathbb{N}^c$ one has $L = \mathbb{Z}^c$, $C = \mathbb{R}_{\geq 0}^c$, $v_0 = (1, \dots, 1)$, $P_{v_0} = [0, 1]^c$ and $\text{Vol}_L(P_{v_0}) = 1$, so $|[x, y]| \sim \tau^c$, recovering [Theorem C.6](#); and the interval-growth dimension $r_{\text{mon}} = c$ agrees with $q_{\text{alg}} = c$ of [Theorem 2.14](#).*

Proof. Specialise [Theorem C.7](#): the unit cube has unit lattice volume, and $\tau^c + O(\tau^{c-1}) \asymp \tau^c$. $\square$

Remark C.10 (Why the intrinsic lattice is essential). The volume in [Theorem C.7](#) (ii) must be normalised against $L = \text{Gr}(\mathcal{M})$, not against any ambient lattice in which $\mathcal{M}$ happens to be written. For instance $\mathcal{M} = \langle (1, 0), (1, 2) \rangle \subset \mathbb{Z}^2$ has $\text{Gr}(\mathcal{M}) = \mathbb{Z} \times 2\mathbb{Z}$ of index 2, and the balanced interval count grows like $\frac{1}{2} \text{Area}(P_{v_0}) \tau^2$, exactly half the ambient Euclidean value: the leading coefficient sees only the predictive lattice the monoid actually generates. This is the order/volume analogue of measuring predictive content intrinsically rather than through a presentation.

Remark C.11 (Order is not signature). The order gives a causal skeleton and a time separation. It does not by itself give an indefinite inner product. The positive CP sector remains Hilbert-positive even when the predictive quotient is partially ordered. This distinction is essential: order supplies causal comparability, while a Lorentzian signature requires a fundamental symmetry.

D The deterministic product-order continuum

D.1 The renewal continuum ladder

The assembly theorem identifies the finite kinematic data. We next ask what their large-scale order geometry becomes in the most regular case,

$$\mathcal{W}_{\text{CP}} \cong \mathbb{N}^c.$$

This subsection gives a deliberately sharp answer. The deterministic commuting model has a continuum limit in every dimension, and the limit is a perfectly good measured Lorentzian pre-length space. But its cone is the product cone. Hence, except in the two-dimensional null-coordinate case, the deterministic product order gives a polyhedral Lorentzian continuum rather than a smooth Minkowski cone. This is not a failure of the construction; it is the reason the later Clifford-isotropisation step is needed.

D.1.1 The discrete object and its rescaling

We first isolate the purely order-theoretic continuum limit, before introducing any Clifford data. Throughout this subsection the predictive monoid is freely generated by c commuting reset channels with no further relations, so that $\mathcal{W}_{\text{CP}} \cong \mathbb{N}^c$ carries the product predictive order of [Theorem C.6](#). Let $\ell_1, \dots, \ell_c > 0$ be calibrated generator lengths, so that the quotient length is the linear rank

$$\Lambda_{\min}(x) = \sum_{i=1}^c \ell_i x_i, \quad x \in \mathbb{N}^c,$$

and let μ be counting measure.

Definition D.1 (Calibrated rescaling). For $R > 0$ let

$$\iota_R : \mathbb{N}^c \longrightarrow [0, \infty)^c, \quad \iota_R(x) = R^{-1}(\ell_1 x_1, \dots, \ell_c x_c),$$

and let

$$\mu_R = \left(\prod_i \ell_i \right) R^{-c} (\iota_R)_* \mu.$$

On $[0, \infty)^c$ write $X \leq Y$ for componentwise order and define the geometric-mean time separation

$$\tau(X, Y) = \begin{cases} (\prod_{i=1}^c (Y_i - X_i))^{1/c}, & X \leq Y, \\ 0, & \text{otherwise.} \end{cases}$$

This separation is positive exactly for chronologically related pairs $X_i < Y_i$ for all i ; it vanishes on the null faces of the product cone.

The rescaling keeps three pieces of data visible at once: the order, the rank/time function, and counting measure. The first elementary observation is that these already form a measured causal set.

Proposition D.2 (Renewal measured causal set). *The tuple $(\mathcal{W}_{\text{CP}}, \preceq, \Lambda_{\min}, \mu)$ is a measured causal set: the rank $\Lambda_{\min}$ is strictly increasing along strict chains, $(\mathcal{W}_{\text{CP}}, \preceq)$ is locally finite, every order interval has finite μ -mass, and level sets of $\Lambda_{\min}$ are antichains.*

Proof. If $x \prec y$ strictly then $x_i \leq y_i$ for all i and strict inequality holds for at least one i , so

$$\Lambda_{\min}(y) - \Lambda_{\min}(x) = \sum_i \ell_i(y_i - x_i) > 0.$$

An interval is the finite box

$$[x, y] = \prod_i \{x_i, x_i + 1, \dots, y_i\},$$

with $\mu([x, y]) = \prod_i (y_i - x_i + 1) < \infty$. Thus intervals are finite and the order is locally finite. If two comparable points have the same rank, the displayed strict-increase identity forces them to be equal; hence rank level sets are antichains. $\square$

Theorem D.3 (Polyhedral Lorentzian continuum). *The ordered space $([0, \infty)^c, \leq, \tau)$ is a Lorentzian pre-length space with compact diamonds on bounded regions. On chronologically related triples it satisfies the reverse triangle inequality. Moreover, on every bounded diamond, the rescaled renewal geometries*

$$(\iota_R(\mathbb{N}^c), \leq, \tau, \mu_R)$$

converge locally to $([0, \infty)^c, \leq, \tau, \text{Leb})$ in the order/length/measure sense, and interval volumes converge:

$$\mu_R([\iota_R(x_R), \iota_R(y_R)]) \longrightarrow \text{Leb}([X, Y]) = \prod_{i=1}^c (Y_i - X_i)$$

whenever $\iota_R(x_R) \rightarrow X$, $\iota_R(y_R) \rightarrow Y$, and $X \leq Y$.

Proof. Let

$$m(v) = \left(\prod_i v_i \right)^{1/c}$$

on $(0, \infty)^c$. The function m is concave and homogeneous of degree one; hence it is superadditive. Therefore, for $X \leq Y \leq Z$ with all coordinate gaps positive,

$$\tau(X, Z) = m((Y - X) + (Z - Y)) \geq m(Y - X) + m(Z - Y) = \tau(X, Y) + \tau(Y, Z).$$

The same inequality extends to boundary cases by continuity, with possible zero time separation on null faces. Diamonds in bounded regions are compact boxes.

For volume convergence, write $X_i = R^{-1}\ell_i(x_R)_i$ and $Y_i = R^{-1}\ell_i(y_R)_i$. Then

$$\begin{aligned} \mu_R([\iota_R(x_R), \iota_R(y_R)]) &= \left(\prod_i \ell_i \right) R^{-c} \prod_i ((y_R)_i - (x_R)_i + 1) \\ &\longrightarrow \prod_i (Y_i - X_i), \end{aligned}$$

which is the Lebesgue volume of the limiting box. Finally, the grids $\iota_R(\mathbb{N}^c)$ are $O(R^{-1})$ -dense on bounded boxes, preserve order exactly, and distort τ by $o(1)$ on compact diamonds. This gives local order/length convergence. $\square$

Remark D.4 (Two times on one order). The product order carries two natural time functions. The longest-chain or rank time is the linear function $\sum_i (Y_i - X_i)$, while τ is the order-plus-volume time separation selected by the identity

$$\text{Leb}([X, Y]) = \tau(X, Y)^c.$$

They share the same causal cones but are not the same metric datum. In the two-dimensional case τ becomes the ordinary Minkowski proper time in null coordinates.

The product cone becomes a round Lorentz cone only in the exceptional case $c = 2$. This is the familiar fact that two-dimensional Minkowski space is most naturally described by two null coordinates.

Theorem D.5 (Two-dimensional Minkowski emergence). *For $c = 2$, the limit $([0, \infty)^2, \leq, \tau, \text{Leb})$ is isometric, up to the constant normalization of volume, to a causally convex region of $(1 + 1)$ -dimensional Minkowski space. In null coordinates $u = X_1$, $v = X_2$ and inertial coordinates*

$$t = \frac{u + v}{2}, \quad \xi = \frac{v - u}{2},$$

the product order is the Minkowski causal order, and

$$\tau(X, Y) = \sqrt{\Delta u \Delta v} = \sqrt{\Delta t^2 - \Delta \xi^2}.$$

Proof. The change of variables gives

$$\Delta u \Delta v = \Delta t^2 - \Delta \xi^2.$$

The condition $\Delta u \geq 0$ and $\Delta v \geq 0$ is exactly $\Delta t \geq |\Delta \xi|$, the future cone of the flat metric $-dt^2 + d\xi^2$. The time separation τ is therefore the usual proper time. Also $du dv = 2 dt d\xi$, so counting volume fixes the conformal factor up to the displayed constant normalization. $\square$

Theorem D.6 (Polyhedral obstruction to smooth higher-dimensional emergence). *For $c \geq 3$, the deterministic product-order cone $[0, \infty)^c$ is not linearly isomorphic to a round Lorentz cone*

$$L_c = \{x_0 \geq |(x_1, \dots, x_{c-1})|\}.$$

Consequently the deterministic finite commuting renewal model does not produce a smooth higher-dimensional Minkowski causal cone from product order and counting volume alone.

Proof. The orthant $[0, \infty)^c$ is a salient closed convex cone with exactly c extreme rays, the coordinate axes. The Lorentz cone L_c has as extreme rays its null generators, the cone over S^{c-2} . For $c \geq 3$ this set is infinite, indeed a continuum. Linear isomorphisms preserve extreme rays. Hence no linear isomorphism carries the orthant cone to the round Lorentz cone. Since the tangent cones of a smooth Lorentzian manifold are round Lorentz cones, the deterministic product-order continuum is polyhedral rather than smooth Lorentzian in dimensions $c \geq 3$. $\square$

Corollary D.7 (Deterministic lattice Lorentz violation). *For $c \geq 3$, the regular commuting renewal continuum selects the c generator directions as distinguished null-like axes. Thus the deterministic finite-generator model has a preferred polyhedral cone structure and does not possess the full Lorentz symmetry of a smooth Minkowski cone.*

D.1.2 Routes beyond the deterministic product cone

The obstruction above is only an obstruction to the deterministic finite commuting product cone. There are two standard ways to escape it. One is to replace the lattice by a Lorentz-invariant point process; the other, used in the rest of this section, is to keep the renewal origin but round the signed Dirac symbol by isotropic Clifford coarse-graining.

Theorem D.8 (Stochastic renewal sprinkling, established input). *If the deterministic lattice is replaced by a Poisson sprinkling into a causally convex region of $(1+k)$ -dimensional Minkowski space, with intensity proportional to Minkowski volume, then the construction is Lorentz covariant: applying a Lorentz transformation to the region carries the law of the sprinkled causal set to the law of the transformed sprinkled causal set. In Minkowski diamonds, the induced longest-chain function recovers Minkowski proper time in probability after the standard dimension-dependent normalisation [73, 74]. Hence the obstruction of Theorem D.6 is an obstruction to the deterministic commuting realisation, not to renewal-type point processes in general.*

Remark D.9. The renewal input in Theorem D.8 is the point-process realisation and its intensity. Lorentz covariance and longest-chain convergence are classical causal-set inputs, not new consequences of the finite commuting predictive quotient.

Remark D.10 (Clifford isotropisation). The Clifford/isotropic route is proved in Section 5.1. The key point is that positive averaging alone cannot create Lorentzian signature: a positive second-moment tensor is positive semidefinite. Signature is supplied by the signed/Krein sector, while isotropic coarse-graining rounds the principal-symbol cone.

Remark D.11 (What the continuum ladder establishes). In the regular commuting regime, renewal data yield a measured causal set and an unconditional polyhedral Lorentzian continuum with convergent interval volumes. Smooth Minkowski space emerges from product order and counting volume alone only for $c = 2$, where the product coordinates are null coordinates. For $c \geq 3$, the deterministic finite-generator model is polyhedral and Lorentz-violating. Higher-dimensional smooth Lorentzian spacetime therefore requires either stochastic sprinkling or an additional isotropic/Clifford coarse-graining mechanism. No claim is made here of Einstein dynamics, matter coupling, or curvature equations.

Remark D.12 (Signature is not an extra coordinate). The product order on $\mathbb{N}^c$ is already a c -dimensional causal order: its rank is a single time function and its generic antichains are $(c-1)$ -dimensional spacelike slices. The signed sector converts the positive order into Lorentzian signature through the fundamental symmetry J_χ ; it does not add a coordinate. Thus the regular model should be read as a $c = 1 + (c-1)$ dimensional order geometry equipped with a signed/Krein Lorentzian structure.

E Internal holonomy and conditional dimension selectors

E.1 Internal holonomy realization of the spatial block

Definition E.1 (Coherent recurrent internal holonomy). Let U_e be the coherent internal transport attached to an oriented recurrent edge, acting on the same internal Hilbert space and recurrent component as the renewal channel. For a recurrent cycle α , set $U_\alpha = \prod_{e \in \alpha} U_e$ and define the internally dressed revision operator

$$R_\alpha^{\text{int}} = \tilde{S}_\alpha \otimes U_\alpha.$$

Theorem E.2 (Nonabelian internal holonomy supplies a spatial Clifford entry). *Suppose two recurrent spatial classes $\alpha, \beta \in H$ have coherent internal holonomies satisfying the scalar group-commutator identity*

$$U_\alpha U_\beta U_\alpha^{-1} U_\beta^{-1} = -\mathbf{1}.$$

Then the internally dressed revisions satisfy

$$\omega(\alpha, \beta) = 1.$$

More generally, a family of recurrent holonomies whose scalar commutators realise a nondegenerate alternating form on a spatial subspace produces exactly that spatial block of ω . Combined with the forced temporal row, a nondegenerate rank-four restriction yields the primitive $\text{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C})$ revision factor appropriate to a $3 + 1$ endpoint.

Proof. The commutator of the dressed revisions factorises into the flat sheet/base commutator and the internal group commutator. The former is trivial by [Theorem 4.7](#); the latter is $-\mathbf{1}$ by hypothesis. Thus the total scalar commutator is $-\mathbf{1}$, which is the statement $\omega(\alpha, \beta) = 1$. The final assertion follows by applying the same factorisation entrywise and then the rank criterion to a nondegenerate four-dimensional restriction. $\square$

Remark E.3 (One coherent transport family upstream and downstream). The operators U_e in [Theorem E.1](#) are the coherent recurrent transports of the Lorentzian construction. Any downstream matter construction using colour holonomy must restrict this same family on the same recurrent component, rather than introduce an analogous second family. With that literal identification, spatial Clifford dimension and complex colour can be traced to one coherent internal transport system.

Lemma E.4 (Gauge invariance of the spatial commutator form). *Suppose the coherent recurrent transports form a scalar projective representation on the internal recurrent sector,*

$$U_\alpha U_\beta = \tau_H(\alpha, \beta) U_{\alpha+\beta}, \quad \tau_H(\alpha, \beta) \in \text{U}(1),$$

whose scalar commutator takes values in $\{\pm 1\}$. Then

$$(-1)^{\omega_H(\alpha, \beta)} := U_\alpha U_\beta U_\alpha^{-1} U_\beta^{-1} = \tau_H(\alpha, \beta) \tau_H(\beta, \alpha)^{-1}$$

depends only on the projective cohomology class $[\tau_H]$. In particular it is unchanged under scalar regauging $U_\alpha \mapsto \mu(\alpha) U_\alpha$.

Proof. A scalar regauging replaces τ_H by $\tau_H \delta \mu$. Since the coboundary $\delta \mu(\alpha, \beta) = \mu(\alpha) \mu(\beta) \mu(\alpha + \beta)^{-1}$ is symmetric on the elementary abelian group H , its alternation is trivial. Hence the commutator bicharacter, and therefore ω_H , is gauge invariant. $\square$

Theorem E.5 (Nonzero spatial block forces nonabelian recurrent transport). *Work in the local, covariant, internal scalar-projective class in which the spatial form is realised by the coherent recurrent transports U_α as in [Theorem E.4](#). Then:*

- (i) *if the recurrent internal transport is abelian, then $\omega|_{H \times H} = 0$;*
- (ii) *equivalently, $\omega(\alpha, \beta) = 1$ forces $U_\alpha U_\beta \neq U_\beta U_\alpha$ in every scalar gauge;*
- (iii) *if $\text{rank}_{\mathbb{F}_2} \omega_H = 2m$, every faithful irreducible internal realisation has dimension 2^m , and every realisation has internal dimension at least 2^m on the nondegenerate sector.*

Thus a nonzero spatial modular block cannot be carried by the abelian deck algebra or by one-dimensional internal transport.

Proof. If all U_α commute, every group commutator is the identity, hence $\omega_H = 0$. The contrapositive gives (ii), and [Theorem E.4](#) makes the conclusion gauge invariant. For (iii), quotient H by $\text{rad}(\omega_H)$. The resulting $2m$ -dimensional symplectic $\mathbb{F}_2$ -space defines the finite Heisenberg, equivalently extraspecial 2-group, central extension. Finite Stone–von Neumann gives a unique faithful irreducible projective representation of dimension 2^m ; any representation containing that nondegenerate commutator sector has dimension at least 2^m [[75](#), [76](#)]. $\square$

Theorem E.6 (Internal-holonomy normal form). *Let $\omega_H = \omega|_{H \times H}$ be any alternating $\mathbb{F}_2$ -form of rank $2m$ occurring in a local, covariant, internal revision sector. After factoring the radical and applying a scalar gauge, its nondegenerate spatial revision sector is unitarily equivalent to the Heisenberg–Clifford transport on $\mathbb{C}^{2^m}$:*

$$U_\alpha U_\beta U_\alpha^{-1} U_\beta^{-1} = (-1)^{\omega_H(\alpha, \beta)} \mathbf{1}.$$

The representation on the nondegenerate quotient is unique up to unitary equivalence, while $\text{rad}(\omega_H)$ contributes precisely commuting central transport directions. Consequently, within this scalar-projective class,

$$\omega|_{H \times H} \neq 0 \iff \text{the recurrent internal transport is nonabelian.}$$

Proof. Choose a symplectic basis on $H/\text{rad}(\omega_H)$. The associated finite Heisenberg extension has commutator pairing $(-1)^{\omega_H}$ and its unique faithful irreducible representation has dimension 2^m by finite Stone–von Neumann. Pulling this representation back to H realises the given form; the radical acts through commuting central characters. Conversely, [Theorem E.5](#) shows that a nonzero spatial form cannot come from abelian transport. The scalar gauge freedom changes the cocycle by a coboundary but not its alternation. $\square$

Remark E.7 (Precise scope of the equivalence). The equivalence in [Theorem E.6](#) concerns the coherent internal transport that realises the scalar projective spatial revision sector. It does not assert that every arbitrary nonabelian unitary holonomy automatically produces a nonzero $\mathbb{F}_2$ -valued block: a nonabelian family may have noncentral commutators and therefore lie outside the scalar-bicharacter regime. What is proved is the exact normal form relevant to the modular revision datum of this paper.

E.2 The common cycle-rank invariant

The positive predictive sector, the recurrent crystal, and the signed revision sector introduce dimension by different constructions. In general the resulting numbers need not agree. The metric link is genuinely independent: contracting UCP memories can have positive algebraic growth but zero channel box dimension. The remaining algebraic, modular, and heat-kernel links become consequences of one regular topological-crystal package.

Definition E.8 (Finite displacement semigroupoid and accessed lattice). Let $Q = (V, A)$ be a finite strongly connected directed control graph with base state s_0 . Each oriented edge e carries a displacement $\xi(e) \in \Gamma$, where $\Gamma \cong \mathbb{Z}^{r_{\text{mon}}}$, and an additive positive degree $\ell(\xi(e)) > 0$. For states s, t , write

$$\mathcal{M}_{st} := \left\{ \sum_{e \in p} \xi(e) : p : s \rightarrow t \right\}.$$

These sets form a semigroupoid: $\mathcal{M}_{st} + \mathcal{M}_{tu} \subseteq \mathcal{M}_{su}$. The return lattice is

$$\Gamma_{\text{ret}} := \text{Gr}(\mathcal{M}_{s_0 s_0}) \subseteq \Gamma,$$

and the spatial accessed lattice is

$$L := \ker(\ell|_{\Gamma_{\text{ret}}}), \quad b_1^{\text{eff}} := \text{rank}_{\mathbb{Z}} L.$$

Assumption E.1 (Full-dimensional affine reachability). There is a finitely generated affine monoid $\mathcal{M} \subseteq \Gamma$ such that:

- (a) every element of $\mathcal{M}$ is realised by a path starting at s_0 ;
- (b) $\text{Gr}(\mathcal{M}) = \Gamma \cong \mathbb{Z}^{r_{\text{mon}}}$ and $\text{span}_{\mathbb{R}} \mathcal{M} = \Gamma \otimes \mathbb{R}$;
- (c) the quotient length is comparable to a strictly positive additive degree $\ell : \Gamma \rightarrow \mathbb{Z}$;
- (d) $N_{\text{CP}}(R) \asymp R^{r_{\text{mon}}}$.

Normal affine monoids of [Theorem C.7](#) satisfy the growth clause, but normality is not required for the return-rank argument below.

Assumption E.2 (Finite primitive control). The control graph Q is finite, strongly connected, and aperiodic.

Theorem E.9 (Full-rank return completion). *Under [Assumptions E.1](#) and [E.2](#),*

$$\text{rank}_{\mathbb{Z}} \Gamma_{\text{ret}} = r_{\text{mon}}.$$

Consequently

$$\boxed{b_1^{\text{eff}} = r_{\text{mon}} - 1, \quad q_{\text{alg}} = 1 + b_1^{\text{eff}}.}$$

Proof. For each state s , choose a fixed return path $q_s : s \rightarrow s_0$, and let $R_s \in \Gamma$ be its displacement. Since V is finite, $K := \max_s \|R_s\| < \infty$. If $x \in \mathcal{M}$ is realised by a path $p : s_0 \rightarrow s$, then pq_s is a closed walk at s_0 , so $x + R_s \in \Gamma_{\text{ret}}$. Hence every $x \in \mathcal{M}$ lies within distance K of the real span $W := \mathbb{R}\Gamma_{\text{ret}}$.

Choose $u_1, \dots, u_{r_{\text{mon}}} \in \mathcal{M}$ spanning $\Gamma \otimes \mathbb{R}$. Because $\mathcal{M}$ is a monoid, $nu_i \in \mathcal{M}$ for every $n \geq 1$. Therefore

$$n \text{ dist}(u_i, W) = \text{dist}(nu_i, W) \leq K$$

for all n , which forces $u_i \in W$. Thus $W = \Gamma \otimes \mathbb{R}$ and $\text{rank} \Gamma_{\text{ret}} = r_{\text{mon}}$. Since ℓ is nonzero on Γ_{ret} , its kernel L has rank $r_{\text{mon}} - 1$. Finally $q_{\text{alg}} = r_{\text{mon}}$ by [Assumption E.1](#) (d). $\square$

Remark E.10 (What the theorem does and does not use). The proof uses bounded return completion and full-dimensional scalable reachability, not saturation of the affine monoid. Normality remains useful for the Ehrhart formula and for eliminating lattice holes, but it is not the central hypothesis in [Theorem E.9](#). No claim is made that every vector in $\ker \ell \cap \Gamma$ is represented by a single prescribed difference of equal-degree closed walks; only the finite-index rank statement is needed.

For $R \geq 1$, let

$$S_R := \{\Phi_v : \Lambda_{\min}([v]) \leq R\}, \quad N_{\text{CP}}(R) := |S_R|,$$

and let $N_{\text{cb}}(\delta)$ be the least number of cb-metric balls of radius δ covering the channel closure.

Assumption E.3 (Two-sided channel coding). There are constants $C \geq c > 0$ such that S_R is a C/R -net of the channel closure and distinct elements of S_R are separated by at least c/R in cb-distance.

Theorem E.11 (Metric-predictive coincidence). *Under [Assumption E.3](#),*

$$N_{\text{cb}}(C/R) \leq N_{\text{CP}}(R) \leq N_{\text{cb}}(c/(3R)), \quad \boxed{q_{\text{met}} = q_{\text{alg}}}.$$

Proof. The net property gives the first inequality. A ball of radius $c/(3R)$ contains at most one point of the c/R -separated set S_R , giving the second. Taking logarithms and limsups along reciprocal integer scales proves the equality. $\square$

Proposition E.12 (Metric collapse without non-collapse). *There are genuine UCP renewal memories with $q_{\text{alg}} > 0$ and $q_{\text{met}} = 0$. Specifically, on $M_d(\mathbb{C})$ let $\Psi(x) = d^{-1} \text{tr}(x)\mathbf{1}$ and*

$$\Phi_a = \frac{1}{2}\Psi + \frac{1}{2}\text{id}, \quad \Phi_b = \frac{1}{3}\Psi + \frac{2}{3}\text{id}.$$

Then the predictive quotient is $\mathbb{N}^2$, so $q_{\text{alg}} = 2$, while all degree- n channels lie within $O(2^{-n})$ of Ψ , whence $N_{\text{cb}}(\delta) = O((\log(1/\delta))^2)$ and $q_{\text{met}} = 0$.

Proof. A word with contents (i, j) gives $\Phi_{i,j} = (1 - 2^{-i}3^{-j})\Psi + 2^{-i}3^{-j}\text{id}$. Unique factorisation makes the classes distinct, while exponential contraction to Ψ gives the stated covering bound. $\square$

The modular comparison must use the actual accessed lattice, not a larger finite-index lattice before reduction modulo 2.

Definition E.13 (Accessible revision block). Let

$$\bar{\rho} : H_1(G_0; \mathbb{F}_2) \rightarrow L/2L$$

be the mod-2 reduction of recurrent holonomy. Universal coefficients give the canonical subspace

$$H_{\text{acc}} := \text{Ann}(\ker \bar{\rho}) \cong \text{Hom}(L/2L, \mathbb{F}_2), \quad \dim_{\mathbb{F}_2} H_{\text{acc}} = b_1^{\text{eff}}.$$

Set

$$H_{\text{full,acc}} := H_{\text{acc}} \oplus \langle t \rangle.$$

Theorem E.14 (Parity-correct accessible modular rank). *Assume the full modular form is non-degenerate on $H_{\text{full,acc}}$. Then*

$$b_1^{\text{eff}} \text{ is odd}, \quad \dim \text{rad}(\omega|_{H_{\text{acc}}}) = 1,$$

the temporal generator pairs nontrivially with that radical, and

$$\boxed{\text{rank}_{\mathbb{F}_2}(\omega|_{H_{\text{full,acc}}}) = 1 + b_1^{\text{eff}}}.$$

If the inaccessible complement is modularly invisible and has no mixed block with $H_{\text{full,acc}}$, the same equality holds for the full form.

Proof. A nondegenerate alternating form exists only in even dimension, so $1 + b_1^{\text{eff}} = \dim H_{\text{full,acc}}$ is even. The restriction to the odd-dimensional space H_{acc} is therefore degenerate. If a vector in its radical also paired trivially with t , it would lie in the radical of the full accessible form. Thus the linear functional $\alpha \mapsto \omega(\alpha, t)$ is injective on the spatial radical, forcing that radical to be one-dimensional. Adding t supplies the missing hyperbolic partner and raises the rank from $b_1^{\text{eff}} - 1$ to $1 + b_1^{\text{eff}}$. $\square$

Assumption E.4 (Topological-crystal realisation). The recurrent cover carries a free cocompact action of L with finite quotient motif, and the renewal time coordinate has bounded local distortion. Thus the recurrent cover is quasi-isometric to $L \cong \mathbb{Z}^{b_1^{\text{eff}}}$, and the space-time crystal is quasi-isometric to $\mathbb{Z}^{b_1^{\text{eff}}} \times \mathbb{N}$.

Assumption E.5 (Crystal heat-kernel regularity). The induced walk is lazy, reversible, uniformly elliptic, and bounded range.

Corollary E.15 (Return-probability spectral dimension). Under [Assumptions E.4](#) and [E.5](#),

$$p_n(x, x) \asymp n^{-(1+b_1^{\text{eff}})/2}, \quad \boxed{d_s = 1 + b_1^{\text{eff}}}.$$

Proof. The free cocompact lattice action gives volume growth $|B(x, r)| \asymp r^{1+b_1^{\text{eff}}}$. Volume doubling and the scale-invariant Poincaré inequality are stable for such bounded-geometry crystals; Delmotte's theorem [77] gives the two-sided Gaussian estimate and hence the diagonal exponent. $\square$

Here d_s denotes the return-probability, or heat-kernel, spectral dimension of the regular crystal. It is not the zeta/Weyl dimension of [Theorem 2.15](#), nor a claim about dimensional flow in causal dynamical triangulations or other quantum-gravity ensembles; its equality with q_{alg} below is a theorem only under the stated crystal and channel non-collapse hypotheses.

Theorem E.16 (Dimension coincidence in the regular Lorentzian crystal). Assume [Assumptions E.1](#) to [E.5](#), and suppose ω is nondegenerate on $H_{\text{full,acc}}$, with inaccessible classes modularly invisible. Then

$$\boxed{q_{\text{met}} = q_{\text{alg}} = d_s = \text{rank}_{\mathbb{F}_2} \omega = 1 + b_1^{\text{eff}}, \quad d_{\text{Cl}} = b_1^{\text{eff}} = r_{\text{mon}} - 1.}$$

Without two-sided channel coding, the algebraic, accessible modular, and heat-kernel dimensions still coincide, but the channel-metric equality is not universal.

Proof. [Theorem E.9](#) gives $q_{\text{alg}} = 1 + b_1^{\text{eff}}$. [Theorem E.11](#) gives $q_{\text{met}} = q_{\text{alg}}$. [Theorem E.14](#) gives $\text{rank } \omega = 1 + b_1^{\text{eff}}$ on the accessible block, and on the full form under the stated invisibility condition. [Theorem E.15](#) gives $d_s = 1 + b_1^{\text{eff}}$. Finally $d_{\text{Cl}} = \dim H_{\text{acc}} = b_1^{\text{eff}}$. $\square$

Remark E.17 (Logical scope). The growth-cycle equality is derived from full-dimensional affine reachability and finite primitive return completion. The mod-2 comparison is made with the actual accessed lattice L , so no finite-index substitution is used. The crystal quasi-isometry and the cb-metric non-collapse remain distinct phase conditions: rank equality alone does not imply either one.

Proposition E.18 (Cycle covariance has full accessed rank). Let the finite motif chain be irreducible, aperiodic, and reversible, and let $\xi : E(G_0) \rightarrow L \otimes \mathbb{R}$ be an antisymmetric displacement cocycle. If every nonzero $a \in \text{Hom}(L, \mathbb{R})$ has a nonzero period on some closed directed cycle, then the centred winding process has covariance Q with

$$\boxed{\text{rank } Q = b_1^{\text{eff}}}.$$

Proof. The finite-state Markov-additive central limit theorem applies. A scalar projection has zero asymptotic variance precisely when its edge cocycle is exact, equivalently when all closed-cycle periods vanish. The hypothesis excludes this for every nonzero projection. $\square$

Corollary E.19 (Full regular-phase chain). Under the hypotheses of [Theorems E.16](#) and [E.18](#),

$$\boxed{q_{\text{met}} = q_{\text{alg}} = d_s = \text{rank}_{\mathbb{F}_2} \omega = 1 + b_1^{\text{eff}}, \quad d_{\text{Cl}} = \text{rank } Q = b_1^{\text{eff}}}.$$

E.3 Survival of the volume-dual response as a primitive first-order observable

This selection statement cannot be made fully unconditional without an explicit coarse-graining operator. The following gives the sharpest rigorous reduction: the algebraic degree statement (unconditional), a *survival criterion* that converts the vague “survives coarse-graining” into a concrete parity/chirality condition on reset statistics (unconditional), and two transparent closure routes – renormalisation-group relevance and minimal field content – each replacing the abstract closure principle by a recognisable hypothesis.

E.3.1 The algebraic core, restated

Let $V_{\text{sp}} \cong \mathbb{R}^{d_{\text{Cl}}}$ be the spatial revision module, $\Gamma : V_{\text{sp}} \rightarrow \text{Cl}(V_{\text{sp}})$ the Clifford embedding, and

$$\beta_{\text{ren}} : \bigwedge^2 V_{\text{sp}} \rightarrow \text{Cl}(V_{\text{sp}}), \quad \beta_{\text{ren}}(u \wedge v) = \frac{1}{2}[\Gamma(u), \Gamma(v)] = \Gamma(u \wedge v),$$

the canonical two-reset response, an $SO(d_{\text{Cl}})$ -equivariant isomorphism onto the bivector sector. With the oriented volume element $\Omega_{\text{ren}} = \Gamma(e_1) \cdots \Gamma(e_{d_{\text{Cl}}})$, the volume-dual response is $\mathcal{H}_{\text{ren}} = \Omega_{\text{ren}}^{-1} \beta_{\text{ren}} : \bigwedge^2 V_{\text{sp}} \xrightarrow{\sim} \bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}$. The preceding analysis proves $\text{Im } \mathcal{H}_{\text{ren}} \subseteq V_{\text{sp}}$ (degree one) iff $d_{\text{Cl}} = 3$. We take this as given and ask what survives coarse-graining.

E.3.2 Survival is oriented circulation

Definition E.20 (Oriented reset circulation). Let ν be the coarse-grained reset-pair law with joint weights $p_{ab} \geq 0$ on ordered direction pairs $(\hat{\theta}_a, \hat{\theta}_b)$. Its *oriented circulation* is the bivector

$$\Xi(\nu) := \sum_{a,b} p_{ab} \hat{\theta}_a \wedge \hat{\theta}_b = \sum_{a,b} p^{[ab]} \hat{\theta}_a \wedge \hat{\theta}_b \in \bigwedge^2 V_{\text{sp}}, \quad p^{[ab]} := \frac{1}{2}(p_{ab} - p_{ba}).$$

The law is *parity- (time-reversal-) symmetric* if $p_{ab} = p_{ba}$, i.e. $\Xi(\nu) = 0$.

Theorem E.21 (Survival criterion). *The coarse-grained two-reset response equals*

$$\beta_{\text{ren}}[\nu] := \sum_{a,b} p_{ab} \frac{1}{2}[\Gamma(\hat{\theta}_a), \Gamma(\hat{\theta}_b)] = \Gamma(\Xi(\nu)),$$

and its volume dual is $\mathcal{H}_{\text{ren}}[\nu] = \Omega_{\text{ren}}^{-1} \Gamma(\Xi(\nu))$. Consequently the coarse-grained volume-dual response is a nonzero one-reset (degree-one) observable if and only if

$d_{\text{Cl}} = 3 \quad \text{and} \quad \Xi(\nu) \neq 0.$

In particular it vanishes identically for every parity-symmetric reset law, regardless of dimension.

Proof. Since $\hat{\theta}_a \wedge \hat{\theta}_b$ is antisymmetric in (a, b) and $\frac{1}{2}[\Gamma(\hat{\theta}_a), \Gamma(\hat{\theta}_b)] = \Gamma(\hat{\theta}_a \wedge \hat{\theta}_b)$, summing against p_{ab} keeps only the antisymmetric part: $\beta_{\text{ren}}[\nu] = \sum_{a,b} p^{[ab]} \Gamma(\hat{\theta}_a \wedge \hat{\theta}_b) = \Gamma(\Xi(\nu))$ by linearity of Γ on bivectors. As Γ and Ω_{ren}^{-1} are injective, $\mathcal{H}_{\text{ren}}[\nu] = \Omega_{\text{ren}}^{-1} \Gamma(\Xi(\nu)) \neq 0$ iff $\Xi(\nu) \neq 0$, and its image lies in $\bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}$, which is the degree-one sector V_{sp} iff $d_{\text{Cl}} - 2 = 1$. If $p_{ab} = p_{ba}$ then $p^{[ab]} = 0$ and $\Xi(\nu) = 0$, so $\beta_{\text{ren}}[\nu] = 0$. $\square$

Remark E.22 (Why survival was conditional). [Theorem E.21](#) explains the conditionality precisely: the two-reset response is built from the *antisymmetric* part of the reset-pair statistics, so it is killed by parity/time-reversal symmetry. “Survival under coarse-graining” is therefore equivalent to the reset law possessing nonzero oriented circulation – a handedness/chirality of the renewal dynamics. This is a concrete, measurable condition on microscopic reset statistics, replacing the opaque survival postulate. Existence of laws with $\Xi \neq 0$ is immediate ([Theorem E.23](#)).

Example E.23 (A chiral reset law with $\Xi \neq 0$, and a symmetric one with $\Xi = 0$). Take $d_{\text{Cl}} = 3$ and three coplanar directions $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3$ at 120° in the $e_1 e_2$ -plane, with a cyclic (chiral) ordered pair law $p_{12} = p_{23} = p_{31} = \frac{1}{3}$, $p_{ba} = 0$. Then $\Xi = \frac{1}{3}(\hat{\theta}_1 \wedge \hat{\theta}_2 + \hat{\theta}_2 \wedge \hat{\theta}_3 + \hat{\theta}_3 \wedge \hat{\theta}_1) = \frac{1}{2} e_1 \wedge e_2 \neq 0$, so the volume-dual response is the nonzero one-reset observable $\mathcal{H}_{\text{ren}}[\nu] = \Omega_{\text{ren}}^{-1} \Gamma(\Xi) \propto \Gamma(e_3)$. Symmetrising the pair law, $p_{ab} \mapsto \frac{1}{2}(p_{ab} + p_{ba})$, gives $\Xi = 0$ and kills the response. Thus chirality of the reset ordering is exactly what survives.

E.3.3 Route B: Clifford-grade closure and the RG obstruction

The oriented volume-dual response has a simple algebraic dimension count. The band-bottom curvature is a two-form. Its Hodge dual belongs to $\Lambda^{d_{\text{Cl}}-2} V_{\text{sp}}$, and therefore closes into the elementary vector generator sector precisely when

$$d_{\text{Cl}} - 2 = 1, \quad \text{that is,} \quad d_{\text{Cl}} = 3.$$

Thus, if the oriented response is required to survive in the principal grade-one Clifford sector, the minimal nondegenerate odd endpoint is selected.

This should not be confused with a canonical renormalisation-group relevance argument. A tempting strengthening would be to assign the degree- k Clifford sector an engineering eigenvalue relative to a one-reset generator and to infer a factor $b^{3-d_{\text{Cl}}}$. That conclusion is not obtained from ordinary field-theoretic engineering dimensions alone. Once the response is coupled to a fermion bilinear, the bilinear has engineering dimension d_{Cl} in $(1 + d_{\text{Cl}})$ -dimensional spacetime, independently of the inserted Clifford grade. A gauge-potential-type coefficient must therefore have dimension one in every spatial dimension. Canonical power counting is grade-blind.

Theorem E.24 (Engineering power counting does not select the dimension). *The coupling of the volume-dual oriented response to a fermion bilinear is a dimension-one, gauge-potential-type coupling and is marginal by canonical engineering power counting in every spatial dimension. Hence ordinary power counting does not produce a universal eigenvalue proportional to $b^{3-d_{\text{Cl}}}$, and it does not by itself suppress the higher-grade currents present for $d_{\text{Cl}} > 3$. The special role of $d_{\text{Cl}} = 3$ is therefore an exterior-algebra and Clifford-grade matching statement, not a consequence of canonical renormalisation-group relevance.*

Proof. A first-order Dirac coupling has the form of a Clifford-valued one-particle operator inserted between fermions. In $(1 + d_{\text{Cl}})$ -dimensional spacetime, the fermion field has engineering dimension $d_{\text{Cl}}/2$, so a fermion bilinear has dimension d_{Cl} , independently of which Clifford grade is inserted in the matrix acting on spinors. Since the action density has dimension $d_{\text{Cl}} + 1$, the coefficient of a first-order gauge-potential-type coupling has dimension one in every d_{Cl} . Thus canonical engineering dimension does not distinguish the Clifford grades.

The algebraic distinction is separate. The curvature response is a two-form, and

$$\star : \Lambda^2 V_{\text{sp}} \longrightarrow \Lambda^{d_{\text{Cl}}-2} V_{\text{sp}}.$$

This lands in the vector sector $\Lambda^1 V_{\text{sp}}$ only when $d_{\text{Cl}} - 2 = 1$. Therefore $d_{\text{Cl}} = 3$ is selected by Hodge degree and Clifford-grade closure, not by canonical power counting. $\square$

Corollary E.25 (Dimensional status after the RG audit). *Primitive modular nondegeneracy forces odd spatial rank and excludes the lower degenerate endpoints. Therefore $d_{\text{Cl}} = 3$ remains the unconditional minimal nondegenerate endpoint. If, in addition, the oriented signed-cluster response is required to close into and remain within the principal grade-one generator sector, then*

$d_{\text{Cl}} = 3$ is selected among the odd endpoints. The present paper does not derive this principal-grade survival condition from canonical power counting: higher-grade couplings are not removed by engineering dimension alone. A genuine uniqueness theorem among higher odd endpoints would require an interacting anomalous-dimension separation or an independent microscopic operator-closure theorem.

Remark E.26 (What remains of Route B). The robust conclusion is the grade-closure statement: the oriented response has Hodge degree $d_{\text{Cl}} - 2$, and hence can be identified with a primitive grade-one Dirac generator only for $d_{\text{Cl}} = 3$. A special block-renewal model may still generate additional anomalous scaling that suppresses higher grades, but that would be an extra dynamical theorem about the interacting renewal flow, not a consequence of canonical engineering power counting.

E.3.4 Route C: minimal continuum field content

Theorem E.27 (Minimal-field-content closure). *Suppose the continuum endpoint's one-reset observable space is exactly V_{sp} – the metric coframe and a first-order Dirac spinor, with no independently propagating higher-form sector. If the oriented volume-dual response is nonzero (equivalently $\Xi \neq 0$, [Theorem E.21](#)), then it can be a one-reset observable only if its Clifford degree is one, i.e.*

$$d_{\text{Cl}} - 2 = 1, \quad \boxed{d_{\text{Cl}} = 3}.$$

Proof. By the preceding construction's degree computation, $\text{Im } \mathcal{H}_{\text{ren}} = \bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}$. Under the minimal-field-content hypothesis the only one-reset observables are degree one, so a nonzero $\mathcal{H}_{\text{ren}}[\nu]$ is a one-reset observable iff $d_{\text{Cl}} - 2 = 1$. $\square$

E.3.5 Net status of the survival problem

Remark E.28 (Exactly what is and is not discharged). The degree equivalence ($\text{Im } \mathcal{H}_{\text{ren}} = \bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}$, elementary iff $d_{\text{Cl}} = 3$) and the survival criterion (survival $\Leftrightarrow d_{\text{Cl}} = 3$ and $\Xi \neq 0$, [Theorem E.21](#)) are unconditional. Together they reduce the four-part principal-symbol-closure principle of the preceding construction to two concrete questions: (i) does the reset law have nonzero oriented circulation Ξ ? – answered by a parity test, with chiral examples; (ii) does the response enter the principal symbol? – answered by *either* RG relevance ([Theorem E.32](#), on $\mathcal{R}$) *or* minimal field content ([Theorem E.27](#),). The irreducible residue is exactly the construction of the block-renewal/OPE map needed to settle (ii) unconditionally; absent it, $d_{\text{Cl}} = 3$ uniqueness from this route remains conditional, consistent with the stated hypotheses. What *has* changed is that the conditional content is now a sharp, checkable nonvanishing condition plus one transparent closure route, rather than an opaque “survives, is principal, is primitive, is not an independent field” bundle.

E.4 An explicit block-renewal coarse-graining map

the survival analysis made the renormalisation-group selection of $d_{\text{Cl}} = 3$ conditional on the existence of a block-renewal map with engineering eigenvalue $b^{3-d_{\text{Cl}}}$. We now construct one such map explicitly in the graded-Clifford renewal class and prove the eigenvalue with a controlled band error. This is not a universal RG theorem; it is one mathematically controlled model, which is what is needed to upgrade the engineering argument to a theorem.

E.4.1 Joint-degree reset-interference observables

Work in the graded-Clifford renewal memory: $\mathcal{A}_{\text{rev}} = \text{Cl}_n(\mathbb{C})$, $n = 1 + d_{\text{Cl}}$, with Hermitian generators $\Gamma_0, \dots, \Gamma_{d_{\text{Cl}}}$, spatial Clifford embedding $\Gamma : V_{\text{sp}} \rightarrow \text{Cl}(V_{\text{sp}})$, and the volume element $\Omega_{\text{ren}} = \Gamma(e_1) \cdots \Gamma(e_{d_{\text{Cl}}})$. Fix a reset-direction system $\{\hat{\theta}_a\} \subset \mathbb{S}^{d_{\text{Cl}}-1}$ and the symmetric reset differences $\delta_{\hat{\theta}_a, h}$ with Fourier multiplier $s_{\hat{\theta}_a, h}(\xi) = h^{-1} \sin(h \hat{\theta}_a \cdot \xi)$.

Definition E.29 (Degree- k reset-interference observable). For $1 \leq k \leq d_{\text{Cl}}$, the canonical degree- k reset-interference observable is the k -fold antisymmetrised reset difference

$$\mathcal{O}_k^{(h)} = \sum_{a_1 < \dots < a_k} \Gamma(\hat{\theta}_{a_1} \wedge \dots \wedge \hat{\theta}_{a_k}) \delta_{\hat{\theta}_{a_1}, h} \cdots \delta_{\hat{\theta}_{a_k}, h},$$

with Fourier symbol, valued in $\text{Cl}(V_{\text{sp}})$,

$$\sigma_k^{(h)}(\xi) = \sum_{a_1 < \dots < a_k} \Gamma(\hat{\theta}_{a_1} \wedge \dots \wedge \hat{\theta}_{a_k}) i^k s_{\hat{\theta}_{a_1}, h}(\xi) \cdots s_{\hat{\theta}_{a_k}, h}(\xi).$$

Its *joint degree* is k : it carries Clifford degree k and momentum homogeneity k . The continuum symbol $\sigma_k(\xi) = \lim_{h \rightarrow 0} \sigma_k^{(h)}$ is homogeneous of degree k in ξ , $\sigma_k(\lambda \xi) = \lambda^k \sigma_k(\xi)$.

Remark E.30 (The volume-dual response is a degree- $(d_{\text{Cl}} - 2)$ observable). By Hodge duality in $\text{Cl}(V_{\text{sp}})$, $\Omega_{\text{ren}}^{-1} \Gamma(u \wedge v) = \pm \Gamma(\star(u \wedge v))$, a Clifford element of degree $d_{\text{Cl}} - 2$. Hence the oriented volume-dual response $\mathcal{H}_{\text{ren}} = \Omega_{\text{ren}}^{-1} \beta_{\text{ren}}$ lands in the degree- $(d_{\text{Cl}} - 2)$ sector and its natural renewal realisation is the degree- $(d_{\text{Cl}} - 2)$ observable $\mathcal{O}_{d_{\text{Cl}}-2}^{(h)}$ of [Theorem E.29](#): it is Γ of a $(d_{\text{Cl}} - 2)$ -form, realised by $d_{\text{Cl}} - 2$ antisymmetrised reset differences, of joint degree $d_{\text{Cl}} - 2$. This is what makes its engineering dimension $d_{\text{Cl}} - 2$, not 2.

E.4.2 The block-renewal map and its eigenvalue

Construction E.31 (Block-renewal coarse-graining). Fix an integer block size $b \geq 2$. The block-renewal map $\mathcal{R}_b$ acts on $\text{Cl}(V_{\text{sp}})$ -valued Fourier symbols by mesh dilation $h \mapsto bh$ followed by the momentum rescaling $\xi \mapsto \xi/b$ and the normalisation prefactor b chosen to fix the degree-one Dirac symbol:

$$(\mathcal{R}_b \sigma)(\xi) := b \sigma^{(bh)}(\xi/b),$$

where $\sigma^{(bh)}$ is the symbol recomputed at the block mesh bh (block reset differences $\delta_{\hat{\theta}_a, bh}$). The prefactor is fixed by requiring $\mathcal{R}_b$ to fix the degree-one continuum symbol $\sigma_1(\xi) = \sum_a \Gamma(\hat{\theta}_a) i (\hat{\theta}_a \cdot \xi)$: indeed $\mathcal{R}_b \sigma_1(\xi) = b \sigma_1(\xi/b) = b \cdot b^{-1} \sigma_1(\xi) = \sigma_1(\xi)$.

Theorem E.32 (Engineering eigenvalue of the block-renewal map). *On the continuum symbols, $\mathcal{R}_b$ acts diagonally on joint degree with exact eigenvalue*

$$\boxed{\mathcal{R}_b(\sigma_k) = b^{1-k} \sigma_k} \quad (1 \leq k \leq d_{\text{Cl}}).$$

On the lattice, for every momentum band $|\xi| \leq \Lambda$ there is $C_k = C_k(\Lambda)$ with

$$\left\| \mathcal{R}_b(\sigma_k^{(h)}) - b^{1-k} \sigma_k^{(h)} \right\|_{|\xi| \leq \Lambda} \leq C_k b^{1-k} (bh)^2 \Lambda^2 \xrightarrow{h \rightarrow 0} 0.$$

Consequently the volume-dual response, of joint degree $k = d_{\text{Cl}} - 2$ ([Theorem E.30](#)), has eigenvalue

$$\boxed{\lambda_{d_{\text{Cl}}} = b^{1-(d_{\text{Cl}}-2)} = b^{3-d_{\text{Cl}}}.$$

Proof. Continuum. σ_k is homogeneous of degree k , so $\sigma_k(\xi/b) = b^{-k}\sigma_k(\xi)$ and $\mathcal{R}_b\sigma_k(\xi) = b b^{-k}\sigma_k(\xi) = b^{1-k}\sigma_k(\xi)$.

Lattice error. Each scalar factor obeys $|s_{\hat{\theta},bh}(\eta) - \hat{\theta} \cdot \eta| \leq \frac{1}{6}(bh)^2|\eta|^3$. Writing $\eta = \xi/b$ and using $|\hat{\theta} \cdot \eta| \leq |\eta| \leq \Lambda/b$ on the rescaled band, the k -fold product symbol $\sigma_k^{(bh)}(\xi/b)$ differs from its leading homogeneous term $\prod_l(\hat{\theta}_{a_l} \cdot \xi/b) = b^{-k} \prod_l(\hat{\theta}_{a_l} \cdot \xi)$ by a relative error $O((bh)^2(\Lambda/b)^2)$ per factor, hence $O((bh)^2\Lambda^2b^{-2})$ overall after the k bounded Clifford factors. Multiplying by the prefactor b and the homogeneity factor b^{-k} gives, on the band,

$$\|\mathcal{R}_b(\sigma_k^{(h)}) - b^{1-k}\sigma_k^{(h)}\| \leq C_k b \cdot b^{-k} \cdot (bh)^2\Lambda^2b^{-2} \cdot b^2 = C_k b^{1-k}(bh)^2\Lambda^2,$$

the final b^2 restoring the comparison to the unrescaled lattice symbol $\sigma_k^{(h)}$ on $|\xi| \leq \Lambda$; the Clifford prefactors are unit norm and contribute the constant C_k . Letting $h \rightarrow 0$ at fixed b, Λ sends the error to zero. The volume-dual statement is the case $k = d_{\text{Cl}} - 2$. $\square$

E.4.3 Dimensional selection from the eigenvalue

Corollary E.33 (Relevance trichotomy). *With $\lambda_{d_{\text{Cl}}} = b^{3-d_{\text{Cl}}}$ and $b \geq 2$:*

$$\begin{aligned} d_{\text{Cl}} = 3 : \lambda = 1 \text{ (marginal/principal);} \quad & d_{\text{Cl}} \geq 5 : \lambda < 1 \text{ (irrelevant);} \\ d_{\text{Cl}} = 1 : \text{no bivector domain.} \end{aligned}$$

Hence, in the graded-Clifford renewal class with the explicit block-renewal map $\mathcal{R}_b$, the oriented volume-dual response is a principal first-order observable only at $d_{\text{Cl}} = 3$, is RG-irrelevant for every higher odd spatial rank, and has no two-reset domain at $d_{\text{Cl}} = 1$.

Proof. $b^{3-d_{\text{Cl}}} = 1$ iff $d_{\text{Cl}} = 3$; $b^{3-d_{\text{Cl}}} < 1$ iff $d_{\text{Cl}} > 3$ (in particular for odd $d_{\text{Cl}} \geq 5$); for $d_{\text{Cl}} = 1$, $\bigwedge^2 V_{\text{sp}} = 0$ so $\mathcal{O}_{d_{\text{Cl}}-2} = \mathcal{O}_{-1}$ is empty. Marginality of a $\lambda = 1$ operator is precisely its persistence in the principal symbol. $\square$

Remark E.34 (Precise scope of CXIII). The eigenvalue b^{1-k} and the band error are theorems in the graded-Clifford renewal model with the explicit map $\mathcal{R}_b$ of [Theorem E.31](#). The engineering dimension is assigned by joint degree, which is justified because the natural renewal realisation of a degree- k Clifford observable is the k -fold reset difference $\mathcal{O}_k$, of momentum homogeneity k ([Theorem E.30](#)). This is *one* controlled model, not a universal RG theorem; the selection of $d_{\text{Cl}} = 3$ by RG relevance therefore remains a model-class statement. Within that class, the eigenvalue and quantitative band estimate are proved rather than inferred from engineering bookkeeping.

E.5 From interference postulate to the canonical volume-dual response

The preceding results determine the common spatial Clifford rank

$$d_{\text{Cl}} = b_1^{\text{eff}} = q_{\text{alg}} - 1$$

in the regular spacetime-crystal phase and prove that primitivity makes d_{Cl} odd. They do not, by themselves, exclude the higher odd values $5, 7, \dots$. We now sharpen the earlier interference-closure discussion in three directions. First, we show by explicit models that the current renewal axioms do not imply closure. Second, we identify the two-reset commutator as a canonical bivector-valued renewal response in every odd dimension. Third, after choosing an orientation, the Clifford volume element gives a canonical Hodge dual of that response; this dual is a nonzero elementary revision exactly in three spatial dimensions.

The relevant endpoint is not merely the abstract matrix algebra $\mathcal{A}_{\text{rev}} \cong \text{Cl}_{1+d_{\text{Cl}}}(\mathbb{C})$. It is the *marked renewal-Clifford datum*

$$(\mathcal{A}_{\text{rev}}, V_{\text{full}}, q, \Gamma, V_{\text{sp}}),$$

where $\Gamma : V_{\text{full}} \rightarrow \mathcal{A}_{\text{rev}}$ is the Clifford embedding supplied by the primitive revision construction and $V_{\text{sp}} \subset V_{\text{full}}$ is the spatial subspace selected by the modular time direction. The abstract matrix algebra alone does not canonically remember a degree-one subspace; all statements below are relative to this marked endpoint.

Set

$$V_{\text{sp}} := V_{\text{sp}} \cong \mathbb{R}^{d_{\text{Cl}}}, \quad \mathcal{I} := \text{span}_{\mathbb{R}}\{\Gamma(e_i)\Gamma(e_j) : i < j\}.$$

Lemma E.35 (The canonical two-reset response and the Lorentz symmetric pair). *The antisymmetric two-reset response*

$$\beta_{\text{ren}} : \bigwedge^2 V_{\text{sp}} \longrightarrow \mathcal{A}_{\text{rev}}, \quad \beta_{\text{ren}}(x \wedge y) := \frac{1}{2}[\Gamma(x), \Gamma(y)]$$

takes values in $\mathcal{I}$, is $O(V_{\text{sp}})$ -equivariant, and is a linear isomorphism

$$\beta_{\text{ren}} : \bigwedge^2 V_{\text{sp}} \xrightarrow{\sim} \mathcal{I}.$$

Under the associative commutator,

$$\mathcal{I} \cong \mathfrak{spin}(d_{\text{Cl}}) \cong \mathfrak{so}(d_{\text{Cl}}), \quad \dim_{\mathbb{R}} \mathcal{I} = \binom{d_{\text{Cl}}}{2},$$

and

$$[\mathcal{I}, \mathcal{I}] \subseteq \mathcal{I}, \quad [\mathcal{I}, V_{\text{sp}}] \subseteq V_{\text{sp}}, \quad [V_{\text{sp}}, V_{\text{sp}}] \subseteq \mathcal{I}.$$

With $K_i = \frac{1}{2}\Gamma(e_i)$ and $J_{ij} = -\frac{1}{2}\Gamma(e_i)\Gamma(e_j)$, the real Lie algebra $\mathcal{I} \oplus V_{\text{sp}}$ is isomorphic to $\mathfrak{so}(d_{\text{Cl}}, 1)$.

Proof. For orthonormal $e_i \neq e_j$, the Clifford relations give

$$\frac{1}{2}[\Gamma(e_i), \Gamma(e_j)] = \Gamma(e_i)\Gamma(e_j).$$

The ordered degree-two Clifford monomials are linearly independent in the faithful full spinor representation, so β_{ren} identifies $\bigwedge^2 V_{\text{sp}}$ with $\mathcal{I}$. Equivariance follows from naturality of the Clifford action. Direct computation gives

$$\left[\frac{1}{2}\Gamma(e_i)\Gamma(e_j), \Gamma(e_k) \right] = \delta_{jk}\Gamma(e_i) - \delta_{ik}\Gamma(e_j),$$

together with the standard $\mathfrak{so}(d_{\text{Cl}})$ relations among bivectors. Moreover

$$[K_i, K_j] = -J_{ij}, \quad [J_{ij}, K_k] = \delta_{ik}K_j - \delta_{jk}K_i,$$

which are the Cartan-decomposition relations of $\mathfrak{so}(d_{\text{Cl}}, 1)$. Jacobi is inherited from the associative commutator. $\square$

The automatic Lie algebra therefore lives on $\mathcal{I} \oplus V_{\text{sp}}$, not on V_{sp} alone. A bracket on the spatial revision module appears only after a degree-lowering identification of interferences with revisions.

Theorem E.36 (Closure is satisfiable and falsifiable within the present model class). *Let $\mathbf{A}$ denote the standing package of finite fibres and bounded increments, aperiodic local finiteness, primitivity, double recurrent ergodicity, weak isotropy, and bounded geometry. Then:*

- (i) *there is a model of $\mathbf{A}$ with $d_{\text{Cl}} = 3$ for which oriented interference closure holds;*
- (ii) *for every odd $d_{\text{Cl}} \geq 5$ there is a model of $\mathbf{A}$ for which it fails.*

Hence closure is satisfiable and falsifiable within the present model class; in this model-theoretic sense it is independent of, and in particular not implied by, $\mathbf{A}$.

Proof. Use the bouquet graded-Clifford renewal memory with the cross-polytope reset frame. The construction has a finite reset alphabet and unit length increments, is locally finite and aperiodic, has bounded degree and unit propagation, and its cross-polytope second moment is $M = d_{\text{Cl}}^{-1}I$. For every odd d_{Cl} , the full rank $1 + d_{\text{Cl}}$ is even and the off-diagonal symplectic form used in the primitive examples is nonsingular over $\mathbb{F}_2$; the associated Clifford representation is irreducible, giving the required primitive and doubly ergodic endpoint. Thus the same axiom package is realized simultaneously in every odd spatial rank. At $d_{\text{Cl}} = 3$, the Hodge map constructed below is an equivariant isomorphism $\bigwedge^2 V_{\text{sp}} \cong V_{\text{sp}}$. For odd $d_{\text{Cl}} \geq 5$,

$$\dim \bigwedge^2 V_{\text{sp}} = \binom{d_{\text{Cl}}}{2} \neq d_{\text{Cl}} = \dim V_{\text{sp}},$$

so no closure isomorphism exists. □

Remark E.37 (Why the current axioms do not see closure). The standing axioms constrain the graph and length grading, irreducibility and nondegeneracy of the primitive revision form, and the symmetric second moment $M \in \text{Sym}^2 V_{\text{sp}}$. None is itself a condition on a map $\bigwedge^2 V_{\text{sp}} \rightarrow V_{\text{sp}}$. This grading separation explains the preceding models, but the explicit models, rather than this observation alone, establish the non-implication.

Assumption E.6 (Oriented spatial phase). The determinant line

$$\det V_{\text{sp}} = \bigwedge^{d_{\text{Cl}}} V_{\text{sp}}$$

is trivialised compatibly with recurrent holonomy, so the acting structure group reduces to

$$\text{Hol}^+ := \text{Hol} \cap SO(V_{\text{sp}}).$$

A time orientation alone does not provide this datum.

Proposition E.38 (Local controllability implies full oriented holonomy). *Suppose the recurrent transports extend to one-parameter oriented holonomy families with infinitesimal generators $X_e \in \mathfrak{so}(V_{\text{sp}})$, and*

$$\text{Lie}\langle X_e \rangle = \mathfrak{so}(V_{\text{sp}}).$$

Then

$$\overline{\text{Hol}^+} = SO(V_{\text{sp}}).$$

Proof. The closed subgroup $\overline{\text{Hol}^+} \subseteq SO(V_{\text{sp}})$ contains every $\exp(tX_e)$. By the closed-subgroup theorem it is a Lie subgroup whose Lie algebra contains $\text{Lie}\langle X_e \rangle = \mathfrak{so}(V_{\text{sp}})$. Since $SO(V_{\text{sp}})$ is connected, a closed subgroup with the full Lie algebra is the whole group. □

Remark E.39 (Scope of controllability). The proposition is established under the stated one-parameter-family hypothesis. The microscopic renewal transports are discrete, so the existence of such families is not automatic. For a purely discrete transport system one must instead prove density of the generated subgroup. Thus controllability replaces a global maximal-isotropy postulate by a local rank condition, but does not derive continuous holonomy from the renewal axioms.

Proposition E.40 (The oriented volume element of the marked endpoint). *Fix the marked Clifford embedding Γ and an orientation of V_{sp} . For any oriented orthonormal basis $(e_1, \dots, e_{d_{\text{Cl}}})$, define*

$$\Omega_{\text{ren}} := \Gamma(e_1) \cdots \Gamma(e_{d_{\text{Cl}}}).$$

Then Ω_{ren} is independent of the chosen oriented orthonormal basis, reversing orientation sends $\Omega_{\text{ren}} \mapsto -\Omega_{\text{ren}}$, and

$$g\Omega_{\text{ren}}g^{-1} = \Omega_{\text{ren}} \quad (g \in \text{Hol}^+).$$

Moreover, for odd d_{Cl} ,

$$\Omega_{\text{ren}}^2 = (-1)^{d_{\text{Cl}}(d_{\text{Cl}}-1)/2} \mathbf{1},$$

and Ω_{ren} is central in the spatial Clifford algebra. It need not be a scalar on the full $\text{Cl}_{1+d_{\text{Cl}}}(\mathbb{C})$ -spinor module.

Proof. Two oriented orthonormal bases differ by an element of $SO(V_{\text{sp}})$. The ordered Clifford product represents the top exterior power, on which the change of basis acts by its determinant, hence by 1; an orientation reversal contributes -1 . The same determinant argument proves oriented-holonomy invariance. The square and centrality are the standard odd-dimensional Clifford volume identities. Since the temporal Clifford generator anticommutes with each spatial generator and d_{Cl} is odd, it anticommutes with Ω_{ren} , so Ω_{ren} is not central in the full spacetime Clifford algebra and therefore need not act as one scalar on the full spinor module. $\square$

Definition E.41 (Oriented volume-dual two-reset response). Define

$$\mathcal{H}_{\text{ren}} := \Omega_{\text{ren}}^{-1} \beta_{\text{ren}} : \bigwedge^2 V_{\text{sp}} \longrightarrow \text{Cl}(V_{\text{sp}}).$$

This is the canonical $SO(V_{\text{sp}})$ -equivariant degree-complementing map supplied by the marked Clifford metric and orientation. Reversing orientation changes $\mathcal{H}_{\text{ren}}$ by an overall sign and therefore does not change its image or its dimension-selection property.

Proposition E.42 (Degree and equivariance of the volume-dual response). *The map $\mathcal{H}_{\text{ren}}$ is Hol^+ -equivariant and is a linear isomorphism*

$$\mathcal{H}_{\text{ren}} : \bigwedge^2 V_{\text{sp}} \xrightarrow{\sim} \bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}.$$

Proof. Equivariance follows from equivariance of β_{ren} and invariance of Ω_{ren} . For basis bivectors,

$$\Omega_{\text{ren}}^{-1} \Gamma(e_i) \Gamma(e_j)$$

cancels the two factors e_i, e_j from the oriented top product, leaving, up to sign, the ordered Clifford product of the remaining $d_{\text{Cl}} - 2$ basis vectors. This is the Clifford realization of Hodge duality $\bigwedge^2 V_{\text{sp}} \rightarrow \bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}$. It is invertible, with inverse the opposite Hodge-degree map; equivalently both spaces have dimension $\binom{d_{\text{Cl}}}{2} = \binom{d_{\text{Cl}}}{d_{\text{Cl}}-2}$. $\square$

Principle E.43 (Oriented interference closure). The oriented volume-dual two-reset response is a nonzero elementary spatial revision:

$$\mathcal{H}_{\text{ren}} \neq 0, \quad \text{Im } \mathcal{H}_{\text{ren}} \subseteq V_{\text{sp}}.$$

Equivalently, pairwise interference, after the canonical metric-and-orientation duality, introduces neither missing nor excess primitive revision modes.

Theorem E.44 (Oriented volume-dual closure selects three spatial dimensions). *Assume a primitive marked renewal–Clifford endpoint and the oriented phase [Assumption E.6](#). Then*

$$\mathcal{H}_{\text{ren}} \neq 0 \text{ and } \text{Im } \mathcal{H}_{\text{ren}} \subseteq V_{\text{sp}} \iff d_{\text{Cl}} = 3.$$

In particular, [Theorem E.43](#) implies

$$\boxed{d_{\text{Cl}} = 3}.$$

For $d_{\text{Cl}} = 1$, $\bigwedge^2 V_{\text{sp}} = 0$ and $\mathcal{H}_{\text{ren}} = 0$; for every odd $d_{\text{Cl}} \geq 5$, $\mathcal{H}_{\text{ren}} \neq 0$ but its image has Clifford degree $d_{\text{Cl}} - 2 > 1$.

Proof. By [Theorem E.42](#),

$$\text{Im } \mathcal{H}_{\text{ren}} = \bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}.$$

At $d_{\text{Cl}} = 1$, the domain $\bigwedge^2 V_{\text{sp}}$ vanishes. For odd $d_{\text{Cl}} \geq 3$, β_{ren} is an isomorphism and Ω_{ren} is invertible, hence $\mathcal{H}_{\text{ren}} \neq 0$. Its image is elementary, i.e. degree one, exactly when

$$d_{\text{Cl}} - 2 = 1,$$

which is equivalent to $d_{\text{Cl}} = 3$. □

Proposition E.45 (Realisation and induced bracket at $d_{\text{Cl}} = 3$). *At $d_{\text{Cl}} = 3$,*

$$\mathcal{H}_{\text{ren}}(x \wedge y) = \star(x \wedge y),$$

up to the orientation-dependent overall sign. In an oriented orthonormal basis,

$$\mathcal{H}_{\text{ren}}(e_i \wedge e_j) = \epsilon_{ijk} e_k.$$

The induced bracket

$$[x, y]_{\star} := \mathcal{H}_{\text{ren}}(x \wedge y)$$

satisfies Jacobi and gives

$$(V_{\text{sp}}, [\cdot, \cdot]_{\star}) \cong \mathfrak{so}(3) \cong \mathfrak{su}(2).$$

If the oriented holonomy acts irreducibly on V_{sp} , every Hol^+ -equivariant closure map is a nonzero real multiple of $\mathcal{H}_{\text{ren}}$.

Proof. For $\Omega_{\text{ren}} = \Gamma(e_1)\Gamma(e_2)\Gamma(e_3)$,

$$\Omega_{\text{ren}}^{-1}\Gamma(e_i)\Gamma(e_j) = \pm\epsilon_{ijk}\Gamma(e_k),$$

which is precisely the Hodge cross product. The vector triple-product identity proves Jacobi. The resulting structure constants are those of $\mathfrak{so}(3)$, and the compact real form is isomorphic to $\mathfrak{su}(2)$. For uniqueness, a real irreducible three-dimensional representation cannot be of complex or quaternionic type, so its commutant is $\mathbb{R}$; Schur's lemma then gives uniqueness up to a real scalar. □

Proposition E.46 (Clifford-internal form). *At $d_{\text{Cl}} = 3$, the spatial volume multiplier satisfies*

$$\Omega_{\text{ren}}^2 = -\mathbf{1}, \quad \Gamma(e_i)\Gamma(e_j) = \epsilon_{ijk}\Omega_{\text{ren}}\Gamma(e_k).$$

It is central and invertible in the spatial algebra $\text{Cl}_3(\mathbb{C})$, but anticommutes with the temporal generator. On the full spacetime spinor module it acts with the two opposite square roots of -1 on the two irreducible spatial summands and is not a scalar.

Proof. These are the odd-dimensional Clifford volume identities. The temporal generator anticommutes with each of the three spatial generators, hence with their product. Restricting the full even-dimensional spinor representation to $\text{Cl}_3(\mathbb{C}) \cong M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$ yields the two simple summands, on which the central spatial volume element acts by opposite square roots of -1 . $\square$

Remark E.47 (Why the Hilbert–Schmidt projection does not work). On the full spacetime spinor module, every spatial bivector is Hilbert–Schmidt orthogonal to every spatial vector:

$$\text{Tr}\left(\Gamma(e_k)^\dagger\Gamma(e_i)\Gamma(e_j)\right) = 0.$$

Thus orthogonal projection of β_{ren} onto V_{sp} vanishes in every odd dimension, including $d_{\text{Cl}} = 3$. The correct degree-lowering map is multiplication by Ω_{ren}^{-1} , not projection. The identity at $d_{\text{Cl}} = 3$ is “bivector equals volume times vector,” not “bivector equals vector” on the full module.

Remark E.48 (Why seven dimensions do not survive). Although a norm-compatible cross product exists on $\mathbb{R}^7$, its linearisation is only a surjection

$$\bigwedge^2 \mathbb{R}^7 \longrightarrow \mathbb{R}^7$$

with a 14-dimensional kernel:

$$\bigwedge^2 \mathbb{R}^7 \cong \mathbb{R}^7 \oplus \mathfrak{g}_2.$$

It is not an isomorphism and is only G_2 -equivariant, not $SO(7)$ -equivariant.

Theorem E.49 (Full-isotropy route). *If $\overline{\text{Hol}}^+ = SO(V_{\text{sp}})$, then a nonzero Hol^+ -equivariant map*

$$\bigwedge^2 V_{\text{sp}} \longrightarrow V_{\text{sp}}$$

exists only for $d_{\text{Cl}} = 3$. At $d_{\text{Cl}} = 3$ the space of such maps is one-dimensional.

Proof. Dense-subgroup equivariance extends by continuity to $SO(V_{\text{sp}})$. For $d_{\text{Cl}} \neq 3$, the defining representation does not occur in the adjoint module $\bigwedge^2 V_{\text{sp}}$; equivalently

$$\text{Hom}_{SO(d_{\text{Cl}})}\left(\bigwedge^2 V_{\text{sp}}, V_{\text{sp}}\right) = 0.$$

For $d_{\text{Cl}} = 3$, the Hodge star gives a nonzero map, and irreducibility together with Schur’s lemma makes the intertwiner space one-dimensional. The seven-dimensional cross product does not contradict this because it is only G_2 -equivariant. $\square$

Principle E.50 (Primitive first-order symbol closure). The oriented volume-dual response survives the continuum scaling limit with nonzero coefficient, contributes to the principal rather than subprincipal symbol, lies in the primitive one-reset observable sector, and does not become an independent higher-form or internal field.

Corollary E.51 (Dimension consequence of primitive-symbol closure). *Under [Theorem E.50](#), with $d_{\text{Cl}} \geq 3$,*

$$\boxed{d_{\text{Cl}} = 3}.$$

Proof. The primitive first-order Dirac symbol has Clifford degree one, whereas [Theorem E.42](#) places the nonzero volume-dual response in degree $d_{\text{Cl}} - 2$. The principle requires these degrees to agree, hence $d_{\text{Cl}} - 2 = 1$. $\square$

Remark E.52 (Logical scope). The canonical response β_{ren} , its Hodge degree $\bigwedge^{d_{\text{Cl}}-2} V_{\text{sp}}$, the model-theoretic non-implication, and the equivalence in [Theorem E.44](#) are established. What remains conditional is the physical requirement that the volume-dual response be an elementary revision, or equivalently the stronger continuum claim [Theorem E.50](#). The latter isolates four open requirements: survival under scaling, appearance in the principal symbol, membership in the primitive one-reset sector, and exclusion of a separate higher-form field.

Corollary E.53 (Dimension statement with hard conditional uniqueness). *Within the primitive class,*

$$d_{\text{Cl}} = 3 \text{ is minimal unconditionally and unique when the nonzero oriented volume-dual response closes into the } \mathbb{R}.$$

In the regular spacetime-crystal phase this is equivalently

$$b_1^{\text{eff}} = 3, \quad q_{\text{alg}} = q_{\text{met}} = \text{rank } \omega = 4,$$

and, under the optional heat-kernel assumptions, $d_s = 4$.

E.6 Alternative conditional selection mechanisms

[Theorem 4.41](#) places a sharp ceiling on any unconditional dimension theorem: every odd spatial rank is realised by a primitive revision datum. Hence no consequence of primitivity, locality, exact covariance, the derived sheet/internal ergodicity, or isotropy alone can exclude $d_{\text{Cl}} = 5, 7, \dots$. Uniqueness therefore requires an additional structural principle formulated inside the revision sector.

The structural interference-closure criterion has now been proved in full in [Section E.5](#). The remaining criteria in this subsection are retained only as independent comparisons: the real-even division condition and the access-based principles provide different conditional routes, but they are not needed for [Theorem E.44](#).

Theorem E.54 (Real-even division condition). *Fix the Euclidean real Clifford convention carried by the signed spatial sector. If one requires the real even revision algebra to be a division algebra, then among odd ranks it permits only $d_{\text{Cl}} = 1$ and $d_{\text{Cl}} = 3$:*

$$\text{Cl}_1^0(\mathbb{R}) \cong \mathbb{R}, \quad \text{Cl}_3^0(\mathbb{R}) \cong \mathbb{H},$$

whereas for odd $d_{\text{Cl}} \geq 5$ the even algebra is a nontrivial matrix algebra and contains zero divisors. Spatial nondegeneracy excludes $d_{\text{Cl}} = 1$, leaving $d_{\text{Cl}} = 3$.

Proof. The standard even-algebra identity reduces the classification to the real Clifford table. Here the spatial convention is $(\gamma^i)^2 = +\mathbf{1}$, so $\text{Cl}_1^0(\mathbb{R}) \cong \mathbb{R}$ and $\text{Cl}_3^0(\mathbb{R}) \cong \text{Cl}_{0,2}(\mathbb{R}) \cong \mathbb{H}$. With this convention the higher odd cases are matrix algebras of size at least two over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$; such matrix algebras contain nonzero zero divisors. See [\[78, 79\]](#). $\square$

Corollary E.55 (Concordant native selection). *Interference closure and the real-even division condition are logically distinct conditions on the same signed revision sector, and both select*

$$d_{\text{Cl}} = 3$$

uniquely among spatially nondegenerate odd ranks. They strengthen the dimension argument by replacing a monotone cost preference with two exact, intrinsic and concordant criteria. They remain selection principles rather than consequences of the renewal axioms.

E.7 Why isotropy cannot make the selection unconditional

Theorem E.56 (Isotropy is dimension-blind). *For every $d_{\text{Cl}} \geq 1$, the cross-polytope frame $\{\pm e_1, \dots, \pm e_{d_{\text{Cl}}}\}$ with equal weights $1/(2d_{\text{Cl}})$ has second moment*

$$M = \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}.$$

Therefore isotropic self-averaging is available in every spatial dimension and cannot imply interference closure or select $d_{\text{Cl}} = 3$.

Proof. Directly,

$$\sum_{i=1}^{d_{\text{Cl}}} \frac{1}{2d_{\text{Cl}}} (e_i e_i^\top + (-e_i)(-e_i)^\top) = \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}.$$

Since the construction works in every rank, isotropy excludes none of the higher odd primitive endpoints. $\square$

Proposition E.57 (Symmetric and alternating sectors are independent). *The isotropic second moment belongs to $\text{Sym}^2 V_{\text{sp}}$, whereas revision interference belongs to $\bigwedge^2 V_{\text{sp}}$. These are inequivalent natural tensor sectors. The Dirac-square contraction between metric and spin connection reduces to the dimension-independent Lichnerowicz scalar term and does not transfer a rank-selection condition from one sector to the other.*

Remark E.58 (The dimension frontier). The failure of isotropy to select the rank is a no-go, not merely an unfinished proof. Together with [Theorem 4.41](#), it shows that unconditional uniqueness of $3 + 1$ cannot follow from the present renewal, primitive and isotropic axioms. The rigorous headline remains unconditional *minimality*; uniqueness is supported by several concordant native principles, but remains conditional unless a new representation-level axiom is derived from a deeper layer.

E.8 Alternative conditional uniqueness: the access fixed point

Before stating the selection criterion, we emphasise what the paper actually relies on. The central dimension result is the *unconditional* minimality theorem [Theorem 4.40](#): in the primitive class $2 + 1$ is impossible, $1 + 1$ is spatially degenerate, and $3 + 1$ is the minimal spatially nondegenerate Lorentzian Dirac endpoint. Nothing downstream of this subsection feeds back into that theorem. The access principle introduced here, and the access-efficiency functional of the next subsection, are *optional* selection layers: they upgrade minimality to uniqueness among the higher odd primitive ranks, but they are extra modelling choices, not theorems about renewal memory, and a reader may stop at [Theorem 4.40](#) with the headline intact. We keep them explicit precisely so that the conditional content is never silently absorbed into the word “emergence.”

The preceding subsection leaves a genuine gap: primitivity permits every higher odd spatial rank. To turn minimality into uniqueness, an additional selection criterion is required. The first such criterion is an access self-consistency principle. It asks that the dimension realised by the signed renewal component agree with the integer dimension preferred by the same signed budget.

Once the primitive modular revision algebra supplies the full Clifford grading, the signed budget is the number of noncentral holonomy degrees. Write $B_{\text{sign}} = b_1 \log 2$.

Proposition E.59 (Additive access functional). *Among the three elementary ways of spending a signed budget $e^{B_{\text{sign}}}$ over m candidate axes – factored, replicated, and additively partitioned – only the additive partition has a finite interior optimum. Its maximal access is*

$$\mathcal{R}_m(B_{\text{sign}}) = m(B_{\text{sign}} - \log m),$$

attained at the isotropic allocation.

Proof. For a factored allocation $\prod_i W_i = e^{B_{\text{sign}}}$, the logarithmic access is constant in m . For a replicated allocation $W_i = e^{B_{\text{sign}}}$, it is mB_{sign} and hence runs away with m . For an additive allocation $\sum_i W_i = e^{B_{\text{sign}}}$, the arithmetic–geometric mean inequality gives $\prod_i W_i \leq (e^{B_{\text{sign}}}/m)^m$, with equality at equal allocation. This gives the displayed functional. $\square$

Principle E.60 (Access self-consistency). Among primitive signed renewal components, the realised spatial rank is an access fixed point: the dimension present equals the integer dimension that the additive access functional prefers for the same signed budget generated by the component,

$$d_{\text{Cl}} = \arg \max_{m \geq 1} \mathcal{R}_m(d_{\text{Cl}} \log 2).$$

This is a selection principle, not a consequence of primitivity alone.

The resulting fixed-point problem is elementary but useful. Among odd primitive ranks it leaves only the one-axis endpoint and the 3-axis endpoint.

Lemma E.61 (Odd access fixed points). *For $\mathcal{R}_m(B) = m(B - \log m)$, the access self-consistency equation*

$$d_{\text{Cl}} = \arg \max_{m \geq 1} \mathcal{R}_m(d_{\text{Cl}} \log 2)$$

has, among odd positive integers, exactly the solutions $d_{\text{Cl}} = 1$ and $d_{\text{Cl}} = 3$.

Proof. For fixed B , the increment is

$$\mathcal{R}_{n+1}(B) - \mathcal{R}_n(B) = B - ((n+1) \log(n+1) - n \log n),$$

and the bracketed term is strictly increasing in n . Equivalently, the real stationary point satisfies $d_* = e^{B-1}$. At the self-consistent budget $B = d_{\text{Cl}} \log 2$, this is $m_* = 2^{d_{\text{Cl}}}/e$. For $d_{\text{Cl}} = 1$ the integer maximizer is 1; for $d_{\text{Cl}} = 3$, $8/e$ lies in the integer cell of 3. For every odd $d_{\text{Cl}} \geq 5$, $2^{d_{\text{Cl}}}/e > d_{\text{Cl}} + 1/2$, so the integer maximizer is strictly larger than d_{Cl} . Thus the only odd fixed points are 1 and 3. $\square$

Theorem E.62 (Conditional access selection of the nondegenerate endpoint). *Assume the signed renewal component is primitive, the finite-shell modular form stabilises, and the access self-consistency principle [Theorem E.60](#) holds. Then the only primitive access-self-consistent endpoints are 1+1 and 3+1. The first is spatially degenerate by [Theorem 4.39](#). Therefore the unique nondegenerate access-self-consistent primitive endpoint is 3+1.*

Proof. By [Theorem 4.24](#), primitivity gives a Clifford revision algebra on H_{full} . The rank parity argument in [Theorem 4.40](#) forces d_{Cl} odd. By [Theorem E.61](#), access self-consistency leaves only $d_{\text{Cl}} = 1$ and $d_{\text{Cl}} = 3$. Excluding the spatially degenerate one-axis endpoint leaves $d_{\text{Cl}} = 3$. The corresponding 3+1 Lorentzian Dirac endpoint is the $d_{\text{Cl}} = 3$ specialisation of the isotropic Clifford rounding and operator-convergence theorems. $\square$

E.9 A secondary optional functional: access efficiency

The access fixed-point principle is one way to rank the odd primitive endpoints. We now record a second, more transparent functional built directly from two quantities already present in the construction: the operator dimension of the primitive Clifford revision algebra and the smallest isotropic reset frame needed for the spatial symbol. This criterion is useful because it gives the same conditional conclusion by a simpler monotonicity argument.

The functional ranks the odd primitive endpoints as

$$d_{\text{Cl}} = 1 > d_{\text{Cl}} = 3 > d_{\text{Cl}} = 5 > d_{\text{Cl}} = 7 > \dots,$$

and therefore selects $d_{\text{Cl}} = 3$ once the independently established spatial-nondegeneracy condition removes $d_{\text{Cl}} = 1$. As above, this is a selection criterion rather than an algebraic exclusion: the higher odd endpoints of [Theorem 4.41](#) remain admissible.

Lemma E.63 (Primitive complexity gap). *For odd candidate spatial Clifford rank m , a primitive revision datum has $\mathcal{A}_{\text{rev}} \cong M_{2^{(m+1)/2}}(\mathbb{C})$, equivalently $\mathcal{A}_{\text{rev}} \cong \text{Cl}_{m+1}(\mathbb{C})$, by [Theorem 4.24](#); hence operator dimension*

$$\dim_{\mathbb{C}} \mathcal{A}_{\text{rev}}(m) = \dim_{\mathbb{C}} \text{Cl}_{m+1} = 2^{m+1}.$$

Consequently

$$\frac{\dim_{\mathbb{C}} \mathcal{A}_{\text{rev}}(m+2)}{\dim_{\mathbb{C}} \mathcal{A}_{\text{rev}}(m)} = \frac{2^{m+3}}{2^{m+1}} = 4.$$

Higher odd endpoints are not forbidden, but each step $m \mapsto m+2$ multiplies the primitive revision-algebra dimension by four.

Proof. Primitivity is full $\mathbb{F}_2$ -rank of the modular commutator form ([Theorem 4.24](#)), giving $\mathcal{A}_{\text{rev}} \cong M_{2^{(m+1)/2}}(\mathbb{C})$, equivalently the even complex Clifford algebra on $m+1$ generators. Its complex vector-space dimension is 2^{m+1} . The ratio is $2^{(m+3)-(m+1)} = 2^2 = 4$. $\square$

Lemma E.64 (Minimal isotropic access frame). *Among finite, antipodal, equal-weight direction systems ν on $\mathbb{S}^{d_{\text{Cl}}-1}$ whose second moment is isotropic, $M(\nu) = \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}$, the minimal cardinality is*

$$N_{\min}(d_{\text{Cl}}) = 2d_{\text{Cl}},$$

attained, uniquely up to an orthogonal change of spatial frame, by the cross-polytope frame $\{\pm e_1, \dots, \pm e_{d_{\text{Cl}}}\}$.

Proof. An antipodal equal-weight system is a union of k antipodal pairs $\{\pm u_1, \dots, \pm u_k\}$ of unit vectors, of cardinality $2k$ and equal weight $1/(2k)$. Its second moment is

$$M(\nu) = \sum_{a=1}^k \frac{1}{2k} (u_a u_a^\top + (-u_a)(-u_a)^\top) = \frac{1}{k} \sum_{a=1}^k u_a u_a^\top.$$

Isotropy $M(\nu) \propto I_{d_{\text{Cl}}}$ forces $\sum_a u_a u_a^\top \propto I_{d_{\text{Cl}}}$, i.e. the u_a form a tight frame for $\mathbb{R}^{d_{\text{Cl}}}$, which requires $k \geq d_{\text{Cl}}$. With $k = d_{\text{Cl}}$ unit vectors, a tight frame must be an orthonormal basis, giving $\sum_a e_a e_a^\top = I_{d_{\text{Cl}}}$ and $M(\nu) = \frac{1}{d_{\text{Cl}}} I_{d_{\text{Cl}}}$ (trace 1 as required). Hence the minimum is $k = d_{\text{Cl}}$, i.e. $N_{\min}(d_{\text{Cl}}) = 2d_{\text{Cl}}$, realised by $\{\pm e_i\}$. This is the d_{Cl} -dimensional cross-polytope frame; for $d_{\text{Cl}} = 3$ it is the octahedral frame of [Theorem 5.10](#). $\square$

The two preceding lemmas isolate the numerator and denominator of the efficiency ratio: isotropic spatial access divided by primitive revision cost.

Definition E.65 (Primitive access efficiency). For a candidate spatial Clifford rank m , let

$$C_{\text{rev}}(m) := \dim_{\mathbb{C}} \mathcal{A}_{\text{rev}}(m) = 2^{m+1}$$

be the primitive revision cost, and let

$$A_{\text{iso}}(m) := N_{\min}(m) = 2m$$

be the minimal isotropic access count. The primitive access efficiency is

$$\eta(m) := \frac{A_{\text{iso}}(m)}{C_{\text{rev}}(m)} = \frac{2m}{2^{m+1}} = \frac{m}{2^m}.$$

Thus $\eta(m)$ measures isotropically accessible spatial directions per primitive Clifford revision mode for rank m ; the denominator is the operator-theoretic cost of the primitive revision algebra.

The monotonicity of this efficiency is immediate, but it provides a useful second reading of why the first nondegenerate odd endpoint is preferred.

Theorem E.66 (Access-efficiency selection of the nondegenerate endpoint). *With $\eta(m) = m/2^m$ of [Theorem E.65](#):*

- (i) $\eta(1) = \eta(2) = \frac{1}{2}$, and η is strictly decreasing for integer $d_{\text{Cl}} \geq 2$ (since $\eta(m+1)/\eta(m) = (m+1)/(2m) < 1$ iff $m > 1$); in particular η is strictly decreasing along the odd integers $m = 1, 3, 5, \dots$, with

$$\eta(1) = \frac{1}{2} > \eta(3) = \frac{3}{8} > \eta(5) = \frac{5}{32} > \dots;$$

- (ii) among spatially nondegenerate odd primitive endpoints ($d_{\text{Cl}} \geq 3$ odd, [Theorem 4.39](#)), η is uniquely maximised at $d_{\text{Cl}} = 3$;
- (iii) read as a selection functional, the unique spatially nondegenerate primitive endpoint of maximal access efficiency is

$$\boxed{d_{\text{Cl}} = 3},$$

while $d_{\text{Cl}} = 5, 7, \dots$ remain algebraically admissible ([Theorem 4.41](#)) but strictly less efficient.

Proof. (i) $\eta(1) = 1/2$, $\eta(2) = 2/4 = 1/2$, and

$$\eta(m+1)/\eta(m) = ((m+1)/2^{m+1})(2^m/m) = \frac{m+1}{2m},$$

which is < 1 exactly when $m+1 < 2m$, i.e. $m > 1$. Hence η is strictly decreasing for $d_{\text{Cl}} \geq 2$, and the displayed odd chain follows. (ii) Spatial nondegeneracy excludes $d_{\text{Cl}} = 1$ ([Theorem 4.39](#)); on the remaining odd $d_{\text{Cl}} \geq 3$, (i) gives a strictly decreasing sequence, so the maximum is the smallest admissible value $d_{\text{Cl}} = 3$. (iii) is the selection reading of (ii), with the explicit values $\eta(5) = 5/32$, $\eta(7) = 7/128$ exhibiting the strict loss; admissibility of higher ranks is [Theorem 4.41](#). $\square$

Remark E.67 (Exactly what is, and is not, selected). Four points are needed for the statement to be mathematically precise.

- (a) $d_{\text{Cl}} = 1$ is the unconstrained maximum. Since $\eta(1) = \frac{1}{2} > \eta(3) = \frac{3}{8}$, efficiency *by itself* is maximised at $d_{\text{Cl}} = 1$, not $d_{\text{Cl}} = 3$. The restriction to $d_{\text{Cl}} \geq 3$ is the spatial-nondegeneracy condition of [Theorem 4.39](#) (the $d_{\text{Cl}} = 1$ exclusion already used in [Theorem 4.40](#)), established independently of any cost. Thus η selects $d_{\text{Cl}} = 3$ only *after* nondegeneracy removes $d_{\text{Cl}} = 1$; it does not make $d_{\text{Cl}} = 3$ the globally most efficient rank.
- (b) *The selection is cost-dependent.* It uses the full Clifford module dimension $2^{d_{\text{Cl}}+1}$ as the revision cost. With the logarithmic cost $\log_2 \dim \text{Cl}_{d_{\text{Cl}}+1} = d_{\text{Cl}} + 1$, the efficiency $2d_{\text{Cl}}/(d_{\text{Cl}} + 1)$ is strictly increasing toward 2 and selects nothing. The exponential cost is justified operator-theoretically: the spectral package acts on the entire Clifford module, so its dimension is the natural operator-level cost. This is nevertheless a modelling choice, not an algebraic theorem. Hence η is an internal selection criterion, tagged , exactly like [Theorem E.60](#).
- (c) $d_{\text{Cl}} = 5$ is not defeated. [Theorem 4.41](#) shows that higher odd ranks remain a valid nondegenerate primitive endpoint; η only ranks it lower. This is consistent with [Theorem 4.42](#): no algebraic exclusion of higher odd Clifford ranks is claimed or implied.
- (d) *Corroboration, not independence.* Both η and the access self-consistency principle place the first two odd primitive endpoints $d_{\text{Cl}} = 1$ and $d_{\text{Cl}} = 3$ ahead of all higher odd Clifford ranks; after the independent spatial-nondegeneracy exclusion of $d_{\text{Cl}} = 1$, both criteria select $d_{\text{Cl}} = 3$. They are two readings of one exponential-cost backbone, so they reinforce the conditional $d_{\text{Cl}} = 3$ selection rather than confirming it from logically independent premises.

Remark E.68 (Net effect on the dimension headline). The unconditional headline is unchanged: by [Theorem 4.40](#), $3 + 1$ is the minimal nondegenerate primitive endpoint and $2 + 1$ is impossible. What the efficiency functional adds is a second, transparent *conditional* selection that agrees with the access self-consistency selection of [Theorem E.62](#): $3 + 1$ is the unique spatially nondegenerate primitive endpoint of maximal access efficiency $d_{\text{Cl}}/2^{d_{\text{Cl}}}$. The strengthened conditional statement is therefore

$$3 + 1 = \text{minimal nondegenerate \textbf{and} uniquely access-efficient primitive endpoint},$$

with the $d_{\text{Cl}} = 3$ *uniqueness* remaining, irreducibly, conditional on an intrinsic cost choice — two such choices now agreeing.

E.10 Why decimation does not discharge the access principle

One might hope that the access principle is not really an additional principle, but merely the fixed point of an ordinary renewal coarse-graining. This subsection explains why that hope fails for quotient-type decimations. Such decimations identify predictive classes, contract or identify edges, delete transient structure, or pass to a coarser shift congruence. They do not create new independent cycles.

Lemma E.69 (Quotient decimation is non-increasing in b_1). *For any quotient-type decimation $\mathcal{D}$,*

$$b_1(\mathcal{D}G) \leq b_1(G).$$

Proof. A quotient-type decimation identifies vertices and/or edges and kills or identifies cycles. The induced map on first homology over $\mathbb{Z}/2$ is therefore surjective from the original cycle space onto the decimated cycle space, so the rank cannot increase. $\square$

Proposition E.70 (Quotient decimation cannot derive access selection). *Quotient-type decimation cannot implement the access fixed-point map*

$$F(b) = \arg \max_{d' \geq 1} \mathcal{R}_{d'}(b \log 2).$$

Indeed quotient decimation is non-increasing in b_1 , whereas $F(b)$ grows like $2^b/e$ and satisfies $F(b) > b$ for all sufficiently large b (in particular for $b \geq 5$). Thus a rule sending b to $F(b)$ is not an ordinary renewal coarse-graining; it is precisely an additional access optimisation principle.

Proof. The first assertion is [Theorem E.69](#). For the access map, the continuous maximizer of $\mathcal{R}_{d'}(b \log 2)$ is $2^b/e$, and integer rounding therefore grows exponentially in b . For $b \geq 5$, $2^b/e > b + 1/2$, hence $F(b) > b$. A non-increasing quotient operation cannot produce this increasing map. Defining a “decimation” to output $F(b)$ would simply restate the access principle in different language. $\square$

Remark E.71 (Scope of the dimension statement). The unconditional headline is: *primitivity forbids $2+1$ and makes $3+1$ the minimal nondegenerate Lorentzian Dirac endpoint*. The preferred hard conditional uniqueness theorem is [Theorem E.44](#): oriented interference closure excludes every higher odd endpoint. The access fixed point and efficiency functional below remain alternative, softer conditional criteria and are not consequences of ordinary coarse-graining.

E.11 Interfaces with downstream constructions

The primitive $3+1$ result remains kinematic and operator-theoretic. It does not produce matter multiplets, field equations, or phenomenology. It says that the primitive signed renewal sector supplies the Clifford revision algebra and that $3+1$ is the minimal nondegenerate endpoint, with hard conditional uniqueness under oriented interference closure and optional alternative access criteria. There are four natural downstream interfaces.

Balanced reset statistics. The reset statistics used in the continuum limit may be treated as self-consistent rather than externally prescribed. In a balancing extension one models reset ages by a piecewise-deterministic Markov process and asks for a stationary law whose induced reset frequencies and channel weights reproduce the renewal data. Under covering and feasibility hypotheses such a balanced law exists; when feasibility fails, a separating semidefinite witness gives an obstruction; and under contraction or determined monotonicity hypotheses the balanced law is unique and stable up to the evident predictive gauge freedom. This layer is not used in the proofs of the present article. Its role is to explain how the second-moment fields and predictive measures used in the continuum limit could arise self-consistently; it is not needed to enforce rotational isotropy, since the flat endpoint depends only on the nondegenerate metric defined by the selected tensor.

After predictive collapse, a balanced law pushes forward to a canonical measure μ_{CP} on predictive classes. In conformal regular regimes its information dimension is

$$\text{Dim}_1(\mu_{\text{CP}}) = \frac{h}{\lambda},$$

where h is the entropy rate of the balanced reset law and λ is the corresponding contraction exponent. Thus q_{met} is the unweighted capacity dimension of the available predictive attractor, while h/λ is the dimension actually filled by the stationary renewal dynamics. The gap is a free-energy defect, equivalently a relative entropy from the dimension-maximising measure, and should not be confused with entropy production.

Finite matter geometry. An almost-commutative or finite-real-spectral-triple matter sector can be coupled to the signed renewal geometry by taking a product with the finite internal renewal quotient. Lorentzian versions of the noncommutative Standard Model already show that signature $(1, 3)$, KO-dimension, fermion reduction, and the finite indefinite structure must be treated together [80, 81, 82]. Twisted constructions provide another route from the Euclidean almost-commutative model to Lorentzian fermionic data [18, 19], and finite extensions such as $U(1)_{B-L}$ illustrate the downstream model-building interface [83]. The present paper does not reproduce those matter models. It supplies the external renewal-generated Lorentzian spectral datum; the signed/Krein generator, real structure, and first-order conditions would then constrain which finite matter sectors can be coupled consistently.

Amplitude and charge local systems. The finite $\mathbb{Z}/4$ lift of Section 3.4 is not used to obtain Lorentzian signature; the $\mathbb{Z}/2$ signed cover already does that. It is instead the correct interface for downstream charged local systems. Any matter construction using phases, charged incidence, or square roots of projective signs should therefore be formulated in the amplitude category and then reduced to the real/Krein quotient when only signature is being read. If a central $\mathbb{Z}/6$ screen is introduced in a finite matter sector, the renewal-compatible form is a composite of the Krein parity with an independent three-cycle renewal or generation grading,

$$\mathbb{Z}/2^{\text{Krein}} \times \mathbb{Z}/3 \cong \mathbb{Z}/6,$$

rather than a new primitive assumption. Likewise, neutral or dark twisted sectors constructed from real $\mathbb{Z}/2$ holonomy should be regarded as real shadows until their amplitude, Pin/Majorana, or charged-local-system refinement has been specified. Finally, real-category protection statements for larger ambient cells or exteriors do not by themselves rule out amplitude-level splittings after a $\mathbb{Z}/4$ lift; non-adoption of such ambient cells must rest on independent minimality, projection, or dynamical-selection criteria.

Dynamical gravity. Dynamical gravity requires additional input not present here: a rule identifying variations of reset statistics with a connection, curvature, and an entropy or action principle. The adiabatic curved Dirac theorem of Section 5.4 now supplies the complete geometric Dirac operator through subprincipal order: the renewal-generated coframe determines the discrete torsion-free Cartan connection, whose limit is the Levi-Civita spin connection, and the covariant Hamiltonians converge in strong resolvent sense. It does not derive Einstein equations, a curvature evolution law, or a gravitational variational principle.

Modular and conformal completions. Modular or conformal extensions would require placing the renewal inclusion into a genuine modular-theoretic framework. The signed cover and the renewal resolvent give the correct algebraic ingredients for such a comparison, but the present paper does not construct a conformal net, a string worldsheet theory, or a quantum-gravity model.

Thus the present article should be read as the foundation layer: it proves the positive predictive geometry, the signed/Krein obstruction and classification, the Lorentzian Dirac endpoint, universality and enhanced metric reconstruction, and the primitive modular minimal-endpoint interface. Matter, dynamics, conformal extensions, and phenomenology are logically downstream.

Remark E.72 (Summary of the dimension statement). The logical content of this section is therefore deliberately stratified. Topology supplies signed holonomy but not Clifford anticommutation; modular revision supplies the commutator form. Under full-dimensional affine reachability and finite primitive control, bounded return completion derives the predictive growth–cycle identity. The accessible modular rank then follows from nondegeneracy, while topological-crystal and heat-kernel regularity supply the return dimension. The channel-metric equality remains a separate non-collapse rider, as stated in [Theorem E.16](#). Primitivity then forces even total Clifford rank and hence odd spatial rank; spatial nondegeneracy removes the $1+1$ endpoint; and $3+1$ becomes the minimal nondegenerate primitive endpoint. Uniqueness among higher odd endpoints is conditional: the primary theorem uses oriented interference closure, while the access fixed point and access-efficiency functional are retained only as alternative optional criteria. None is claimed as a consequence of primitivity alone.

E.12 The canonicity theorem

We now consolidate the preceding characterisation results into a single statement that complements the main assembly theorem. The result gathers the modular, symplectic, and Cartan reconstructions under one transparent hypothesis package and isolates the sole remaining datum needed to upgrade dimensional minimality to uniqueness.

Condition E.1 (Canonical renewal hypotheses). A signed renewal memory is *canonical* if:

- (C1) [connected recurrence] the recurrent predictive component is finite, connected, graded with unit length increments, and its signed cover is connected;
- (C2) [modular homogeneity] the elementary resets are analytic modular eigenoperators of a faithful weight with a common eigenvalue β ([Condition 3.2](#));
- (C3) [reversible predictive revision phase] the spatial revision labels are predictive units satisfying [Assumption 4.1](#); hence their action on the finite recurrent factor is by automorphisms and has the automatic projective lift of [Theorem 4.10](#); the optional finite $\mathbb{Z}/4$ square-root completion is retained only at the amplitude level;
- (C4) [nondegenerate covariant second moment] the renewal-generated coframe $e_j^a = \kappa M^{aj}$ is uniformly nondegenerate, $M \succeq m_0 I$, and C_b^4 , and neighbouring second moments obey the transport covariance of [Assumption 5.1](#);
- (C5) [exchange-symmetric translational phase] the resolved two-reset diamond records satisfy [Assumption 5.2](#), so first-moment plaquette closure follows from [Theorem 5.39](#).

Theorem E.73 (Canonicity of the renewal Lorentzian datum). *For a canonical signed renewal memory:*

- (1) [E/P] *the unbounded signed modular operator is uniquely*

$$D_- = J_\chi e^{\beta N}$$

up to an overall scale and a bounded J_χ -self-adjoint perturbation B ; the modular Hamiltonian is affine in the clock, $\log \Delta_\varphi = \beta N + c\mathbf{1}$, with β the common reset modular eigenvalue (the modular characterisation and affine modular-flow results);

- (2) the reversible predictive revision action canonically determines the projective multiplier, alternating form ω , and reduced symplectic revision space

$$H_{\text{prim}} = H_{\text{full}} / \text{rad } \omega, \quad \bar{\omega} \text{ nondegenerate,}$$

with reduced revision algebra of Morita class $[M_{2m}(\mathbb{C})] = [\text{Cl}_{2m}(\mathbb{C})]$, factor representations parametrised by $\widehat{\text{rad } \Omega}$ (the canonical reduction and affine modular-flow results);

- (3) the renewal coframe carries a unique metric-compatible torsion-free transport, with torsion the translational and curvature the rotational part of the reset-channel plaquette defect, converging to the Levi–Civita Dirac transport $\omega^{\text{LC}} + O(h^2)$ (the Cartan and channel-composition results);
- (4) $[E/C]$ the only remaining dimensional-selection datum is the nonzero projection of the oriented two-reset circulation Ξ onto the principal first-order sector; this is nonzero iff $d_{\text{C1}} = 3$ and $\Xi \neq 0$, and is RG-marginal exactly at $d_{\text{C1}} = 3$ in the explicit block-renewal model (the survival criterion and block-renewal coarse-graining theorem).

Proof. (1) combines the modular signed-Dirac characterisation with [Theorem 3.38](#), which proves that the modular Hamiltonian is affine in the renewal clock. (2) follows from the automatic projective lift and canonical symplectic reduction, together with [Theorems 4.22](#) and [4.23](#). (3): connected recurrence and the nondegenerate C_b^4 coframe give the discrete fundamental theorem of the Cartan characterisation; isometric comparison gives metric compatibility; second-order plaquette equivalence gives $T = 0$ via [Theorem 5.41](#); uniqueness then yields Levi–Civita with the $O(h^2)$ rate. (4) combines the survival criterion with the block-renewal eigenvalue [Theorem E.32](#) and [Theorem E.33](#). $\square$

Remark E.74 (Reading of the canonicity theorem). Under [Condition E.1](#), the modular weight, primitive revision sector, and Cartan transport are forced by the displayed hypotheses. Dimensional uniqueness is reduced to the single sharp datum: whether oriented two-reset circulation projects nontrivially onto the principal first-order sector. Unconditionally this gives *minimality* ($3 + 1$ is the minimal nondegenerate primitive endpoint); the *uniqueness* upgrade is the marginal/principal selection, a theorem in the explicit block-renewal model and a principle in general. This is the precise boundary of the foundational layer.

F Further continuum, propagation, and global-topology results

F.1 A least-distortion counterexample

Example F.1 (Polar comparison is torsionful on a curved surface). Take $d_{\text{C1}} = 2$ and the warped coframe $E(x) = \text{diag}(1, f(x^1))$, $f > 0$ non-affine, i.e. metric $ds^2 = (dx^1)^2 + f(x^1)^2(dx^2)^2$ (a surface of revolution, curved for $f'' \neq 0$). The frame Gram is $P = \text{diag}(1, f^2)$, and $EE_{,1}^\top = \text{diag}(0, ff')$ is symmetric, so $\text{Asym}(EE_{,1}^\top) = 0$; the Sylvester equation of [Theorem 5.36](#) gives the polar connection $\omega^{\text{polar}} \equiv 0$. Its torsion is

$$T_{12}^2 = \partial_1 e_2^2 - \partial_2 e_1^2 = f' \neq 0,$$

so the polar (least-distortion) connection has nonzero torsion. By contrast the Levi-Civita spin connection in this frame is γ^{LC} with the only nonzero component $\gamma_{2;12} = f'$, which is metric *and* torsion-free. Hence

$$\omega^{\text{polar}} = 0 \neq \omega^{\text{LC}}.$$

Least distortion supplies metric compatibility but not Levi-Civita; the missing ingredient is plaque closure.

F.2 Joint convergence of propagation geometry

The preceding operator limit determines more than a principal symbol. The same second-moment field controls the characteristic cone, its optical distance, and the macroscopic propagation of the renewal evolution. This subsection records the precise joint statement and separates it from the combinatorial reset-ray order, which depends on a different statistic.

Write

$$\widehat{g}(x) = \kappa^{-2} M(x)^{-2}, \quad g_{\mu\nu}(x) = \text{diag}(1, -\widehat{g}(x)).$$

Uniform ellipticity gives

$$\kappa^{-2} |\vec{v}|^2 \leq \widehat{g}(x)[\vec{v}, \vec{v}] \leq \kappa^{-2} m_0^{-2} |\vec{v}|^2,$$

so $\widehat{g}$ is complete and its distance $d_{\widehat{g}}$ is bi-Lipschitz equivalent to the Euclidean distance. The future cone is

$$\text{cone}_g(x) = \{(v^0, \vec{v}) : v^0 \geq 0, (v^0)^2 \geq \widehat{g}(x)[\vec{v}, \vec{v}]\},$$

whose unit-time slice is the ellipsoid

$$\text{cone}_g(x) \cap \{v^0 = 1\} = \kappa M(x) B^{d_{\text{Cl}}}.$$

Lemma F.2 (Stability of the optical metric and distance). *Let $M, \widetilde{M}$ be symmetric fields with $m_0 I \preceq M, \widetilde{M} \preceq I$. Then*

$$\|M^{-2} - \widetilde{M}^{-2}\| \leq 2m_0^{-4} \|M - \widetilde{M}\|.$$

Consequently, if $M_R \rightarrow M$ locally uniformly in a common uniformly elliptic class, then

$$d_{\kappa^{-2} M_R^{-2}} \longrightarrow d_{\widehat{g}}$$

locally uniformly on $\mathbb{R}^{d_{\text{Cl}}} \times \mathbb{R}^{d_{\text{Cl}}}$.

Proof. The algebraic identity

$$M^{-2} - \widetilde{M}^{-2} = M^{-2}(\widetilde{M}^2 - M^2)\widetilde{M}^{-2}$$

is valid without commutativity, and

$$\widetilde{M}^2 - M^2 = (\widetilde{M} - M)\widetilde{M} + M(\widetilde{M} - M).$$

The displayed operator bound follows from $\|M^{-2}\|, \|\widetilde{M}^{-2}\| \leq m_0^{-2}$ and $\|M\|, \|\widetilde{M}\| \leq 1$.

For the distance statement, fix a compact K_0 . Uniform ellipticity confines all minimizing geodesics between points of K_0 to one larger compact set K , independent of R . For any absolutely continuous curve γ in K ,

$$\left| \sqrt{\widehat{g}_R(\dot{\gamma}, \dot{\gamma})} - \sqrt{\widehat{g}(\dot{\gamma}, \dot{\gamma})} \right| \leq C(m_0, \kappa) \|M_R - M\|_{C^0(K)} |\dot{\gamma}|.$$

Integrating along a minimizing curve for either metric and reversing the roles of the two metrics gives a uniform estimate on $K_0 \times K_0$, which tends to zero. $\square$

Proposition F.3 (Frozen-symbol and cone convergence). *Let $M_R \rightarrow M$ locally uniformly and $h_R \rightarrow 0$. On every compact $K \subset \mathbb{R}^{d_{\text{Cl}}}$ and momentum ball $|\xi| \leq \Lambda$,*

$$\sup_{x \in K, |\xi| \leq \Lambda} \|\sigma_{\text{fr}}(\mathbf{H}_R)(x, \xi) - \kappa A(M(x)\xi)\| \rightarrow 0.$$

Moreover

$$\sup_{x \in K} d_{\text{H}}\left(\kappa M_R(x)B^{d_{\text{Cl}}}, \kappa M(x)B^{d_{\text{Cl}}}\right) \leq \kappa \sup_{x \in K} \|M_R(x) - M(x)\| \rightarrow 0.$$

Proof. The frozen symbol is

$$\sigma_{\text{fr}}(\mathbf{H}_R)(x, \xi) = \kappa \sum_a p_a^{(R)}(x) A(\hat{\theta}_a) \frac{\sin(h_R \hat{\theta}_a \cdot \xi)}{h_R}.$$

Using $|\sin s - s| \leq |s|^3/6$, its difference from $\kappa A(M_R(x)\xi)$ is uniformly $O(h_R^2 \Lambda^3)$; the remaining difference is bounded by $C\Lambda \|M_R - M\|$. The Hausdorff estimate follows by coupling the points $\kappa M_R u$ and $\kappa M u$, $|u| \leq 1$. $\square$

Proposition F.4 (Energy-estimate cone and finite propagation). *Let $u(t) = e^{-it\mathbf{H}_g} u_0$. If u_0 is supported in a closed set S , then*

$$\text{supp } u(t) \subseteq J_t^+(S) := \{\vec{q} : d_{\hat{g}}(\vec{q}, S) \leq t\}.$$

The frozen spatial propagation body is exactly $\kappa M(x)B^{d_{\text{Cl}}}$: its support function in direction n is $\kappa|M(x)n|$, which is both the maximal characteristic speed and the support function of the convex hull of the group velocities.

Proof. The equation is a symmetric hyperbolic system with Hermitian principal coefficients $B^j(x) = \kappa A^i M^{ij}(x)$. In a unit conormal direction n ,

$$\rho\left(\sum_j B^j n_j\right) = \rho(\kappa A(Mn)) = \kappa |Mn|.$$

The standard energy estimate therefore propagates support inside the control cone whose unit-time slice has support function $n \mapsto \kappa |Mn|$, namely $\kappa M B^{d_{\text{Cl}}}$. Integrating this cone field gives precisely the optical arrival-time condition $d_{\hat{g}}(\vec{q}, S) \leq t$.

For the two dispersion branches $\lambda_{\pm}(\xi) = \pm \kappa |M\xi|$,

$$\nabla_{\xi} \lambda_{\pm} = \pm \kappa \frac{M^2 \xi}{|M\xi|}.$$

Their convex hull has the same support function

$$\sup_{\xi \neq 0} n \cdot \kappa \frac{M^2 \xi}{|M\xi|} = \kappa |Mn|,$$

and hence equals the same ellipsoid. $\square$

Theorem F.5 (Macroscopic domain-of-dependence convergence). *Under the hypotheses of Theorem 5.49, fix $u_0 \in L^2$ with compact support S .*

(i) If F is closed and $F \cap J_t^+(S) = \emptyset$, then

$$\|\mathbf{1}_F e^{-it\mathbf{H}_R} u_0\|_{L^2} \rightarrow 0.$$

The same conclusion holds for any L^2 -precompact family $u_{0,R} \rightarrow u_0$; in particular it excludes lattice-scale doubler families.

(ii) For open $U, V \subset \mathbb{R}^{d_{\text{Cl}}}$, if

$$\mathbf{1}_U e^{-it\mathbf{H}_g} \mathbf{1}_V \neq 0,$$

then there exists a fixed $u_0 \in L^2(V)$ such that

$$\liminf_{R \rightarrow \infty} \|\mathbf{1}_U e^{-it\mathbf{H}_R} u_0\|_{L^2} > 0.$$

Thus the outer Lorentzian domain of dependence is unconditional, while inner filling for variable $M(x)$ is exactly conditional on local nonvanishing of the continuum propagator.

Proof. Strong resolvent convergence in [Theorem 5.49](#) gives $e^{-it\mathbf{H}_R} u_0 \rightarrow e^{-it\mathbf{H}_g} u_0$ strongly, uniformly for t in compact sets. By [Theorem F.4](#), $\mathbf{1}_F e^{-it\mathbf{H}_g} u_0 = 0$; hence

$$\|\mathbf{1}_F e^{-it\mathbf{H}_R} u_0\| \leq \|(e^{-it\mathbf{H}_R} - e^{-it\mathbf{H}_g})u_0\| \rightarrow 0.$$

For a precompact family, add and subtract $e^{-it\mathbf{H}_R} u_0$.

For (ii), choose $v \in L^2(V)$ with $c := \|\mathbf{1}_U e^{-it\mathbf{H}_g} v\| > 0$. Then

$$\|\mathbf{1}_U e^{-it\mathbf{H}_R} v\| \geq c - \|(e^{-it\mathbf{H}_R} - e^{-it\mathbf{H}_g})v\|,$$

and the second term tends to zero. $\square$

Remark F.6 (No exact finite-mesh causal order). Each $\mathbf{H}_R$ is bounded and finite range, but $e^{-it\mathbf{H}_R} = \sum_{n \geq 0} (-it\mathbf{H}_R)^n / n!$ contains arbitrarily long paths at every $t > 0$. The discrete evolution therefore has rapidly decaying instantaneous tails rather than an exact light cone. The statement of [Theorem F.5](#) concerns macroscopic mass in the continuum limit, not nonzero amplitude at finite mesh.

F.2.1 Exact constant-coefficient support

Assume now $M(x) \equiv M_0$, with M_0 symmetric positive definite, and set

$$\mathbf{H}_0 = \kappa A(M_0 D), \quad K_t = e^{-it\mathbf{H}_0}.$$

After the linear change $y = M_0^{-1}x$, this is the standard massless Dirac evolution of speed κ .

Lemma F.7 (Forward Dirac kernel and its support). *Let $\mathcal{K}_t$ be the distribution kernel of K_t . Then*

$$\mathcal{K}_t = \partial_t W_t \mathbf{1} - i\mathbf{H}_0 W_t, \quad \widehat{W}_t(\xi) = \frac{\sin(t\kappa|M_0\xi|)}{\kappa|M_0\xi|},$$

and

$$\text{supp } \mathcal{K}_t = \begin{cases} \{x : |M_0^{-1}x| \leq \kappa|t|\}, & d_{\text{Cl}} = 1 \text{ or } d_{\text{Cl}} \text{ even,} \\ \{x : |M_0^{-1}x| = \kappa|t|\}, & d_{\text{Cl}} \geq 3 \text{ odd.} \end{cases}$$

In particular, every neighbourhood of a point in the displayed support carries a test spinor on which $\mathcal{K}_t$ is nonzero.

Proof. Since $H_0^2 = \kappa^2 |M_0 D|^2 \mathbf{1}$, functional calculus gives

$$K_t = \cos(t\kappa |M_0 D|) \mathbf{1} - i \frac{\sin(t\kappa |M_0 D|)}{\kappa |M_0 D|} H_0 = \partial_t W_t \mathbf{1} - i H_0 W_t.$$

The inverse Fourier transform of W_t is the standard wave sine kernel after the invertible linear change $y = M_0^{-1}x$. Its distributional support is the closed ball for $d_{C1} = 1$ and for even d_{C1} , and the sphere for odd $d_{C1} \geq 3$. Constant-coefficient differentiation cannot enlarge support. It also does not remove an open component of the interior support in the even case: taking the matrix trace eliminates the Clifford-odd term $H_0 W_t$, so

$$\text{tr } \mathcal{K}_t = (\dim V) \partial_t W_t.$$

The standard wave kernel has distributional support equal to the same closed ball, and $\partial_t W_t$ is nonzero on every nonempty open subset of its interior. In odd dimension both terms are supported on the characteristic sphere, and the nonzero wave front amplitude gives the full sphere as support. The last assertion is precisely the definition of distributional support. $\square$

Theorem F.8 (Constant-coefficient inner characterization). *Let $U, V \subset \mathbb{R}^{d_{C1}}$ be nonempty open sets and $t > 0$. If*

$$(U - V) \cap \text{supp } \mathcal{K}_t \neq \emptyset,$$

then

$$\mathbf{1}_U e^{-itH_0} \mathbf{1}_V \neq 0,$$

and the corresponding propagation transfers to the renewal approximants by [Theorem F.5\(ii\)](#). Consequently:

(i) *for $d_{C1} = 1$ and even d_{C1} , every open pair satisfying*

$$\inf_{u \in U, v \in V} |M_0^{-1}(u - v)| < \kappa t$$

has nonzero forward transfer;

(ii) *for odd $d_{C1} \geq 3$, forward transfer at the fixed time t occurs whenever the difference set meets the Huygens shell $|M_0^{-1}(u - v)| = \kappa t$;*

(iii) *the convex hull of $\text{supp } \mathcal{K}_t$ is the full causal ellipsoid $\kappa t M_0 B^{d_{C1}}$, so the characteristic cone is attained in every direction.*

Proof. Choose $(u_*, v_*) \in U \times V$ with $u_* - v_* \in \text{supp } \mathcal{K}_t$. By the definition of distributional support, there is a matrix-valued test function ϕ , supported in a sufficiently small neighbourhood of $u_* - v_*$, with $\langle \mathcal{K}_t, \phi \rangle \neq 0$. Shrinking the neighbourhood, write ϕ as a finite sum of convolutions $\sum_j f_j * \tilde{g}_j$ with $f_j \in C_c^\infty(U, V)$ and $g_j \in C_c^\infty(V, V)$, where $\tilde{g}(x) = g(-x)^*$. Then for some j ,

$$\langle f_j, e^{-itH_0} g_j \rangle \neq 0,$$

which proves the operator nonvanishing. The parity statements follow from [Theorem F.7](#). The final convex-hull statement is immediate: the closed ball is already convex, while the convex hull of the characteristic sphere is the ball. $\square$

Remark F.9 (Variable coefficients). For general $M(x)$, Green hyperbolicity yields only $\text{supp } E^+ \subseteq J^+$; it does not imply equality of supports or local kernel nonvanishing at every causally related pair. Conjugate points, polarization transport, and subprincipal terms may create zeros or tails, and Huygens behaviour is exceptional. Accordingly the general inner statement is the exact transfer theorem [Theorem F.5\(ii\)](#), not universal filling of J^+ .

F.2.2 Ballistic reachability versus dispersive causality

For a constant finite reset frame, put

$$\Theta = \text{conv}\{\hat{\theta}_a : p_a > 0\}.$$

If one allows free concatenation of reset labels and an $O(h)$ endpoint tolerance, the reachable set after N steps is $Nh\Theta + O(h)$. Hence the ballistic travel time converges to the Minkowski gauge

$$d_{\text{bal}}(p, q) = \gamma_{\Theta}(q - p).$$

This depends only on the support of the reset law, whereas the Dirac cone depends on its second moment M .

Proposition F.10 (Drift hull versus dispersion ellipsoid). *For constant reset data and free concatenation, the approximate ballistic travel times converge locally uniformly to the Finsler distance with indicatrix Θ . They agree with the Lorentzian optical distance exactly when*

$$\Theta = \kappa M B^{d_{\text{Cl}}}.$$

For a refining equidistributed family, $\Theta \rightarrow B^{d_{\text{Cl}}}$ and $M \rightarrow d_{\text{Cl}}^{-1}I$; agreement therefore fixes the clock normalization to $\kappa = d_{\text{Cl}}$. For varying direction fields the analogous Finsler convergence requires, in addition, Hausdorff continuity of $x \mapsto \Theta(x)$, a uniform interior-ball condition, and a separate compactness/recovery-sequence argument; it is not used elsewhere in this paper.

Proof. For constant data, the N -step displacement set is the Minkowski sum of N copies of the finite step set. Its convexification is $Nh\Theta$, and the $O(h)$ endpoint tolerance removes the arithmetic lattice obstruction. The minimal Nh therefore converges to the gauge $\gamma_{\Theta}(q - p)$.

The optical distance is the length distance whose infinitesimal unit ball is $\kappa M B^{d_{\text{Cl}}}$. Equality of the two length distances near the diagonal forces equality of their metric derivatives and hence equality of the indicatrices. Conversely equal indicatrices give equal length functionals. The isotropic normalization follows from $M = \int \theta \theta^{\top} d\nu = d_{\text{Cl}}^{-1}I$. $\square$

Theorem F.11 (Joint propagation-geometry convergence). *Under the curved-limit hypotheses, the frozen symbols, ellipsoidal cone fields, optical distances, macroscopic outer domains of dependence, and self-adjoint Hamiltonians converge simultaneously to the data of the single ultrastatic metric*

$$g_{\mu\nu}(x) = \text{diag}(1, -\kappa^{-2}M(x)^{-2}).$$

Inner propagation channels are inherited wherever the limiting propagator is locally nonzero; for constant M_0 this nonvanishing is characterized exactly by [Theorems F.7](#) and [F.8](#). Ballistic reset-ray reachability is a distinct drift-hull geometry and coincides with this Lorentzian propagation geometry only under the indicatrix equality of [Theorem F.10](#).

Proof. Combine [Theorems 5.49](#), [F.2](#) to [F.5](#), [F.8](#) and [F.10](#). $\square$

F.3 Universality and metric reconstruction

We finish the continuum section by recording what the limiting construction forgets and what it remembers. It forgets much of the microscopic renewal presentation: after scaling, only the second-moment field of the reset directions enters the principal symbol. This is the universality statement. But if the second-moment field is included as part of the scaling data, then the signed spectral package reconstructs the limiting Lorentzian principal symbol up to the unavoidable gauge and spatial-frame choices. This is the reconstruction statement.

Definition F.12 (Enhanced signed spectral package). An *enhanced signed renewal spectral package* consists of the signed predictive package

$$(\mathcal{B}_{\text{CP},0}, \mathcal{H}, D_{\text{CP}}, J_\chi, D_-, \mathcal{A}_{\text{rev}})$$

together with the reset-direction scaling datum

$$\Theta : x \mapsto \nu_x, \quad M_\Theta(x) = \int_{\mathbb{S}^{d_{\text{Cl}}-1}} \hat{\theta} \hat{\theta}^\top d\nu_x(\hat{\theta}),$$

up to the evident flat gauge on the signed cover and orthogonal changes of the spatial Clifford frame. The adjective *enhanced* is important: the bare package reconstructs the ordered signed predictive monoid and the holonomy class, while the continuum metric also uses the scaling statistics encoded by Θ .

Lemma F.13 (Continuum factorisation through the second moment). *The continuum Hamiltonian $H_g[M]$ of Section 5.4.2 depends on the microscopic renewal data only through the normalisation κ , the fixed Krein–Clifford datum, and the limiting second-moment field $M(x)$ together with its first derivatives. In particular, two reset fields with the same $M(x)$ and the same clock normalisation give the same continuum operator.*

Proof. By Section 5.4.2,

$$H_g[M] = \kappa A^i M^{ij}(x)(-i\partial_j) - \frac{i\kappa}{2} A^i(\partial_j M^{ij})(x).$$

The direction measure ν_x appears only through $M(x) = \int \hat{\theta} \hat{\theta}^\top d\nu_x$; no higher moment or microscopic presentation of the reset alphabet enters the limiting operator. $\square$

Theorem F.14 (Universality of Lorentzian Dirac continua). *Let two primitive signed renewal memories satisfy the adiabatic hypotheses of Theorem 5.49. Suppose that, after the standard continuum scaling, they have the same clock normalisation κ , the same fixed Krein–Clifford datum, and the same uniformly elliptic limiting second-moment field*

$$M(x) = M'(x) \succeq m_0 I.$$

Then their rescaled discrete renewal Hamiltonians converge in strong resolvent sense to the same variable-coefficient Lorentzian Dirac Hamiltonian $H_g[M]$. Consequently

$$e^{-itH_R} \longrightarrow e^{-itH_g[M]} \longleftarrow e^{-itH'_R}$$

strongly, uniformly for t in compact intervals. Thus the Lorentzian continuum endpoint depends only on the renormalised second-moment field, not on the microscopic renewal presentation.

Proof. Apply Theorem 5.49 to each family. The first family converges to $H_g[M]$, and the second to $H_g[M']$. By Theorem F.13, the two limits coincide when $M = M'$, with common κ and common Clifford datum. The propagator statement follows from strong resolvent convergence by the Trotter–Kato theorem. $\square$

Theorem F.15 (Inverse theorem for the Lorentzian principal symbol). *Let two enhanced signed renewal spectral packages have adiabatic scaling limits with uniformly elliptic second-moment fields M and M' . Suppose a unitary intertwines the diagonal predictive algebra, descended shifts, length operator, fundamental symmetry, modular revision algebra, and reset-direction scaling datum.*

Then the unitary induces a graph isomorphism $\varphi : G \rightarrow G'$ and a spatial frame change $O(x) \in O(d_{\text{Cl}})$ such that

$$M'(\varphi(x)) = O(x)M(x)O(x)^\top,$$

and hence the limiting Lorentzian inverse metrics are isometric:

$$g'(\varphi(x)) = (1 \oplus O(x))g(x)(1 \oplus O(x))^\top.$$

Thus the Lorentzian principal symbol is an invariant of the enhanced signed renewal spectral package, up to flat gauge and spatial-frame equivalence.

Proof. The signed reconstruction theorem [Theorem 3.43](#) gives an isomorphism φ of graded ordered predictive monoids and identifies the sign class. Intertwining the reset-direction datum means that the direction measure at x is carried to the direction measure at $\varphi(x)$ up to the orthogonal relabelling $O(x)$ of the spatial Clifford frame. Taking second moments gives

$$M'(\varphi(x)) = \int \hat{\theta} \hat{\theta}^\top d\nu'_{\varphi(x)} = O(x) \left(\int \hat{\theta} \hat{\theta}^\top d\nu_x \right) O(x)^\top.$$

The metric formula follows from $g = \text{diag}(1, -\kappa^2 M^2)$ and the common clock normalisation. $\square$

Corollary F.16 (Lorentzian principal-symbol reconstruction). *The enhanced signed renewal spectral package of [Theorem F.12](#) determines the limiting Lorentzian inverse metric*

$$g^{\mu\nu}(x) = \text{diag}(1, -\kappa^2 M_\Theta(x)^2)$$

as an invariant, up to signed-cover flat gauge and orthogonal spatial Clifford-frame equivalence $O(x) \in O(d_{\text{Cl}})$. That is, the assignment

$$(\mathcal{B}_{\text{CP},0}, \mathcal{H}, D_{\text{CP}}, J_\chi, D_-, \mathcal{A}_{\text{rev}}, \Theta) \longmapsto [g^{\mu\nu}(x)]$$

is well defined on unitary-equivalence classes of enhanced packages with the common clock normalisation κ and Krein-Clifford datum, and is complete for the principal symbol: two enhanced packages have isometric Lorentzian principal symbols if and only if they are intertwined as in [Theorem F.15](#).

Proof. By [Theorem F.15](#) an intertwining unitary yields $\varphi : G \rightarrow G'$ and $O(x) \in O(d_{\text{Cl}})$ with $M'_\Theta(\varphi(x)) = O(x)M_\Theta(x)O(x)^\top$. Substituting into $g^{\mu\nu} = \text{diag}(1, -\kappa^2 M_\Theta^2)$ and using $O(x)M_\Theta^2 O(x)^\top = (OM_\Theta O^\top)^2$ gives $g'^{\mu\nu}(\varphi(x)) = (1 \oplus O(x))g^{\mu\nu}(x)(1 \oplus O(x))^\top$, an isometry of inverse metrics fixing the time slot. Conversely, equal M_Θ up to $O(d_{\text{Cl}})$ frame and equal κ give equal $g^{\mu\nu}$ up to the same frame, and the package data determine M_Θ via its definition in [Theorem F.12](#). The timelike slot is frame-independent because it is the image of the Krein-odd generator γ^0 ([Theorem 5.5](#)). $\square$

Remark F.17 (Why the enhancement is necessary). The unenhanced signed package reconstructs the discrete predictive monoid and its $\mathbb{Z}/2$ -holonomy class. It does not by itself determine the continuum second-moment field, since different microscopic direction measures can have the same signed quotient but different scaling limits. The reconstruction theorem therefore uses the enhanced package of [Theorem F.12](#); with this datum included, the Lorentzian principal symbol is recovered exactly up to the unavoidable frame and gauge choices.

Remark F.18 (Summary of the continuum passage). The logical path of this section is therefore layered. Product renewal order gives a measured polyhedral continuum; signed Clifford isotropisation rounds the principal symbol to a Lorentzian cone; the Hamiltonian formulation gives the operator-level limit; slow spatial variation gives the curved adiabatic extension; and the enhanced package records exactly the second-moment data needed to reconstruct the limiting metric. None of these steps adds field equations or matter couplings. They identify the Lorentzian Dirac continuum which the renewal geometry supplies kinematically.

F.4 A marked flat-torus correspondence carrying spin structure

The local continuum results above identify the Lorentzian principal symbol but, on a contractible chart, cannot retain the cohomology class of the signed cover. A genuine two-sided statement therefore requires a global topology on which the sign class survives. We give such a statement on a marked flat torus. The result is deliberately band-limited: naive symmetric differences have high-momentum lattice doublers, so no global norm-resolvent or global kernel statement is asserted.

Let

$$\mathbb{T}^{d_{C1}} = \mathbb{R}^{d_{C1}} / \mathbb{Z}^{d_{C1}}$$

carry its standard marking and flat parallelisation, and fix a reference spin structure $\mathfrak{s}_0$. Relative to $\mathfrak{s}_0$, the marked spin structures are indexed by

$$\rho \in H^1(\mathbb{T}^{d_{C1}}, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^{d_{C1}}, \quad \mathfrak{s}_\rho = \mathfrak{s}_0 \otimes L_\rho,$$

where the flat real line bundle L_ρ has holonomy $(-1)^{\rho_k}$ around the k -th marked generator. We realise its spinors as equivariant functions on the universal cover,

$$L_\rho^2(\mathbb{T}^{d_{C1}}, V) = \left\{ \psi \in L_{\text{loc}}^2(\mathbb{R}^{d_{C1}}, V) : \psi(x + e_k) = (-1)^{\rho_k} \psi(x) \right\}.$$

Their physical Fourier momenta are

$$\xi \in 2\pi(\mathbb{Z} + \rho/2)^{d_{C1}}.$$

For constant $M \succ 0$, $\text{tr } M = 1$, define

$$\mathbf{H}_g^\rho[M] = \kappa A^i M^{ij} (-i\partial_j) \quad \text{on } L_\rho^2(\mathbb{T}^{d_{C1}}, V).$$

The marked deck transformations

$$(\tilde{T}_k \psi)(x) = \psi(x + e_k)$$

act as the scalars

$$\tilde{T}_k = (-1)^{\rho_k} \text{id}.$$

Hence the full marked class ρ is retained by the deck representation. The bare spectrum always distinguishes the trivial class through its zero modes,

$$0 \in \text{spec } \mathbf{H}_g^\rho[M] \iff \rho = 0,$$

but need not distinguish all nonzero classes; the complete class belongs to the operator together with its marked deck holonomies.

Lemma F.19 (Exact integer-stencil decomposition). *Every positive definite $M \in \text{Sym}_{d_{\text{Cl}}}(\mathbb{R})$ with $\text{tr } M = 1$ admits a finite decomposition*

$$M = \sum_{a=1}^m p_a \hat{\theta}_a \hat{\theta}_a^\top, \quad \hat{\theta}_a = \frac{v_a}{|v_a|}, \quad v_a \in \mathbb{Z}^{d_{\text{Cl}}} \setminus \{0\}, \quad p_a > 0, \quad \sum_a p_a = 1.$$

Proof. The rank-one projections $uu^\top$, $u \in \mathbb{S}^{d_{\text{Cl}}-1}$, are the extreme points of the compact convex set

$$\mathcal{Q}_{d_{\text{Cl}}} = \{S \in \text{Sym}_{d_{\text{Cl}}}(\mathbb{R}) : S \succeq 0, \text{tr } S = 1\}.$$

Because $M \succ 0$, it lies in the relative interior of $\mathcal{Q}_{d_{\text{Cl}}}$. Choose finitely many rank-one projections whose convex hull contains M in its relative interior. The set of directions $v/|v|$, with $v \in \mathbb{Z}^{d_{\text{Cl}}} \setminus \{0\}$, is dense in the sphere, so the chosen vertices may be perturbed to integer-direction projections. Since M has positive distance from every facet of the original finite polytope, sufficiently small vertex perturbations preserve the strict containment of M . The resulting positive barycentric coordinates give the stated decomposition. $\square$

For $h_R = N_R^{-1}$, define the integer-stencil symmetric difference

$$(\delta_{v_a, R} u)(x) = \frac{u(x + h_R v_a) - u(x - h_R v_a)}{2h_R |v_a|}.$$

The twist is imposed intrinsically through the boundary condition

$$u(n + N_R e_k) = (-1)^{\rho_k} u(n), \quad n \in \mathbb{Z}^{d_{\text{Cl}}},$$

equivalently by a flat cellular $\mathbb{Z}/2$ -cocycle on the cubical torus whose plaquette holonomies vanish and whose marked winding holonomies are $(-1)^{\rho_k}$. This convention also fixes the sign of a long stencil shift that crosses one or more fundamental boundaries.

With an exact decomposition from [Theorem F.19](#), set

$$H_R^\rho = \kappa \sum_a p_a A(\hat{\theta}_a) (-i\delta_{v_a, R}^\rho).$$

Let

$$V_* = \max_a |v_a|, \quad \mathcal{B}_\Lambda(\rho) = \left\{ \xi \in 2\pi(\mathbb{Z} + \rho/2)^{d_{\text{Cl}}} : |\xi| \leq \Lambda \right\},$$

and let P_Λ be the Fourier projection onto this finite set. The smallest twisted momentum is

$$\min_{\xi \in 2\pi(\mathbb{Z} + \rho/2)^{d_{\text{Cl}}}} |\xi| = \pi \sqrt{|\rho|_0},$$

where $|\rho|_0$ is the number of nonzero components of ρ .

Theorem F.20 (Band-limited twisted convergence). *Fix $\Lambda > 0$. For all sufficiently large R , the discrete and continuum physical bands are canonically identified by twisted Fourier modes, and*

$$\|P_\Lambda (J_R H_R^\rho J_R^* - H_g^\rho[M]) P_\Lambda\| \leq \frac{\kappa}{6} V_*^2 h_R^2 \Lambda^3 \longrightarrow 0.$$

Consequently the resolvents converge in norm on the fixed band, and the band eigenvalues converge with multiplicities. If

$$\delta_\Lambda(\rho) := \kappa \min_{\substack{\xi \in \mathcal{B}_\Lambda(\rho) \\ \xi \neq 0}} |M\xi|, \quad \min \emptyset := +\infty,$$

then, whenever

$$\frac{\kappa}{6} V_*^2 h_R^2 \Lambda^3 < \delta_\Lambda(\rho),$$

the band contains no additional zero modes beyond the genuine continuum zero mode. In particular, for fixed $0 < \varepsilon < \delta_\Lambda(\rho)$ and large R ,

$$\dim \mathbf{1}_{[-\varepsilon, \varepsilon]}(P_\Lambda H_R^\rho P_\Lambda) = \dim V \mathbf{1}[\rho = 0].$$

Proof. On a physical mode ξ , the multiplier of the a -th stencil is

$$s_{v_a, R}(\xi) = \frac{\sin(h_R v_a \cdot \xi)}{h_R |v_a|}.$$

Using $|\sin t - t| \leq |t|^3/6$,

$$|s_{v_a, R}(\xi) - \hat{\theta}_a \cdot \xi| \leq \frac{h_R^2 |v_a|^2 |\xi|^3}{6}.$$

Since $\|A(\hat{\theta}_a)\| = 1$ and $\sum_a p_a = 1$, taking the supremum over $\mathcal{B}_\Lambda(\rho)$ gives the displayed operator-norm estimate. The second resolvent identity gives norm-resolvent convergence on the finite band, and finite-dimensional spectral stability gives convergence with multiplicities.

The condition $\pi N_R/V_* > \Lambda$ excludes the first nonzero sine zeros of every individual stencil from the physical band. This alone does not exclude a cancellation in the total matrix symbol. The latter is ruled out by the uniform estimate: away from the genuine zero mode the continuum symbol has spectral gap $\delta_\Lambda(\rho)$, so any perturbation of norm strictly smaller than that gap remains invertible. Hence no additional band zero can occur for sufficiently large R . $\square$

Definition F.21 (Strictly marked Dirac datum). A strictly marked Dirac datum is

$$\mathfrak{D} = (\pi, \{\tilde{T}_k\}, c, \mathbf{H}),$$

consisting of the multiplication representation, the marked deck holonomies, the Clifford representation, and the Dirac Hamiltonian. A strict equivalence intertwines all four structures and fixes the marking. A marking-preserving geometric equivalence may additionally act by

$$O \in O(d_{\text{Cl}}, \mathbb{Z}) := O(d_{\text{Cl}}) \cap GL_{d_{\text{Cl}}}(\mathbb{Z}),$$

the signed-permutation isometry group of the standard marked torus.

Theorem F.22 (Classification of marked flat-torus data). *For constant $M, M' \succ 0$, $\text{tr } M = \text{tr } M' = 1$, and $\rho, \rho' \in (\mathbb{Z}/2)^{d_{\text{Cl}}}$:*

1. *the strictly marked data of $H_g^\rho[M]$ and $H_g^{\rho'}[M']$ are equivalent if and only if*

$$M' = M, \quad \rho' = \rho;$$

2. *they are equivalent through a marking-preserving torus isometry $O \in O(d_{\text{Cl}}, \mathbb{Z})$ if and only if*

$$M' = OMO^\top, \quad \rho' \equiv O\rho \pmod{2}.$$

No transformation law for the symmetric-gauge matrix M under a general $GL_{d_{\text{Cl}}}(\mathbb{Z})$ mapping class is asserted: the geometric co-metric transforms tensorially, whereas its symmetric square root does not, and the fixed normalisation $\text{tr } h^{1/2} = \kappa$ is not preserved by a general nonorthogonal mapping class.

Proof. A strict intertwiner of the multiplication representation is a matrix-valued multiplier $U(x)$. Intertwining the irreducible constant Clifford representation forces $U(x) = e^{i\alpha(x)}\mathbf{1}$. Conjugation of the Dirac operator then adds the multiplication term

$$\kappa A^i M^{ij} \partial_j \alpha.$$

Equality with another pure constant-coefficient operator forces $d\alpha = 0$, hence U is constant and equality of principal symbols gives $M' = M$. Intertwining the deck scalars gives $\rho' = \rho$. For a marking-preserving isometry, the simultaneous coordinate and Clifford rotation is restricted to $O(d_{\text{Cl}}, \mathbb{Z})$, yielding the displayed transformations. $\square$

Theorem F.23 (Marked torus realization and retention). *For every constant trace-normalised flat co-metric*

$$h \succ 0, \quad \text{tr } h^{1/2} = \kappa,$$

and every marked spin structure $\rho \in (\mathbb{Z}/2)^{d_{\text{Cl}}}$, there is an integer-stencil signed renewal torus whose winding class is ρ and whose Hamiltonians converge in norm resolvent on every fixed physical momentum band to

$$H_\rho^\rho[M], \quad M = \kappa^{-1} h^{1/2}.$$

The continuum strictly marked datum retains

$$h = \kappa^2 M^2$$

through its principal symbol and retains the full spin class ρ through its deck holonomies. The bare spectrum always detects the trivial class by its harmonic spinors but does not determine the full nonzero class in general. Thus

$$(M, [\chi]) \longmapsto (h = \kappa^2 M^2, \mathfrak{s}_0 \otimes L_{[\chi]})$$

is a bijection of strictly marked enriched moduli, while the microscopic integer stencil realising M remains non-unique.

Proof. The integer-stencil decomposition is [Theorem F.19](#); the prescribed spin class is imposed by the twisted periodicity, whose winding signs equal the deck holonomies; the band-limited convergence is [Theorem F.20](#); metric and holonomy retention are the principal-symbol and deck-representation statements above; and the strict classification is [Theorem F.22](#). Non-uniqueness of the microscopic frame is the continuum factorisation of [Theorem F.13](#). $\square$

Remark F.24 (Scope). The theorem is global but flat. A periodic variable field $M(x)$ would combine the local adiabatic limit with twisted boundary conditions, while a general closed spin manifold would require a separate graph-Dirac-to-manifold-Dirac convergence theorem. Neither extension is used here.

G Examples and sharp model cases

This appendix collects elementary models realising the main mechanisms of the paper. The examples are organised by role. The first group illustrates the positive predictive geometry; the second isolates signed holonomy and its separation from spectral deficiency; the third separates finite polyhedral order cones from Lorentzian Clifford symbols; the last group records the primitive modular patterns behind the $3 + 1$ dimension statement.

Example type	Purpose	Reference
Positive predictive examples	Separate algebraic distinguishability, cb-metric dimension, and order-volume growth.	Section G.1
Signed holonomy examples	Show when non-removable Krein sectors exist, and why holonomy is not the same as spectral-radius deficiency.	Section G.2
Continuum examples	Separate finite polyhedral order cones from Lorentzian Clifford symbols, and show how isotropy is only a special representative of the flat endpoint.	Section G.3
Primitive modular examples	Realise the 3+1 primitive shell and the higher-rank obstruction to unconditional uniqueness.	Section G.4

G.1 Positive predictive geometry

Example G.1 (Separation of algebraic and metric predictive dimension). Let the reset channels act, in a finite affine chart, by similarities of contraction ratios $r_1, \dots, r_m$ satisfying the open set condition. If the products remain exactly distinguishable as words, then the algebraic predictive quotient has exponential word growth, so the additive degree triple is not finitely summable. Nevertheless the cb-metric closure is a self-similar attractor, and the metric predictive dimension is the unique solution s of

$$\sum_{i=1}^m r_i^s = 1.$$

Thus q_{alg} and q_{met} measure genuinely different features: exact predictive distinguishability and metric resolution of the channel attractor.

Example G.2 (Free commuting $\mathbb{N}^c$: dimension and order volume). For $\mathcal{M} = \mathbb{N}^c$, the channels are indexed by exponent vectors. The number of classes of rank at most R satisfies

$$N_{\text{CP}}(R) \asymp R^c,$$

so $q_{\text{alg}} = c$ by [Theorem 2.14](#). The predictive order is the product order, and balanced intervals have volume growth

$$|[x, y]| \sim \tau^c$$

by [Theorem C.6](#). This is the basic manifold-like regime: algebraic dimension, order-growth dimension, and interval-volume exponent all equal c .

G.2 Signed covers and holonomy

Example G.3 (Tree graph: no non-removable signed sector). If the recurrent predictive graph G is a tree up to orientation, then

$$b_1(G) = 0, \quad H^1(G, \mathbb{Z}/2) = 0.$$

By [Theorems 3.7](#) and [3.8](#), there is one gauge class of compatible signed covers and no non-removable Krein sector. A tree therefore carries no intrinsic signed holonomy datum; a cycle is needed.

Example G.4 (One-cycle graph: one non-removable Krein class). If $b_1(G) = 1$, then

$$H^1(G, \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Hence there are 2 signed gauge classes and exactly one non-removable class. The nontrivial class is the loop-holonomy cocycle. This is the minimal renewal carrier of a signed/Krein sector.

Example G.5 (Half-period sign: nontrivial holonomy without spectral deficiency). On the two-vertex cycle, take a sign cocycle χ with holonomy -1 around the loop. The unsigned adjacency matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

has spectral radius $\rho(A) = 1$, while the signed transfer operator

$$A_\chi = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

has eigenvalues $\pm i$, hence $\rho(A_\chi) = 1$. Thus

$$[\chi] \neq 0 \in H^1(G, \mathbb{Z}/2), \quad r_\chi := \frac{\rho(A_\chi)}{\rho(A)} = 1.$$

The class is topologically nontrivial but spectrally silent. This illustrates why the signed sector is classified by holonomy, not by spectral-radius deficiency.

Example G.6 (The smallest nontrivial amplitude lift). Let G be a single cycle. Then

$$H^1(G, \mathbb{Z}/2) \cong \mathbb{Z}/2, \quad H^1(G, \mathbb{Z}/4) \cong \mathbb{Z}/4.$$

The nontrivial signed cover sends the generator to -1 . Its amplitude lifts send the generator to i or $-i$, and both square to -1 . Thus the two amplitude lifts have the same real/Krein shadow but distinct fourth-root phase data. This is the local model for the general distinction between the signed cover and its amplitude completion.

G.3 Polyhedral order cones and Lorentzian Clifford symbols

Example G.7 (Finite $c \geq 3$ product cone: the polyhedral obstruction). For $\mathcal{M} = \mathbb{N}^c$ with $c \geq 3$, the product-order cone

$$[0, \infty)^c$$

has exactly c extreme rays, the coordinate axes. A round Lorentz cone

$$L_c = \{x_0 \geq |(x_1, \dots, x_{c-1})|\}$$

has the continuum of null generators over S^{c-2} . Linear isomorphisms preserve extreme rays. Hence no deterministic finite c -generator product order can be linearly equivalent to a smooth Lorentz cone in dimension $c \geq 3$. This is the obstruction of [Theorem D.6](#): finite deterministic order cones remain polyhedral, even when a separate signed Clifford symbol has a Lorentzian continuum limit.

Example G.8 (Finite tight frames round the symbol in a chosen frame). A finite isotropic tight frame can round the Dirac *symbol* exactly without rounding the order cone. In spatial rank $d_{\text{Cl}} = 3$, take the six octahedral directions

$$\{\pm e_1, \pm e_2, \pm e_3\} \subset S^2$$

with equal weights. Then

$$M(\nu) = \frac{1}{3}I_3,$$

and with $\kappa = 3$ the signed Clifford symbol has the exact Minkowski characteristic cone

$$\xi_0^2 = \xi_1^2 + \xi_2^2 + \xi_3^2.$$

However, the order cone generated by the six null rays $(1, \pm e_i)$ has octahedral cross-section, hence remains polyhedral. Thus finite isotropic frames round the spectral cone in the chosen renewal frame, while order-cone rounding requires an equidistributing limit. This is a special isotropic representative of the general nondegenerate symbol theorem: non-isotropic positive-definite second moments give linearly isometric flat endpoints after choosing the corresponding vielbein.

Example G.9 (Non-isotropic second moment and flat isometry). Let the signed reset directions in spatial rank 3 have positive-definite second moment

$$M_* = \text{diag}(a_1, a_2, a_3), \quad a_i > 0,$$

with not all a_i equal. In the original renewal frame the characteristic quadric is anisotropic:

$$\xi_0^2 = \kappa^2(a_1^2\xi_1^2 + a_2^2\xi_2^2 + a_3^2\xi_3^2).$$

Equivalently, with the fixed spatial vielbein

$$V = \kappa M_*,$$

the spatial covector $\eta = V\xi$ brings the symbol to the standard flat form

$$\xi_0^2 = |\eta|^2.$$

Thus isotropy of the reset law is not required for the flat Lorentzian Dirac isometry class. Isotropy merely makes the standard Euclidean frame coincide with the original renewal frame.

Example G.10 (Equidistributed directions: order-cone rounding). Let $\nu_R \rightarrow \nu_{\text{unif}}$ weak-* on $\mathbb{S}^{d_{\text{Cl}}-1}$, with directions becoming dense. Then

$$M(\nu_R) \longrightarrow \frac{1}{d_{\text{Cl}}}I_{d_{\text{Cl}}},$$

and with $\kappa = d_{\text{Cl}}$, the spectral cones converge to the round Lorentz cone in the original renewal frame. Moreover, the conical hulls of the null rays $(1, \hat{\theta}_a^{(R)})$ converge in Hausdorff distance on rays to the solid future cone

$$\{x_0 \geq |\vec{x}|\}.$$

This realises order-cone rounding. It is stronger than what is needed for the flat Dirac endpoint, where nondegenerate second moments already determine a Lorentzian symbol up to linear isometry.

G.4 Primitive modular dimension patterns

Example G.11 (Primitive $3 + 1$ modular revision pattern). Let

$$H_{\text{full}} = H^1(G, \mathbb{Z}/2) \oplus \langle t \rangle \cong (\mathbb{Z}/2)^4$$

with basis (t, e_1, e_2, e_3) . Define the modular commutator form by

$$\Omega = J_4 - I_4 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_2.$$

Since 4 is even, this matrix is nonsingular over $\mathbb{F}_2$. Hence

$$\text{rank}_{\mathbb{F}_2} \Omega = 4,$$

and the modular revision algebra is a full matrix factor, equivalently the even complex Clifford algebra

$$\mathcal{A}_{\text{rev}} \cong \text{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C}).$$

The spatial 3×3 block has a radical generated by $e_1 + e_2 + e_3$, and the temporal generator t pairs with this class to make the full 4×4 form nondegenerate. This is the minimal primitive pattern realising the $3 + 1$ endpoint.

Example G.12 (Primitive $d_{\text{Cl}} = 5$ shell: no unconditional uniqueness). Let

$$H_{\text{full}} \cong (\mathbb{Z}/2)^6$$

and take the same off-diagonal form

$$\Omega = J_6 - I_6.$$

Again 6 is even, so Ω is nonsingular over $\mathbb{F}_2$. Thus

$$\text{rank}_{\mathbb{F}_2} \Omega = 6,$$

and the corresponding modular revision algebra is the full matrix factor

$$\mathcal{A}_{\text{rev}} \cong \text{Cl}_6(\mathbb{C}) \cong M_8(\mathbb{C}).$$

This gives a primitive revision datum with spatial Clifford rank $d_{\text{Cl}} = 5$. Therefore primitivity alone does not exclude the higher odd ranks $d_{\text{Cl}} = 5, 7, \dots$. The selection of $d_{\text{Cl}} = 3$ is consequently a minimality statement unconditionally, and a uniqueness statement only after imposing the stated oriented-circulation or volume-response closure; access and efficiency principles are optional alternative comparisons.

Remark G.13 (Why these two primitive examples are useful). The $3 + 1$ example realises the endpoint selected by the minimality theorem. The $d_{\text{Cl}} = 5$ example prevents overclaiming: it shows concretely why primitive modular nondegeneracy cannot by itself prove uniqueness of $3 + 1$. Together they make the status of the dimension theorem transparent.

References

- [1] A. Connes. *Noncommutative Geometry*. Academic Press, 1994.
- [2] A. Connes. Compact metric spaces, Fredholm modules, and hyperfiniteness. *Ergodic Theory and Dynamical Systems*, 9(2):207–220, 1989.
- [3] E. Christensen and C. Ivan. Spectral triples for AF C^* -algebras and metrics on the Cantor set. *Journal of Operator Theory*, 56(1):17–46, 2006.
- [4] J. Pearson and J. Bellissard. Noncommutative Riemannian geometry and diffusion on ultrametric Cantor sets. *Journal of Noncommutative Geometry*, 3(3):447–480, 2009.
- [5] J. Kellendonk and J. Savinien. Spectral triples and characterization of aperiodic order. *Proceedings of the London Mathematical Society*, 104(1):123–157, 2012.
- [6] J. Bognár. *Indefinite Inner Product Spaces*. Springer, 1974.
- [7] A. Strohmaier. On noncommutative and pseudo-riemannian geometry. *Journal of Geometry and Physics*, 56:175–195, 2006.
- [8] Mario Paschke and Andrzej Sitarz. Equivariant lorentzian spectral triples. *arXiv e-prints*, 2006.
- [9] K. van den Dungen, M. Paschke, and A. Rennie. Pseudo-riemannian spectral triples and the harmonic oscillator. *Journal of Geometry and Physics*, 73:37–55, 2013.
- [10] N. Franco. Temporal lorentzian spectral triples. *Reviews in Mathematical Physics*, 26:1430007, 2014.
- [11] N. Franco and M. Eckstein. An algebraic formulation of causality for noncommutative geometry. *Classical and Quantum Gravity*, 30:135007, 2013.
- [12] K. van den Dungen. Krein spectral triples and the fermionic action. *Mathematical Physics, Analysis and Geometry*, 19(1):4, 2016.
- [13] N. Bizi, C. Brouder, and F. Besnard. Space and time dimensions of algebras with application to lorentzian noncommutative geometry and quantum electrodynamics. *Journal of Mathematical Physics*, 59(6):062303, 2018.
- [14] A. Connes and H. Moscovici. Type III and spectral triples. In *Traces in Number Theory, Geometry and Quantum Fields*, volume E38 of *Aspects of Mathematics*, pages 57–71. Vieweg, 2008.
- [15] Nicolas Franco. Temporal lorentzian spectral triples. *Reviews in Mathematical Physics*, 26(8):1430007, 2014.
- [16] Nicolas Franco and Michał Eckstein. Noncommutative geometry, lorentzian structures and causality. In Michał Eckstein, Michał Heller, and Sebastian J. Szybka, editors, *Mathematical Structures of the Universe*, pages 315–340. Copernicus Center Press, 2014.
- [17] Fabien Besnard. On the definition of spacetimes in noncommutative geometry, part ii. *arXiv e-prints*, 2016.

- [18] Agostino Devastato, Shane Farnsworth, Fedele Lizzi, and Pierre Martinetti. Lorentz signature and twisted spectral triples. *Journal of High Energy Physics*, 2018(3):089, 2018.
- [19] Agostino Devastato, Manuele Filaci, Pierre Martinetti, and Devashish Singh. Actions for twisted spectral triple and the transition from the euclidean to the lorentzian. *International Journal of Geometric Methods in Modern Physics*, 17(supp01):2030001, 2020.
- [20] G. Nieuviarts. Emergence of time from a twisted spectral triple in almost-commutative geometry. *Journal of Noncommutative Geometry*, 2026. To appear; arXiv:2512.15450.
- [21] G. Nieuviarts. Emergence of lorentz symmetry from an almost-commutative twisted spectral triple. *International Journal of Geometric Methods in Modern Physics*, 2026. To appear; arXiv:2502.18105.
- [22] W. Feller. *An Introduction to Probability Theory and its Applications, Vol. II*. Wiley, 1971.
- [23] C. R. Shalizi and J. P. Crutchfield. Computational mechanics: pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001.
- [24] Nicholas F. Travers and James P. Crutchfield. Equivalence of history and generator ϵ -machines. *Physical Review E*, 89:062107, 2014.
- [25] Sarah E. Marzen and James P. Crutchfield. Structure and randomness of continuous-time, discrete-event processes. *Journal of Statistical Physics*, 169(2):303–315, 2017.
- [26] Sarah Marzen. Intrinsic computation of a monod–wyman–changeux molecule. *Entropy*, 20(8):599, 2018.
- [27] Paul M. Riechers and James P. Crutchfield. Spectral simplicity of apparent complexity, part i: The nondiagonalizable metadynamics of prediction. *Chaos*, 28:033115, 2018.
- [28] Paul M. Riechers and James P. Crutchfield. Spectral simplicity of apparent complexity, part ii: Exact complexities and complexity spectra. *Chaos*, 28:033116, 2018.
- [29] Samuel P. Loomis and James P. Crutchfield. Strong and weak optimizations in classical and quantum models of stochastic processes. *Journal of Statistical Physics*, 176(6):1317–1342, 2019.
- [30] A. Connes and C. Rovelli. Von neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories. *Classical and Quantum Gravity*, 11:2899–2917, 1994.
- [31] A. L. Carey, J. Phillips, and A. Rennie. Twisted cyclic theory and an index theory for the gauge invariant KMS state on the Cuntz algebra $\mathcal{O}_n$. *Journal of K-Theory*, 6(2):339–380, 2010.
- [32] A. L. Carey, S. Neshveyev, R. Nest, and A. Rennie. Twisted cyclic theory, equivariant KK-theory and KMS states. *Journal für die reine und angewandte Mathematik (Crelle’s Journal)*, 650:161–191, 2011.
- [33] N. S. Mankoč Borštnik and H. B. Nielsen. Why odd-space and odd-time dimensions in even-dimensional spaces? *Physics Letters B*, 486:314–321, 2000.

- [34] H. van Dam and Y. Jack Ng. Why 3+1 metric rather than 4+0 or 2+2? *Physics Letters B*, 520:159–162, 2001.
- [35] Holger Bech Nielsen and Svend Erik Rugh. Why do we live in 3+1 dimensions?, 1994.
- [36] T. Jacobson. Thermodynamics of spacetime: the einstein equation of state. *Physical Review Letters*, 75:1260–1263, 1995.
- [37] E. Verlinde. On the origin of gravity and the laws of newton. *Journal of High Energy Physics*, 2011(4):029, 2011.
- [38] M. Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42:2323–2329, 2010.
- [39] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59:521–524, 1987.
- [40] S. Surya. The causal set approach to quantum gravity. *Living Reviews in Relativity*, 22:5, 2019.
- [41] T. Konopka, F. Markopoulou, and S. Severini. Quantum graphity: a model of emergent locality. *Physical Review D*, 77:104029, 2008.
- [42] J. Ambjørn, J. Jurkiewicz, and R. Loll. Quantum gravity, or the art of building spacetime. In D. Oriti, editor, *Approaches to Quantum Gravity*, pages 341–359. Cambridge University Press, 2009.
- [43] D. Oriti, editor. *Approaches to Quantum Gravity: Toward a New Understanding of Space, Time and Matter*. Cambridge University Press, 2009.
- [44] A. H. Chamseddine and A. Connes. The spectral action principle. *Communications in Mathematical Physics*, 186:731–750, 1997.
- [45] R. P. Stanley. *Enumerative Combinatorics, Vol. 1*. Cambridge University Press, 2011.
- [46] A. Jaffe, A. Lesniewski, and K. Osterwalder. Quantum K -theory. I. The Chern character. *Communications in Mathematical Physics*, 118(1):1–14, 1988.
- [47] J. Cuntz. Simple C^* -algebras generated by isometries. *Communications in Mathematical Physics*, 57:173–185, 1977.
- [48] D. Pask and A. Rennie. The noncommutative geometry of graph C^* -algebras I: The index theorem. *Journal of Functional Analysis*, 233(1):92–134, 2006.
- [49] A. L. Carey, M. Marcolli, and A. Rennie. Modular index invariants of Mumford curves. In *Noncommutative Geometry, Arithmetic, and Related Topics*, pages 31–73. Johns Hopkins University Press, 2011. arXiv:0905.3157.
- [50] A. Connes. On the spectral characterization of manifolds. *Journal of Noncommutative Geometry*, 7:1–82, 2013.
- [51] M. Reed and B. Simon. *Methods of Modern Mathematical Physics I: Functional Analysis*. Academic Press, revised and enlarged edition, 1980.

- [52] H. Cornean, H. Garde, and A. Jensen. Discrete approximations to dirac operators and norm resolvent convergence. *Journal of Spectral Theory*, 12:1201–1244, 2022.
- [53] K. M. Schmidt and T. Umeda. Continuum limits for discrete dirac operators on two-dimensional square lattices. *Analysis and Mathematical Physics*, 13:31, 2023.
- [54] K. G. Wilson. Quarks and strings on a lattice. In A. Zichichi, editor, *New Phenomena in Subnuclear Physics*, pages 69–142. Plenum, New York, 1977. Erice, 1975.
- [55] H. B. Nielsen and M. Ninomiya. Absence of neutrinos on a lattice. i. proof by homotopy theory. *Nuclear Physics B*, 185:20–40, 1981. Erratum: Nuclear Physics B 195 (1982), 541.
- [56] H. B. Nielsen and M. Ninomiya. Absence of neutrinos on a lattice. ii. intuitive topological proof. *Nuclear Physics B*, 193:173–194, 1981.
- [57] P. R. Chernoff. Essential self-adjointness of powers of generators of hyperbolic equations. *Journal of Functional Analysis*, 12:401–414, 1973.
- [58] Alexandra M. Jurgens and James P. Crutchfield. Divergent predictive states: The statistical complexity dimension of stationary, ergodic hidden markov processes. *Chaos*, 31(8):083114, 2021.
- [59] J. E. Hutchinson. Fractals and self similarity. *Indiana University Mathematics Journal*, 30:713–747, 1981.
- [60] K. Falconer. *Fractal Geometry: Mathematical Foundations and Applications*. Wiley, 1990.
- [61] K. Falconer. *Techniques in Fractal Geometry*. Wiley, 1997.
- [62] R. Bowen. *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms*, volume 470 of *Lecture Notes in Mathematics*. Springer, 1975.
- [63] Nicholas H. Bingham, Charles M. Goldie, and Jozef L. Teugels. *Regular Variation*, volume 27 of *Encyclopedia of Mathematics and Its Applications*. Cambridge University Press, Cambridge, 1987.
- [64] Adriano M. Garsia and John Lamperti. A discrete renewal theorem with infinite mean. *Commentarii Mathematici Helvetici*, 37:221–234, 1962.
- [65] K. Bruce Erickson. Strong renewal theorems with infinite mean. *Transactions of the American Mathematical Society*, 151:263–291, 1970.
- [66] Ronald A. Doney. One-sided local large deviation and renewal theorems in the case of infinite mean. *Probability Theory and Related Fields*, 107(4):451–465, 1997.
- [67] Francesco Caravenna and Ron Doney. Local large deviations and the strong renewal theorem. *Electronic Journal of Probability*, 24:1–48, 2019. Paper No. 72.
- [68] Yves Pomeau and Paul Manneville. Intermittent transition to turbulence in dissipative dynamical systems. *Communications in Mathematical Physics*, 74(2):189–197, 1980.
- [69] Jon Aaronson. *An Introduction to Infinite Ergodic Theory*, volume 50 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 1997.

- [70] Omri M. Sarig. Phase transitions for countable markov shifts. *Communications in Mathematical Physics*, 217(3):555–577, 2001.
- [71] W. Bruns and J. Gubeladze. *Polytopes, Rings, and K-Theory*. Springer, 2009.
- [72] M. Beck and S. Robins. *Computing the Continuous Discretely: Integer-Point Enumeration in Polyhedra*. Undergraduate Texts in Mathematics. Springer, 2007.
- [73] Jan Myrheim. Statistical geometry. Technical Report CERN-TH-2538, CERN, 1978.
- [74] G. Brightwell and R. Gregory. Structure of random discrete spacetime. *Physical Review Letters*, 66:260–263, 1991.
- [75] G. Karpilovsky. *Projective Representations of Finite Groups*. Marcel Dekker, 1985.
- [76] George W. Mackey. *Unitary Group Representations in Physics, Probability, and Number Theory*. Number 55 in Mathematics Lecture Note Series. Benjamin/Cummings Publishing Company, Reading, MA, 1978.
- [77] Thierry Delmotte. Parabolic harnack inequality and estimates of markov chains on graphs. *Revista Matemática Iberoamericana*, 15(1):181–232, 1999.
- [78] Michael F. Atiyah, Raoul Bott, and Arnold Shapiro. Clifford modules. *Topology*, 3:3–38, July 1964.
- [79] H. B. Lawson and M.-L. Michelsohn. *Spin Geometry*. Princeton University Press, 1989.
- [80] John W. Barrett. A lorentzian version of the non-commutative geometry of the standard model of particle physics. *Journal of Mathematical Physics*, 48:012303, 2007.
- [81] Arkadiusz Bochniak and Andrzej Sitarz. Finite pseudo-riemannian spectral triples and the standard model. *Physical Review D*, 97:115029, 2018.
- [82] Arkadiusz Bochniak, Andrzej Sitarz, and Paweł Zalecki. Spectral geometry for the standard model without fermion doubling. *Physical Review D*, 101:075038, 2020.
- [83] Fabien Besnard and Christian Brouder. Noncommutative geometry, the lorentzian standard model and its B-L extension. *Physical Review D*, 103:035003, 2021.

Layer	Result in this paper	Status
Positive predictive geometry	The predictive quotient carries a positive spectral triple, zeta dimension, residue volume, metric predictive dimension, predictive order, and renewal resolvent.	Proved under finite-fibre, bounded-increment, and regular-growth hypotheses.
Positivity boundary	The completely positive sector is intrinsically Hilbert-positive and cannot functorially supply a nontrivial Lorentzian fundamental symmetry.	Proved as a structural obstruction.
Signed/Krein enrichment	The minimal renewal-positive indefinite enrichment over a connected recurrent component is a principal $\mathbb{Z}/2$ -cover; its gauge classes lie in $H^1(G, \mathbb{Z}/2)$, and the sheet label gives the Krein fundamental symmetry.	Proved in the renewal-positive edge-local class; label-local variants may restrict to H_{loc}^1 .
Modular signed Dirac	The signed cover and renewal clock determine the modular signed operator $D_- = J_\chi e^{\beta N} + B$, giving a twisted Krein spectral triple under the bounded twisted-commutator hypotheses.	Proved; nontrivial differential content is carried by the bounded renewal correction B .
Flat continuum and self-averaging	Nondegenerate Clifford coarse-graining gives a Lorentzian characteristic quadric. Under stationary ergodic reset statistics with $M_* \succ 0$, the flat endpoint is deterministic and unitarily equivalent to the standard Minkowski massless Dirac Hamiltonian.	Proved at symbol level and for the flat Hamiltonian; no rotational-isotropy hypothesis is required.
Adiabatic and curved endpoint	In the slowly varying regime, channel plaquette composition separates translational torsion from rotational curvature; second-order predictive equivalence gives the discrete Levi–Civita transport, and the covariant Hamiltonians converge through subprincipal order, up to a separated bounded real scalar renewal potential.	Proved under uniform nondegeneracy and regularity assumptions.
Propagation and marked tori	The same second-moment field controls frozen symbols, optical distance, and macroscopic domains of dependence. Flat marked tori with marked spin structure admit integer-stencil signed-renewal realisations with fixed-band norm-resolvent convergence.	Proved in the stated flat/adiabatic regimes; curved periodic tori and general spin manifolds remain open.
Dimension and primitive interface	Full-dimensional affine reachability with finite primitive return completion gives $q_{\text{alg}} = 1 + b_1^{\text{eff}}$. Primitive stabilised $\mathbb{F}_2$ -commutator data generate the Clifford revision factor and make $3 + 1$ the minimal spatially nondegenerate Lorentzian Dirac endpoint.	Minimality proved in the primitive class; uniqueness among higher odd endpoints requires the stated oriented-circulation or volume-response closure.

Table 1: Summary of the theorem chain and logical status of each layer.